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**Report prepared for the WHO annual consultation on
the composition of influenza vaccine for the
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Influenza Epidemiology in the WHO European Region

The epidemiological section of this report was prepared using data reported to the European Surveillance System (TESSy) and presented in Flu News Europe (<http://www.flunewseurope.org>) the joint ECDC-WHO Europe weekly influenza update, with the assistance of members of the Influenza and Other Respiratory Pathogens Group at the WHO European Office and the ECDC Influenza Surveillance Group:

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Epidemiology

Influenza activity in the WHO European Region, October 2019 – August 2020

Note

The novel coronavirus disease 2019 (**COVID-19**) pandemic, caused by severe acute respiratory syndrome coronavirus 2 (**SARS-CoV-2**), adversely affected healthcare presentations and testing capacities of Member States and areas in the Region. This resulted in negative impacts on influenza surveillance/detection and reporting of influenza epidemiologic and virologic data. As such, the data presented here must be interpreted with caution.

Timing and spread

The 2019-2020 influenza season (weeks 40/2019 to 20/2020) in the WHO European Region is considered to have started in week 47/2019, one week later than the 2016-2017 season, and one to four weeks earlier than those of 2015-2016, 2017-2018 and 2018-2019 (**Figure E1-A**). This was the first week when more than 10% of ILI and/or ARI specimens from sentinel primary care sites in the Region tested positive for influenza viruses (**Figure E1-B**). The proportion of specimens testing positive for influenza on a weekly basis reached a peak for the season of 55% in week 05/2020, which is comparable to peaks of 53-60% in the previous four seasons. High activity, in which $\geq 40\%$ of sentinel specimens tested positive for influenza, lasted for 10 weeks, compared to an average of 10.5 weeks (range: 9-14 weeks) for the previous four seasons, before falling below 10% in week 13/2019 (**Figure E1-A**). Based on weekly reporting on the intensity of influenza activity by each of the Member States and areas, activity first increased in the west of the WHO European Region – notably in Ireland and the United Kingdom (Northern Ireland, Scotland, and Wales) – as well as Georgia and Latvia in the eastern part of the Region (**Figure E2**).

Distribution of viruses

Between week 40/2019 and week 20/2020, nearly 163,000 influenza detections from sentinel and non-sentinel primary care sources were reported to The European Surveillance System (TESSy) (**Table E1-A**), a significant decrease (of approximately 42,000) compared to the 2018-2019 season despite a reported increase in the number of specimens tested (of approximately 64,000) in the 2019-2020 season. Combined data from sentinel and non-sentinel sources showed that most viruses detected (73%) were type A and 27% were type B. This ratio was generally consistent for Southern Europe, Western Europe, Northern Europe and Western Asia, but with lower proportions of type A virus detections reported in Eastern Europe (62%) and Central Asia (52%) (**Figure E1-A**). Combined sentinel and non-sentinel data for type A viruses showed a slight majority (56%) to be A(H1N1)pdm09 viruses, with 44% being A(H3N2) viruses. However, there was much greater variation in these proportions when stratified by sub-region. Majorities of A(H1N1)pdm09 viruses, ranging from 63% to 97%, occurred in Southern Europe, Western Europe, Eastern Europe, and West Asia. In contrast, A(H3N2) viruses predominated in Northern Europe (72%) and in Central Asia (52%) (**Figure E1-A**). Only 10% of type B viruses reported were ascribed to a lineage, primarily

from Southern, Western and Northern Europe, of which 97-100% were B/Victoria and up to 3% were B/Yamagata.

In general, the regional distribution of influenza by type was not consistent when analysis was restricted to sentinel source data only. From sentinel sources there were lower proportions of type A viruses in Southern Europe (56% vs. 70%), Northern Europe (70% vs. 80%) and Central Asia (37% vs 52%), compared to combined data from sentinel and non-sentinel sources. In Eastern Europe, the opposite trend was observed, and there was a higher proportion of type A viruses (81% vs. 62%) in sentinel sources compared to the combined data. The regional distribution of influenza by subtype was more consistent between sentinel sources and combined sentinel and non-sentinel data, although there was a higher proportion of A(H3N2) viruses in Western Europe (42% vs. 34%), Eastern Europe (44% vs. 25%) and Central Asia (80% vs. 52%) compared to combined data from sentinel and non-sentinel sources (**Table E1-B**).

Of influenza B virus detections from sentinel source data, the majority (98-100%) were of the B/Victoria-lineage. The subtype distribution proportions for influenza type A viruses detected, from both sentinel and non-sentinel sources, were relatively consistent for most weeks in the season (**Figure E1-B/C**).

A dominant virus (sub)type (comprising at least 60% of influenza virus detections) could be assigned by 45 Member States and areas within the WHO European Region, with only five having insufficient numbers (ten or fewer) of detections. Twenty-five Member States and areas reported predominance of influenza type A virus, while eight reported predominance of influenza type B viruses, and 12 reported co-dominance of type A and type B viruses (**Figure E3**). Two reported dominance of type A without specifying a subtype, and 23 were able to ascribe dominance (the proportion of detections of an influenza subtype was 60% or greater) or co-dominance (between 41% and 59%) of influenza A virus subtypes for the season: 11 reported A(H1N1)pdm09 dominance, five reported A(H3N2) dominance, and seven reported co-dominance of A(H1N1)pdm09 and A(H3N2) viruses. Eight Member States or areas reported dominance of type B without specifying a lineage, but very few B/Yamagata-lineage viruses were detected, as was seen worldwide.

Since the end of the 2019-2020 influenza season (weeks 21-34/2019), a total of 24 influenza detections from non-sentinel sources across the Region have been reported. Of these, 13 were type A viruses and 11 were type B. Of the three type A viruses subtyped, one was A(H1N1)pdm09 and two were A(H3N2). None of the type B viruses detected were ascribed to a lineage. Over the same post-season period, there was just one detection of an influenza virus from sentinel sources; it was influenza type B and not ascribed to a lineage.

Influenza virus characterisation

Based on HA gene sequencing 2,707 influenza viruses, collected between weeks 40/2019 to 20/2020, were ascribed to genetic clades, as defined at the start of the 2019-2020 season, and reported to TESSy:

- Among 982 A(H1N1)pdm09 viruses characterised, 904 (92%) fell in the A/Norway/3433/2018 clade (6B.1A5A), 41 (4%) in the A/ Switzerland/3330/2018 clade (6B.1A5B), 19 (2%) in the A/Slovenia/1489/2019 clade (6B.1A7) and 11 (1%) in the A/Brisbane/02/2018 clade (6B.1A1);
- Among 1,048 A(H3N2) viruses attributed to a clade, 487 (46%) fell in the 3C.2a1b group [subgroups T131K (n = 342: 33%), T135K-A (n = 64: 6%) and T135K-B (n = 81: 8%)] and 560 (54%) in clade 3C.3a;

- 649 were B/Victoria-lineage viruses (represented by B/Brisbane/60/2008, clade 1A), with 19 (3%) of these falling in a subclade of viruses with a two amino acid deletion in HA1 (1A(Δ 2); represented by the vaccine component B/Colorado/06/2017) and 630 (97%) falling in a subclade of viruses with a three amino acid deletion and K136E substitution in HA1 (1A(Δ 3)B; represented by B/Washington/02/2019, a vaccine component for the southern hemisphere 2020 season);
- 26 (93%) of 28 influenza B/Yamagata-lineage viruses belonged to clade 3, represented by the vaccine component B/Phuket/3073/2013.

Neuraminidase inhibitor susceptibility was assessed for 2,292 influenza viruses with collection dates since week 40/2019: 942 A(H1N1)pdm09, 794 A(H3N2), and 556 type B viruses. Very few viruses (<1%) showed any reduction in susceptibility. Three A(H1N1)pdm09 viruses carried amino acid substitution H275Y in NA, with one of the three also carrying an N295S substitution, both of which are indicative of highly reduced inhibition (HRI) by oseltamivir. An additional two A(H1N1)pdm09 viruses showed reduced inhibition (RI) by oseltamivir in the sialidase inhibition (phenotypic) assay. One of these viruses also showed RI by zanamivir in the sialidase inhibition assay. Further, one A(H3N2) virus carried an R292K substitution in NA, indicative of HRI by oseltamivir and RI by zanamivir, which was confirmed by phenotypic assay, and 1 B/Victoria-lineage virus showed evidence of HRI by oseltamivir and RI by zanamivir in the sialidase inhibition assay.

Hospitalised cases

Between week 40/2019 and week 20/2020, 11 Member States reported a total of 11,251 hospitalised laboratory-confirmed influenza cases. Of these, 4,016 cases were admitted to intensive care units (ICUs) with three Member States accounting for over 85% of the cases: United Kingdom (England, 45%), France (22%), and Spain (19%). The remaining 7,235 cases were reported from other hospital wards with two Member States accounting for 90% of these cases; Ireland (58%) and Spain (32%). Among cases admitted to ICUs, the majority (90%) of influenza viruses detected were type A, and of the 1,273 viruses subtyped, 59% were A(H1N1)pdm09 and 41% were A(H3N2). More than one third (37%) of the 2,201 influenza-associated ICU admissions with a recorded age were in patients aged 65 years or older, and 29% were 15-64 years of age (**Figure E4-A**).

Eighteen Member States and areas (Albania, Armenia, Azerbaijan, Belarus, Bosnia and Herzegovina, Georgia, Kazakhstan, Kosovo*, Kyrgyzstan, Montenegro, North Macedonia, Republic of Moldova, Russian Federation, Serbia, Tajikistan, Turkmenistan, Ukraine and Uzbekistan) with surveillance systems for severe acute respiratory infection (SARI) reported a total of 38,914 cases between week 40/2019 and week 20/2020. In eight of the Member States and areas, more than half of the SARI cases were among children aged 0-4 years, while in five Member States or areas, more than half of the SARI cases were aged 15-64 years (**Figure E4-B**). Of 9,743 SARI cases tested for influenza virus over the period, 2,917 (30%) were positive. Approximately half (56%) were infected with influenza type A viruses and, of the 1,376 subtyped viruses, 61% were A(H1N1)pdm09 and 39% were A(H3N2). Of the influenza positive cases infected with type B viruses, 682 were ascribed to a lineage, and 99% of these were B/Victoria-lineage. Information on distributions of influenza types and subtypes or lineages by age group are not available for these data.

* in accordance with. United Nations Security Council. Resolution 1244 (1999).

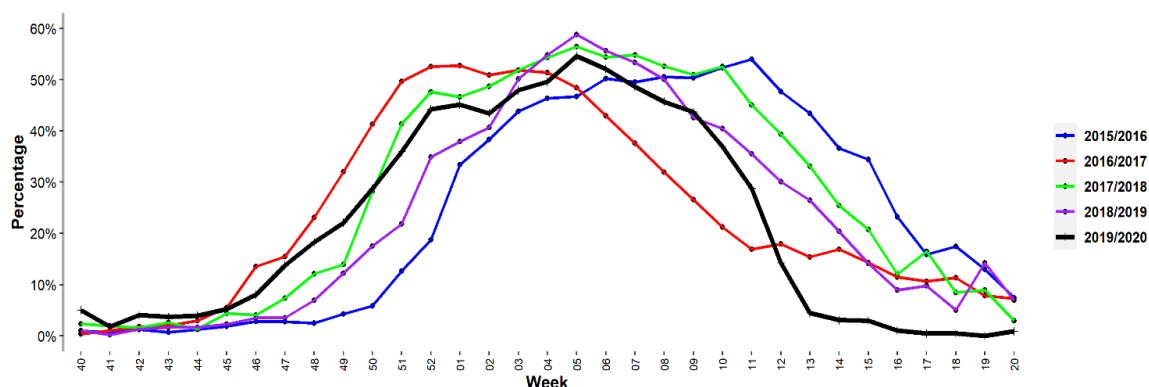
Mortality data

Pooled estimates of all-cause excess mortality from up to 24 European countries or sub-national regions participating in the EuroMOMO network for the 2019/20 season [Austria, Belgium, Berlin (Germany), Denmark, England (United Kingdom), Estonia, Finland, France, Greece, Hesse (Germany), Hungary, Ireland, Italy, Luxembourg, Malta, the Netherlands, Northern Ireland (United Kingdom), Norway, Portugal, Scotland (United Kingdom), Spain, Sweden, Switzerland and Wales (United Kingdom)] showed a period of substantial excess mortality, observed in some countries, beginning in week 10/2020.

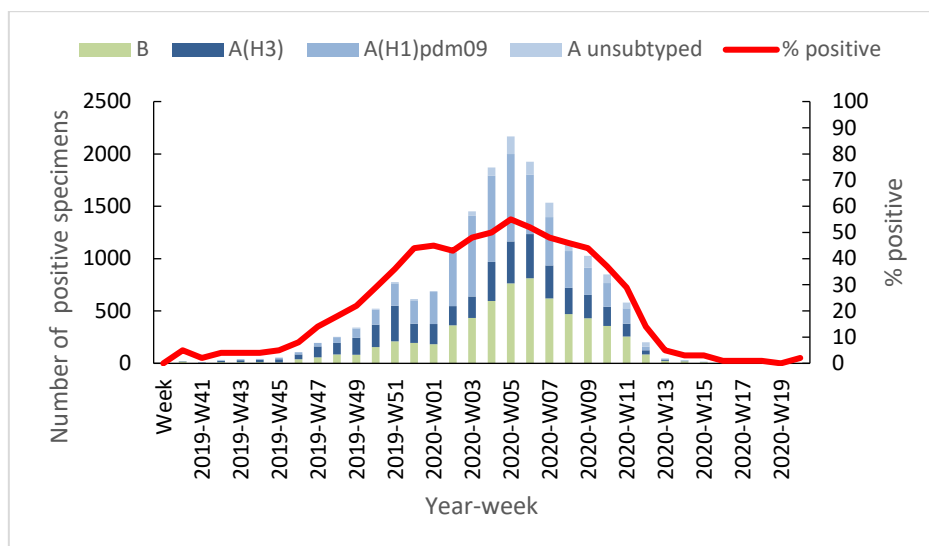
For the EuroMOMO network, from week 10 through week 20 of 2020, there were over 159,000 excess deaths estimated in total and, of these, 143,000 were in the age group ≥ 65 years and 13,000 in the 15-64 years age group. These weeks cover the later part of the influenza season as well as the start of the COVID-19 pandemic which is ongoing. A significant majority of these deaths are probably related to COVID-19 as excess mortality was low when influenza circulation was at its highest.

Figure E1. Seasonal (weeks 40–20) influenza activity in the WHO European Region

A. Percentage of sentinel ILI and ARI specimens testing positive for influenza by season and week, 2015-2020 (five seasons)



B. Sentinel ILI and ARI specimens testing positive for influenza by week, (sub)type and percentage positive, 2019–2020 season, weeks 40 through 20



C. Non-sentinel ILI and ARI specimens testing positive for influenza by week and (sub)type, 2019–2020 season, weeks 40 through 20

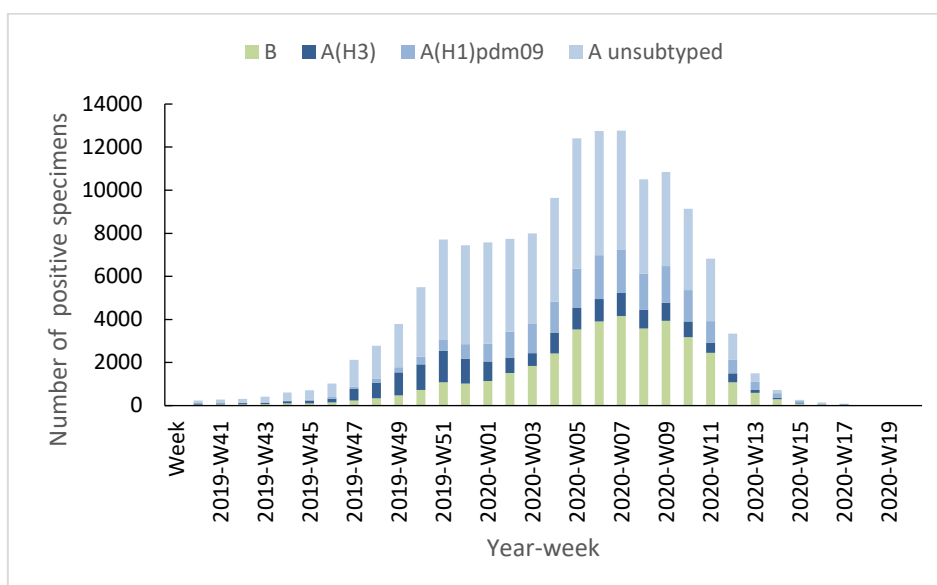
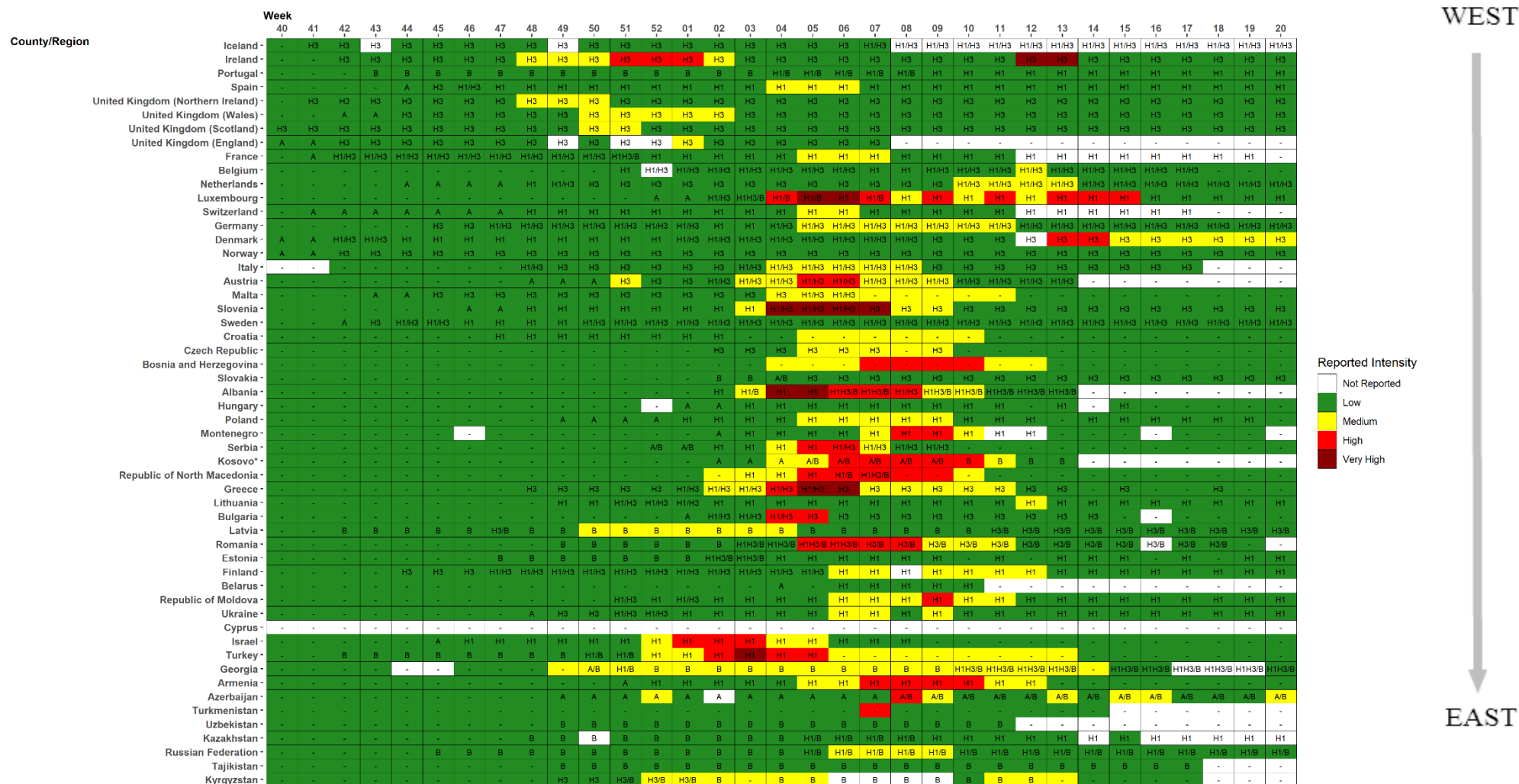


Figure E2. Intensity of influenza activity and dominant virus (sub)type in the WHO European Region by week and Member State or area, 2019–2020 season (weeks 40 through 20)

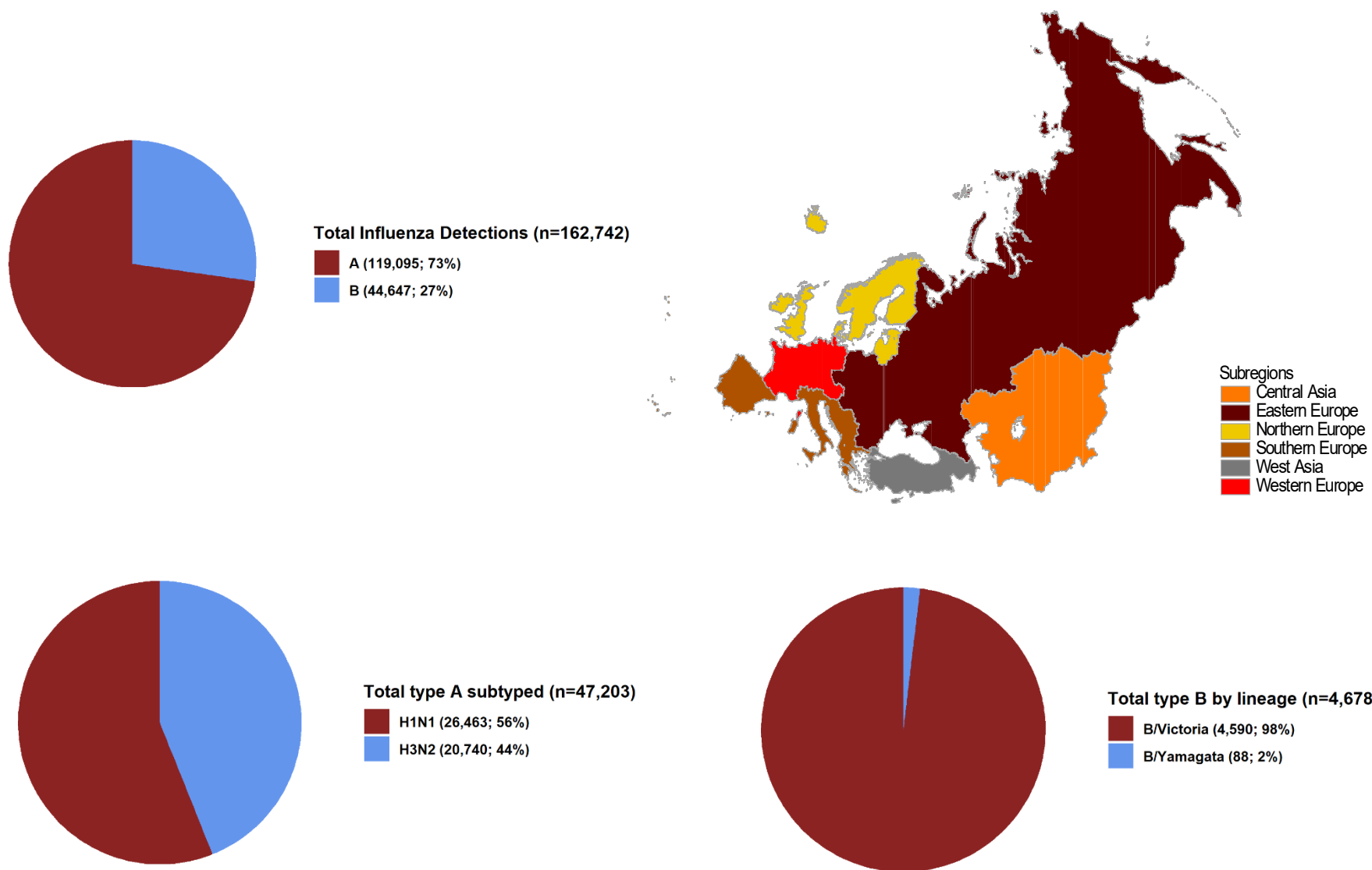


* In accordance with. United Nations Security Council. Resolution 1244 (1999).

Low intensity influenza activity is the level at which clinical influenza activity remains throughout the summer and most of the winter.
For a virus (sub)type to be defined as dominant it must account for as at least 60% of detections.

Table E1. Distribution of Influenza Detections in the WHO European Region², 2019–2020 season (weeks 40/2019–06/2020)

Laboratory confirmed influenza from sentinel and non-sentinel sources by type and subtype/lineage



² WHO European Region Member States grouped into sub-regions according to United Nations geographic composition. Available from: <http://unstats.un.org/UNSD/METHODS/M49/M49REGIN.HTM>

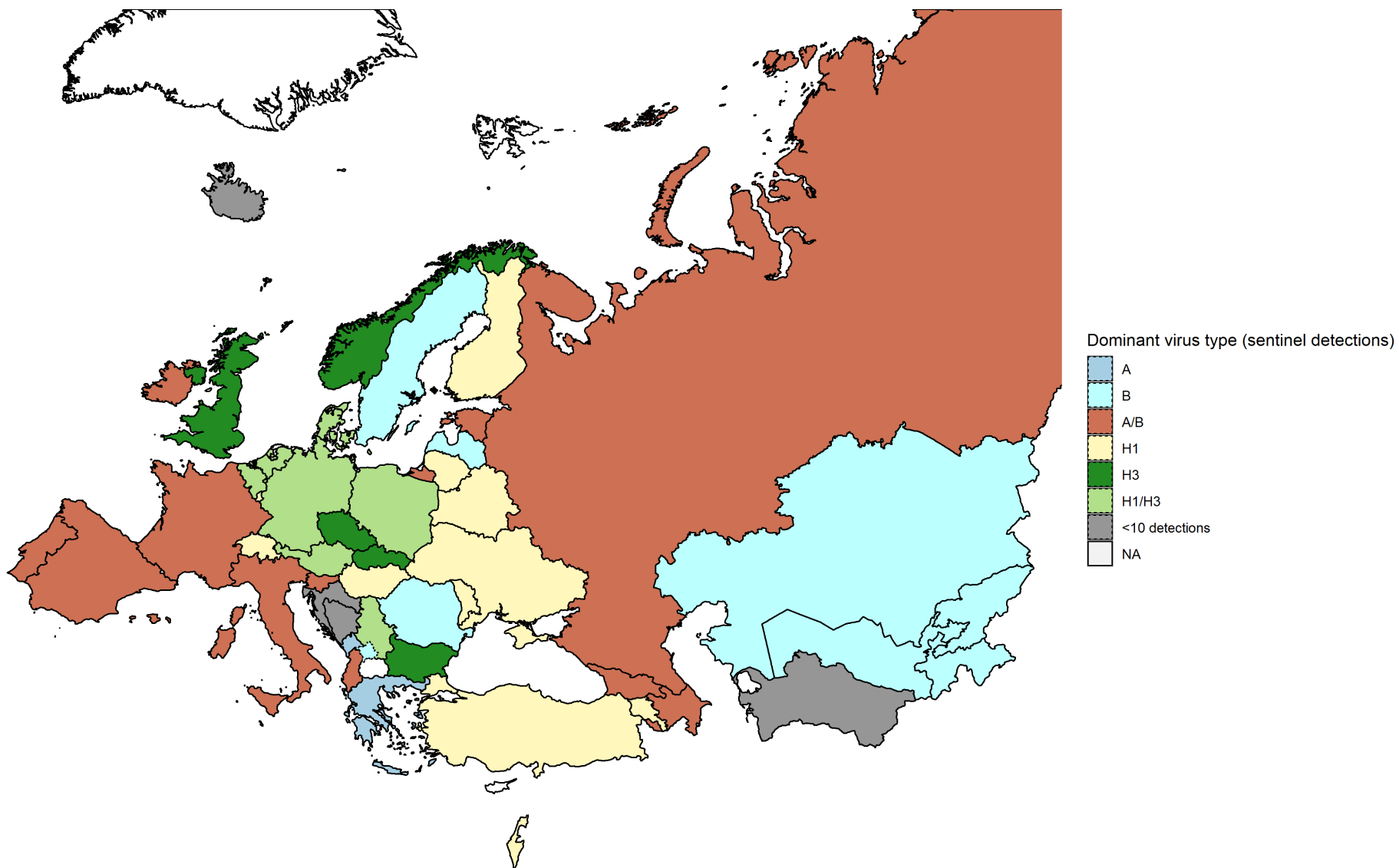
A. Laboratory confirmed influenza from sentinel and non-sentinel sources by geographic sub-region and influenza type and subtype/lineage

Influenza type and subtype/lineage	Southern Europe	Western Europe	Eastern Europe	Northern Europe	West Asia	Central Asia	WHO European Region
Influenza A	17,930 (70%)	28,582 (71%)	14,791 (62%)	54,030 (80%)	3,171 (67%)	591 (52%)	119,095 (73%)
Influenza A subtyped	9,087	6,315	11,701	16,533	3,002	565	47,203
A(H1N1)pdm09	5,735 (63%)	4,174 (66%)	8,795 (75%)	4,588 (28%)	2,901 (97%)	270 (48%)	26,463 (56%)
A(H3N2)	3,352 (37%)	2,141 (34%)	2,906 (25%)	11,945 (72%)	101 (3%)	295 (52%)	20,740 (44%)
Influenza B	7,866 (30%)	11,804 (29%)	9,056 (38%)	13,781 (20%)	1,596 (33%)	544 (48%)	44,647 (27%)
Influenza B lineage determined	1,371	992	287	1,870	153	0	4,678
B/Yamagata-lineage	4 (<1%)	16 (2%)	6 (2%)	62 (3%)	0 (0%)	0(0%)	88 (2%)
B/Victoria-lineage	1,368 (>99%)	976 (98%)	281 (98%)	1,808 (97%)	153 (100%)	4(100%)	4,590 (98%)
Total	25,796	40,386	23,847	67,811	4,767	1,135	162,742

B. Laboratory confirmed influenza from sentinel site ILI and ARI specimens by geographic sub-region and influenza type and subtype/lineage

Influenza type and subtype/lineage	Southern Europe	Western Europe	Eastern Europe	Northern Europe	West Asia	Central Asia	WHO European Region
Influenza A	3,113 (56%)	3,053 (69%)	1,241 (81%)	2,018 (70%)	1,723 (63%)	198 (37%)	11,346 (64%)
Influenza A subtyped	2664	2,973	865	1,972	1,691	184	10,349
A(H1N1)pdm09	1,741 (65%)	1,721 (58%)	483 (56%)	565 (29%)	1,603 (95%)	37 (20%)	6,150 (59%)
A(H3N2)	923 (35%)	1,252 (42%)	382 (44%)	1,407 (71%)	88 (5%)	147 (80%)	4,199 (41%)
Influenza B	2,402 (44%)	1,350 (31%)	413 (19%)	857 (30%)	1,031 (37%)	331 (63%)	6,348 (36%)
Influenza B lineage determined	857	977	119	440	135	4	2,532
B/Yamagata-lineage	1 (<1%)	15 (2%)	0 (0%)	7 (2%)	0 (0%)	0 (0%)	23 (1%)
B/Victoria-lineage	856 (>99%)	962 (98%)	119 (100%)	433 (98%)	135 (100%)	4 (100%)	2,509 (99%)
Total	5,515	4,403	1,654	2,875	2,754	529	17,730

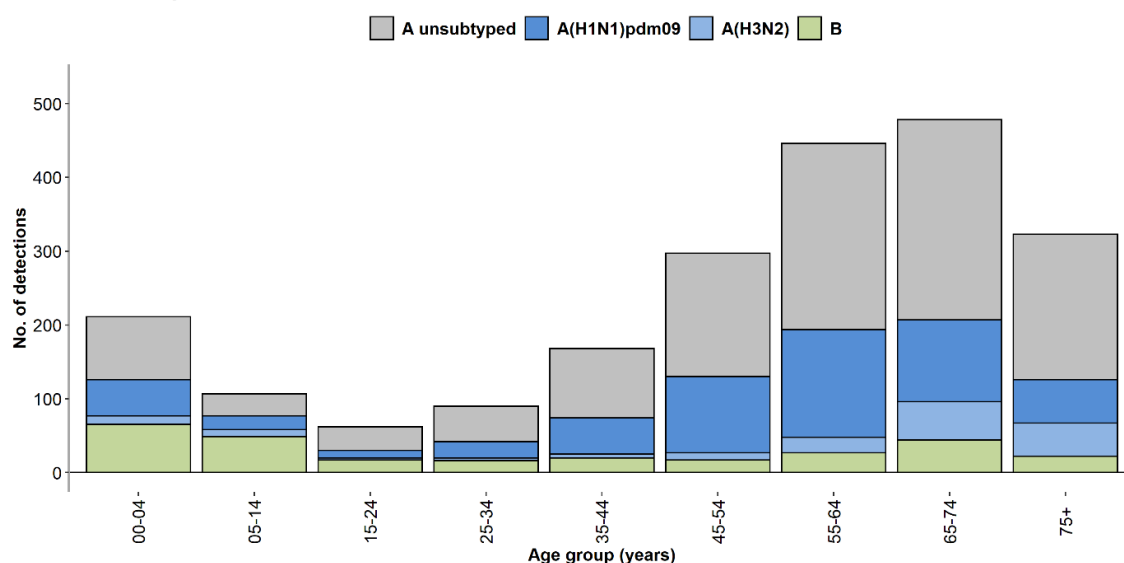
Figure E3. Influenza circulation³ in the WHO European Region by (sub)type and Member State or area, 2019–2020 season (weeks 40–20)



³ Dominance of (sub)type is based on at least 60% of detections and co-dominance is based on between 40% and 60% of detections. Calculations are only considered valid if at least 10 viruses were detected.

Figure E4. Hospitalised influenza cases in the WHO European Region, 2019–2020 season (weeks 40–20)

A. Distribution of virus (sub)type in laboratory-confirmed hospitalised cases in intensive care units^a by age group



^a Five reporting Member States account for 94% of the reported cases: France (40%), Spain (34%), Sweden (8%), and Ireland (9%) and Czechia (7%).

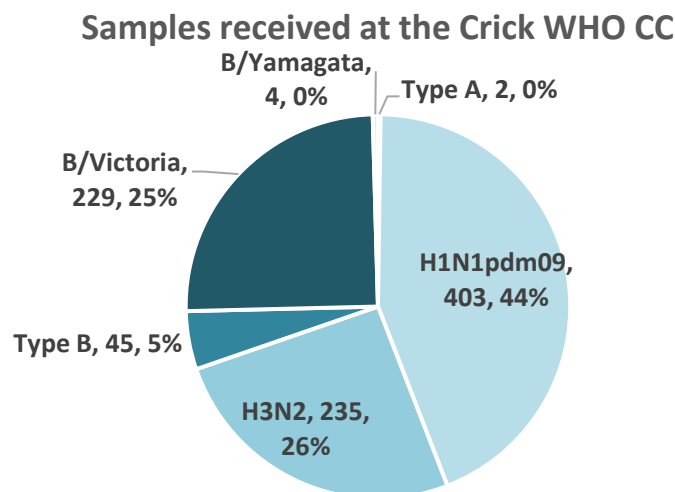
B. Percentage distribution of severe acute respiratory infection (SARI) cases by age group and Member State or area, and influenza positives, 2019-2020 season (weeks 40-20)

Member State or area	SARI cases with age data	Percentage of total cases					All-age % positive
		0-4 years	5-14 years	15-29 years	30-64 years	65+ years	
Albania	1274	12	10	21	35	22	31
Armenia	3934	62	22	7	6	2	30
Azerbaijan	904	1	0	36	52	11	0
Belarus	352	24	8	18	39	10	2
Bosnia and Herzegovina	571	13	11	11	39	27	37
Georgia	2777	57	22	8	6	6	40
Kazakhstan	15603	55	16	14	11	4	38
Kyrgyzstan	2739	80	11	5	3	1	44
Moldova, Republic of	2320	69	8	4	10	9	18
Montenegro	32	47	6	9	22	16	13
North Macedonia	709	46	12	9	20	12	43
Russian Federation	3139	23	11	24	28	14	15
Serbia	981	13	7	11	37	32	37
Tajikistan	346	59	38	0	0	3	22
Turkmenistan	155	70	25	1	0	4	2
Ukraine	2318	16	14	35	30	6	20
Uzbekistan	265	58	22	14	6	0	16
Kosovo ^b	139	27	22	8	30	13	15

^b In accordance with. United Nations Security Council. Resolution 1244 (1999).

Influenza specimens received at the Francis Crick Institute.

Seasonal influenza virus samples with collection dates after 2020-01-31, totalling 918, have been received from 34 countries in Europe, Africa, the Middle East and Central Asia. Influenza A viruses represented 70% of the samples, of which A(H1N1)pdm09 and A(H3N2) viruses were received in a ratio of ~2:1 (63% v 37%). Influenza B viruses represented 30% of the samples received with B/Victoria lineage viruses outnumbering those of the B/Yamagata lineage at a ratio of >60:1 (**Table 1**). Receipts are summarised in the pie chart below.



Of the influenza-positive samples derived from individual patients (n = 918) received, 331 (36%) were virus (V) isolates, 471 (51%) were clinical specimens (CS) and 116 (13%) were sent as CS/V pairs.

Table 2 shows the recovery/isolation rates from samples received with collection dates after 2020-01-31. Generally, viruses were recovered efficiently. For samples sent as clinical specimens, that were taken for culture, virus recoveries of 73%, 65% and 80% were achieved for influenza type B, A(H1N1pdm09) and A(H3N2) viruses, respectively. Recoveries of samples sent as viruses were $\geq 95\%$. For A(H3N2) viruses, 98 of 116 (~85%) could be assessed by HI assay, representing a marked increase over recent years, the other viruses (12/116, 10%) had low HA titres (<4 HA units in the presence of 20nM oseltamivir) or were unable to agglutinate red blood cells.

Table 3 summarises the reactivity of A(H1N1)pdm09, A(H3N2) and viruses of the B/Victoria and B/Yamagata lineages with post-infection ferret antisera raised against vaccine viruses or reference viruses in HI assays. For A(H3N2) viruses, the reactivity of test viruses in plaque reduction neutralisation assays (PRNA) is also summarised. Numbers and percentages of viruses showing ≤ 2 -, 4- and ≥ 8 -fold reductions in titres relative to the titres of the antisera with their respective homologous reference/vaccine viruses are shown.

For A(H1N1)pdm09 test viruses the HI titres obtained with antisera raised against egg-propagated vaccine viruses A/Michigan/45/2015, A/Brisbane/02/2018 and A/Guangdong-

Maonan/SWL1536/2019, and cell culture-propagated A/Denmark/3280/2019, a virus carrying an N156K substitution in HA1, were compared to the respective homologous titres.

For A(H3N2) test viruses the HI titres of antisera raised against egg-propagated vaccine viruses A/South Australia/34/2019 (3C.2a1b+T131K subgroup), A/Hong Kong/2671/2019 (3C.2a1b+T135K-B subgroup) and A/Kansas/14/2017 (clade 3C.3a), and the cell culture-propagated cultivars of A/Hong Kong/2671/2019 and A/Kansas/14/2017 were compared to the titres of the antisera with their homologous viruses. For the A(H3N2) test viruses assessed by PRNA the 50% neutralising titres of antisera raised against tissue culture-propagated cultivars of A/Norway/3275/2018 (3C.2a1b+T131K), A/Denmark/3264/2019 (3C.2a1b+T135K-A, with the additional substitutions A138S, G186D, D190N, F193S and S198P) and A/Kansas/14/2017 (clade 3C.3a), egg-propagated A/South Australia/34/2019 (3C.2a1b+T131K) and A/Paris/2554/2029 (3C.2a1b+T135K-A with the substitutions A138S, G186D, D190N, F193S and S198P), and both tissue culture- and egg-propagated cultivars of A/Hong Kong/2671/2019 (subclade 3C.2a1b+T135K-B) were compared to the titres of the antisera with their homologous viruses.

For Influenza B viruses of the B/Victoria lineage the HI titres of test viruses with antisera raised against both tissue culture- and egg-propagated cultivars of vaccine viruses B/Colorado/06/2017 (subclade 1A(Δ 2)) and B/Washington/02/2019 (subclade 1A(Δ 3)B: with the additional HA1 amino acid substitutions G133R and K136E) were compared to the titres of the antisera with their homologous viruses.

For Influenza B viruses of the B/Yamagata lineage the HI titres of test viruses with antisera raised against cell culture- and egg-propagated cultivars of the vaccine virus, B/Phuket/3073/2013, were compared to the titres of the antisera with their homologous viruses.

HA and NA phylogenies for all viruses with collection dates after 2020-01-31 received at the Francis Crick Institute, for which genetic sequencing was performed successfully as of 2020-09-11, are presented in the report Annex for A(H1N1)pdm09, A(H3N2) and B/Victoria lineage viruses (**Figures A-1 to A-6**). The phylogenies for B/Yamagata lineage viruses are shown in **Figures 16 and 18**. The following clade distribution of sequences originating from individual patient specimens was observed:

- A(H1N1)pdm09 – total 309 of which there were 301 (97.4%) 6B.1A5A, 5 (1.6%) 6B.1A5B, 3 (1%) 6B.1A7 viruses;
- A(H3N2) – total 205 of which there were 119 (58.0%) clade 3C.3a, 38 (18.5%) 3C.2a1b+T135K-A, 36 (17.6%) 3C.2a1b+T131K, 12 (5.9%) 3C.2a1b+T135K-B viruses;
- B/Victoria lineage – total 189 of which there were 185 (98%) 1A(Δ 3)B and 4 (2%) 1A(Δ 2) viruses;
- B/Yamagata lineage – total 3, all of which were clade 3 viruses.

Table 1. Summary of clinical samples and virus isolates received, with collection dates after 2020-01-31, by country*

MONTH	TOTAL RECEIVED Seasonal	A		H1N1pdm09		H3N2		B		B/Victoria-lineage		B/Yamagata-lineage	
Country		Number received	Number propagated ¹	Number received	Number propagated ¹	Number received	Number propagated ²	Number received	Number propagated ¹	Number received	Number propagated ¹	Number received	Number propagated ¹
2020													
FEBRUARY													
Algeria	15			7	0	4	1	2	0	2	2		
Austria	2	1	0					1	0				
Belgium	50			28	28	17	17			5	5		
Bosnia and Herzegovia	25			17	6	5	2			3	3		
Bulgaria	26			8	7	15	10			3	3		
Cote d'Ivoire	10			9	in process	1	in process						
Cyprus	38			6	5	21	12			11	in process		
Denmark	9					6	4			3	3		
Estonia	12			5	4	2	2			5	4		
Finland	8			5	5	1	1			2	2		
France	32			14	14	7	7			9	9	2	1
Germany	25			13	13	7	5			5	5		
Hungary	9			3	3	4	3			2	2		
Iceland	10			4	4	2	2			4	4		
Ireland	12			1	0			3	0	8	6		
Italy	25			7	7	12	12			6	6		
Kyrgyzstan	2					2	2						
Lebanon	37			16	11	10	2			11	5		
Macedonia	18			1	0	11	9	1	0	5	1		
Moldova	7			6	2	1	0						
Morocco	11					10	0	1	0				
Mozambique	1			1	in process								
Norway	13			4	2	5	2			3	0	1	1
Poland	22			8	2	11	4	1	0	2	in process		
Portugal	32			29	19	2	1			1	1		
Russia	8			6	6					2	2		
Slovakia	8			2	2	4	4			2	2		
Slovenia	15			3	3	9	6			3	3		
South Africa	25			21	21					4	4		
Spain	35	1	0	23	17	3	2			8	7		
Sweden	18			8	8	5	3			5	5		
Switzerland	15			5	0	2	2			8	8		
Ukraine	32			30	in process			2	in process				
UK England	7			1	1	2	2			4	4		
UK Scotland	7							7	in process				
MARCH													
Belgium	6			4	4					2	2		
Bulgaria	9			2	2	1	1			6	6		
Cote d'Ivoire	10			3	in process	2	in process			5	in process		
Cyprus	9			1	1	1	1			7	in process		
Estonia	5			1	1	3	2			1	1		
Finland	4					2	2			2	2		
France	19			8	8	4	3			7	7		
Germany	5			2	2	2	2			1	1		
Hungary	1					1	1						
Iceland	13			4	4	6	6			3	3		
Ireland	14			3	3	1	1			6	4		
Lebanon	3			1	1	2	1						
Macedonia	2									2	2		
Moldova	5					1	1			2	2		
Mozambique	16			16	in process			2	0				
Norway	37			5	4	18	15			14	9		
Poland	4			4	2								
Portugal	4			3	2	1	1						
Russia	41			16	15	3	3			21	21	1	1
Slovenia	8					2	2			6	6		
South Africa	17			16	16					1	1		
Spain	7			4	3	1	1	1	0	1	1		
Switzerland	14			5	3	3	2			6	6		
Ukraine	3			2	in process			1	in process				
UK England	1									1	1		
UK Scotland	18							17	in process	1	in process		
APRIL													
Cote d'Ivoire	1			1	in process								
Iceland	1									1	1		
Mozambique	9			7	in process			2	in process				
Norway	1									1	0		
South Africa	1			1	1								
MAY													
Cote d'Ivoire	2			1	in process					1	in process		
JUNE													
Cote d'Ivoire	3			2	in process					1	in process		
JULY													
Cote d'Ivoire	2									2	in process		
AUGUST													
Cote d'Ivoire	2									2	in process		
34 Countries	918	2	0	403	262	235	162	45	0	229	172	4	3
		0.22%		43.90%		25.60%	18	4.90%		24.95%		0.44%	
				69.7%						30.3%			

1. Propagated to sufficient titre to perform HI assay (the totalled number does not include any from batches that are in process)

2. Propagated to sufficient titre to perform HI assay in the presence of 20nM oseltamivir (the totalled number does not include any from batches that are in process)

Numbers in red indicate viruses recovered but with insufficient HA titre to permit HI assay

Includes clinical samples sent either in lysis-mix or as purified RNA, for which genetic characterisation only can be performed

* As of 2020-09-08

Table 2. Isolation rates for specimens collected after 2020-01-31

	Clinical specimen			Virus isolate		
	Number processed	Number propagated ¹	Percent recovered	Number processed	Number propagated ¹	Percent recovered
H1	119	77	65%	212	204	96%
H3	88	64 ²	73%	116	98 ²	84%
		6 ³	80% (91%) ⁴ 7%		12 ³	95% (89%) ⁴ 10%
B Victoria	92	68	74%	110	109	99%
B Yamagata	2	1	50%	2	2	100%

1. Propagated to sufficient HA titre to perform HI assay
2. Propagated to sufficient HA titre in the presence of 20nM oseltamivir to allow HI assay
3. Viruses recovered (NA activity by MUNANA-based assay) but with insufficient HA titre to allow HI assay
4. The percentages for viruses recovered that could be assayed by HI are shown in parentheses

Table 3. Summary of isolates with collection dates after 2020-01-31 showing reduction in reactivity with antisera raised against reference viruses in HI & plaque-reduction neutralisation assays (PRNA)

Virus type/subtype	Number tested	≤2-fold ^a	4-fold ^a	≥8-fold ^a
H1 ^{1a}	281	203 (72.2%)	10 (3.6%)	68 (24.2%)
H1 ^{1b}	281	190 (67.6%)	20 (7.1%)	71 (25.3%)
H1 ^{1c}	281	49 (17.4%)	14 (5.0%)	218 (77.6%)
H1 ^{1d}	281	207 (73.7%)	5 (1.8%)	69 (24.6%)
H3 ^{2a}	156	1 (0.6%)	10 (6.4%)	145 (92.9%)
H3 ^{2b}	162	2 (1.2%)	11 (6.8%)	149 (92.0%)
H3 ^{2c}	162	22 (13.8%)	23 (14.4%)	115 (71.9%)
H3 ^{2d}	162	9 (5.6%)	53 (32.7%)	100 (61.7%)
H3 ^{2e}	162	108 (66.7%)	15 (9.3%)	39 (24.1%)
H3 ^{3a}	61	17 (27.9%)	1 (1.6%)	43 (70.5%)
H3 ^{3b}	61	0	1 (1.6%)	60 (98.4%)
H3 ^{3c}	61	10 (16.4%)	26 (42.6%)	25 (41.0%)
H3 ^{3d}	61	1 (1.6%)	3 (5.0%)	57 (93.4%)
H3 ^{3e}	61	19 (31.1%)	15 (24.6%)	27 (44.3%)
H3 ^{3f}	61	0	2 (3.3%)	59 (96.7%)
H3 ^{3g}	61	17 (27.9%)	5 (8.2%)	39 (63.9%)
B/Victoria ^{4a}	177	8 (4.5%)	33 (18.6%)	136 (76.8%)
B/Victoria ^{4b}	177	3 (1.7%)	12 (6.8%)	162 (91.5%)
B/Victoria ^{4c}	177	150 (84.7%)	4 (2.3%)	23 (13.0%)
B/Victoria ^{4d}	177	117 (66.1%)	28 (15.8%)	32 (18.1%)
B/Yamagata ^{5a}	3	0	1 (33.3%)	2 (66.7%)
B/Yamagata ^{5b}	3	3 (100%)	0	0

a. Reduction in titre, compared to the homologous titre with post-infection ferret antisera indicated below:

1a. Compared to homologous titres for ferret antiserum raised against egg-propagated A/Michigan/45/2015 (vaccine virus: genetic clade 6B.1)

1b. Compared to homologous titres for ferret antiserum raised against egg-propagated A/Brisbane/02/2018 (vaccine virus: genetic subgroup 6B.1A1)

1a. Compared to homologous titres for ferret antiserum raised against A/Denmark/3280/2019 (tissue culture-propagated virus: genetic subgroup 6B.1A5A)

1a. Compared to homologous titres for ferret antiserum raised against egg-propagated A/Guangdong-Maonan/SWL1536/2019 (vaccine virus: genetic subgroup 6B.1A5A)

2a. Compared to homologous titres for ferret antiserum raised against egg-propagated A/South Australia/34/2019 (vaccine virus: genetic subgroup 3C.2a1b+T131K)

2b. Compared to homologous titres for ferret antiserum raised against egg-propagated A/Hong Kong/2671/2019 (vaccine virus: genetic subgroup 3C.2a1b+T135K-B)

2c. Compared to homologous titres for ferret antiserum raised against A/Hong Kong/2671/2019 (tissue culture-propagated virus: genetic subgroup: 3C.2a1b+T135K-B)

2d. Compared to homologous titres for ferret antiserum raised against egg-propagated NYMC X-327 [A/Kansas/14/2017] (vaccine virus: genetic clade 3C.3a)

2e. Compared to homologous titres for ferret antiserum raised against A/Kansas/14/2017 (tissue culture-propagated virus: genetic clade 3C.3a)

3a. Compared to homologous titres for ferret antiserum raised against tissue culture-propagated A/Norway/3275/2018 (genetic subgroup 3C.2a1b+T131K) in PRNA (2-fold)

3b. Compared to homologous titres for ferret antiserum raised against egg-propagated A/South Australia/34/2019 (genetic subgroup 3C.2a1b+T131K) in PRNA (2-fold)

3c. Compared to homologous titres for ferret antiserum raised against tissue culture-propagated A/Denmark/3264/2019 (genetic subgroup 3C.2a1b+T135K-A) in PRNA (2-fold)

3d. Compared to homologous titres for ferret antiserum raised against egg-propagated A/Paris/2554/2019 (genetic subgroup 3C.2a1b+T135K-A) in PRNA (2-fold)

3e. Compared to homologous titres for ferret antiserum raised against tissue culture-propagated A/Hong Kong/2671/2019 (genetic subgroup 3C.2a1b+T135K-B) in PRNA (2-fold)

3f. Compared to homologous titres for ferret antiserum raised against egg-propagated A/Hong Kong/2671/2019 (genetic subgroup 3C.2a1b+T135K-B) in PRNA (2-fold)

3g. Compared to homologous titres for ferret antiserum raised against tissue culture-propagated A/Kansas/14/2017 (genetic clade 3C.3a) in PRNA (2-fold)

4a. Compared to homologous titres for ferret antisera raised against tissue culture-propagated B/Colorado/06/2017 - Clade 1A(Δ2)

4b. Compared to homologous titres for ferret antisera raised against egg-propagated B/Colorado/06/2017 - Clade 1A(Δ2)

4c. Compared to homologous titres for ferret antisera raised against tissue culture-propagated B/Washington/02/2019 - Clade 1A(Δ3)B

4d. Compared to homologous titres for ferret antisera raised against egg-propagated B/Washington/02/2019 - Clade 1A(Δ3)B

5a. Compared to homologous titres for ferret antiserum raised against tissue culture-propagated vaccine virus B/Phuket/3073/2013

5b. Compared to homologous titres for ferret antiserum raised against egg-propagated vaccine virus B/Phuket/3073/2013

Influenza A(H1N1)pdm09 virus analyses.

Influenza A(H1N1)pdm09 viruses have been received from NICs in 30 countries or administrative regions in four WHO Regions: African (4: Algeria, Cote d'Ivoire, Mozambique and South Africa), Eastern Mediterranean (1: Lebanon) and European (25) (**Table 1**). **Table 4** summarises antigenic (HI) results for the recognition of viruses, with collection dates after 2020-01-31, by antisera raised against egg-propagated viruses A/Guangdong-Maonan/SWL1563/2019 (the recommended vaccine virus for the Northern Hemisphere 2020-2021 influenza season), A/Michigan/45/2015 and A/Brisbane/02/2018 (former vaccine viruses), and against cell culture-propagated A/Denmark/3280/2019.

HI analysis

Tables 5-1 to 5-14 show the results of HI assays of A(H1N1)pdm09 viruses carried out since the February 2020 report. Test viruses in each table are sorted by both date of collection and by genetic group and viruses with collection dates after 2020-01-31 are shown below red lines in **Tables 5-5 to 5-14**. A summary of the HI results for viruses with collection dates after 2020-01-31 is shown in **Table 5-15** including a breakdown by the genetic group of the HA where known.

The majority (75%) of viruses collected after 2020-01-31 were antigenically similar to the egg-propagated vaccine virus A/Guangdong-Maonan/SWL1563/2019 recommended for use in the Northern Hemisphere 2020-2021 influenza season, being recognised at titres within 4-fold of the titre of the antiserum with the homologous virus (**Table 4** and **Table 5-15**) with 74% being recognised at titres within 2-fold of the homologous titre of the antiserum. Similar majorities were recognised by the antisera raised against the former vaccine viruses egg-propagated A/Michigan/45/2015 and A/Brisbane/02/2018 at titres within 4-fold of the respective homologous titres of the antisera. The antisera raised against egg-propagated A/Ireland/87733/2019 and A/Switzerland/3330/2017 also recognised ~75% of the test viruses at titres within 4-fold of their homologous titres. However, a sizable proportion ~25% of viruses were recognised poorly by these five antisera.

Antisera raised against cell culture-propagated reference viruses (A/Paris/1447/2017, A/Norway/3433/2018, A/Ireland/84630/2018 and the cell culture-propagated cultivar of A/Guangdong-Maonan/SWL1563/2019) recognised ~75% of test viruses with collection dates after 2020-01-31 at titres within 4-fold of the titres of the antisera with their respective homologous viruses, with between 56% and 76% being recognised at titres within 2-fold of the respective homologous titres (**Table 5-15**). The antiserum raised against cell culture-propagated A/Denmark/3280/2019, with substitutions **K130N**, **N156K**, **L161I**, **V250A** in **HA1** and **E179D** in **HA2** recognised 22.5% of the test viruses at titres within 4-fold of the homologous titre and 18% within 2-fold. Egg-propagated A/Hong/Kong/4394/2019 has the same amino acid substitutions as A/Denmark/3280/2019 with an additional **HA1** substitution of **K209M** and egg-adaptation substitutions of **HA1 D187N** and **Q223R**: antiserum raised against this virus recognised A/Denmark/3280/2019 at a titre of 640 but recognised only one of 13 test

viruses at a titre of at least 160 – this virus, A/Norway/2341/2020, has **HA1** substitutions **E112K**, **K130N**, **N156K**, **L161I**, **K209M** and **V250A** (Table 5-13).

The vast majority (98%) of viruses analysed by HI had HA genes in subclade 6B.1A5A with small numbers falling in subclades 6B.1A5B and 6B.1A7.

Genetic analysis

All test viruses with collection dates after 2020-01-31, for which HA gene sequences were known, fell into subclade 6B.1A which is defined by the amino acid substitutions **S74R**, **S164T** (altering a glycosylation motif introduced at residues 162-164 in clade 6B.1 viruses) and **I295V** in **HA1**, compared with the clade 6B.1 prototype A/Michigan/45/2019 virus. The vast majority (98%) of A(H1N1)pdm09-positive clinical specimens and viruses analysed had HA genes in subclade 6B.1A5A with small numbers falling in subclades 6B.1A5B and 6B.1A7.

Figure 1 shows a phylogenetic tree for HA genes of representative A(H1N1)pdm09 viruses sequenced at the Francis Crick Institute and other representative viruses. A phylogenetic tree of the HA genes of all viruses with collection dates after 2020-01-31 analysed genetically at the Francis Crick Institute is shown in the Annex, **Figure A-1**.

Within subclade 6B.1A genetic clusters of viruses encoding a range of **HA1** amino acid substitutions have emerged. Most test viruses carry the **S183P** amino acid substitution in **HA1** and the major groups in circulation recently are:

- 6B.1A5A, defined by **N129D**, **T185I** and **N260D** substitutions in **HA1**, with two major genetic groups encoding either **D187A** and **Q189E** substitutions in **HA1**, or **K130N**, **N156K**, **L161I** and **V250A** substitutions in **HA1** with **E179D** substitution in **HA2**,
- 6B.1A5B, with HAs defined by **E235D** and **N260D** with **K130N**, **K160M**, **T216K** and **H296N** in **HA1**, and **V193A** in **HA2** (**V520A**),
- 6B.1A7, with HAs defined by **E68D**, **S121N**, **T120A**, **L161L** and **K302T** in **HA1** and **I76M**, **N169S** and **E179D** in **HA2** (**I404M**, **N496S** and **E506D**).

Viruses with the substitutions **K130N**, **N156K**, **L161I** and **V250A** in **HA1** and **E179D** in **HA2** represented 21.5% of the recently collected viruses analysed, which represents a large increase since the February 2020 report.

An amino acid sequence alignment of the HAs for representative A(H1N1)pdm09 viruses is shown in **Figure 2**; the vaccine viruses are shown in red, reference viruses are coloured blue, glycosylation motifs are highlighted, the genetic group/subgroup is shown in green and the HA1/HA2 proteolytic-processing site is highlighted.

Figure 3a shows H1pdm09-HA structures with the positions of amino acid substitutions that differentiate virus genetic groups indicated. Those differentiating subclade 6B.1A viruses (**S74R**, **S164T** and **I295V** in **HA1**) from viruses in clade 6B.1 are indicated in red. Those that define subclade 6B.1A5A viruses (**N129D**, **S183P**, **T185I** and **N260D** in **HA1**) are indicated in blue and within this subclade the substitutions **D187A** and **Q197E**,

defining the predominant genetic group in recent months, are indicated in green. **Figure 3b** again shows **HA1** substitutions defining subclade 6B.1A and 6B.1A5A viruses, with those in **HA1** (**K130N**, **N156K**, **L161I** and **V250A**) that define the emerging antigenically distinct genetic group within subclade 6B.1A5A shown in green.

Figure 4 shows a phylogenetic tree for the NA genes of representative A(H1N1)pdm09 viruses sequenced at the Francis Crick Institute and other representative viruses. A phylogenetic tree of the NA genes of all viruses with collection dates after 2020-01-31 analysed genetically at the Francis Crick Institute is shown in the Annex, **Figure A-2**. The vast majority encoded the amino acid substitutions **Q51K**, **F74S**, **I389K**, **D416N** and **T452I** compared with the NA of A/Brisbane/02/2018, many also encode **S52N**, resulting in the loss of a glycosylation site in the stalk domain. Those with HA genes in genetic group 6B.1A5A with the HA substitutions **K130N**, **N156K**, **L161I** and **V250A** in **HA1** and **E179D** in **HA2** carry NA genes encoding substitutions **M19T**, **N22K** and **Y66F**.

Figure 5 shows a NA amino acid alignment for the same representative set of A(H1N1)pdm09 viruses as used in **Figure 2**, annotated as **Figure 2** but lacking a proteolytic-processing site.

Antiviral resistance.

Phenotypic testing to assess susceptibility to oseltamivir and zanamivir was performed on 276 viruses with collection dates after 2020-01-31. All but a single virus (>99.5%) showed normal inhibition (NI) by both neuraminidase inhibitors (**Table 13**). The exception was A/Lund/4/2020 showing highly reduced inhibition (HRI) by oseltamivir but normal inhibition (NI) by zanamivir. The NA gene encoded the substitution H275Y.

Of the 359 influenza A(H1N1)pdm09 viruses that yielded full-length PA gene sequences all encoded isoleucine at position 38 (I38) and showed no evidence of other substitutions associated with reduced susceptibility to baloxavir marboxil.

Conclusion.

All A(H1N1)pdm09 viruses received fell within subclade 6B.1A encoding the **HA1** amino acid substitutions **S74R**, **S164T** (which alters the glycosylation motif at residues **162 to 164**) and **I295V**. A majority of viruses in this subclade carry **HA1 S183P** amino acid substitution. The predominant genetic group was 6B.1A5A representing 98% of the viruses tested with collection dates after 2020-01-31; however, within this subclade, the proportion of viruses with the substitutions **K130N**, **N156K**, **L161I**, **V250A** in **HA1** and **E179D** in **HA2** has been increasing. The majority (~75%) of test viruses in each genetic subclade remained antigenically similar to the currently recommended vaccine virus, A/Guangdong-Maonan/SWL1563/2019, as determined by assays using panels of post-infection ferret antisera. However, viruses with the substitutions **K130N**, **N156K**, **L161I**, **V250A** in **HA1** and **E179D** in **HA2** were poorly recognised by this antiserum but recognised better by an antiserum raised against a reference virus with the same substitutions. The vast majority (>99%) of test viruses retained susceptibility to oseltamivir and zanamivir and all assessed genetically were susceptible to the PA inhibitor baloxavir marboxil.

Table 4. Summary of influenza A(H1N1)pdm09 viruses collected after 2020-01-31, analysed by HI

MONTH	H1N1pdm09													
Country	Number received ¹	Number propagated ²	Post infection ferret antisera raised against											
			Clade 6B.1 virus			Clade 6B.1A1 virus			Clade 6B.1A5A virus			Clade 6B.1A5A virus		
			≤2 fold ³	4 fold ³	≥8 fold ³	≤2 fold ⁴	4 fold ⁴	≥8 fold ⁴	≤2 fold ⁵	4 fold ⁵	≥8 fold ⁵	≤2 fold ⁶	4 fold ⁶	≥8 fold ⁶
2020														
FEBRUARY														
Algeria	7	0												
Belgium	28	28	23	1	4	21	1	6	4		24	22	1	5
Bosnia and Herzegovina	17	6	5	1		5	1				6	6		
Bulgaria	8	7	7			6	1				7	7		
Cote d'Ivoire	9	0												
Cyprus	6	5	3		2	2	1	2	2		3	3		2
Estonia	5	4	4			4					4	4		
Finland	5	5	2		3	2		3	2	1	2	2		3
France	14	14	10		4	10		4	3		11	10		4
Germany	13	13	7		6	7		6	5	1	7	7		6
Hungary	3	3	2		1	1	1	1	1		2	2		1
Iceland	4	4			4			4	4					4
Ireland	1	0												
Italy	7	7	5		2	5		2			7	5		2
Lebanon	16	11	10		1	10		1	1		10	10		1
Macedonia	1	0												
Moldova	6	2	2			2					2	2		
Mozambique	1	0												
Norway	4	2	1	1		1	1				2	2		
Poland	8	2	2			2					2	2		
Portugal	29	19	16		3	15	1	3	3		16	15	1	3
Russia	6	6	4	2		4	2				6	5	1	
Slovakia	2	2	2			2					2	2		
Slovenia	3	3	2		1	1	1	1	1		2	2		1
South Africa	21	21	14		7	14		7	3	4	14	14		7
Spain	23	17	14		3	13	1	3	3		14	14		3
Sweden	8	8	3		5	2	1	5	3	1	4	3		5
Switzerland	5	0												
Ukraine	30	19	17	1	1	18		1	1		18	17	1	1
United Kingdom	1	1	1			1					1	1		
MARCH														
Belgium	4	4	4			4					4	4		
Bulgaria	2	2	2			2					2	2		
Cote d'Ivoire	3	0												
Cyprus	1	1	1				1				1	1		
Estonia	1	1			1			1	1					1
France	8	8	5		3	5		3	3		5	5		3
Germany	2	2	2			2					2	2		
Iceland	4	4	3		1	1	2	1		1	3	3		1
Ireland	3	3			3			3	3					3
Lebanon	1	1	1			1					1	1		
Mozambique	16	0												
Norway	5	4	2	1	1	2	1	1		1	3	3		1
Poland	4	2	2			1	1				2	2		
Portugal	3	2	2			2					2	2		
Russia	16	15	12	3		12	2	1			15	14	1	
South Africa	16	16	8		8	8		8	4	4	8	8		8
Spain	4	3	2		1	1	1	1	1		2	2		1
Switzerland	5	3	1		2	1		2	1	1	1	1		2
Ukraine	2	0												
APRIL														
Cote d'Ivoire	1	0												
Mozambique	7	0												
South Africa	1	1			1			1			1			1
MAY														
Cote d'Ivoire	1	0												
JUNE														
Cote d'Ivoire	2	0												
JULY														
AUGUST														
30 Countries	403	281	203	10	68	190	20	71	49	14	218	207	5	69
			72.2%	3.6%	24.2%	67.6%	7.1%	25.3%	17.4%	5.0%	77.6%	73.7%	1.8%	24.6%

1 Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared

2 Propagated to sufficient titre to perform HI assay

3 Reduction in HI titre with post-infection ferret antiserum raised against A/Michigan/45/2015 (egg-propagated vaccine virus)

4 Reduction in HI titre with post-infection ferret antiserum raised against A/Brisbane/02/2018 (egg-propagated vaccine virus)

5 Reduction in HI titre with post-infection ferret antiserum raised against A/Denmark/3280/2019 (cell culture-propagated virus)

6 Reduction in HI titre with post-infection ferret antiserum raised against A/Guangdong-Maonan/SWL1536/2019 (egg-propagated vaccine virus)

Numbers highlighted in blue show the number of viruses subjected to HI assay for 'completed' sample sets. Under a 'sequence first' virus characterisation scheme: where multiple viruses shared the same HA sequence only a selection were propagated to allow assay by HI

Table 5-1. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2020-02-11)

Haemagglutination inhibition titre															
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera											
				A/Bayern	A/Lviv	A/Mich	A/Slov	A/Paris	A/Swit	A/Bris	A/Norway	A/Swit	A/Ire	A/HK	
				69/09	N6/09	45/15	2903/2015	1447/17	2656/17	02/18	3433/18	3330/17	84630/18	110/19	
				MDCK	MDCK	Egg	Egg	MDCK	Egg	Egg	MDCK	Egg	MDCK	MDCK	
Passage history		F09/15 ¹	F13/18 ¹	F31/16 ¹	F48/16 ¹	F03/18 ²	F20/18 ¹	F09/19 ¹	F04/19 ¹	F23/18 ¹	F08/19 ¹	F28/19			
Ferret number															
Genetic group				6B.1		6B.1	6B.1A	6B.1A	6B.1A1	6B.1A5A	6B.1A5B	6B.1A6	6B.1A2		
REFERENCE VIRUSES															
A/Bayern/69/2009	G155E	2009-07-01	MDCK5/MDCK1	320	320	80	<	160	160	80	320	80	40	ND	
A/Lviv/N6/2009	G155E, D222G	2009-10-27	MDCK4/SIAT1/MDCK3	640	640	320	160	640	320	320	1280	320	160	ND	
A/Michigan/45/2015	6B.1	2015-09-07	E3/E3	320	160	640	1280	1280	1280	640	1280	640	1280	ND	
A/Slovenia/2903/2015	clone 37	2015-10-26	E4/E2	320	320	1280	1280	1280	1280	640	2560	640	1280	ND	
A/Paris/1447/2017	6B.1A	2017-10-20	MDCK1/MDCK3	160	80	640	640	1280	1280	640	1280	640	1280	ND	
A/Switzerland/2656/2017	6B.1A	2017-12-21	E5/E3	640	640	2560	2560	2560	2560	1280	2560	1280	2560	ND	
A/Brisbane/02/2018	6B.1A1	2018-01-04	E3/E1	320	320	2560	2560	2560	2560	1280	2560	1280	2560	40	
A/Norway/3433/2018	6B.1A5A	2018-10-30	MDCK3	80	80	320	320	640	640	320	1280	320	640	ND	
A/Switzerland/3330/2017	clone 35	2017-12-20	E6/E1	320	160	640	640	1280	1280	640	2560	1280	1280	ND	
A/Ireland/84630/2018	6B.1A6	2018-11-28	MDCK1/MDCK3	160	80	1280	1280	1280	1280	640	1280	640	1280	ND	
A/Hong Kong/110/2019	6B.1A2	2019-01-01	MDCK1/MDCK2	ND	ND	ND	ND	ND	ND	ND	ND	ND	2560		
TEST VIRUSES															
A/Friuli Venezia Giulia/234/2019	No seq	2019-11-26	MDCK1/MDCK1	160	160	640	640	640	640	320	2560	320	640	ND	
A/Lisboa/111/2019	6B.1A5A	2019-11-14	SIAT1/MDCK1	160	160	640	640	640	640	640	1280	320	640	ND	
A/Lisboa/109/2019	6B.1A5A	2019-11-19	SIAT1/MDCK1	160	160	320	320	320	320	320	1280	160	320	ND	
A/Parma/389/2019	6B.1A5A	2019-11-20	MDCK1/MDCK1	80	40	320	160	320	320	160	1280	160	320	<	
A/St. Petersburg/2174V/2019	6B.1A5A	2019-11-22	MDCK1/MDCK1	160	80	640	640	640	640	320	1280	320	640	ND	
A/St. Petersburg/2185V/2019	6B.1A5A	2019-11-23	MDCK1/MDCK1	320	160	640	640	1280	640	640	2560	640	640	ND	
A/England/433/2019	6B.1A5A	2019-11-23	SIAT1/MDCK1	160	80	640	320	640	320	320	1280	320	320	ND	
A/Beirut/567/2019	6B.1A5A	2019-11-27	MDCK1/MDCK1/MDCK1	160	80	640	640	640	640	320	1280	320	640	ND	
A/St. Petersburg/2219V/2019	6B.1A5A	2019-11-27	MDCK1/MDCK1	320	160	640	1280	1280	640	640	2560	640	640	ND	
A/St. Petersburg/2263V/2019	6B.1A5A	2019-12-03	MDCK1/MDCK1	320	320	1280	1280	1280	1280	640	2560	640	1280	ND	
A/England/687/2019	6B.1A5A	2019-12-05	SIAT1/MDCK1	160	160	160	80	160	80	160	640	160	80	<	
A/England/690/2019	6B.1A5A	2019-12-06	SIAT1/MDCK1	320	160	1280	1280	1280	640	640	2560	640	640	ND	
A/Lisboa/114/2019	6B.1A5A	2019-12-09	MDCK3/MDCK1	80	80	320	320	640	320	320	1280	320	320	ND	
A/Moscow/228/2019	6B.1A5A	2019-12-09	MDCK1/MDCK1	160	80	640	640	1280	640	640	2560	640	640	ND	
A/Beirut/578/2019	6B.1A5A	2019-12-10	MDCK1/MDCK1/MDCK1	160	80	320	320	640	320	320	1280	320	320	ND	
A/Palermo/31/2019	6B.1A5A	2019-12-10	MDCK3/MDCK1	160	160	640	640	1280	640	640	2560	640	640	ND	
A/Moscow/230/2019	6B.1A5A	2019-12-11	MDCK1/MDCK1	320	160	1280	640	1280	1280	640	2560	640	1280	ND	
A/Lithuania/MB41308/2019	6B.1A5A	2019-12-12	MDCK2/MDCK1	320	320	1280	1280	1280	1280	640	2560	640	1280	ND	
A/England/692/2019	6B.1A5A	2019-12-16	MDCK1/MDCK1	160	160	640	640	640	640	320	1280	320	640	ND	
A/Moscow/232/2019	6B.1A5A	2019-12-17	MDCK1/MDCK1	160	160	640	640	640	640	640	1280	640	640	ND	
A/Lisboa/115/2019	6B.1A5A	2019-12-18	MDCK3/MDCK1	320	160	640	1280	1280	640	640	2560	640	640	ND	
A/Lithuania/MB42127/2019	6B.1A5A	2019-12-21	MDCK1/MDCK1	320	160	1280	640	1280	1280	640	2560	640	1280	ND	
A/Parma/390/2019	6B.1A5A	2019-12-24	MDCK1/MDCK1	80	40	320	320	320	320	160	640	160	320	<	
A/Lithuania/MB42385/2019	6B.1A5A	2019-12-25	MDCK2/MDCK1	40	<	<	<	<	<	<	80	<	<	320	
A/Lithuania/MB42388/2019	6B.1A5A	2019-12-26	MDCK1/MDCK1	320	160	1280	640	1280	640	640	2560	640	1280	ND	
A/Nordrhein-Westfalen/142/2019	6B.1A5A	2019-12-28	C1/MDCK3	40	<	<	<	<	<	<	80	<	<	320	
A/Palermo/01/2020	6B.1A5A	2020-01-02	MDCK2/MDCK1	160	80	640	640	640	640	320	1280	320	640	ND	
A/Montenegro/174/2020	6B.1A5A	2020-01-03	MDCKx/MDCK1	160	160	640	640	640	640	320	1280	320	640	ND	
A/Rheinland-Pfalz/2/2020	6B.1A5A	2020-01-04	C1/MDCK1	160	80	320	640	640	320	320	1280	320	320	ND	
A/Montenegro/179/2020	6B.1A5A	2020-01-05	MDCKx/MDCK1	320	160	1280	1280	1280	1280	640	2560	640	1280	ND	
A/Estonia/125640/2020	6B.1A5A	2020-01-08	SIAT1/MDCK2	40	<	40	<	80	40	<	160	40	<	640	
A/Estonia/125685/2020	6B.1A5A	2020-01-09	SIAT1/MDCK1	160	160	640	640	1280	640	320	1280	320	640	ND	
A/Palencia/12/2020	6B.1A5A	2020-01-09	MDCK1/MDCK1	320	80	1280	1280	1280	1280	640	2560	640	1280	ND	
A/Leon/14/2020	6B.1A5A	2020-01-09	MDCK1/MDCK1	320	160	640	1280	1280	640	640	2560	640	640	ND	
A/Leon/10/2020	6B.1A5A	2020-01-09	MDCK1/MDCK1	160	160	640	640	640	640	640	1280	320	640	ND	
A/Czech Republic/82/2020	6B.1A5A	2020-01-09	P1/MDCK1	320	80	1280	1280	2560	1280	640	2560	1280	1280	ND	
A/Czech Republic/81/2020	6B.1A5A	2020-01-09	P1/MDCK1	160	<	640	640	640	640	320	1280	320	640	ND	
A/Bucuresti/256171/2020	6B.1A5A	2020-01-10	SIAT1/MDCK1	320	160	1280	640	1280	1280	640	2560	640	1280	ND	
A/Bulgaria/083/2020	6B.1A5A	2020-01-12	MDCK1	160	160	640	640	640	640	320	1280	320	640	ND	
A/Bulgaria/092/2020	6B.1A5A	2020-01-13	MDCK2	320	160	640	640	1280	640	640	2560	640	1280	ND	
A/Bulgaria/084/2020	6B.1A5A	2020-01-13	MDCK2	160	160	640	640	640	640	320	1280	320	640	ND	
A/Bucuresti/256303/2020	6B.1A5A	2020-01-13	MDCK1/MDCK1	80	<	320	160	320	160	160	640	160	320	<	
A/Estonia/125790/2020	6B.1A5A	2020-01-14	SIAT1/MDCK1	160	160	640	640	1280	640	640	2560	640	640	ND	
A/Estonia/125789/2020	6B.1A5A	2020-01-14	SIAT1/MDCK1	40	<	40	<	<	40	<	320	<	<	<	
A/Valladolid/17/2020	6B.1A5B	2020-01-10	MDCK1/MDCK1	320	160	1280	2560	2560	2560	1280	2560	1280	1280	ND	
A/Rheinland-Pfalz/1/2020	6B.1A5B	2020-01-10	C1/MDCK1	160	40	640	640	1280	640	320	1280	160	640	ND	
A/Brandenburg/1/2020	6B.1A7	2020-01-06	C1/MDCK1	160	160	1280	1280	1280	1280	640	2560	640	1280	ND	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80; ND =Not Done

Vaccine
NH 2018-19
SH 2019Vaccine
NH 2019-20
SH 2020

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 5-2. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2020-02-18)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				A/Bayern/69/09	A/Lviv/6/09	A/Mich/45/15	A/Slov/2903/2015	A/Paris/1447/17	A/Swit/2656/17	A/Bris/02/18	A/Norway/3433/18	A/Swit/3330/17	A/Ire/84630/18	A/HK/110/19
				MDCK	MDCK	Egg	Egg	MDCK	Egg	Egg	MDCK	Egg	MDCK	MDCK
	Passage history			F09/15 ¹	F13/18 ¹	F31/16 ¹	NIB F48/16 ¹	F03/18 ²	F20/18 ¹	F09/19 ¹	F04/19 ¹	F23/18 ¹	F08/19 ¹	F28/19
	Ferret number													
	Genetic group					6B.1	6B.1	6B.1A	6B.1A	6B.1A1	6B.1A5A	6B.1A5B	6B.1A6	6B.1A2
REFERENCE VIRUSES														
A/Bayern/69/2009	G155E	2009-07-01	MDCK5/MDCK1	160	160	80	40	160	80	80	320	80	40	ND
A/Lviv/6/2009	G155E, D222G	2009-10-27	MDCK4/SIAT1/MDCK3	640	640	320	160	640	320	320	640	160	160	ND
A/Michigan/45/2015	clone 37	2015-09-07	E3/E3	160	160	640	640	1280	640	640	1280	640	640	ND
A/Slovenia/2903/2015		2015-10-26	E4/E3	320	320	1280	1280	2560	1280	1280	2560	1280	1280	ND
A/Paris/1447/2017		2017-10-20	MDCK1/MDCK3	160	80	640	640	1280	640	640	1280	640	640	ND
A/Switzerland/2656/2017		2017-12-21	E5/E3	640	640	2560	1280	5120	2560	1280	2560	1280	1280	ND
A/Brisbane/02/2018		2018-01-04	E3/E1	320	320	2560	2560	2560	2560	1280	2560	1280	1280	40
A/Norway/3433/2018		2018-10-30	MDCK3	80	40	320	320	640	320	320	1280	320	320	ND
A/Switzerland/3330/2017	clone 35	2017-12-20	E6/E2	160	80	320	320	640	640	320	1280	640	640	ND
A/Ireland/84630/2018		2018-11-28	MDCK1/MDCK3	320	160	1280	1280	2560	1280	1280	2560	1280	1280	ND
A/Hong Kong/110/2019		2019-01-01	MDCK1/MDCK2	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	2560
TEST VIRUSES														
A/Cote d'Ivoire/1419/2019	6B.1A5A	2019-07-24	MDCK1	320	160	640	640	1280	640	640	1280	640	640	ND
A/La Reunion/2126/2019	6B.1A5A	2019-08-22	MDCK2/MDCK2	40	<	40	<	80	<	<	160	40	<	<
A/La Reunion/2128/2019	6B.1A5A	2019-09-17	MDCK2/MDCK1	80	80	320	320	640	320	160	1280	320	320	<
A/Catalonia/2145999NS/2019	6B.1A5A	2019-10-11	P0/MDCK1	80	80	320	320	640	320	320	1280	320	320	ND
A/Catalonia/3510341NS/2019	6B.1A5A	2019-10-17	P0/MDCK1	80	80	320	320	320	320	320	640	320	320	ND
A/Catalonia/3510633NS/2019	6B.1A5A	2019-10-29	P0/MDCK1	80	80	320	320	320	320	320	1280	320	320	ND
A/Lisboa/102/2019	6B.1A5A	2019-10-31	SIAT3/MDCK1	160	80	320	320	640	320	320	1280	320	320	ND
A/Lisboa/105/2019	6B.1A5A	2019-11-01	MDCK1/MDCK1	160	160	640	640	640	640	320	1280	320	640	ND
A/Lisboa/103/2019	6B.1A5A	2019-11-04	SIAT3/MDCK1	160	80	320	640	640	640	320	1280	320	320	ND
A/Lisboa/107/2019	6B.1A5A	2019-11-10	SIAT2/MDCK1	160	80	320	320	640	320	320	640	320	320	ND
A/Lisboa/106/2019	6B.1A5A	2019-11-11	SIAT1/MDCK1	160	80	320	640	1280	640	320	1280	320	320	ND
A/Lisboa/110/2019	6B.1A5A	2019-11-14	SIAT2/MDCK1	40	40	160	160	160	160	80	640	160	160	<
A/Catalonia/3511346NS/2019	6B.1A5A	2019-11-20	P0/MDCK1	160	160	640	640	1280	640	320	1280	320	640	ND
A/Netherlands/01659/2019	6B.1A5A	2019-11-24	hCK1/MDCK1	160	80	640	320	640	640	320	1280	320	640	ND
A/Rennes/2646/2019	6B.1A5A	2019-11-25	MDCK1/MDCK1	160	80	320	320	640	320	320	1280	160	320	ND
A/Lyon/2067/2019	6B.1A5A	2019-11-25	MDCK2/MDCK1	80	40	160	160	320	160	160	640	160	160	<
A/Poitiers/2102/2019	6B.1A5A	2019-11-26	MDCK2/MDCK1	80	80	320	320	640	320	320	1280	320	320	ND
A/Marseille/2089/2019	6B.1A5A	2019-11-27	MDCK2/MDCK1	160	160	640	640	1280	640	640	2560	640	640	ND
A/Schleswig-Holstein/6/2019	6B.1A5A	2019-11-28	C1/MDCK1	160	80	320	320	1280	640	320	1280	320	640	ND
A/Berlin/59/2019	6B.1A5A	2019-11-29	C1/MDCK1	80	80	320	320	640	320	320	640	160	320	ND
A/Catalonia/3511745NS/2019	6B.1A5A	2019-11-30	P0/MDCK1	160	80	640	640	640	640	320	1280	320	640	ND
A/Netherlands/10029/2019	6B.1A5A	2019-12-03	hCK1/MDCK1	160	160	640	640	1280	640	640	2560	640	640	ND
A/Catalonia/12003S/2019	6B.1A5A	2019-12-05	P0/MDCK1	160	80	640	640	640	640	320	1280	320	640	ND
A/Brandenburg/29/2019	6B.1A5A	2019-12-11	C2/MDCK1	160	80	320	640	640	640	320	1280	320	320	ND
A/Sachsen/151/2019	6B.1A5A	2019-12-11	C1/MDCK1	80	80	320	320	640	320	160	1280	160	320	<
A/Clermont Ferrand/2280/2019	6B.1A5A	2019-12-12	MDCK3/MDCK1	80	80	320	320	640	320	320	1280	320	320	ND
A/Lyon/2266/2019	6B.1A5A	2019-12-12	MDCK2/MDCK1	160	80	320	320	640	320	320	1280	320	320	ND
A/Pays de Loire/2796/2019	6B.1A5A	2019-12-17	MDCK1	160	80	320	320	640	320	320	1280	160	320	ND
A/Hessen/73/2019	6B.1A5A	2019-12-17	C1/MDCK1	160	80	640	320	640	320	320	1280	320	320	ND
A/Estonia/125382/2019	6B.1A5A	2019-12-19	SIAT1/MDCK1	160	80	320	320	640	640	320	1280	320	640	ND
A/Bretagne/2804/2019	6B.1A5A	2019-12-20	MDCK1	160	80	640	320	640	320	320	1280	320	320	ND
A/Mecklenburg-Vorpommern/13/2019	6B.1A5A	2019-12-23	C1/MDCK1	40	<	40	<	80	40	40	320	40	<	40
A/Thuringen/125/2019	6B.1A5A	2019-12-27	C2/MDCK1	160	80	320	320	640	320	320	1280	160	320	ND
A/Catalonia/11996S/2019	6B.1A5B	2019-12-05	P0/MDCK1	160	40	1280	1280	1280	1280	640	2560	640	1280	ND

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80; ND =Not Done

Vaccine
NH 2018-19
SH 2019Vaccine
NH 2019-20
SH 2020

 < 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 5-3. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2020-03-10)

Haemagglutination inhibition titre																				
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera																
				A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Mich 45/15 Egg	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Bris 02/18 Egg	A/Norway 3433/18 MDCK	A/Denmark 3280/19 MDCK	CNIC-1909 A/G-M/1536/19 Egg	A/HK 2655/19 Egg	A/Swit 3330/17 Egg	A/Ire 84630/18 MDCK	A/HK 110/19 MDCK			
				F09/15 ⁻¹	F13/18 ⁻¹	F31/16 ⁻¹	F48/16 ⁻¹	F03/18 ⁻²	F20/18 ⁻¹	F09/19 ⁻¹	F04/19 ⁻¹	F08/20 ⁻¹	F08/19 ⁻¹	F08/19 ⁻¹	F23/18 ⁻¹	F08/19 ⁻¹	F28/19			
				Genetic group							6B.1	6B.1	6B.1A	6B.1A	6B.1A1	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5B	6B.1A6
REFERENCE VIRUSES																				
A/Bayern/69/2009	G155E	2009-07-01	MDCK5/MDCK1	320	160	80	<	160	80	80	320	ND	<	40	80	40	ND			
A/Lviv/N6/2009	G155E, D222G	2009-10-27	MDCK4/SIAT1/MDCK3	640	640	320	160	640	320	320	640	ND	40	160	320	160	ND			
A/Michigan/45/2015	clone 37	2015-09-07	E3/E3	320	320	1280	1280	1280	1280	640	1280	ND	640	1280	640	1280	ND			
2015-10-26		E4/E3	320	320	1280	1280	2560	1280	640	2560	ND	640	640	640	1280	ND				
2017-10-20		MDCK1/MDCK3	320	80	1280	1280	2560	1280	640	2560	ND	640	640	640	1280	ND				
A/Switzerland/2656/2017	clone 35	2017-12-21	E5/E4	320	320	1280	1280	1280	1280	1280	640	2560	ND	640	1280	640	1280	ND		
2018-01-04		E3/E1	320	320	1280	1280	2560	1280	1280	2560	ND	1280	1280	1280	1280	ND				
2018-10-30		MDCK3	160	40	320	320	640	320	320	1280	ND	320	640	320	640	ND				
2019-06-17		E2/E8/E1	ND	ND	ND	ND	ND	ND	ND	ND	ND	640	1280	ND	ND	ND				
2019-06-18		E2	ND	ND	ND	ND	ND	ND	ND	ND	ND	1280	1280	ND	ND	ND				
2017-12-20		E6/E2	320	160	640	640	1280	1280	640	1280	ND	640	640	1280	1280	ND				
2018-11-28		MDCK1/MDCK3	320	160	1280	640	2560	1280	640	2560	ND	640	640	640	1280	ND				
2019-01-01		MDCK1/MDCK2	160	80	80	<	160	160	80	160	1280	ND	ND	80	40	1280				
TEST VIRUSES																				
A/Qatar/10-VI-19-0070537/2019		6B.1A5A	2019-11-10	MDCK1	640	320	1280	1280	2560	1280	1280	2560	ND	1280	1280	1280	1280	ND		
A/Qatar/34-VI-19-0071429/2019	6B.1A5A	2019-11-13	MDCK1	160	80	640	320	640	320	320	1280	ND	320	640	320	640	ND			
A/Qatar/16-VI-19-0071385/2019	6B.1A5A	2019-11-13	MDCK1	80	40	80	40	160	80	80	320	<	40	160	80	40	<			
A/Ireland/92962/2019	6B.1A5A	2019-11-20	MDCK2	320	160	640	640	1280	640	640	2560	ND	640	1280	640	1280	ND			
A/Ireland/93509/2019	6B.1A5A	2019-11-21	MDCK2	160	80	320	640	640	640	320	1280	ND	640	640	320	640	ND			
A/Ireland/93555/2019	6B.1A5A	2019-11-22	MDCK2	160	80	640	640	1280	640	640	1280	ND	320	640	320	640	ND			
A/Ireland/94203/2019	6B.1A5A	2019-11-25	MDCK2	320	160	640	640	1280	640	640	2560	ND	320	640	640	640	ND			
A/Ireland/93911/2019	6B.1A5A	2019-11-25	MDCK2	320	160	640	640	1280	640	640	1280	ND	320	320	320	640	ND			
A/Ireland/93802/2019	6B.1A5A	2019-11-25	MDCK2	160	160	640	640	1280	640	640	1280	ND	640	640	320	640	ND			
A/Ireland/94393/2019	6B.1A5A	2019-11-26	MDCK2	320	160	640	640	1280	640	320	160	ND	320	640	320	640	ND			
A/Iran/202699/2019	6B.1A5A	2019-11-27	MDCK1	160	80	640	640	1280	640	640	1280	ND	320	640	320	640	ND			
A/Iran/203838/2019	6B.1A5A	2019-11-28	MDCK1	160	80	640	640	1280	640	640	1280	ND	640	640	640	640	ND			
A/Iran/203381/2019	6B.1A5A	2019-11-28	MDCK3	80	80	320	320	640	320	160	640	<	160	320	160	320	<			
A/Austria/1207285/2019	6B.1A5A	2019-12-11	MDCK1	160	80	320	320	640	320	320	1280	ND	320	320	320	640	ND			
A/Austria/1208476/2019	6B.1A5A	2019-12-16	MDCK1	320	160	640	640	1280	640	320	1280	ND	640	640	320	640	ND			
A/Bosnia and Herzegovina/67/2019	6B.1A5A	2019-12-17	MDCK1	160	160	640	640	1280	640	640	2560	ND	640	640	640	640	ND			
A/Valladolid/2/2020	6B.1A5A	2019-12-30	MDCK1/MDCK1	320	160	640	640	1280	640	640	1280	ND	640	640	640	640	ND			
A/Valladolid/4/2020	6B.1A5A	2020-01-01	MDCK1/MDCK1	160	80	640	640	1280	640	640	1280	ND	320	640	320	640	ND			
A/Valladolid/3/2020	6B.1A5A	2020-01-01	MDCK1/MDCK1	160	160	640	640	640	640	320	1280	ND	320	640	320	640	ND			
A/Valladolid/18/2020	6B.1A5A	2020-01-03	MDCK1/MDCK1	160	160	640	640	1280	640	640	2560	ND	320	640	320	1280	ND			
A/Valladolid/5/2020	6B.1A5A	2020-01-03	MDCK1/MDCK2	80	<	40	<	80	40	80	160	320	40	80	40	40	320			
A/Valladolid/9/2020	6B.1A5A	2020-01-07	MDCK1/MDCK1	320	160	640	640	1280	640	640	1280	ND	640	640	640	640	ND			
A/Valladolid/6/2020	6B.1A5A	2020-01-07	MDCK1/MDCK2	160	80	320	320	320	320	160	640	<	160	320	160	320	<			
A/Valladolid/19/2020	6B.1A5A	2020-01-08	MDCK1/MDCK1	320	160	640	640	640	640	320	1280	ND	640	640	320	640	ND			
A/Valladolid/13/2020	6B.1A5A	2020-01-08	MDCK1/MDCK2	160	80	640	640	1280	640	320	1280	ND	640	640	320	640	ND			
A/Athens.GR/37/2020	6B.1A5A	2020-01-09	MDCK1	160	160	640	640	1280	640	640	1280	ND	640	640	320	640	ND			
A/Athens.GR/61/2020	6B.1A5A	2020-01-13	MDCK1	160	160	640	640	1280	640	640	2560	ND	640	640	640	640	ND			
A/Athens.GR/129/2020	6B.1A5A	2020-01-16	MDCK1	320	160	1280	640	1280	1280	640	2560	ND	1280	1280	640	1280	ND			
A/Athens.GR/156/2020	6B.1A5A	2020-01-17	MDCK1	320	160	1280	640	1280	1280	640	2560	ND	640	1280	640	1280	ND			
A/Valladolid/301/2019	6B.1A5B	2019-12-13	MDCK1/MDCK1	320	80	1280	1280	2560	1280	640	2560	ND	640	640	640	1280	ND			
A/Valladolid/302/2019	6B.1A5B	2019-12-15	MDCK1/MDCK2	320	80	1280	640	1280	1280	640	1280	ND	640	640	640	1280	ND			
A/Valladolid/306/2019	6B.1A5B	2019-12-23	MDCK1/MDCK1	320	80	1280	640	1280	1280	640	1280	ND	640	640	640	1280	ND			
A/Valladolid/305/2019	6B.1A5B	2019-12-23	MDCK1/MDCK1	320	80	1280	1280	1280	1280	640	2560	ND	640	1280	640	1280	ND			
A/Valladolid/307/2019	6B.1A5B	2019-12-26	MDCK1/MDCK1	160	80	640	640	1280	640	640	1280	ND	320	640	640	640	ND			
A/Valladolid/311/2019	6B.1A5B	2019-12-30	MDCK1/MDCK2	160	40	320	320	320	640	320	640	ND	320	640	320	320	ND			

Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80; ND =Not Done

Vaccine
NH 2018-19
SH 2019Vaccine
NH 2019-20
SH 2020Vaccine
NH 2020-21

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 5-4. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2020-03-17)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre							
					Post-infection feret antisera							
					A/Mich 45/15 Egg	A/Paris 1447/17 MDCK	A/Bris 02/18 Egg	A/Norway 3433/18 MDCK	A/Denmark 3280/19 MDCK	CNIC-1909 A/G-MSWL1536/19 Egg	A/Swit 3330/17 Egg	A/Ire 84630/18 MDCK
					F31/16 [†]	F03/18 [†]	F09/19 [†]	F04/19 [†]	F08/20 [†]	CNIC F1014/19 [†]	F23/18 [†]	F08/19 [†]
	Passage history				6B.1	6B.1A	6B.1A1	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5B	6B.1A6
	Feret number											
	Genetic group											
REFERENCE VIRUSES												
A/Michigan/45/2015		6B.1	2015-09-07	E3/E3	640	1280	640	1280	40	320	640	640
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	640	1280	640	1280	<	320	640	640
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	1280	2560	1280	2560	40	640	640	1280
A/Norway/3433/2018		6B.1A5A	2018-10-30	MDCK3	320	320	160	640	<	160	160	320
A/Denmark/3280/2019	N156K	6B.1A5A	2019-11-10	MDCK4/MDCK3	40	80	40	160	640	40	40	<
CNIC-1909 (A/Guangong-Maonan/SWL1536/2019)	D187A, Q189E	6B.1A5A	2019-06-17	E2/E8/E1	1280	1280	640	2560	40	640	640	1280
A/Guangong-Maonan/SWL1536/2019	D187A, Q189E	6B.1A5A	2019-06-17	C2/MDCK1	640	1280	640	2560	<	640	640	640
A/Switzerland/3330/2017	clone 35	6B.1A5B	2017-12-20	E6/E2	640	640	320	1280	<	320	640	640
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK3	640	1280	640	1280	<	320	640	640
TEST VIRUSES												
A/Turkmenistan/11/2019		6B.1A5A	2019-02-25	MDCK1	320	640	320	1280	<	320	320	640
A/Turkmenistan/288/2019		6B.1A5A	2019-04-15	MDCK1	320	640	320	1280	<	320	160	320
A/Turkmenistan/21/2019		6B.1A5A	2019-04-15	MDCK1	320	640	320	1280	<	320	320	640
A/Netherlands/10258/2019		6B.1A5A	2019-10-31	MDCK2	320	640	320	1280	<	320	320	640
A/England/353/2019		6B.1A5A	2019-11-06	MDCK1/MDCK1	640	1280	320	1280	<	320	320	640
A/England/430/2019		6B.1A5A	2019-11-15	SIAT2/MDCK1	640	1280	320	1280	<	640	320	640
A/England/431/2019		6B.1A5A	2019-11-21	MDCK1/MDCK1	640	1280	640	2560	<	640	640	640
A/Yaroslavl/234/2019		6B.1A5A	2019-12-02	MDCK1/MDCK1	320	640	320	640	<	160	160	320
A/Moscow/226/2019		6B.1A5A	2019-12-04	MDCK1/MDCK1	640	640	320	1280	<	320	320	320
A/South Lebanon_Saida/9/2019		6B.1A5A	2019-12-11	MDCK2	<	<	<	80	320	<	<	<
A/Beirut/594/2019		6B.1A5A	2019-12-14	MDCK1	640	1280	640	2560	<	640	640	640
A/Beirut/601/2019		6B.1A5A	2019-12-17	MDCK1	640	640	320	1280	<	320	320	640
A/Khmelnytskyi/373/2019		6B.1A5A	2019-12-19	MDCK1	320	640	320	1280	<	320	320	320
A/Ukraine/8364/2019		6B.1A5A	2019-12-20	MDCK1	320	640	320	1280	40	320	320	640
A/Kazakhstan/232/2019		6B.1A5A	2019-12-22	MDCK1	640	1280	640	2560	<	320	640	640
A/Kazakhstan/233/2019		6B.1A5A	2019-12-23	MDCK1	<	<	<	80	320	<	<	<
A/Ukraine/8367/2019		6B.1A5A	2020-01-01	MDCK1	640	1280	640	2560	<	320	640	640
A/Khmelnytskyi/12/2020		6B.1A5A	2020-01-01	MDCK1	640	1280	640	1280	<	640	320	640
A/Kazakhstan/270/2020		6B.1A5A	2020-01-01	MDCK1	320	640	320	640	<	320	160	320
A/Khmelnytskyi/15/2020		6B.1A5A	2020-01-02	MDCK1	640	1280	640	2560	<	640	640	640
A/Ukraine/8370/2019		6B.1A5A	2020-01-03	MDCK1	640	1280	320	1280	40	320	320	640
A/Khmelnytskyi/10/2020		6B.1A5A	2020-01-03	MDCK1	40	80	40	320	<	40	40	<
A/Kazakhstan/278/2020		6B.1A5A	2020-01-06	MDCK1	320	640	320	1280	<	320	320	640
A/Athens.GR/145/2020		6B.1A5A	2020-01-17	MDCK2	<	<	<	80	320	<	40	<
A/Targu-Mures/256559/2020		6B.1A5A	2020-01-20	MDCK1	320	640	320	640	<	320	160	320
A/Lesvos Island.GR/4/2020		6B.1A5A		MDCK3	640	1280	640	2560	<	640	640	640

[†] Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80; ND =Not Done

Vaccine
NH 2018-19
SH 2019

Vaccine
NH 2019-20
SH 2020

Vaccine
NH 2020-21

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 5-5. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2020-03-24)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre							
					Post-infection fereret antisera							
					A/Mich 45/15 Egg	A/Paris 1447/17 MDCK	A/Bris 02/18 Egg	A/Norway 3433/18 MDCK	A/Denmark 3280/19 MDCK	CNIC-1909 A/G-MSWL1536/19 Egg	A/Swit 3330/17 Egg	A/Ire 84630/18 MDCK
					F31/16 [†]	F03/18 [‡]	F09/19 [†]	F04/19 [†]	F08/20 [†]	CNIC F1014/19	F23/18 [†]	F08/19 [†]
					6B.1	6B.1A	6B.1A1	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5B	6B.1A6
REFERENCE VIRUSES												
A/Michigan/45/2015		6B.1	2015-09-07	E3/E3	640	160	640	1280	<	640	640	640
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	1280	2560	1280	2560	<	640	1280	1280
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	1280	2560	640	2560	<	640	640	1280
A/Norway/3433/2018		6B.1A5A	2018-10-30	MDCK3	320	640	320	1280	<	320	320	320
A/Denmark/3280/2019	N156K	6B.1A5A	2019-11-10	MDCK4/MDCK3	40	40	40	160	640	40	40	<
CNIC-1909 (A/Guangong-Maonan/SWL1536/2019)	D187A, Q189E	6B.1A5A	2019-06-17	E2/E8/E1	2560	2560	1280	2560	80	1280	1280	2560
A/Switzerland/3330/2017	clone 35	6B.1A5B	2017-12-20	E6/E2	640	1280	640	1280	<	320	640	640
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK3	1280	2560	1280	2560	<	640	1280	2560
TEST VIRUSES												
A/Mauritius/I-2963/2019		6B.1A5A	2019-11-13	MDCK2/MDCK1	640	640	320	1280	<	320	320	640
A/Mauritius/I-2993/2019		6B.1A5A	2019-11-18	MDCK1/MDCK1	640	1280	640	2560	<	640	640	640
A/Lithuania/MB39709/2019		6B.1A5A	2019-11-20	MDCK2/MDCK1	320	640	320	640	<	320	320	640
A/Lithuania/MB39710/2019		6B.1A5A	2019-11-27	MDCK2/MDCK1	320	640	160	640	<	320	160	320
A/Lithuania/MB39704/2019		6B.1A5A	2019-12-01	MDCK2/MDCK1	640	640	320	1280	<	640	320	640
A/Mauritius/I-3030/2019		6B.1A5A	2019-12-02	MDCK1/MDCK1	640	1280	640	1280	<	640	320	640
A/Lithuania/MB40148/2019		6B.1A5A	2019-12-04	MDCK2/MDCK1	320	640	160	640	<	320	160	320
A/Norway/2970/2019		6B.1A5A	2019-12-24	MDCK1/MDCK2	<	<	<	80	640	<	<	<
A/Norway/2969/2019		6B.1A5A	2019-12-24	MDCK1	640	1280	640	1280	<	640	320	640
A/Mauritius/I-11/2020		6B.1A5A	2020-01-01	MDCKx/MDCK1	40	<	<	80	640	<	<	<
A/Hessen/2/2020		6B.1A5A	2020-01-13	C1/MDCK1	640	1280	640	2560	40	640	640	640
A/Sachsen/3/2020		6B.1A5A	2020-01-14	C1/MDCK1	80	160	40	640	80	40	80	40
A/Baden-Wurttemberg/11/2020		6B.1A5A	2020-01-20	C1/MDCK1	640	640	320	1280	<	640	320	640
A/Rheinland-Pfalz/12/2020		6B.1A5A	2020-01-30	C1/MDCK1	640	640	320	1280	<	320	320	320
A/Norway/2906/2019		6B.1A7	2019-12-16	MDCK1/MDCK1	1280	2560	1280	2560	<	640	640	1280
A/Norway/2886/2019		6B.1A7	2019-12-18	MDCK1	640	2560	1280	2560	<	640	640	1280
A/Norway/3118/2019		6B.1A7	2019-12-29	MDCK1	640	1280	640	2560	<	640	320	640
A/Thuringen/17/2020	N156K, K130N, L161I, V47A, R113S, H138R, K209M, V250A	6B.1A5A	2020-02-01	C1/MDCK1	<	<	<	40	320	<	<	<
A/Berlin/28/2020	N156K, K130N, L161I, V250A	6B.1A5A	2020-02-01	C1/MDCK1	40	<	40	160	1280	40	40	<
A/Bremen/5/2020		6B.1A5A	2020-02-04	C1/MDCK1	1280	2560	640	2560	<	1280	640	1280
A/Baden-Wurttemberg/73/2020		6B.1A5A	2020-02-11	C1/MDCK1	640	640	320	1280	<	640	320	640
A/Mecklenburg-Vorpommern/3/2020		6B.1A5A	2020-02-11	C1/MDCK1	1280	1280	640	2560	<	1280	640	1280

[†] Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80; ND =Not Done

Vaccine
NH 2018-19
SH 2019

Vaccine
NH 2019-20
SH 2020

Vaccine
NH 2020-21

< 4-fold
4-fold
8-fold
> 8-fold
not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 5-6. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2020-06-09)

					Haemagglutination inhibition titre										
					Post-infection fereret antisera										
Viruses	Other information	Passage history	Collection date	Passage history	NEW					NEW					
					A/Mich 45/15 Egg	A/Paris 1447/17 MDCK	A/Bris 02/18 Egg	A/Norway 3433/18 MDCK	A/Denmark 3280/19 MDCK	A/Ire 87733/19 Egg	A/G-M SWL1536/19 Egg	A/G-M SWL1536/19 Egg	A/G-M SWL1536/19 MDCK	A/Swit 3330/17 Egg	A/Ire 84630/18 MDCK
					F31/16 [†]	F03/18 ²	F09/19 [†]	F04/19 [†]	F08/20 [†]	St Jude's F18/20 [†]	F12/20 [†]	St Jude's F19/20 [†]	F09/20 [†]	F23/18 [†]	F08/19 [†]
					Genetic group	6B.1	6B.1A	6B.1A1	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5B
REFERENCE VIRUSES															
A/Michigan/45/2015		6B.1	2015-09-07	E3/E3	640	1280	640	1280	40	640	640	640	1280	640	640
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	1280	2560	1280	2560	<	1280	1280	2560	2560	1280	2560
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	1280	2560	1280	2560	40	1280	1280	2560	1280	1280	1280
A/Norway/3433/2018		6B.1A5A	2018-10-30	MDCK3	320	640	320	1280	<	320	640	1280	640	320	640
A/Denmark/3280/2019	N156K	6B.1A5A	2019-11-10	MDCK4/MDCK3	80	80	80	80	1280	80	40	320	160	40	40
A/Ireland/87733/2019	D187A, Q189E	6B.1A5A	2019-11-03	E3	640	1280	640	2560	40	1280	1280	2560	1280	640	1280
A/Guangong-Maonan/SWL1536/2019	D187A, Q189E	6B.1A5A	2019-06-17	E3/E1	640	1280	640	2560	<	640	1280	2560	1280	640	640
A/Guangong-Maonan/SWL1536/2019	D187A, Q189E	6B.1A5A	2019-06-17	C2/MDCK1	640	1280	640	2560	<	640	1280	2560	1280	640	1280
A/Switzerland/3330/2017	clone 35	6B.1A5B	2017-12-20	E6/E2	640	1280	640	2560	<	320	640	1280	640	640	640
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK3	640	1280	640	1280	<	640	1280	1280	640	640	1280
TEST VIRUSES															
A/Bulgaria/886/2020		6B.1A5A	2020-02-03	SIAT2/MDCK1	640	1280	640	1280	<	640	1280	1280	1280	640	640
A/Slovenia/973/2020		6B.1A5A	2020-02-03	MDCKx/MDCK1	640	1280	320	1280	<	320	640	1280	640	320	320
A/Portugal/GG64/2020		6B.1A5A	2020-02-03	MDCK2/MDCK1	640	1280	640	1280	<	640	1280	2560	1280	320	640
A/Portugal/SU283/2020		6B.1A5A	2020-02-04	MDCK3/MDCK1	640	1280	640	1280	<	640	1280	1280	1280	320	640
A/Bulgaria/885/2020		6B.1A5A	2020-02-05	SIAT2/MDCK1	640	1280	640	1280	<	640	640	1280	640	320	320
A/Bulgaria/918/2020		6B.1A5A	2020-02-06	SIAT2/MDCK1	1280	2560	640	2560	<	1280	2560	2560	1280	640	1280
A/Pays de Loire/1017/2020		6B.1A5A	2020-02-07	MDCK1/MDCK1	640	1280	640	2560	<	640	1280	1280	1280	640	640
A/Portugal/EVA260/2020		6B.1A5A	2020-02-08	MDCK3/MDCK1	1280	2560	1280	2560	<	640	1280	1280	1280	640	1280
A/Portugal/EVA259/2020		6B.1A5A	2020-02-08	MDCK1/MDCK1	1280	1280	640	1280	<	640	1280	1280	640	320	640
A/Portugal/SU280/2020		6B.1A5A	2020-02-09	MDCK2/MDCK1	640	1280	640	1280	<	320	640	1280	640	320	640
A/Portugal/GG67/2020		6B.1A5A	2020-02-10	MDCK2/MDCK1	640	1280	640	1280	<	640	640	1280	1280	320	640
A/Slovenia/1261/2020	N156K, K130N, L161I, V24I, I116V, K209M, V250A	6B.1A5A	2020-02-11	SIATx/MDCK1	40	<	40	80	640	40	40	80	40	40	<
A/Sachsen-Anhalt/20/2020		6B.1A5A	2020-02-13	C1/MDCK1	1280	1280	1280	2560	<	640	2560	2560	2560	640	1280
A/Portugal/EVA277/2020		6B.1A5A	2020-02-13	MDCK3/MDCK1	640	1280	640	2560	<	640	1280	1280	1280	640	640
A/Portugal/GG69/2020		6B.1A5A	2020-02-13	MDCK1/MDCK1	1280	1280	640	2560	<	640	1280	2560	1280	640	1280
A/Bulgaria/988/2020		6B.1A5A	2020-02-14	SIAT2/MDCK1	640	640	320	1280	<	640	640	1280	640	320	320
A/Niedersachsen/52/2020	N156K, K130N, L161I, K209M, V250A	6B.1A5A	2020-02-14	C1/MDCK1	40	<	<	80	320	<	40	80	40	<	<
A/Saarland/6/2020	N156K, K130N, L161I, K209M, V250A	6B.1A5A	2020-02-17	C1/MDCK1	40	40	40	160	640	40	40	160	80	40	40
A/Haute Normandie/1230/2020	E112D, V47A, R205G	6B.1A5A	2020-02-18	MDCK1/MDCK1	80	<	40	160	640	40	80	160	80	40	40
A/Portugal/EVA316/2020		6B.1A5A	2020-02-18	MDCK1/MDCK1	640	1280	640	2560	<	640	1280	2560	1280	640	640
A/Bulgaria/1030/2020		6B.1A5A	2020-02-19	SIAT2/MDCK1	640	1280	640	2560	40	640	1280	2560	2560	640	1280
A/Sachsen/49/2020		6B.1A5A	2020-02-20	C1/MDCK1	1280	1280	640	2560	<	640	1280	2560	1280	640	1280
A/Centre/1537/2020		6B.1A5A	2020-02-24	MDCK1/MDCK1	40	40	40	160	80	80	80	40	40	40	40
A/Slovenia/1564/2020		6B.1A5A	2020-02-24	SIATx/MDCK1	640	1280	640	1280	40	640	1280	2560	1280	320	640
A/Brandenburg/8/2020	N156K, K130N, L161I, V250A	6B.1A5A	2020-02-24	C1/MDCK1	40	40	40	80	640	40	40	80	40	40	<
A/Mecklenburg-Vorpommern/4/2020	N156K, K130N, L161I, P137S, V250A	6B.1A5A	2020-02-24	C1/MDCK1	40	<	40	80	640	40	40	80	40	40	40
A/Portugal/EVA311/2020		6B.1A5A	2020-02-24	MDCK2/MDCK1	640	1280	640	1280	<	640	1280	1280	1280	320	640
A/Baden-Wuerttemberg/116/2020		6B.1A5A	2020-02-28	C1/MDCK1	1280	1280	640	2560	40	640	1280	2560	1280	640	640
A/Schleswig-Holstein/8/2020		6B.1A5A	2020-02-28	C1/MDCK1	640	1280	640	1280	40	640	1280	1280	640	320	640
A/Hessen/32/2020		6B.1A5A	2020-03-02	C1/MDCK1	640	1280	640	1280	40	640	1280	1280	1280	320	640
A/Thuringen/44/2020		6B.1A5A	2020-03-02	C1/MDCK1	640	1280	640	1280	40	640	1280	1280	640	640	640
A/Bulgaria/1243/2020		6B.1A5A	2020-03-04	SIAT2/MDCK1	640	1280	640	1280	<	640	1280	2560	1280	640	640
A/Portugal/EVA239/2020		6B.1A5B	2020-02-02	MDCK1/MDCK1	1280	2560	1280	1280	80	640	1280	1280	1280	640	1280

[†] Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80; ND =Not Done

Vaccine
NH 2018-19
SH 2019Vaccine
NH 2019-20
SH 2020Vaccine
NH 2020-21
NH 2020-21

< 4-fold

4-fold

8-fold

> 8-fold

< not recognised by the antiserum

≥ 160 (when homologous titre ≥ 2560)

≥ 160 (no homologous titre)

Table 5-7. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2020-06-19)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/Mich 45/15 Egg	A/Paris 1447/17 MDCK	A/Bris 02/18 Egg	A/Norway 3433/18 MDCK	A/Denmark 3280/19 MDCK	A/Ire 87733/19 Egg	A/G-M SWL1536/19 Egg	A/G-M SWL1536/19 MDCK	A/Swit 3330/17 Egg	A/Ire 84630/18 MDCK
					F31/16 ¹	F03/18 ²	F09/19 ¹	F04/19 ¹	F08/20 ¹	St Jude's F18/20	F12/20 ¹	F09/20 ¹	F23/18 ¹	F08/19 ¹
		Genetic group			6B.1	6B.1A	6B.1A1	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5B	6B.1A6
REFERENCE VIRUSES														
A/Michigan/45/2015		6B.1	2015-09-07	E3/E3	640	1280	640	1280	40	640	640	1280	640	1280
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	1280	2560	1280	2560	<	1280	1280	2560	1280	2560
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	1280	2560	1280	2560	40	1280	1280	1280	1280	1280
A/Norway/3433/2018		6B.1A5A	2018-10-30	MDCK3	320	640	320	1280	<	640	640	640	320	640
A/Denmark/3280/2019	N156K	6B.1A5A	2019-11-10	MDCK4/MDCK3	80	80	80	80	640	80	40	160	40	40
A/Ireland/87733/2019		6B.1A5A	2019-11-03	E3	1280	1280	640	2560	40	1280	1280	1280	1280	1280
A/Guangong-Maonan/SWL1536/2019	D187A, Q189E	6B.1A5A	2019-06-17	E3/E1	640	640	640	2560	<	640	1280	1280	640	640
A/Guangong-Maonan/SWL1536/2019	D187A, Q189E	6B.1A5A	2019-06-17	C2/MDCK1	640	1280	1280	2560	<	640	1280	1280	640	1280
A/Switzerland/3330/2017	clone 35	6B.1A5B	2017-12-20	E6/E2	640	1280	1280	2560	<	320	640	640	640	640
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK3	640	1280	640	1280	<	640	1280	640	1280	1280
TEST VIRUSES														
A/Centre/675/2020		6B.1A5A	2020-02-03	MDCK1/MDCK2	640	1280	640	2560	<	640	1280	1280	640	1280
A/England/91/2020		6B.1A5A	2020-02-06	SIAT1/MDCK1	1280	1280	1280	2560	<	1280	1280	1280	1280	1280
A/Belgium/S1052/2020		6B.1A5A	2020-02-04	MDCK1/MDCK1	1280	1280	640	2560	<	640	1280	1280	640	640
A/Belgium/S0547/2020		6B.1A5A	2020-02-04	MDCK1/MDCK1	1280	1280	640	2560	<	640	1280	1280	640	640
A/Belgium/S0546/2020		6B.1A5A	2020-02-04	MDCK1/MDCK1	640	1280	640	1280	<	320	1280	1280	640	640
A/Belgium/S0535/2020		6B.1A5A	2020-02-04	MDCK1/MDCK1	640	1280	640	1280	<	320	1280	1280	320	640
A/Belgium/G0204/2020	S157L, R74K, P137S	6B.1A5A	2020-02-04	MDCK1/MDCK1	160	320	160	640	<	80	160	640	80	80
A/Belgium/S1046/2020		6B.1A5A	2020-02-07	MDCK1/MDCK1	1280	1280	640	1280	<	640	1280	1280	640	640
A/Belgium/S0707/2020		6B.1A5A	2020-02-07	MDCK1/MDCK1	1280	1280	640	2560	<	640	1280	2560	640	640
A/Belgium/S1271/2020		6B.1A5A	2020-02-09	MDCK1/MDCK1	1280	1280	640	1280	<	640	1280	1280	640	640
A/Banska Bystrica/525/2020		6B.1A5A	2020-02-10	MDCK1/MDCK1	640	1280	640	1280	<	640	1280	1280	640	1280
A/Belgium/G0255/2020		6B.1A5A	2020-02-10	MDCK1/MDCK1	640	1280	640	1280	<	640	1280	1280	640	640
A/Belgium/G0237/2020		6B.1A5A	2020-02-10	MDCK1/MDCK1	640	1280	640	1280	<	640	1280	1280	640	640
A/Belgium/S0960/2020		6B.1A5A	2020-02-11	MDCK1/MDCK1	640	1280	640	1280	<	640	640	1280	640	640
A/Belgium/G0290/2020	S157L, R74K, P137S	6B.1A5A	2020-02-11	MDCK1/MDCK1	320	320	160	640	<	160	320	640	160	80
A/Belgium/S1292/2020	N156K, K130N, L161I, I5M, V19I, K209M, V250A	6B.1A5A	2020-02-13	MDCK1/MDCK1	<	<	<	40	320	<	<	<	<	<
A/Belgium/G0311/2020		6B.1A5A	2020-02-14	MDCK1/MDCK1	640	1280	320	1280	<	320	640	640	320	640
A/Belgium/S0993/2020		6B.1A5A	2020-02-16	MDCK1/MDCK1	1280	1280	640	2560	<	640	1280	1280	320	1280
A/Belgium/S0863/2020		6B.1A5A	2020-02-16	MDCK1/MDCK1	640	1280	640	1280	<	320	640	1280	320	640
A/Trnava/294/2020		6B.1A5A	2020-02-17	MDCKx/MDCK1	1280	1280	640	1280	40	640	1280	1280	640	1280
A/Belgium/G0354/2020	N156K, K130N, L161I, K209M, V250A	6B.1A5A	2020-02-17	MDCK1/MDCK1	<	<	<	40	320	<	<	<	<	<
A/Belgium/G0338/2020		6B.1A5A	2020-02-18	MDCK1/MDCK1	640	1280	640	1280	<	640	1280	1280	640	640
A/Belgium/S1115/2020		6B.1A5A	2020-02-19	MDCK1/MDCK1	1280	1280	1280	2560	<	640	2560	1280	640	1280
A/Belgium/G0353/2020		6B.1A5A	2020-02-19	MDCK1/MDCK1	640	1280	640	1280	<	640	640	1280	640	640
A/Belgium/G0350/2020		6B.1A5A	2020-02-19	MDCK1/MDCK1	1280	1280	640	2560	<	640	1280	1280	640	1280
A/Iceland/04/2020	N156K, K130N, L161I, P137S, V250A	6B.1A5A	2020-02-20	MDCK1/MDCK1	40	<	<	80	640	40	40	40	40	<
A/Belgium/S1146/2020		6B.1A5A	2020-02-23	MDCK1/MDCK1	640	640	640	1280	<	320	1280	1280	320	640
A/Iceland/11/2020	N156K, K130N, L161I, V250A	6B.1A5A	2020-02-24	MDCK1/MDCK1	80	80	80	160	640	80	80	80	80	40
A/Iceland/10/2020	N156K, K130N, L161I, K154E, D187V, V250A	6B.1A5A	2020-02-24	MDCK1/MDCK1	<	<	40	40	640	<	40	<	<	<
A/Belgium/G0406/2020		6B.1A5A	2020-02-24	MDCK1/MDCK1	1280	1280	640	1280	<	640	1280	1280	640	1280
A/Belgium/S1089/2020	N156K, K130N, L161I, V47I, V250A	6B.1A5A	2020-02-25	MDCK1/MDCK1	40	<	40	80	640	40	40	40	40	<
A/Belgium/S1308/2020		6B.1A5A	2020-02-26	MDCK1/MDCK1	1280	1280	640	1280	<	640	1280	1280	640	1280
A/Belgium/S1155/2020	N156K, K130N, L161I, V250A	6B.1A5A	2020-02-26	MDCK1/MDCK1	80	80	80	160	640	80	80	80	80	40
A/Belgium/H0024/2020		6B.1A5A	2020-02-26	MDCK1/MDCK1	1280	1280	640	2560	<	640	1280	1280	640	1280
A/Belgium/G0418/2020		6B.1A5A	2020-02-26	MDCK1/MDCK1	1280	1280	640	2560	<	640	1280	1280	640	1280
A/Iceland/15/2020	N156K, K130N, L161I, V250A	6B.1A5A	2020-02-27	MDCK1/MDCK1	40	<	40	80	640	40	40	40	40	<
A/Belgium/S1200/2020		6B.1A5A	2020-03-01	MDCK1/MDCK1	640	640	640	1280	<	320	640	640	320	640
A/Belgium/S1317/2020		6B.1A5A	2020-03-03	MDCK1/MDCK1	1280	1280	640	1280	<	640	1280	1280	640	640
A/Lipetsk/04/2020		6B.1A5A	2020-03-04	MDCK3/MDCK1	160	320	160	640	<	160	320	160	160	160
A/Belgium/S1336/2020		6B.1A5A	2020-03-05	MDCK1/MDCK1	640	1280	640	1280	<	640	1280	1280	320	640
A/Belgium/G0467/2020		6B.1A5A	2020-03-05	MDCK1/MDCK1	640	1280	640	1280	<	640	1280	1280	320	640
A/Iceland/22/2020	N156K, K130N, L161K, P137S, V250A	6B.1A5A	2020-03-07	MDCK1/MDCK1	<	<	<	80	160	<	40	40	<	<
A/Iceland/24/2020		6B.1A5A	2020-03-08	MDCK1/MDCK1	640	640	320	640	<	320	640	640	320	640
A/S. Petersburg/R11-179/2020		6B.1A5A	2020-03-10	MDCK2/MDCK1	640	640	640	1280	<	320	640	1280	640	640
A/Iceland/34/2020		6B.1A5A	2020-03-22	MDCK1/MDCK1	640	640	320	1280	<	320	640	640	320	640
A/Iceland/36/2020		6B.1A5A	2020-03-25	MDCK1/MDCK1	640	1280	640	1280	<	640	1280	1280	160	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80; ND =Not Done

Vaccine

NH 2018-19

SH 2019

Vaccine

NH 2019-20

SH 2020

Vaccine

NH 2020-21

< 4-fold

4-fold

8-fold

> 8-fold

< not recognised by the antiserum

≥ 160 (when homologous titre ≥ 2560)

≥ 160 (no homologous titre)

Table 5-8. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2020-06-26)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/Mich 45/15 Egg	A/Paris 1447/17 MDCK	A/Bris 02/18 Egg	A/Norway 3433/18 MDCK	A/Denmark 3280/19 MDCK	A/Ire 87733/19 Egg	A/G-M SWL1536/19 Egg	A/G-M SWL1536/19 MDCK	A/Swit 3330/17 Egg	A/Ire 84630/18 MDCK
					F31/16 ¹	F03/18 ²	F09/19 ¹	F04/19 ¹	F08/20 ¹	St Jude's F18/20	F12/20 ¹	F09/20 ¹	F23/18 ¹	F08/19 ¹
		Passage history			6B.1	6B.1A	6B.1A1	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5B	6B.1A6
REFERENCE VIRUSES														
A/Michigan/45/2015			2015-09-07	E3/E3	640	1280	640	1280	40	640	640	1280	640	1280
A/Paris/1447/2017			2017-10-20	MDCK1/MDCK3	1280	2560	1280	2560	<	1280	1280	2560	1280	2560
A/Brisbane/02/2018			2018-01-04	E3/E1	1280	2560	1280	2560	40	1280	1280	1280	1280	1280
A/Norway/3433/2018			2018-10-30	MDCK3	320	640	640	1280	<	640	640	640	320	640
A/Denmark/3280/2019	N156K		2019-11-10	MDCK4/MDCK3	80	80	80	160	1280	80	80	160	80	40
A/Ireland/87733/2019			2019-11-03	E4	1280	1280	640	2560	40	1280	1280	1280	1280	1280
A/Guangong-Maonan/SWL1536/2019	D187A, Q189E		2019-06-17	E3/E1	640	640	640	2560	<	640	1280	1280	640	640
A/Guangong-Maonan/SWL1536/2019	D187A, Q189E		2019-06-17	C2/MDCK1	640	1280	1280	2560	<	640	1280	1280	640	1280
A/Switzerland/3330/2017	clone 35		2017-12-20	E6/E2	640	1280	1280	2560	<	320	640	640	640	640
A/Ireland/84630/2018			2018-11-28	MDCK1/MDCK3	640	1280	640	1280	<	640	1280	1280	1280	1280
TEST VIRUSES														
A/Finland/173/2020			2020-01-25	MDCK1/MDCK1	320	1280	320	1280	<	320	640	320	320	320
A/Finland/177/2020			2020-01-29	MDCK1/MDCK1	640	1280	640	2560	<	640	1280	1280	640	640
A/Finland/151/2019			2019-12-12	MDCK1/MDCK1	640	2560	640	1280	80	640	1280	1280	640	1280
A/Stockholm/14/2020	N156K, K130N, L161I, K209M, V250A		2020-02-03	SIAT0/MDCK1	80	<	40	160	640	40	40	80	40	<
A/Lund/3/2020			2020-02-04	SIAT0/MDCK1	640	1280	640	2560	40	640	1280	1280	640	640
A/Linköping/3/2020	N156K, K130N, L161I, K209M, V250A		2020-02-08	SIAT0/MDCK1	40	<	<	160	640	<	40	40	40	<
A/Milano/47/2020			2020-02-08	MDCK2/MDCK1	1280	2560	1280	2560	<	640	1280	1280	640	1280
A/Friuli Venezia Giulia/237/2020			2020-02-08	MDCK2/MDCK1	640	1280	640	1280	<	640	1280	1280	640	640
A/Gavle/1/2020	N156K, K130N, L161I, I179V, K209M, V250A		2020-02-10	SIAT0/MDCK1	<	<	<	40	160	<	<	<	<	<
A/Stockholm/29/2020			2020-02-11	SIAT0/MDCK1	640	640	320	1280	<	320	1280	640	320	640
A/Finland/189/2020			2020-02-11	MDCK1/MDCK1	1280	1280	640	2560	<	640	1280	1280	640	1280
A/Milano/59/2020			2020-02-11	MDCK3/MDCK1	1280	2560	1280	2560	<	640	1280	1280	640	1280
A/Halmstad/1/2020	N156K, K130N, L161I, K209M, V250A		2020-02-13	SIAT0/MDCK1	40	<	40	160	640	40	40	40	40	<
A/Lund/4/2020			2020-02-17	SIAT0/MDCK1	640	640	640	2560	40	640	1280	1280	640	640
A/Finland/191/2020	N156K, K130N, L161I, V250A		2020-02-17	MDCK1/MDCK1	80	80	40	160	640	40	80	80	40	<
A/Milano/79/2020			2020-02-17	MDCK3/MDCK1	1280	2560	1280	2560	<	640	1280	1280	640	1280
A/Finland/208/2020	N156K, K130N, L161I, V250A		2020-02-18	MDCK1/MDCK1	40	<	40	80	320	40	40	40	40	<
A/Parma/27/2020			2020-02-18	MDCK2/MDCK1	<	<	<	40	160	<	<	<	<	<
A/Finland/206/2020			2020-02-19	MDCK1/MDCK1	640	640	640	2560	<	640	1280	1280	640	640
A/Milano/77/2020			2020-02-19	MDCK3/MDCK1	640	1280	640	2560	<	640	1280	1280	640	1280
A/Parma/32/2020			2020-02-20	MDCK2/MDCK1	<	<	<	40	160	<	<	<	<	<
A/Visby/1/2020	N156K, F117L, K130N, L161I, V250A		2020-02-22	SIAT0/MDCK2	<	<	<	80	320	<	40	40	40	<
A/Finland/212/2020	N156K, K130N, L161I, K209M, V250A		2020-02-26	MDCK1/MDCK1	40	<	<	160	640	<	40	40	40	<

¹ Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80; ND =Not Done

Vaccine
NH 2018-19
SH 2019

Vaccine
NH 2019-20
SH 2020

Vaccine
NH 2020-21

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 5-9. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2020-07-14)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/Mich 45/15 Egg	A/Paris 1447/17 MDCK	A/Bris 02/18 Egg	A/Norway 3433/18 MDCK	A/Denmark 3280/19 MDCK	A/Ire 87733/19 Egg	A/G-M SWL1536/19 Egg	A/G-M SWL1536/19 MDCK	A/Swit 3330/17 Egg	A/Ire 84630/18 MDCK
					F31/16 ⁻¹	F03/16 ⁻²	F09/19 ⁻¹	F04/19 ⁻¹	F08/20 ⁻¹	St Jude's F18/20	F12/20 ⁻¹	F09/20 ⁻¹	F23/19 ⁻¹	F08/19 ⁻¹
					6B.1	6B.1A	6B.1A1	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5B	6B.1A6
REFERENCE VIRUSES														
A/Michigan/45/2015			2015-09-07	E3/E3	640	1280	640	1280	40	640	1280	1280	640	1280
A/Paris/1447/2017			2017-10-20	MDCK1/MDCK3	1280	2560	1280	2560	<	1280	1280	2560	1280	2560
A/Brisbane/02/2018			2018-01-04	E3/E1	1280	2560	1280	1280	40	640	1280	1280	1280	1280
A/Norway/3433/2018			2018-10-30	MDCK3	320	640	640	1280	<	640	640	640	320	640
A/Denmark/3280/2019	N156K		2019-11-10	MDCK4/MDCK3	80	80	80	160	1280	80	80	160	80	40
A/Ireland/87733/2019			2019-11-03	E4	1280	1280	1280	2560	40	1280	1280	1280	1280	1280
A/Guangong-Maonan/SWL1536/2019	D187A, Q189E		2019-06-17	E3/E1	640	640	640	2560	<	640	1280	1280	640	640
A/Guangong-Maonan/SWL1536/2019	D187A, Q189E		2019-06-17	C2/MDCK1	1280	1280	1280	2560	<	640	1280	1280	640	1280
A/Switzerland/3330/2017	clone 35		2017-12-20	E6/E2	640	1280	1280	2560	<	320	640	640	640	640
A/Ireland/84630/2018			2018-11-28	MDCK1/MDCK3	640	1280	640	1280	<	640	1280	1280	1280	1280
TEST VIRUSES														
A/South Africa/9499/2019			2019-07-30	MDCK1/MDCK1	640	640	640	1280	<	320	640	640	320	640
A/South Africa/0337/2020			2020-01-10	MDCK1/MDCK1	640	640	640	1280	<	320	640	640	640	640
A/South Africa/0871/2020			2020-01-28	MDCK1/MDCK1	640	640	1280	1280	<	640	1280	1280	640	1280
A/Estonia/126130/2020			2020-01-31	SIAT1/MDCK1	640	1280	640	2560	<	640	1280	1280	640	1280
A/South Africa/9182/2019			2019-07-22	MDCK1/MDCK1	640	640	640	1280	<	320	1280	640	640	1280
A/South Africa/1277/2020		No seq	2020-02-04	MDCK1/MDCK1	640	640	640	1280	<	640	1280	1280	640	1280
A/South Africa/3610/2020		No seq	2020-03-10	MDCK2/MDCK1	40	<	40	80	320	<	40	<	<	<
A/South Africa/3958/2020		No seq	2020-03-17	MDCK2/MDCK1	40	<	<	160	640	<	40	40	40	<
A/Estonia/126178/2020			2020-02-03	SIAT1/MDCK1	640	640	640	1280	<	320	640	640	320	640
A/Estonia/126128/2020			2020-02-04	SIAT1/MDCK1	1280	1280	1280	2560	<	1280	1280	2560	640	1280
A/Estonia/126062/2020			2020-02-04	SIAT1/MDCK1	640	1280	1280	2560	<	640	1280	1280	640	1280
A/Estonia/126205/2020			2020-02-04	SIAT1/MDCK1	1280	640	1280	2560	<	640	1280	1280	640	1280
A/South Africa/1192/2020			2020-02-04	MDCK1/MDCK1	640	640	640	1280	40	640	1280	1280	640	320
A/South Africa/1190/2020			2020-02-04	MDCK1/MDCK1	1280	1280	1280	2560	40	1280	2560	2560	1280	2560
A/South Africa/1406/2020			2020-02-06	MDCK1/MDCK1	1280	1280	2560	2560	80	1280	2560	2560	1280	2560
A/South Africa/1320/2020			2020-02-06	MDCK1/MDCK1	640	640	1280	2560	40	640	1280	1280	640	1280
A/South Africa/1448/2020			2020-02-07	MDCK1/MDCK1	640	640	1280	2560	40	640	1280	1280	640	1280
A/South Africa/1453/2020			2020-02-09	MDCK1/MDCK1	640	1280	640	2560	<	640	1280	1280	640	1280
A/South Africa/1519/2020			2020-02-10	MDCK1/MDCK1	640	1280	640	1280	<	640	1280	1280	320	640
A/South Africa/1515/2020	N156K, K130N, L161I, K209M, V250A		2020-02-10	MDCK1/MDCK1	80	<	80	160	640	40	40	80	40	40
A/South Africa/1524/2020	N156K, K130N, L161I, K209M, V250A		2020-02-11	MDCK1/MDCK1	<	80	40	80	320	<	40	<	40	<
A/South Africa/1716/2020	N156K, K130N, L161I, K209M, V250A		2020-02-14	MDCK1/MDCK1	40	80	40	80	640	<	40	40	40	<
A/South Africa/1798/2020			2020-02-17	MDCK1/MDCK1	640	1280	640	2560	40	640	1280	1280	640	1280
A/South Africa/2020/2020			2020-02-20	MDCK1/MDCK1	640	1280	640	1280	<	640	1280	1280	320	640
A/South Africa/1938/2020	N156K, K130N, L161I, K209M, V250A		2020-02-20	MDCK1/MDCK1	80	160	40	160	640	40	40	<	40	40
A/South Africa/1873/2020	N156K, K130N, L161I, T216A, V250A		2020-02-20	MDCK1/MDCK1	<	<	40	80	320	<	<	<	<	<
A/South Africa/2506/2020			2020-02-26	MDCK1/MDCK1	1280	2560	1280	2560	<	1280	1280	2560	1280	1280
A/South Africa/2390/2020	N156K, K130N, L161I, K209M, V250A		2020-02-26	MDCK1/MDCK1	40	80	40	80	320	40	40	40	80	<
A/South Africa/2473/2020			2020-02-27	MDCK1/MDCK1	640	1280	640	1280	<	640	1280	640	320	640
A/South Africa/2459/2020	N156K, K130N, L161I, K209M, V250A, S323P		2020-02-27	MDCK1/MDCK1	<	<	<	80	320	<	<	<	<	<
A/South Africa/2503/2020			2020-02-28	MDCK1/MDCK1	640	1280	640	2560	40	640	1280	1280	640	640
A/South Africa/2476/2020			2020-02-28	MDCK1/MDCK1	1280	1280	640	2560	<	640	1280	1280	640	1280
A/South Africa/2732/2020	N156K, K130N, L161I, V250A		2020-03-04	MDCK2/MDCK1	40	80	40	160	320	40	40	40	40	<
A/South Africa/2727/2020			2020-03-05	MDCK1/MDCK1	1280	2560	1280	2560	40	640	1280	1280	640	1280
A/South Africa/3643/2020			2020-03-09	MDCK1/MDCK1	640	1280	640	2560	<	640	1280	1280	640	1280
A/South Africa/3635/2020	N156K, K130N, L161I, K209M, V250A		2020-03-09	MDCK1/MDCK1	80	80	40	160	640	40	40	40	40	<
A/South Africa/3625/2020			2020-03-10	MDCK1/MDCK1	640	1280	640	1280	<	640	640	640	320	640
A/South Africa/3622/2020			2020-03-11	MDCK1/MDCK1	1280	2560	1280	2560	<	640	1280	1280	640	1280
A/South Africa/3830/2020			2020-03-12	MDCK1/MDCK1	640	1280	640	1280	<	320	640	640	320	640
A/South Africa/3796/2020			2020-03-12	MDCK1/MDCK1	1280	2560	1280	2560	<	1280	1280	1280	1280	1280
A/South Africa/6019/2020	N156K, K130N, L161I, K209M, V250A		2020-03-17	MDCK1/MDCK1	<	<	<	80	320	<	<	<	<	<
A/South Africa/3997/2020	N156K, K130N, L161I, K209I, V250A		2020-03-17	MDCK1/MDCK1	<	<	<	80	640	<	40	40	<	<
A/South Africa/3993/2020	N156K, K130N, L161I, K209M, V250A		2020-03-18	MDCK1/MDCK1	40	80	40	160	640	<	40	40	40	<
A/South Africa/6188/2020	N156K, K130N, L161I, K209M, V250A		2020-03-25	MDCK1/MDCK2	<	<	<	80	320	<	<	<	<	<
A/South Africa/6185/2020			2020-03-25	MDCK1/MDCK1	1280	2560	1280	2560	<	1280	1280	640	640	1280
A/South Africa/6174/2020			2020-03-25	MDCK1/MDCK1	1280	2560	1280	2560	<	1280	1280	640	640	1280
A/South Africa/6433/2020	N156K, K130N, L161I, K209I, V250A, T310K		2020-04-01	MDCK1/MDCK1	<	<	<	80	160	<	<	<	40	<

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80; ND =Not Done

Vaccine
NH 2018-19
SH 2019Vaccine
NH 2019-20
SH 2020Vaccine
NH 2020-21

 < 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 5-10. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2020-07-17)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection fereret antisera									
					A/Mich 45/15 Egg	A/Paris 1447/17 MDCK	A/Bris 02/18 Egg	A/Norway 3433/18 MDCK	A/Denmark 3280/19 MDCK	A/Ire 87733/19 Egg	A/G-M SWL1536/19 Egg	A/G-M SWL1536/19 MDCK	A/Swit 3330/17 Egg	A/Ire 84630/18 MDCK
					F31/16 ¹	F03/18 ²	F09/19 ¹	F04/19 ¹	F08/20 ¹	St Jude's F18/20	F12/20 ¹	F09/20 ¹	F23/18 ¹	F08/19 ¹
	Passage history	Genetic group			6B.1	6B.1A	6B.1A1	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5B	6B.1A6
REFERENCE VIRUSES														
A/Michigan/45/2015		6B.1	2015-09-07	E3/E3	640	1280	640	1280	40	640	1280	1280	640	1280
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	1280	2560	1280	2560	<	1280	1280	2560	640	2560
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	1280	2560	1280	1280	40	640	1280	1280	1280	1280
A/Norway/3433/2018		6B.1A5A	2018-10-30	MDCK3	320	640	640	1280	<	640	640	640	320	640
A/Denmark/3280/2019	N156K	6B.1A5A	2019-11-10	MDCK4/MDCK3	80	80	80	160	1280	80	80	80	80	40
A/Ireland/87733/2019		6B.1A5A	2019-11-03	E4	1280	1280	1280	2560	40	1280	1280	1280	1280	1280
A/Guangong-Maonan/SWL1536/2019	D187A, Q189E	6B.1A5A	2019-06-17	E3/E1	640	640	640	2560	<	640	1280	1280	640	1280
A/Guangong-Maonan/SWL1536/2019	D187A, Q189E	6B.1A5A	2019-06-17	C2/MDCK1	640	640	640	2560	<	640	640	1280	640	640
A/Switzerland/3330/2017	clone 35	6B.1A5B	2017-12-20	E6/E2	640	1280	1280	1280	<	320	640	640	640	640
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK3	640	1280	640	1280	<	640	1280	1280	1280	1280
TEST VIRUSES														
A/Portugal/EVA265/2020		No seq	2020-02-03	MDCK3	40	<	<	80	640	<	<	40	<	<
A/Paris/1066/2020		6B.1A5A	2020-02-11	MDCK1	640	1280	640	1280	<	320	640	640	320	640
A/Saint-Etienne/2546/2020		6B.1A5A	2020-02-12	MDCK2/MDCK1	640	1280	1280	2560	40	640	1280	1280	640	1280
A/Cyprus/F1116/2020		6B.1A5A	2020-02-16	MDCK1	640	1280	640	2560	40	640	1280	1280	640	1280
A/Chamagne-Ardenne/1194/2020		6B.1A5A	2020-02-17	MDCK1	640	640	640	1280	<	320	640	640	320	640
A/Saint-Etienne/1762/2020	N156K, K130N, L161I, V250A	6B.1A5A	2020-02-17	MDCK2/MDCK1	40	<	40	160	640	<	40	40	40	<
A/La Rochelle/1578/2020	N156K, K130N, L161I, K209M, V250A	6B.1A5A	2020-02-17	MDCK2/MDCK1	40	<	40	160	640	<	80	40	40	40
A/Portugal/EVA307/2020		6B.1A5A	2020-02-18	MDCK1	1280	1280	640	2560	40	640	1280	640	320	640
A/Cyprus/F1128/2020	N156K, K130N, L161I, G170E, V250A	6B.1A5A	2020-02-19	MDCK1	80	80	80	320	1280	40	80	80	40	40
A/Cyprus/F1163/2020		6B.1A5A	2020-02-21	MDCK1	320	640	320	640	<	320	640	640	160	640
A/Cyprus/F1180/2020		6B.1A5A	2020-02-23	MDCK1	640	1280	1280	2560	40	640	1280	2560	640	1280
A/Guérét/1628/2020		6B.1A5A	2020-02-23	MDCK2/MDCK1	640	1280	1280	2560	40	640	1280	1280	640	1280
A/Poland/241/2020		6B.1A5A	2020-02-24	MDCK2	640	320	640	1280	<	320	640	640	320	640
A/Portugal/SU315/2020	N156K, K130N, L161I, K209M, V250A	6B.1A5A	2020-02-24	MDCK1	40	80	40	160	640	40	40	40	40	<
A/Portugal/SU314/2020		6B.1A5A	2020-02-24	MDCK1	320	320	320	640	<	160	320	320	160	320
A/Portugal/SU304/2020		6B.1A5A	2020-02-24	MDCK1	640	640	640	1280	40	320	640	640	320	640
A/La Rochelle/1577/2020		6B.1A5A	2020-02-24	MDCK2/MDCK1	1280	1280	1280	2560	40	640	1280	1280	640	1280
A/Poland/103/2020		6B.1A5A	2020-02-25	MDCK2	640	640	640	2560	<	320	1280	1280	640	1280
A/Avignon/1629/2020		6B.1A5A	2020-02-25	MDCK2/MDCK1	640	1280	640	2560	<	640	1280	1280	640	1280
A/Pays de Loire/1652/2020		6B.1A5A	2020-02-26	MDCK1	640	640	640	1280	<	320	1280	1280	320	320
A/Portugal/SU306/2020		6B.1A5A	2020-02-27	MDCK1	640	1280	640	1280	<	320	640	640	320	640
A/Cyprus/F1277/2020	N156K, K130N, L161I, S121N, K209M, V250A	6B.1A5A	2020-02-27	MDCK1	40	<	40	160	640	<	40	40	<	<
A/Portugal/SU317/2020	N156K, K130N, L161I, V250A	6B.1A5A	2020-02-29	MDCK1	40	80	40	160	640	40	40	40	40	<
A/Poland/159/2020		6B.1A5A	2020-03-02	MDCK1	640	640	640	1280	<	320	1280	1280	640	640
A/Portugal/EVA343/2020		6B.1A5A	2020-03-02	MDCK1	1280	1280	640	2560	40	640	1280	1280	320	640
A/Poland/169/2020		6B.1A5A	2020-03-03	MDCK2	320	320	320	1280	<	160	640	640	320	640
A/Switzerland/3428/2020	N156K, K130N, L161I, Y198H, K209M, V250A	6B.1A5A	2020-03-04	MDCK1	<	<	<	80	320	<	<	<	<	<
A/Cyprus/F1318/2020		6B.1A5A	2020-03-04	MDCK1	320	640	320	1280	<	320	640	640	160	320
A/Portugal/EVA339/2020		6B.1A5A	2020-03-05	MDCK1	640	1280	640	2560	<	640	1280	1280	320	640
A/Politiors/2589/2020		6B.1A5A	2020-03-05	MDCK3/MDCK1	640	1280	640	2560	<	640	1280	1280	640	1280
A/Paris/2187/2020		6B.1A5A	2020-03-06	MDCK1	640	640	640	2560	40	320	1280	1280	640	640
A/Lyon/1758/2020		6B.1A5A	2020-03-07	MDCK2/MDCK1	640	640	640	1280	<	640	1280	1280	640	640
A/Lyon/CHU-R20.69.78/2020		6B.1A5A	2020-03-09	MDCK2/MDCK1	640	640	640	1280	<	320	640	640	320	640
A/Saint-Etienne/2581/2020	N156K, K130N, L161I, V250A	6B.1A5A	2020-03-09	MDCK3/MDCK1	40	<	40	80	640	40	40	40	40	<
A/Centre/2250/2020		6B.1A5A	2020-03-10	MDCK1	1280	1280	1280	2560	40	640	1280	2560	1280	1280
A/Lyon/1756/2020	N156K, K130N, L161I, V250A	6B.1A5A	2020-03-10	MDCK2/MDCK1	40	<	40	160	640	<	40	40	40	<
A/Switzerland/7603/2020	N156K, K130N, L161I, V250A, P271A	6B.1A5A	2020-03-11	MDCK1	40	40	40	160	640	<	40	40	40	<
A/Switzerland/8628/2020		6B.1A5A	2020-03-12	MDCK1	640	640	640	2560	<	320	1280	1280	640	640
A/Lyon/1799/2020	N156K, K130N, L161I, K209M, V250A	6B.1A5A	2020-03-16	MDCK3/MDCK1	40	<	40	160	640	<	40	40	40	<
A/Portugal/MS71/2020		6B.1A5B	2020-02-05	MDCK1	640	1280	640	1280	40	320	640	640	320	640
A/Libourne/1245/2020		6B.1A7	2020-02-15	MDCK2/MDCK1	640	1280	1280	1280	<	320	1280	1280	640	1280

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80; ND =Not Done

Vaccine
NH 2018-19
SH 2019Vaccine
NH 2019-20
SH 2020Vaccine
NH 2020-21

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 5-11. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2020-07-28)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/Mich 45/15 Egg	A/Paris 1447/17 MDCK	A/Bris 02/18 Egg	A/Norway 3433/18 MDCK	A/Denmark 3280/19 MDCK	A/Ire 87733/19 Egg	A/G-M SWL1536/19 Egg	A/G-M SWL1536/19 MDCK	A/Swit 3330/17 Egg	A/Ire 84630/18 MDCK
					F31/16 ¹	F03/18 ²	F09/19 ¹	F04/19 ¹	F08/20 ¹	St Jude's F18/20	F12/20 ¹	F09/20 ¹	F23/18 ¹	F08/19 ¹
		Genetic group			6B.1	6B.1A	6B.1A1	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5B	6B.1A6
REFERENCE VIRUSES														
A/Michigan/45/2015		6B.1	2015-09-07	E3/E3	1280	1280	640	1280	40	640	1280	1280	640	1280
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	1280	2560	1280	2560	<	1280	1280	2560	640	2560
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	1280	2560	1280	1280	40	640	1280	1280	640	1280
A/Norway/3433/2018		6B.1A5A	2018-10-30	MDCK3	320	640	640	1280	<	640	640	640	320	640
A/Denmark/3280/2019	N156K	6B.1A5A	2019-11-10	MDCK4/MDCK3	80	80	80	160	1280	80	80	80	80	40
A/Ireland/87733/2019		6B.1A5A	2019-11-03	E4	1280	1280	1280	2560	40	1280	1280	1280	1280	1280
A/Guangong-Maonan/SWL1536/2019	D187A, Q189E	6B.1A5A	2019-06-17	E3/E1	640	1280	1280	2560	<	640	1280	1280	640	1280
A/Guangong-Maonan/SWL1536/2019	D187A, Q189E	6B.1A5A	2019-06-17	C2/MDCK1	1280	640	640	2560	<	1280	1280	1280	640	1280
A/Switzerland/3330/2017	clone 35	6B.1A5B	2017-12-20	E6/E2	640	1280	1280	1280	<	320	640	640	640	640
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK3	640	1280	640	1280	<	640	1280	640	1280	1280
TEST VIRUSES														
A/Azerbaijan/284/2019		6B.1A5A	2019-12-24	Cx/MDCK1	640	1280	640	1280	40	640	1280	640	640	640
A/Ukraine/8382/2020		6B.1A5A	2020-01-15	MDCK2/MDCK1	640	2560	640	1280	<	1280	1280	640	640	1280
A/Ukraine/8398/2020		6B.1A5A	2020-01-22	MDCK1/MDCK1	640	1280	640	1280	<	640	1280	640	320	1280
A/Ukraine/8459/2020		6B.1A5A	2020-01-23	MDCK1/MDCK1	320	1280	320	2560	<	320	640	320	320	640
A/Ukraine/8461/2020		6B.1A5A	2020-01-27	MDCK1/MDCK1	1280	2560	1280	1280	<	1280	1280	1280	640	2560
A/Ukraine/8463/2020		6B.1A5A	2020-01-28	MDCK1/MDCK1	640	1280	640	1280	<	640	1280	640	320	1280
A/Ukraine/8765/2020		6B.1A5A	2020-01-31	MDCK1/MDCK1	640	1280	640	1280	<	640	1280	640	320	1280
A/Ukraine/8771/2020		6B.1A5A	2020-02-02	MDCK1/MDCK1	640	1280	640	1280	<	640	1280	1280	320	1280
A/Ukraine/8482/2020		6B.1A5A	2020-02-03	MDCK1/MDCK1	320	640	640	640	<	320	320	320	160	640
A/Ukraine/8481/2020		6B.1A5A	2020-02-05	MDCK1/MDCK1	640	1280	1280	1280	<	640	640	640	640	1280
A/Ukraine/8474/2020		6B.1A5A	2020-02-06	MDCK2/MDCK1	1280	1280	1280	2560	40	1280	1280	1280	640	2560
A/Ukraine/8770/2020		6B.1A5A	2020-02-07	MDCK1/MDCK1	640	640	640	1280	<	640	640	640	320	1280
A/Ukraine/8450/2020		6B.1A5A	2020-02-11	MDCK1/MDCK1	640	2560	1280	1280	<	640	640	640	320	1280
A/Ukraine/8449/2020		6B.1A5A	2020-02-11	MDCK1/MDCK1	640	2560	1280	1280	<	640	640	640	320	1280
A/Ukraine/8772/2020		6B.1A5A	2020-02-14	MDCK1/MDCK1	640	2560	640	1280	<	640	640	640	320	1280
A/Bulgaria/1006/2020		6B.1A5A	2020-02-17	SIAT2/MDCK1	640	1280	1280	1280	<	640	1280	640	320	640
A/Ukraine/8732/2020		6B.1A5A	2020-02-17	MDCK1/MDCK1	640	640	640	1280	<	640	640	640	320	640
A/Ukraine/8670/2020		6B.1A5A	2020-02-17	MDCK1/MDCK1	640	1280	1280	1280	40	1280	1280	1280	640	2560
A/Ukraine/8617/2020		6B.1A5A	2020-02-17	MDCK1/MDCK1	640	2560	1280	1280	<	640	640	640	640	1280
A/Ukraine/8618/2020		6B.1A5A	2020-02-18	MDCK1/MDCK1	1280	1280	1280	2560	40	1280	1280	1280	640	1280
A/Ukraine/8685/2020	N156K, K130N, L161I, V250A	6B.1A5A	2020-02-24	MDCK1/MDCK1	40	80	640	160	640	<	<	40	<	<
A/Ukraine/8684/2020		6B.1A5A	2020-02-24	MDCK1/MDCK1	640	1280	1280	1280	40	1280	1280	1280	640	1280
A/Ukraine/8687/2020		6B.1A5A	2020-02-25	MDCK1/MDCK1	640	640	<	1280	<	640	640	640	320	640
A/Ukraine/8690/2020		6B.1A5A	2020-02-27	MDCK1/MDCK1	640	1280	640	1280	<	640	1280	640	640	640
A/Ukraine/8673/2020		6B.1A5A	2020-02-27	MDCK2/MDCK2	1280	1280	640	2560	40	1280	1280	1280	640	1280
A/Bulgaria/1169/2020		6B.1A5A	2020-02-29	SIAT2/MDCK1	640	1280	1280	1280	<	640	1280	1280	640	640
A/Bulgaria/1257/2020		6B.1A5A	2020-03-04	SIAT2/MDCK1	640	1280	1280	1280	<	640	1280	1280	640	640
A/Ukraine/8768/2020		6B.1A7	2020-02-04	MDCK1/MDCK1	640	1280	1280	1280	<	640	640	640	640	1280
A/Ukraine/8769/2020		6B.1A7	2020-02-05	MDCK1/MDCK1	640	2560	1280	1280	<	640	640	640	640	1280

¹ Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80; ND =Not Done

Vaccine
NH 2018-19
SH 2019Vaccine
NH 2019-20
SH 2020Vaccine
NH 2020-21

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 5-12. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2020-07-31)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/Mich 45/15 Egg	A/Paris 1447/17 MDCK	A/Bris 02/18 Egg	A/Norway 3433/18 MDCK	A/Denmark 3280/19 MDCK	A/Ire 87733/19 Egg	A/G-M SWL1536/19 Egg	A/G-M SWL1536/19 MDCK	A/Swit 3330/17 Egg	A/Ire 84630/18 MDCK
					F31/16 ¹	F03/18 ²	F09/19 ¹	F04/19 ¹	F08/20 ¹	St Jude's F18/20	F12/20 ¹	F09/20 ¹	F23/18 ¹	F08/19 ¹
					6B.1	6B.1A	6B.1A1	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5B	6B.1A6
REFERENCE VIRUSES														
A/Michigan/45/2015		6B.1	2015-09-07	E3/E3	1280	1280	640	1280	<	640	1280	1280	1280	1280
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	1280	2560	1280	2560	<	1280	1280	2560	640	2560
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	1280	2560	1280	1280	40	640	1280	1280	640	1280
A/Norway/3433/2018		6B.1A5A	2018-10-30	MDCK3	320	640	640	1280	<	640	640	640	320	640
A/Denmark/3280/2019	N156K	6B.1A5A	2019-11-10	MDCK4/MDCK3	80	80	80	160	1280	80	80	80	80	40
A/Ireland/87733/2019		6B.1A5A	2019-11-03	E4	1280	1280	1280	2560	40	1280	1280	1280	1280	1280
A/Guangong-Maonan/SWL1536/2019	D187A, Q189E	6B.1A5A	2019-06-17	E3/E1	640	1280	1280	2560	<	640	1280	1280	640	1280
A/Guangong-Maonan/SWL1536/2019	D187A, Q189E	6B.1A5A	2019-06-17	C2/MDCK1	640	640	640	2560	<	640	1280	1280	640	640
A/Switzerland/3330/2017	clone 35	6B.1A5B	2017-12-20	E6/E2	640	640	1280	1280	<	320	640	640	640	640
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK3	640	1280	640	2560	40	640	1280	1280	1280	1280
TEST VIRUSES														
A/Cheboksary/125/2020		6B.1A5A	2020-01-29	MDCK3/MDCK1	1280	1280	1280	1280	40	640	1280	1280	1280	1280
A/Beirut/321/2020		6B.1A5A	2020-02-03	MDCK1	640	640	1280	2560	<	640	1280	1280	640	1280
A/AinWazein/266/2020		6B.1A5A	2020-02-14	MDCK1	1280	1280	1280	2560	<	1280	1280	1280	640	2560
A/AinWazein/264/2020		6B.1A5A	2020-02-14	MDCK1	1280	1280	1280	2560	<	1280	1280	1280	640	2560
A/AinWazein/267/2020		6B.1A5A	2020-02-15	MDCK1	1280	1280	1280	2560	40	1280	1280	1280	640	2560
A/AinWazein/340/2020		6B.1A5A	2020-02-17	MDCK1	1280	1280	1280	2560	40	640	1280	1280	640	1280
A/Orenburg/5/2020		6B.1A5A	2020-02-17	MDCK3/MDCK1/MDCK1	320	640	320	640	<	160	640	320	320	320
A/Beirut/296/2020		6B.1A5A	2020-02-19	MDCK1	1280	1280	1280	2560	<	640	1280	1280	640	1280
A/Saida/364/2020		6B.1A5A	2020-02-23	MDCK1	640	640	640	1280	<	640	640	1280	320	640
A/Beirut/NS38/2020		6B.1A5A	2020-02-24	MDCK1	1280	1280	1280	2560	40	640	1280	1280	640	1280
A/Beirut/NS37/2020	N156K, K130N, L161I, V250A	6B.1A5A	2020-02-24	MDCK2	80	80	40	160	1280	40	40	40	40	40
A/Zahle/357/2020		6B.1A5A	2020-02-25	MDCK1	640	1280	1280	2560	<	640	1280	1280	640	640
A/Moscow/96/2020		6B.1A5A	2020-02-26	MDCK2/MDCK1	640	640	640	1280	<	320	640	640	320	640
A/Moscow/94/2020		6B.1A5A	2020-02-26	MDCK2/MDCK1	1280	1280	1280	2560	40	640	1280	1280	640	1280
A/Moscow/74/2020		6B.1A5A	2020-02-26	MDCK2/MDCK1	320	640	320	640	<	160	320	320	160	320
A/Moscow/92/2020		6B.1A5A	2020-02-26	MDCK2/MDCK1	640	640	640	1280	<	320	640	320	320	640
A/Saida/372/2020		6B.1A5A	2020-02-27	MDCK1	1280	1280	1280	2560	40	640	1280	1280	640	1280
A/Moscow/123/2020		6B.1A5A	2020-02-28	MDCK3/MDCK1	1280	1280	1280	2560	40	640	1280	1280	640	1280
A/Moscow/76/2020		6B.1A5A	2020-03-02	MDCK2/MDCK1	640	1280	640	1280	40	320	1280	640	640	1280
A/Zahle/359/2020		6B.1A5A	2020-03-02	MDCK1	640	640	640	1280	<	320	640	640	640	640
A/Ireland/20619/2020	N156K, K130N, L161I, V250A	6B.1A5A	2020-03-03	MDCK3	80	80	40	160	640	40	80	40	40	40
A/Moscow/99/2020		6B.1A5A	2020-03-03	MDCK2/MDCK1	320	640	320	1280	<	320	640	320	320	640
A/Ireland/21126/2020	N156K, T120I, K130N, L161I, K209M, V250A	6B.1A5A	2020-03-04	MDCK3	40	<	40	80	640	40	40	40	<	<
A/Ireland/20925/2020	N156K, K130N, L161I, V250A	6B.1A5A	2020-03-04	MDCK1	40	80	40	160	640	<	40	40	40	<
A/Moscow/102/2020		6B.1A5A	2020-03-04	MDCK2/MDCK1	640	1280	640	1280	<	640	1280	1280	320	1280
A/Moscow/79/2020		6B.1A5A	2020-03-04	MDCK2/MDCK1	640	1280	640	2560	<	640	1280	1280	640	1280
A/Moscow/78/2020		6B.1A5A	2020-03-04	MDCK2/MDCK1	1280	2560	1280	2560	<	1280	1280	1280	1280	1280
A/Orenburg/10/2020		6B.1A5A	2020-03-05	MDCK3/MDCK1/MDCK1	640	1280	640	2560	<	640	1280	1280	640	1280
A/Cheboksary/115/2020		6B.1A5A	2020-03-10	MDCK1/MDCK1	1280	2560	1280	2560	40	640	1280	1280	640	1280
A/Moscow/83/2020		6B.1A5A	2020-03-10	MDCK2/MDCK1	1280	2560	1280	2560	<	640	1280	1280	640	1280
A/Moscow/104/2020		6B.1A5A	2020-03-12	MDCK2/MDCK1	640	1280	640	2560	<	640	1280	640	640	640
A/Yaroslavl/133/2020		6B.1A5A	2020-03-15	MDCK2/MDCK1	640	1280	640	2560	<	640	1280	640	640	1280
A/Moscow/110/2020		6B.1A5A	2020-03-17	MDCK2/MDCK1	320	640	320	1280	<	320	640	640	320	640
A/Moscow/124/2020		6B.1A5A	2020-03-18	MDCK2/MDCK1	640	640	640	1280	<	320	640	640	320	640
A/Yaroslavl/135-L/2020		6B.1A5A	2020-03-23	MDCK1/MDCK1	640	1280	640	1280	<	320	640	640	320	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80; ND =Not Done

Vaccine
NH 2018-19
SH 2019Vaccine
NH 2019-20
SH 2020Vaccine
NH 2020-21

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 5-13. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2020-08-11)

Viruses	Other information	Passage history	Collection date	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Mich 45/15 Egg	A/Paris 1447/17 MDCK	A/Bris 02/18 Egg	A/Norway 3433/18 MDCK	A/Denmark 3280/19 MDCK	A/HK 4394/19 Egg	A/Ire 87733/19 Egg	A/G-M SWL1536/19 Egg	A/G-M SWL1536/19 MDCK	A/Swit 3330/17 Egg	A/Ire 84630/18 MDCK	
				F31/16 ^{†1}	F03/18 ^{†2}	F09/19 ^{†1}	F04/19 ^{†1}	F08/20 ^{†1}	F16/20 ^{†1}	St Jude's F18/20	F12/20 ^{†1}	F09/20 ^{†1}	F23/18 ^{†1}	F08/19 ^{†1}	
				Genetic group	6B.1	6B.1A	6B.1A1	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5B	6B.1A6
NEW															
REFERENCE VIRUSES															
A/Michigan/45/2015		6B.1	2015-09-07	E3/E3	1280	1280	1280	2560	40	40	640	1280	1280	640	1280
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	640	1280	640	1280	40	<	320	640	640	320	1280
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E2	1280	2560	1280	2560	40	40	640	1280	1280	1280	2560
A/Norway/3433/2018		6B.1A5A	2018-10-30	MDCK3	320	640	320	640	<	<	160	640	320	160	640
A/Denmark/3280/2019	N156K	6B.1A5A	2019-11-10	MDCK4/MDCK2	80	80	80	160	640	640	40	80	80	40	40
A/Ireland/87733/2019		6B.1A5A	2019-11-03	E4	640	1280	640	1280	<	<	640	1280	640	320	640
A/Guangdong-Maonan/SWL1536/2019	D187A, Q189E	6B.1A5A	2019-06-17	E3/E1	640	1280	640	1280	<	40	320	1280	640	320	640
A/Guangdong-Maonan/SWL1536/2019	D187A, Q189E	6B.1A5A	2019-06-17	C2/MDCK1	640	1280	640	1280	<	40	640	1280	1280	320	1280
A/Switzerland/3330/2017	clone 35	6B.1A5B	2017-12-20	E6/E2	640	1280	640	1280	40	40	320	640	640	640	640
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK3	640	1280	640	1280	<	40	640	1280	640	640	1280
TEST VIRUSES															
A/Bosnia and Herzegovina/219/2020		No Seq	2020-02-03	MDCK1	1280	1280	1280	1280	<	40	640	1280	1280	640	1280
A/Bosnia and Herzegovina/213/2020		6B.1A5A	2020-02-03	MDCK2	640	640	640	1280	<	<	320	1280	640	320	640
A/Bosnia and Herzegovina/212/2020		6B.1A5A	2020-02-03	MDCK2	1280	1280	1280	2560	<	40	640	1280	1280	640	1280
A/Bosnia and Herzegovina/261/2020		6B.1A5A	2020-02-10	MDCK1	1280	1280	1280	2560	<	40	640	1280	1280	640	1280
A/Bosnia and Herzegovina/264/2020		6B.1A5A	2020-02-10	MDCK2	320	320	320	640	<	<	320	640	320	320	640
A/Norway/941/2020		6B.1A5A	2020-02-10	MDCK1	320	320	320	640	<	<	320	640	640	160	640
A/Bosnia and Herzegovina/286/2020		6B.1A5A	2020-02-12	MDCK1	640	640	640	1280	<	<	320	640	640	320	640
A/Norway/2258/2020		6B.1A5A	2020-03-02	MDCK1	640	640	1280	1280	<	40	640	1280	1280	640	1280
A/Estonia/127389/2020	N156K, K130N, L161I, V250A	6B.1A5A	2020-03-11	SIAT1/SIAT2	40	40	40	80	320	<	40	<	<	<	ND
A/Norway/2233/2020		6B.1A5A	2020-03-16	MDCK1	640	320	640	1280	<	<	320	640	640	320	640
A/Norway/2289/2020		6B.1A5A	2020-03-20	MDCK1	320	320	320	1280	<	<	320	640	640	320	640
A/Norway/2341/2020	N156K, E112K, K130N, L161I, K209M, V250A	6B.1A5A	2020-03-21	MDCK1	<	<	<	40	160	160	<	<	<	<	<
A/Norway/729/2020		6B.1A5B	2020-02-04	MDCK2	640	640	640	1280	40	<	320	640	640	320	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80; ND =Not Done

Vaccine
NH 2018-19
SH 2019Vaccine
NH 2019-20
SH 2020Vaccine
NH 2020-21

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 5-14. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2020-09-02)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre										
					Post-infection fereret antisera										
					A/Bayern 69/09 MDCK	A/Mich 45/15 Egg	A/Paris 1447/17 MDCK	A/Bris 02/18 Egg	A/Norway 3433/18 MDCK	A/Denmark 3280/19 MDCK	A/Ire 87733/19 Egg	A/G-M SWL1536/19 Egg	A/G-M SWL1536/19 MDCK	A/Swit 3330/17 Egg	A/Ire 84630/18 MDCK
					F09/15 ⁻¹	F31/16 ⁻¹	F03/18 ⁻²	F09/19 ⁻¹	F04/19 ⁻¹	F08/20 ⁻¹	St Jude's F18/20	F12/20 ⁻¹	F09/20 ⁻¹	F23/18 ⁻¹	F08/19 ⁻¹
					Genetic group	6B.1	6B.1A	6B.1A1	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5A	6B.1A5B	6B.1A6
REFERENCE VIRUSES															
A/Bayern/69/2009	G155E		2020-11-10	MDCK5/MDCK2	160	80	160	80	160	<	<	40	40	40	40
A/Michigan/45/2015		6B.1	2015-09-07	E3/E3	320	640	1280	1280	1280	40	640	1280	640	640	1280
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	320	640	2560	1280	2560	<	640	1280	640	640	1280
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E2	320	1280	2560	1280	2560	40	640	1280	1280	1280	1280
A/Norway/3433/2018		6B.1A5A	2018-10-30	MDCK3	80	320	640	320	1280	40	160	640	320	320	640
A/Denmark/3280/2019	N156K	6B.1A5A	2019-11-10	MDCK4/MDCK4	40	40	80	40	80	320	<	80	40	40	<
A/Ireland/87733/2019		6B.1A5A	2019-11-03	E4	320	640	1280	640	1280	40	640	1280	1280	320	1280
A/Guangdong-Maonan/SWL1536/2019	D187A, Q189E	6B.1A5A	2019-06-17	E3/E1	160	640	1280	640	1280	<	320	1280	640	320	640
A/Guangdong-Maonan/SWL1536/2019	D187A, Q189E	6B.1A5A	2019-06-17	C2/MDCK1	320	640	1280	640	1280	<	640	1280	1280	640	1280
A/Switzerland/3330/2017	clone 35	6B.1A5B	2017-12-20	E6/E2	320	640	1280	640	1280	<	320	1280	640	640	1280
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK3	320	640	1280	1280	1280	<	320	1280	640	640	1280
TEST VIRUSES															
A/Castilla La Mancha/1169/2020		No Seq	2020-02-19	MDCK2	160	320	640	320	1280	<	320	640	640	320	640
A/Moldova/174.06/2020		6B.1A5A	2020-02-05	MDCK2	320	640	1280	640	2560	<	640	1280	1280	640	1280
A/Hungary/71/2020		6B.1A5A	2020-02-11	MDCK1/MDCK1	160	320	1280	320	640	<	320	640	640	320	640
A/Hungary/80/2020		6B.1A5A	2020-02-13	MDCK1/MDCK1	160	640	1280	640	1280	<	320	640	640	320	640
A/Castilla La Mancha/1089/2020		6B.1A5A	2020-02-16	MDCK1	320	1280	2560	1280	2560	40	640	1280	1280	640	1280
A/Aragon/13165/2020		6B.1A5A	2020-02-17	MDCK1	160	640	1280	640	1280	<	320	640	1280	320	640
A/Melilla/995/2020		6B.1A5A	2020-02-17	MDCK1	320	640	1280	640	1280	40	640	640	1280	320	640
A/Hungary/82/2020	N156K, K130N, L161I, K209M, V250A	6B.1A5A	2020-02-18	MDCK1/SIAT1	40	40	80	40	80	320	40	40	40	40	<
A/Pais Vasco/1081/2020		6B.1A5A	2020-02-18	MDCK1	160	640	1280	640	1280	<	640	1280	1280	320	640
A/Pais Vasco/1079/2020		6B.1A5A	2020-02-18	MDCK1	160	640	1280	640	1280	40	320	1280	1280	320	640
A/Castilla La Mancha/1073/2020		6B.1A5A	2020-02-18	MDCK1	320	640	1280	640	1280	<	640	1280	1280	320	1280
A/Navarra/1051/2020		6B.1A5A	2020-02-20	MDCK1	160	320	640	640	1280	<	320	640	640	320	640
A/Castilla La Mancha/1461/2020		6B.1A5A	2020-02-21	MDCK1	160	640	1280	640	1280	<	320	640	640	320	640
A/Castilla La Mancha/1464/2020		6B.1A5A	2020-02-22	MDCK1	160	640	1280	640	1280	<	320	640	640	320	640
A/Castilla La Mancha/1466/2020	N156K, K130N, L161I, V250A	6B.1A5A	2020-02-24	MDCK1	40	40	80	40	80	320	<	40	40	40	<
A/Castilla La Mancha/1465/2020		6B.1A5A	2020-02-24	MDCK1	320	640	1280	640	1280	<	640	1280	1280	320	1280
A/Melilla/1083/2020		6B.1A5A	2020-02-24	MDCK1	320	640	1280	640	1280	<	640	1280	1280	320	1280
A/Castilla La Mancha/1075/2020	N156K, A48T, K130N, L161I, D187X, K209M, V250A, E283X	6B.1A5A	2020-02-24	MDCK1	<	<	<	<	40	160	<	40	<	<	<
A/Melilla/1087/2020		6B.1A5A	2020-02-25	MDCK1	320	640	1280	640	1280	<	320	640	640	320	640
A/Moldova/348.09/2020		6B.1A5A	2020-02-26	MDCK1	160	640	1280	640	1280	<	320	1280	640	320	1280
A/Castilla La Mancha/1815/2020		6B.1A5A	2020-02-27	MDCK1	160	640	1280	640	1280	<	320	1280	1280	320	640
A/Castilla La Mancha/1411/2020	N156K, K130N, L161I, V250A	6B.1A5A	2020-02-29	MDCK1	<	<	<	<	80	320	<	<	<	<	<
A/Castilla La Mancha/1463/2020	N156K, K130N, L161I, D187V, K209M, V250A	6B.1A5A	2020-03-02	MDCK1	<	<	<	40	40	320	<	40	<	<	<
A/Aragon/13177/2020		6B.1A5A	2020-03-03	MDCK1	160	640	1280	640	1280	<	320	640	640	320	640
A/Castilla La Mancha/1698/2020		6B.1A5A	2020-03-04	MDCK1	160	320	640	320	1280	<	320	640	640	320	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80; ND =Not Done

Vaccine
NH 2018-19
SH 2019Vaccine
NH 2019-20
SH 2020Vaccine
NH 2020-21

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 5-15. Antigenic analyses of influenza A(H1N1)pdm09 viruses - Summary

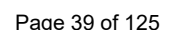
Viruses	Other information		Haemagglutination inhibition titre											
			Post-infection fereret antisera											
			A/Bayern 69/09 MDCK	A/Mich 45/15 Egg	A/Paris 1447/17 MDCK	A/Bris 02/18 Egg	A/Norway 3433/18 MDCK	A/Denmark 3280/19 MDCK	A/HK 4394/19 Egg	A/Ire 87733/19 Egg	A/G-M SWL1536/19 Egg	A/G-M SWL1536/19 MDCK	A/Swit 3330/17 Egg	A/Ire 84630/18 MDCK
			F09/15 ^{*1}	F31/16 ^{*1}	F03/18 ^{*2}	F09/19 ^{*1}	F04/19 ^{*1}	F08/20 ^{*1}	F16/20 ^{*1}	St Jude's F18/20	F12/20 ^{*1}	F09/20 ^{*1}	F23/18 ^{*1}	F08/19 ^{*1}
Genetic group														
REFERENCE VIRUSES														
A/Bayern/69/2009	G155E		160	80	160	80	160	<	ND	<	40	40	40	40
A/Michigan/45/2015		6B.1	320	640	1280	1280	1280	40	40	640	1280	640	640	1280
A/Paris/1447/2017		6B.1A	320	640	2560	1280	2560	<	<	640	1280	640	640	1280
A/Brisbane/02/2018		6B.1A1	320	1280	2560	1280	2560	40	40	640	1280	1280	1280	1280
A/Norway/3433/2018		6B.1A5A	80	320	640	320	1280	40	<	160	640	320	320	640
A/Denmark/3280/2019	N156K	6B.1A5A	40	40	80	40	80	320	640	<	80	40	40	<
A/Ireland/87733/2019		6B.1A5A	320	640	1280	640	1280	40	<	640	1280	1280	320	1280
A/Guangdong-Maonan/SWL1536/2019	D187A, Q189E	6B.1A5A	160	640	1280	640	1280	<	40	320	1280	640	320	640
A/Guangdong-Maonan/SWL1536/2019	D187A, Q189E	6B.1A5A	320	640	1280	640	1280	<	40	640	1280	1280	640	1280
A/Switzerland/3330/2017	clone 35	6B.1A5B	320	640	1280	640	1280	<	40	320	1280	640	640	1280
A/Ireland/84630/2018		6B.1A6	320	640	1280	1280	1280	<	40	320	1280	640	640	1280
TEST VIRUSES														
Genetic group														
All viruses	No tested		25	281	281	281	281	281	13	276	281	276	281	280
		Titre reduced ≤2-fold	20	203	159	190	213	49	1	165	207	202	203	193
		%	80.0	72.2	56.6	67.6	75.8	17.4	7.7	59.8	73.7	73.2	72.2	68.9
		Titre reduced =4-fold	2	10	48	20	1	14		38	5	7	9	16
		%	8.0	3.6	17.1	7.1	0.4	5.0		13.8	1.7	2.5	3.2	5.7
		Titre reduced ≥8-fold	3	68	74	71	67	218	12	73	69	67	69	71
		%	12.0	24.2	26.3	25.3	23.8	77.6	92.3	26.4	24.6	24.3	24.6	25.4
6B.1A5A	No tested		24	269	269	269	269	269	11	264	269	264	269	268
		Titre reduced ≤2-fold	19	194	151	182	204	47	1	158	198	193	194	184
		%	79.2	72.1	56.1	67.7	75.8	17.5	9.1	59.8	73.6	73.1	72.1	68.7
		Titre reduced =4-fold	2	10	47	19	1	13		36	5	7	9	16
		%	8.3	3.7	17.5	7.0	0.4	4.8		13.7	1.9	2.7	3.4	5.9
		Titre reduced ≥8-fold	3	65	71	68	64	209	10	70	66	64	66	68
		%	12.5	24.2	26.4	25.3	23.8	77.7	90.9	26.5	24.5	24.2	24.5	25.4
6B.1A5B	No tested			3	3	3	3	3	1	3	3	3	3	3
		Titre reduced ≤2-fold		3	3	3	3			2	3	3	3	3
		Titre reduced =4-fold								1				
		Titre reduced ≥8-fold						3	1					
6B.1A7	No tested			3	3	3	3	3		3	3	3	3	3
		Titre reduced ≤2-fold		3	3	3	3			2	3	3	3	3
		Titre reduced =4-fold								1				
		Titre reduced ≥8-fold						3						
No Seq	No tested		1	6	6	6	6	6	1	6	6	6	6	6
		Titre reduced ≤2-fold	1	3	2	2	3	2		3	3	3	3	3
		Titre reduced =4-fold			1	1		1						
		Titre reduced ≥8-fold		3	3	3	3	3	1	3	3	3	3	3
			Vaccine NH 2018-19 SH 2019		Vaccine NH 2019-20 SH 2020		Vaccine NH 2020-21							

Reference virus results are taken from an individual table as an example. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed

Substitutions at HA1 position 156



Vaccine virus: Reference virus: 6B Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site



	310	320	330	340	350	360	370	380	390	400
A/Michigan/45/2015 Sep	GKCPKYVKSTKLR	ATGLRNVP	SIQSRGLFGA	ITAGFIEGG	WTGMXDGWYGYHHQNEQSGGYAADLKSTQNAIDKITNKVNSVIEKMNTQFTAVGKEFNHL					1A
A/Brisbane/02/2018 Jan					V					1A1
A/Ukraine/8768/2020 Feb	T				V					1A7
A/Switzerland/3330/2017 Dec					V					1A5B
A/Norway/729/2020 Feb					V					1A5B
A/Portugal/EVA239/2020 Feb					V					1A5B
A/Denmark/3280/2019 Nov					V					1A5A
A/Lyon/1756/2020 Mar					V					1A5A
A/Estonia/127389/2020 Mar					V					1A5A
A/Ireland/20925/2020 Mar					V					1A5A
A/Iceland/22/2020 Mar					V					1A5A
A/South Africa/6433/2020 Apr		K			V					1A5A
A/Thuringen/17/2020 Feb					V					1A5A
A/Switzerland/3428/2020 Mar					V			S		1A5A
A/South Africa/6019/2020 Mar					V					1A5A
A/Norway/3433/2018 Oct					V					1A5A
A/Lund/4/2020 Feb					V					1A5A
A/Lipetsk/04/2020 Mar					V					1A5A
A/Switzerland/8628/2020 Mar					V					1A5A
A/South Africa/6185/2020 Mar					V					1A5A
A/S.Petersburg/RII-179/2020 Ma					V					1A5A
A/Poland/159/2020 Mar					V					1A5A
A/Belgium/S1317/2020 Mar					V					1A5A
A/Iceland/36/2020 Mar		I			V			F		1A5A
CNIC-1909 (A/G-M/SWL1536/2019)					V					1A5A
A/Bulgaria/1243/2020 Mar					V					1A5A
A/South Africa/3796/2020 Mar					V					1A5A
A/Paris/2187/2020 Mar					V					1A5A
A/Hawaii/70/2019 Oct					V					1A5A
A/Cyprus/F1318/2020 Mar					V					1A5A

	410	420	430	440	450	460	470	480	490	500
A/Michigan/45/2015 Sep	EKRIENLNKKVDDGFLDIWTYN	AEALLVLENER	TLDYHDSNVKNLYEKVRNQLKNN	AKEIGNGCFEFYHKCDNTCMESVKN	NGTYDYPKYSEEAKLNREKI					
A/Brisbane/02/2018 Jan										
A/Ukraine/8768/2020 Feb	M								S	
A/Switzerland/3330/2017 Dec										
A/Norway/729/2020 Feb										
A/Portugal/EVA239/2020 Feb										
A/Denmark/3280/2019 Nov										
A/Lyon/1756/2020 Mar										
A/Estonia/127389/2020 Mar										
A/Ireland/20925/2020 Mar								A		
A/Iceland/22/2020 Mar								A		
A/South Africa/6433/2020 Apr										
A/Thuringen/17/2020 Feb				I						
A/Switzerland/3428/2020 Mar										
A/South Africa/6019/2020 Mar										
A/Norway/3433/2018 Oct										
A/Lund/4/2020 Feb										
A/Lipetsk/04/2020 Mar										
A/Switzerland/8628/2020 Mar							N			
A/South Africa/6185/2020 Mar										
A/S.Petersburg/RII-179/2020 Ma										
A/Poland/159/2020 Mar									D	
A/Belgium/S1317/2020 Mar										
A/Iceland/36/2020 Mar										
CNIC-1909 (A/G-M/SWL1536/2019)										
A/Bulgaria/1243/2020 Mar					R			X		
A/South Africa/3796/2020 Mar										
A/Paris/2187/2020 Mar										
A/Hawaii/70/2019 Oct										
A/Cyprus/F1318/2020 Mar										

	510	520	530	540
A/Michigan/45/2015 Sep	DGVKLESTRIYQILAIYSTVASSILVSVSLGAISFWMC	NSLQCRICI		
A/Brisbane/02/2018 Jan				
A/Ukraine/8768/2020 Feb	D			
A/Switzerland/3330/2017 Dec		A		
A/Norway/729/2020 Feb		A		
A/Portugal/EVA239/2020 Feb		A		
A/Denmark/3280/2019 Nov	D			
A/Lyon/1756/2020 Mar	D			
A/Estonia/127389/2020 Mar	D	A		
A/Ireland/20925/2020 Mar	D			
A/Iceland/22/2020 Mar	D			
A/South Africa/6433/2020 Apr	D			
A/Thuringen/17/2020 Feb	D			
A/Switzerland/3428/2020 Mar	D			
A/South Africa/6019/2020 Mar	D			
A/Norway/3433/2018 Oct				
A/Lund/4/2020 Feb				
A/Lipetsk/04/2020 Mar		A		
A/Switzerland/8628/2020 Mar				
A/South Africa/6185/2020 Mar				
A/S.Petersburg/RII-179/2020 Ma	G	X		
A/Poland/159/2020 Mar				
A/Belgium/S1317/2020 Mar				
A/Iceland/36/2020 Mar				
CNIC-1909 (A/G-M/SWL1536/2019)				
A/Bulgaria/1243/2020 Mar				
A/South Africa/3796/2020 Mar				
A/Paris/2187/2020 Mar	K			
A/Hawaii/70/2019 Oct				
A/Cyprus/F1318/2020 Mar				

Vaccine virus: Reference virus: 6B Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 3a. HA amino acid substitutions defining H1N1pdm09 subgroup 6B.1A5A viruses

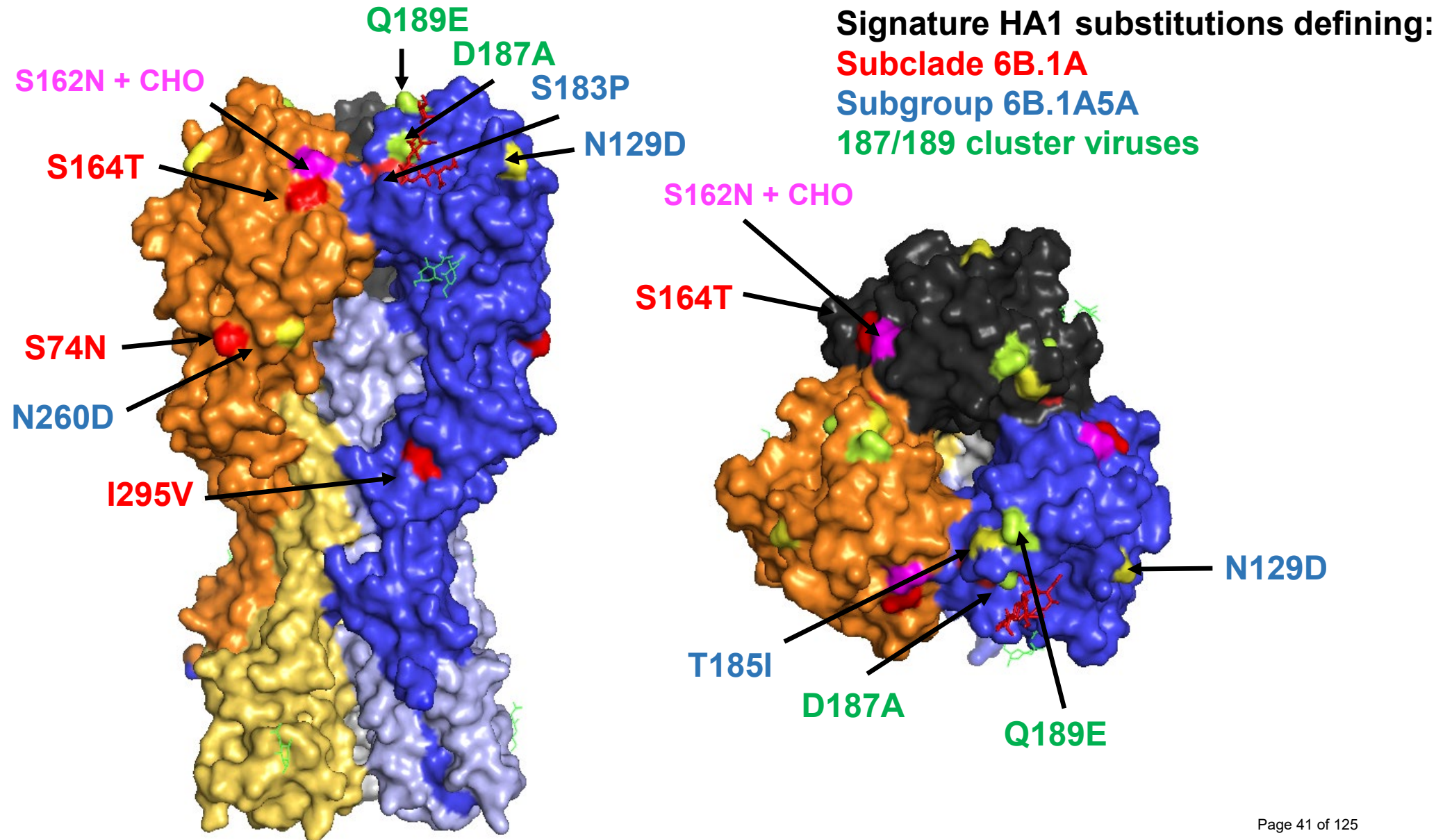


Figure 3b. HA amino acid substitutions defining H1N1pdm09 subgroup 6B.1A5A viruses

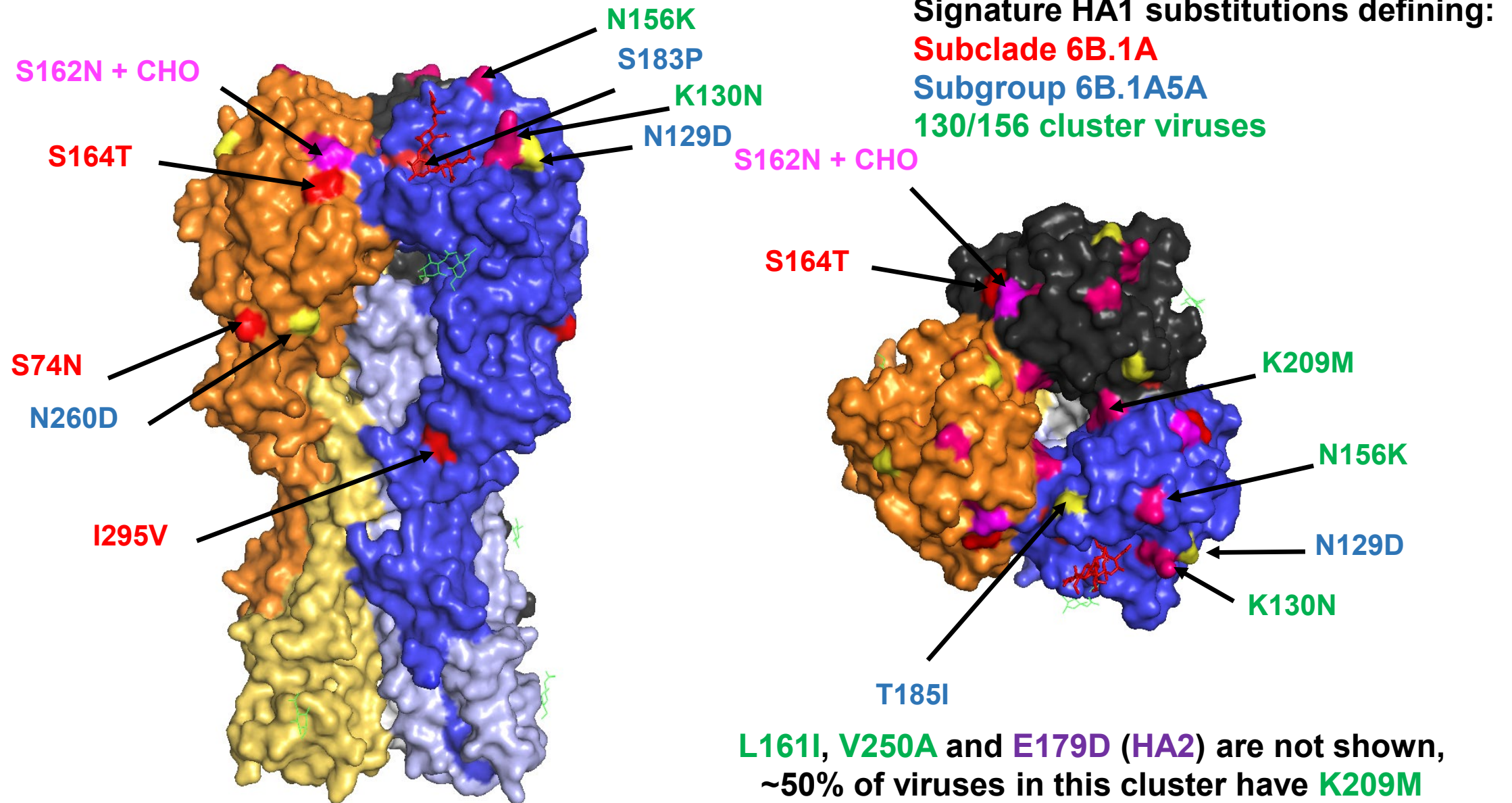
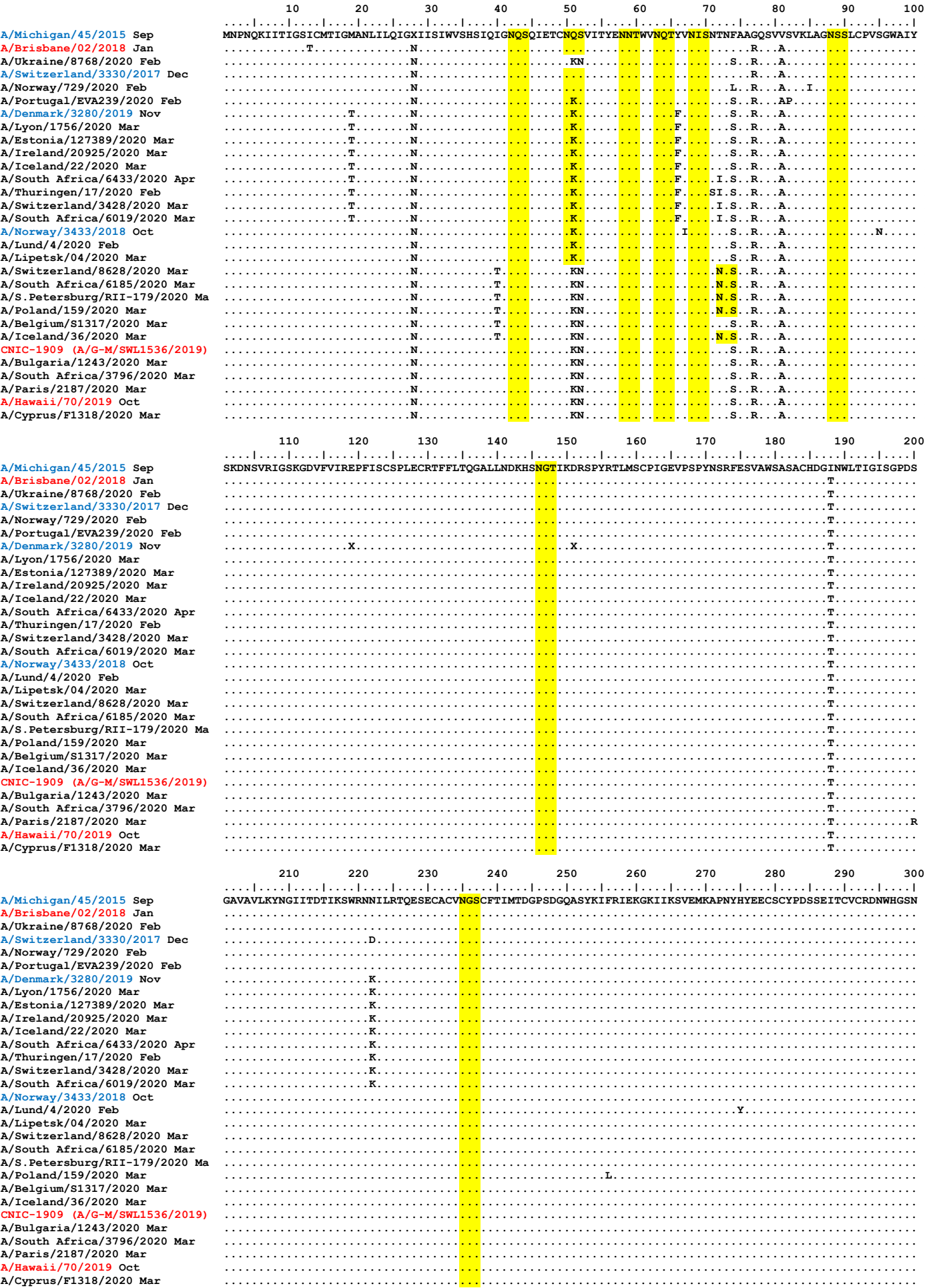


Figure 5. NA protein alignment for A(H1N1)pdm09 viruses

Vaccine virus: Reference virus: 6B Genetic group: Potential N-linked glycosylation site



	310	320	330	340	350	360	370	380	390	400															
A/Michigan/45/2015 Sep	RPWVSFNQ	NLEYQMG	YICSGVF	GDNP	RPNDKT	GSCGPV	SSNGANG	VKGFSF	KYGN	GVWIGR	TKSIS	SRKGF	FEMI	WD	PNGWT	GT	DN	KFSI	KQ	DIV	G	IN	EWS	1A	
A/Brisbane/02/2018 Jan																								1A1	
A/Ukraine/8768/2020 Feb																									1A7
A/Switzerland/3330/2017 Dec																									1A5B
A/Norway/729/2020 Feb																									1A5B
A/Portugal/EVA239/2020 Feb																									1A5B
A/Denmark/3280/2019 Nov																									1A5A
A/Lyon/1756/2020 Mar																									1A5A
A/Estonia/127389/2020 Mar																									1A5A
A/Ireland/20925/2020 Mar																									1A5A
A/Iceland/22/2020 Mar																									1A5A
A/South Africa/6433/2020 Apr																									1A5A
A/Thuringen/17/2020 Feb																									1A5A
A/Switzerland/3428/2020 Mar																									1A5A
A/South Africa/6019/2020 Mar																									1A5A
A/Norway/3433/2018 Oct																									1A5A
A/Lund/4/2020 Feb																									1A5A
A/Lipetsk/04/2020 Mar																									1A5A
A/Switzerland/8628/2020 Mar																									1A5A
A/South Africa/6185/2020 Mar																									1A5A
A/S. Petersburg/RII-179/2020 Ma																									1A5A
A/Poland/159/2020 Mar																									1A5A
A/Belgium/S1317/2020 Mar																									1A5A
A/Iceland/36/2020 Mar																									1A5A
CNIC-1909 (A/G-M/SWL1536/2019)																									1A5A
A/Bulgaria/1243/2020 Mar																									1A5A
A/South Africa/3796/2020 Mar																									1A5A
A/Paris/2187/2020 Mar																									1A5A
A/Hawaii/70/2019 Oct																									1A5A
A/Cyprus/F1318/2020 Mar																									1A5A

	410	420	430	440	450	460
A/Michigan/45/2015 Sep	GYSGSFVQ	HP	ELTGLD	CI	RPCF	FWV
A/Brisbane/02/2018 Jan						
A/Ukraine/8768/2020 Feb						
A/Switzerland/3330/2017 Dec						
A/Norway/729/2020 Feb						
A/Portugal/EVA239/2020 Feb						
A/Denmark/3280/2019 Nov						
A/Lyon/1756/2020 Mar						
A/Estonia/127389/2020 Mar						
A/Ireland/20925/2020 Mar						
A/Iceland/22/2020 Mar						
A/South Africa/6433/2020 Apr						
A/Thuringen/17/2020 Feb						
A/Switzerland/3428/2020 Mar						
A/South Africa/6019/2020 Mar						
A/Norway/3433/2018 Oct						
A/Lund/4/2020 Feb						
A/Lipetsk/04/2020 Mar						
A/Switzerland/8628/2020 Mar						
A/South Africa/6185/2020 Mar						
A/S. Petersburg/RII-179/2020 Ma						
A/Poland/159/2020 Mar						
A/Belgium/S1317/2020 Mar						
A/Iceland/36/2020 Mar						
CNIC-1909 (A/G-M/SWL1536/2019)						
A/Bulgaria/1243/2020 Mar						
A/South Africa/3796/2020 Mar						
A/Paris/2187/2020 Mar						
A/Hawaii/70/2019 Oct						
A/Cyprus/F1318/2020 Mar						

Vaccine virus: Reference virus: 6B Genetic group: Potential N-linked glycosylation site

Influenza A(H3N2) virus analyses.

Influenza A(H3N2) viruses collected since 2020-01-31 have been received from NICs in 30 countries or administrative regions in three WHO Regions: African (2: Algeria and Cote d'Ivoire), Eastern Mediterranean (2: Lebanon and Morocco) and European (26 countries) (**Table 1**).

Influenza A(H3N2) viruses have been difficult to characterise antigenically by HI assay due to either the variable agglutination of red blood cells (RBCs) from guinea pigs, turkeys and humans or the loss of the ability of viruses to agglutinate any of these RBCs. However, since the February 2020 VCM HI analysis was able to be done on 242 viruses that agglutinated guinea pig RBCs (**Tables 7-1 to 7-13**) with 162 (69%) of these viruses being recovered from 235 received samples with collection dates after 2020-01-31. Virus neutralisation by plaque reduction neutralisation assay (PRNA) was done on a selected subset of 61 test viruses with collection dates after 2020-01-31 (**Tables 8-1 to 8-5**).

HI analyses.

A total of 162 viruses with collection dates after 2020-01-31 could be analysed by HI assay and summaries of the HI results are shown in **Tables 2** and **6**. Individual HI results are shown in **Tables 7-1 to 7-13**. Test viruses in each table are sorted by date of collection and genetic clade/subclade for those viruses that have been sequenced, and those below the red horizontal lines in the tables are viruses with collection dates after 2020-01-31 (**Tables 7-4 to 7-13**). The viruses sequenced and analysed by HI with collection dates after 2020-01-31 fell into clade 3C.3a (67%) or clade 3C.2a subclades 3C.2a1b+131K (11%), 3C.2a1b+135K-A (17%) and 3C.2a1b+135K-B (viruses in the 135K subclade with the substitutions S137F, A138S and F193S) (5%).

A summary of the HI results for viruses collected since 2020-01-31 with antisera raised against the egg-propagated cultivars of A/South Australia/34/2019 (subclade 3C.2a1b+T131K), A/Hong Kong/2671/2019 (subclade 3C.2a1b+T135K-B) and A/Kansas/14/2017 (NYMC X-327, clade 3C.3a), recently recommended vaccine viruses, and the cell culture-propagated cultivars of A/Hong Kong/2671/2019 and A/Kansas/14/2017 is shown in **Table 6**. A more detailed summary of the whole panel of antisera is shown in **Table 7-14**.

Viruses in clade 3C.3a:

- antisera raised against cell culture-propagated A/Kansas/14/2017 and A/England/538/2018 recognised viruses in clade 3C.3a very well, >97% at titres within 2-fold of the homologous titres, and they were recognised at generally similar titres by the antiserum raised against NYMC X-327 (derived from A/Kansas/14/2017) although the titre of the antiserum for the homologous virus was higher and recognised only 54% of clade 3C.3a test viruses at titres within 4-fold of the high homologous titre;

- the antiserum raised against cell culture-propagated A/Hong Kong/5738/2014 recognised only 41% of the 3C.3a viruses at titres within 4-fold of the homologous titre of the antiserum, 7.5% within 2-fold; antisera raised against other cell culture-propagated reference viruses in clade 3C.2a performed less well:
 - antisera raised against cell culture-propagated A/La Rioja/2202/2018 (subclade 3C.2a1b + T135K-A) and A/Norway/3275/2018 (subclade 3C.2a1b + T131K) had no homologous titres since the viruses are unable to agglutinate RBCs and recognised all clade 3C.3a test viruses below a titre of 160;
 - the antisera raised against cell culture-propagated A/Hong Kong/2671/2019 and A/Hong Kong/2669/2019 (both subclade 3C.2a1b+T135K-B) recognised 97% and 100%, respectively, of the test viruses poorly;
- the antiserum raised against egg-propagated A/Singapore/INFIMH-16-0019/2016 (subclade 3C.2a1) recognised 88% of the clade 3C.3a viruses at titres within 4-fold of the homologous titre of the antiserum and 39% within 2-fold;
- the antisera raised against the egg-propagated cultivars of A/South Australia/34/2019 (subclade 3C.2a1+T131K) and A/Hong Kong/2671/2019 (subclade 3C.2a1b+135K-B) recognised none of the test viruses at titres within 4-fold of the homologous titres of the antisera.

Viruses in subclade 3C.2a1b+T131K:

- antisera raised against the cell culture-propagated clade 3C.3a viruses A/Kansas/14/2017 and A/England/538/2018 recognised 17% and 56%, respectively, of viruses in subclade 3C.2a1b+T131K at titres within 4-fold of the homologous titres, and 6% and 56%, respectively, at titres within 2-fold. All were recognised poorly by the antiserum raised against NYMC X-327 (derived from A/Kansas/14/2017);
- the antiserum raised against cell culture-propagated A/Hong Kong/5738/2014 (clade 3C.2a) recognised 72% of the 3C.2a1b+T131K viruses at titres within 4-fold of the homologous titre of the antiserum, 61% within 2-fold;
- the antiserum raised against cell culture-propagated A/Norway/3275/2018 (subclade 3C.2a1b+T131K) recognised 14 of 18 (78%) test viruses in subclade 3C.2a1b+T131K at titres of ≥ 160 , and 13 of 18 (72%) were recognised at titres of ≥ 160 by the antiserum raised against A/La Rioja/2202/2018 (subclade 3C.2a1b+T135K-A);
- the antiserum raised against cell culture-propagated A/Hong Kong/2671/2019 (subclade 3C.2a1b+T135K-B) recognised nine of 18 (50%) of the test viruses at titres within 4-fold of the homologous titre, but only a single virus (5.6%) within 2-fold;
- 16 of 23 (89%) test viruses in subclade 3C.2a1b+T131K were recognised at titres within 4-fold, and 78% within 2-fold, of the homologous titre by the antiserum raised against egg-propagated A/Singapore/INFIMH-16-0019/2016;

- the antiserum raised against the vaccine virus egg-propagated A/South Australia/34/2019 (subclade 3C.2a1b+T131K) recognised only 44.5% of test viruses in this subclade at titres within 4-fold of the high homologous titre and none within 2-fold;
- the antiserum raised against egg-propagated A/Hong Kong/2671/2019 (subclade 3C.2a1b+T135K-B) poorly recognised the test viruses in subclade 3C.2a1b+T131K, with only one of the 18 (5.6%) test viruses being recognised at a titre within 4-fold of the homologous titre.

Viruses in subclade 3C.2a1b+T135K-A, with most having additional HA1 amino acid substitutions A138S, G186D, D190N, F193S and S198P:

- antisera raised against the cell culture-propagated clade 3C.3a viruses A/Kansas/14/2017 and A/England/538/2018 recognised 44% and 81%, respectively, of viruses in subclade 3C.2a1b+T135K-A at titres within 4-fold of the homologous titres, but most were recognised less well by the antiserum raised against NYMC X-327 (derived from A/Kansas/14/2017) with only 4 of the 27 test viruses being recognised at a titre within 4-fold of the homologous titre of the antiserum;
- the antiserum raised against cell culture-propagated A/Hong Kong/5738/2014 (clade 3C.2a) recognised 55% of the 3C.2a1b+T135K-A viruses at titres within 4-fold of the homologous titre of the antisera, 26% within 2-fold;
- 19 of 27 (70%) test viruses in subclade 3C.2a1b+T135K-A were recognised at titres of ≥ 160 by the antiserum raised against cell culture-propagated A/La Rioja/2202/2018, in the same genetic group;
- the antiserum raised against cell culture-propagated A/Norway/3275/2018 (subclade 3C.2a1b+T131K) also recognised 16 of 27 (59%) test viruses at titres of ≥ 160 ;
- the antiserum raised against cell culture-propagated A/Hong Kong/2669/2019 (subclade 3C.2a1b+T135K-B) recognised the four test viruses in subclade 3C.211b+T135K-A it was used against at titres within 4-fold of the homologous titre of the antiserum;
- the antiserum raised against the cell culture-propagated cultivar of A/Hong Kong/2671/2019 (subclade 3C.2a1b+T135K-B) recognised 87% of the 26 test viruses at titres within 4-fold of the homologous titre, 42% within 2-fold, but the antiserum raised against the egg-propagated cultivar of A/Hong Kong/2671/2019 recognised only 30% of the test viruses at titres within 4-fold of the homologous titre and only 2 of 27 (7.4%) at titres within 2-fold of the homologous titre;
- all 27 test viruses were recognised at titres within 4-fold of the homologous titre by the antiserum raised against egg-propagated A/Singapore/INFIMH-16-0019/2016 (subclade 3C.2a1), 59% within 2-fold;
- the antiserum raised against the vaccine virus egg-propagated A/South Australia/34/2019 (subclade 3C.2a1b+T131K), recognised only 3 of 27 test viruses at titres within 4-fold of the homologous titre of the antiserum, and a single virus at a titre within 2-fold of the homologous titre.

Viruses in subclade 3C.2a1b+T135K with the substitutions S137F, A138S and F193S (3C.2a1b+T135K-B):

- antisera raised against the cell culture-propagated clade 3C.3a viruses A/Kansas/14/2017 and A/England/538/2018 both recognised two of eight test viruses (25%) in subclade 3C.2a1b+T135K-B at titres within 4-fold of the homologous titres, but all were recognised poorly by the antiserum raised against NYMC X-327 (derived from A/Kansas/14/2017);
- the antiserum raised against cell culture-propagated A/Hong Kong/5738/2014 (clade 3C.2a) recognised four of the eight (50%) 3C.2a1b+T135K-B viruses at titres within 4-fold of the homologous titre of the antiserum, 25% within 2-fold;
- the antisera raised against cell culture-propagated A/Norway/3275/2018 (subclade 3C.2a1b+T131K) and A/La Rioja/2202/2018 (subclade 3C.2a1b+T135K-A) both recognised three (37.5%) test viruses at titres of ≥ 160 ;
- the antiserum raised against A/Hong Kong/2669/2019 (subclade 3C.2a1b+T135K-B) recognised all three viruses with which it was tested at titres within 2-fold of the homologous titre;
- the antiserum raised against the cell culture-propagated cultivar of A/Hong Kong/2671/2019 (subclade 3C.2a1b+T135K-B) recognised all eight test viruses at titres within 4-fold of the homologous titre and seven (87.5%) at titres within 2-fold of the homologous titre, but that raised against the egg-propagated cultivar of A/Hong Kong/2671/2019 recognised only three of the eight test viruses at titres within 4-fold of the homologous titre and none within 2-fold;
- the antiserum raised against egg-propagated A/Singapore/INFIMH-16-0019/2016 (subclade 3C.2a1) recognised six of eight (75%) test viruses in this subclade at titres within 4-fold of the homologous titre, 50% within 2-fold;
- the antiserum raised against the vaccine virus egg-propagated A/South Australia/34/2019 (subclade 3C.2a1b+T131K) recognised none of these test viruses at a titre within 4-fold of the homologous titre of this antiserum.

Virus Neutralisation results

Results of virus neutralisation, assessed by PRNA, are shown in **Tables 8-1 to 8-5** and a summary is presented in **Table 8-6**. Of the viruses analysed 28% were in clade 3C.3a, 33% were in subclade 3C.2a1b+T131K, 28% were in subclade 3C.2a1b+T135K-A and 11% were in subclade 3C.2a1b+T135K-B. A phylogenetic tree of the HA gene of the viruses analysed by PRNA is shown in **Figure 6a**.

Viruses in clade 3C.3a:

- the antiserum raised against the cell culture-propagated cultivar of A/Kansas/14/2017 recognised all test viruses as titres within 2-fold of the homologous titre;
- the antisera raised against viruses in subclades of clade 3C.2a generally recognised these test viruses poorly, but the antiserum raised against egg-

propagated A/Paris/2554/2019 (subclade 3C.2a1b+T135K-A) recognised a single virus at a titre within 4-fold of the homologous titre.

Viruses in subclade 3C.2a1b+T131K:

- antisera raised against cell culture-propagated A/Kansas/14/2007 (clade 3C.3a) recognised all viruses in this subclade poorly;
- the antiserum raised against cell culture-propagated A/Norway/3275/2018, in the same genetic group, recognised the large majority (90%) of test viruses in subclade 3C.2a1b+T131K at titres within 4-fold of the homologous titre of the antiserum, 80% within 2-fold;
- the antiserum raised against cell culture-propagated A/Hong Kong/2671/2019 (subclade 3C.2a1b+T135K-B) recognised only 40% of subclade 3C.2a1b+T131K test viruses at titres within 4-fold of the homologous titre and none at a titre within 2-fold; an antiserum raised against the egg-propagated cultivar of A/Hong Kong/2671/2019 recognised none of the viruses tested;
- the antiserum raised against cell culture-propagated A/Denmark/3264/2019 (subclade 3C.2a1b+T153K-A with the substitutions A138S, G186D, D190N, F193S and S198P) recognised 85% (17 of 20) of the subclade 3C.2a1b+T131K test viruses at titres within 4-fold of the homologous titre, one within 2-fold, but the antiserum raised against egg-propagated A/Paris/2554/2019 (subclade 3C.2a1b+T153K-A with the same subclade substitutions) recognised none of these test viruses at a titre within 4-fold of the homologous titre;
- the antiserum raised against egg-propagated A/South Australia/34/2019 (subclade 3C.2a1b+T131K) recognised the test viruses in this genetic subclade poorly.

Viruses in subclade 3C.2a1b+T135K-A, with all but one having A138S, G186D, D190N, F193S and S198P HA1 substitutions:

- antiserum raised against cell culture-propagated A/Kansas/14/2007 (clade 3C.3a) recognised all viruses but one (5.9%) in this subclade poorly;
- the antiserum raised against cell culture-propagated A/Norway/3275/2018 (subclade 3C.2a1b+T131K) recognised all 17 test viruses poorly;
- the antiserum raised against cell culture-propagated A/Hong Kong/2671/2019 (subclade 3C.2a1b+T135K-B) recognised 8 of 17 (47%) of test viruses in subclade 3C.2a1b+T135K-A at titres within 4-fold of the homologous titre and two (12%) at titres within 2-fold; the antiserum raised against the egg-propagated cultivar of A/Hong Kong/2671/2019 recognised none of the viruses in subclade 3C.2a1b+T135K-A;
- the antiserum raised against cell culture-propagated A/Denmark/3264/2019 (subclade 3C.2a1b+T153K-A with the substitutions A138S, G186D, D190N, F193S and S198P) recognised 82% (14 of 17) of the test viruses in subclade 3C.2a1b+T135K-A at titres within 4-fold of the homologous titre, 29% within 2-fold, but the antiserum raised against egg-propagated A/Paris/2554/2019 (subclade 3C.2a1b+T153K-A with the same subclade substitutions)

recognised none of these test viruses at a titre within 4-fold of the homologous titre;

- the antiserum raised against egg-propagated A/South Australia/34/2019 (subclade 3C.2a1b+T131K) recognised test viruses in subclade 3C.2a1b+T135K-A poorly.

Viruses in subclade 3C.2a1b+T135K-B, defined by HA1 amino acid substitutions S137F, A138S and F193S:

- antisera raised against cell culture-propagated A/Kansas/14/2007 (clade 3C.3a) recognised all viruses in this subclade poorly;
- the antiserum raised against cell culture-propagated A/Norway/3275/2018 (subclade 3C.2a1b+T131K) recognised none of the 18 test viruses in subclade 3C.2a1b+T135K-B at a titre within 4-fold of the homologous titre of the antiserum;
- the antiserum raised against cell culture-propagated A/Denmark/3264/2019 (subclade 3C.2a1b+T135K-A with the substitutions A138S, G186D, D190N, F193S and S198P) recognised 71% (5 of 7) of the test viruses in subclade 3C.2a1b+T135K-A at titres within 4-fold of the homologous titre, 29% within 2-fold, but the antiserum raised against egg-propagated A/Paris/2554/2019 (subclade 3C.2a1b+T153K-A with the same subclade substitutions) recognised only one of the subclade 3C.2a1b+T135K-B viruses at a titre within 4-fold of the homologous titre;
- the antiserum raised against cell culture-propagated A/Hong Kong/2671/2019 (subclade 3C.2a1b+T135K-B) recognised 6 of 7 (86%) test viruses in subclade 3C.2a1b+T135K-B at titres within 2-fold of the homologous titre, but the antiserum raised against the egg-propagated cultivar of A/Hong Kong/2671/2019 recognised all these test viruses poorly;
- the antiserum raised against egg-propagated A/South Australia/34/2019 (3C.2a1b+T131K) recognised none of the test viruses in this genetic subclade at a titre within 4-fold of the homologous titre.

There are two sets of heat maps shown adjacent to the phylogenetic tree in **Figure 6a**. These summarise the VN results and the HI results for the same viruses, where available, and a selected panel of the reference antisera used in the HI assay. The heat map for the VN results illustrates the better recognition of test viruses by antisera raised against viruses within the same clade/subclade and the generally poor recognition of test viruses by antisera raised against egg-propagated reference viruses. The antisera raised against egg-propagated reference viruses perform somewhat better in the HI assay.

Genetic analyses.

Phylogenetic analysis of the HA genes of representative recently circulating A(H3N2) viruses analysed at the Francis Crick Institute and elsewhere (as available in GISAID) is shown in **Figure 6b**. A phylogenetic tree of the HA genes of all viruses with collection dates after 2020-01-31 analysed genetically at the Francis Crick Institute is shown in the Annex, **Figure A-3**. The HA genes fell within clade 3C.3a and clade 3C.2a,

primarily subclade 3C.2a1b. Of those with collection dates after 2020-01-31 58% had HA genes in clade 3C.3a, 18.5% were in subclade 3C.2a1b+T131K, 17.5% in subclade 3C.2a1b+T135K-A and 6% in subclade 3C.2a1b+T135K-B.

Major HA clades/subclades (as illustrated in FigureA-3)

- Clade **3C.3a** viruses differ from older clade **3C.3a** viruses with substitutions **S91N**, **N144K** (with the loss of a glycosylation site, -CHO) and **F193S** in HA1; most also have the substitutions **M17L**, **I149M** and **A2091V** substitutions in HA2.
- Clade **3C.2a** viruses have a series of amino acid substitutions differentiating them from clade 3C.3a viruses: **A128T** (+CHO), **S138A**, **N144S** (-CHO), **S159Y**, **K160T** (+CHO), **Q311H** and **R326K** in HA1;
- Subclade **3C.2a1** viruses have additional substitutions **N121K** and **N171K** in HA1 and **I77V** and **G155E** in HA2 and subclade **3C.2a1b** viruses have the substitutions **E62G**, **K92R** and **H311Q** in HA1:
 - Subclade **3C.2a1b+T131K** viruses have HAs with the substitutions **T131K** in HA1 and **V200I** in HA2 some with **K83E** and **Y94N** in HA1 and **I193M** in HA2, others with **S219F** in HA1 and **V18M** in HA2, many of these also with **Q197R** in HA1 and **E155G** in HA2, some of which also have **K207R** in HA1;
 - Subclade **3C.2a1b+T135K** viruses have the substitutions **T128A** and **T135K** (both resulting in the loss of glycosylation sites), most now have **A138S** and **F193S** in HA1; most of those in subclade **3C.2a1b+T135K-A** now also have the substitutions **G186D**, **D190N** and **S198P** in HA1, and those in subclade **3C.2a1b+T135K-B** have an **S137F** substitution in HA1.

Figure 7 shows an HA amino acid sequence alignment for a representative selection of the viruses used to generate the phylogenies. The vaccine viruses A/Kansas/14/2017 (clade 3C.3a), A/South Australia/34/2019 (subclade 3C.2a1b+T131K), A/Hong Kong/45/2019 and A/Hong Kong/2671/2019 (both subclade 3C.2a1b+T135K-B) are shown in red, glycosylation motifs are highlighted, the reference viruses are coloured blue and the HA1/HA2 proteolytic cleavage site is highlighted. The clade/subclade to which each of the HA sequences belongs is also shown.

Figure 8a shows the location on a 2005 H3 HA structure of amino acid substitutions that define recent subclades of subclade 3C.2a1b viruses compared with the vaccine virus A/Singapore/INFIMH-16-0019/2016. **Figure 8b** shows the location of HA amino acid substitutions found in recently circulating clade 3C.3a viruses compared with A/Switzerland/9715293/2013.

Figure 9 shows a NA phylogenetic tree for representative viruses with their HA genetic groups indicated. A phylogenetic tree of the HA genes of all viruses with collection

dates after 2020-01-31 analysed genetically at the Francis Crick Institute is shown in the Annex, **Figure A-4**. **Figure 10** shows a NA amino acid sequence alignment for a representative selection of these viruses. As above, the vaccine viruses are shown in red, glycosylation motifs are highlighted, the reference viruses are coloured blue and the clade/subclade to which the corresponding HA gene sequences belong is shown.

The NA gene sequences of recent viruses, with HA genes in genetic clusters within clades 3C.3a and 3C.2a and its subclades, show generally similar subclade clustering. All current viruses share the substitutions **S245N** and **S247T** (together creating a glycosylation site), **T267K**, **D339N**, **I380V**, **T392I** and **P468H**. There are several subgroups defined by amino acid substitutions in NA:

- Recent viruses with HA genes in clade **3C.3a** have **NAs** with the substitutions **K75R**, **L140I**, **V149A**, **Y155H**, **N329T** (resulting in the loss of a glycosylation site) and **E344K**; many have the substitutions **I57M**, **N161S** and **P386S**;
- Recently collected viruses with HA genes in subclade **3C.2a1b** share the **NA** amino acid substitutions **P126L**, **K220N**, **V303I** and **N329S** (resulting in the loss of a glycosylation site); many have **E344K** in the **NA** and, of these, some also have substitution **S315R**, which has been seen predominantly in viruses with HA genes in subclade **3C.2a1b+T131K**.

Antiviral resistance

Phenotypic testing to assess susceptibility to oseltamivir and zanamivir was performed on 179 viruses with collection dates after 2020-01-31. None showed reduced or highly reduced inhibition by both oseltamivir and zanamivir (**Table 13**).

Of the 256 influenza A(H3N2) viruses that yielded full-length PA gene sequences all encoded isoleucine at position 38 (I38) and showed no evidence of other substitutions associated with reduced susceptibility to baloxavir marboxil.

Conclusion.

Received A(H3N2) samples with collection dates after 2020-01-31 have HAs in clade 3C.3a and subclade 3C.2a1b. Viruses in clade 3C.3a were recognised well in HI assays and PRNA by antisera raised against A/Kansas/14/2017-like viruses but not well by antisera raised against reference viruses in subclade 3C.2a1b. Viruses in subclade 3C.2a1b fell into three main genetic groups (subclades 3C.2a1b+T131K, 3C.2a1b+T135K-A and 3C.2a1b+T135KB) and all were recognised poorly by antisera raised against clade 3C.3a viruses. Each of these three 3C.2a1b subclades (+T131K, +T135K-A and +T135K-B groups) were best recognised by post-infection ferret antisera raised against viruses within the same group, with those raised against cell culture-propagated viruses giving better recognition than those raised against egg-propagated viruses. All H3N2 viruses were sensitive to the neuraminidase inhibitors and the PA inhibitor baloxavir marboxil.

Table 6. Summary of influenza A(H3N2) viruses collected after 2020-01-31, analysed by HI

MONTH	H3N2																
			Post-infection ferret antisera raised against: fold reduction in titres														
			Egg-propagated A/South Australia/34/2019			Egg-propagated A/Hong Kong/2671/2019			Cell culture-propagated A/Hong Kong/2671/2019			Egg-propagated NYMC X-327 [A/Kansas/14/2017]			Cell culture-propagated A/Kansas/14/2017		
Country	Number received ¹	Number propagated ²	≤2 fold ³	4 fold ³	≥8 fold ³	≤2 fold ⁴	4 fold ⁴	≥8 fold ⁴	≤2 fold ⁵	4 fold ⁵	≥8 fold ⁵	≤2 fold ⁶	4 fold ⁶	≥8 fold ⁶	≤2 fold ⁷	4 fold ⁷	≥8 fold ⁷
2020																	
FEBRUARY																	
Algeria	4	1			1			1	1					1			1
Belgium	17	17			17		1	16		1	16		3	14	16		1
Bosnia and Herzegovina	5	2			2			2			2		2		2		
Bulgaria	15	10			9			10	3	2	5	3	3	4	7		3
Cote d'Ivoire	1	0															
Cyprus	21	12			12			12			12	1	9	2	12		
Denmark	6	4	1		3	1	1	2	2		2		3	1	3		1
Estonia	2	2			2			2			2			2	1	1	
Finland	1	1			1			1			1			1	1		
France	7	7		1	6		1	6		1	6	1	3	3	5		2
Germany	7	5		1	4			5		2	1	1	3	1	4		1
Hungary	4	3		1	2			3	1	1	1		2	1	2		1
Iceland	2	2			2			2			2		1	1	1	1	
Italy	12	12		1	11			11	5	4	3			12	3	1	8
Kyrgyzstan	2	2			2		2	2	2					2		2	
Lebanon	10	2			2			2		1	1		1	1	1		1
Macedonia	11	9			9		2	7	1	2	6	1	5	3	6	3	
Moldova	1	0															
Morocco	10	0															
Norway	5	2			2			2			2			2	2		
Poland	11	4			4			4			4		3	1	4		
Portugal	2	1			1			1			1		1		1		
Slovakia	4	4			4			4			4	2	2		4		
Slovenia	9	6			6			6			6		1	5	6		
Spain	3	2			2			2			2			2	2		
Sweden	5	3			3			3	1		2		1	2	2		1
Switzerland	2	2			2			2			2			2	2		
United Kingdom	2	2			2			2			2		2		2		
MARCH																	
Bulgaria	1	1			1	1			1				1		1		
Cote d'Ivoire	2	0															
Cyprus	1	1			1			1			1			1	1		
Estonia	3	2			2			2			2		1	1	2		
Finland	2	2			2			2			2			2			2
France	4	3			3			3		1	2		1	2	2		1
Germany	2	2			2			2			2		2		2		
Hungary	1	1			1			1			1			1	1		
Iceland	6	6			6		1	5	1		5		1	5	5	1	1
Ireland	1	1		1			1		1					1		1	
Lebanon	2	1			1			1	1					1			1
Moldova	1	1						1			1		1		1		
Norway	18	15		5	10			15		6	9			15	3	2	10
Portugal	1	1			1			1	1					1	1		
Russia	3	3			3		1	2	1	1	1			3	2	1	
Slovenia	2	2			2			2			2		1	1	1		1
Spain	1	1						1			1			1		1	
Switzerland	3	2			2			2		1	1			2			2
APRIL																	
MAY																	
JUNE																	
JULY																	
AUGUST																	
	235	162	1	10	145	2	11	149	22	23	115	9	53	100	108	15	39
30 Countries			0.6%	6.4%	92.9%	1.2%	6.8%	92.0%	13.8%	14.4%	71.9%	5.6%	32.7%	61.7%	66.7%	9.3%	24.1%

1. Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared

2. Propagated to sufficient titre to perform HI assay in the presence of 20nM oseltamivir

3. Compared to homologous titres for ferret antisera raised against egg-propagated A/South Australia/34/2019 (3C.2a1b+T131K)

4. Compared to homologous titres for ferret antisera raised against egg-propagated A/Hong Kong/2671/2019 (3C.2a1b+T131K-B)

5. Compared to homologous titres for ferret antisera raised against tissue culture-propagated A/Hong Kong/2671/2019 (3C.2a1b+T131K-B)

6. Compared to homologous titres for ferret antisera raised against egg-propagated NYMC X-327 [A/Kansas/14/2017] (3C.3a)

7. Compared to homologous titres for ferret antisera raised against tissue culture-propagated A/Kansas/14/2017 (3C.3a)

Numbers highlighted in blue show the number of viruses subjected to HI assay for 'completed' sample sets. Under a 'sequence first' virus characterisation scheme: where multiple viruses shared the same HA sequence only a selection were propagated to allow assay by HI

Table 7-1. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2020-02-14

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/HK 5738/14 MDCK	A/Singapore 0019/16 Egg 10 ⁴	A/Norway 3275/18 SIAT	A/Sth Aus 34/19 Egg	A/La Rioja 2202/18 SIAT	A/HK 2671/19 Egg	A/HK 2669/19 SIAT	A/Eng 538/18 SIAT	NYMC X-327 A/Kansas/14 Egg	A/Kansas 14/17 SIAT
				St Jude's F60/17 ¹	F13/19 ¹	F03/19 ¹	F45/19 ¹	F26/18 ¹	F44/19 ¹	F04/20 ¹	F31/18 ¹	F16/19 ¹	F17/19 ¹
	Passage history			3C.2a	3C.2a1	3C.2a1b+T131K	3C.2a1b+T131K	3C.2a1b+T135K-A	3C.2a1b+T135K-B	3C.2a1b+T135K-B	3C.3a	3C.3a	3C.3a
	Ferret number												
	Genetic group												
REFERENCE VIRUSES													
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK3/SIAT2	160	320	160	160	160	40	40	160	160	160
A/Singapore/INFIMH-16-0019/2016	3C.2a1	2016-04-14	E5/E2	160	320	40	40	160	80	80	80	40	40
A/South Australia/34/2019	3C.2a1b+T131K	2019-02-06	E6/E1	160	320	640	1280	80	320	<	80	80	40
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	2019-06-17	E8/E1	40	160	<	80	40	640	160	160	640	80
A/Hong Kong/2669/2019	3C.2a1b+T135K-B	2019-06-18	MDCK1/SIAT5	80	160	160	80	160	160	320	80	80	80
A/England/538/2018	3C.3a	2018-02-26	MDCK1/SIAT4	40	80	<	<	<	80	<	640	320	320
NYMC X-327 (A/Kansas/14/17)	3C.3a	2017-12-14	Ex/E1	<	40	<	<	<	320	40	320	1280	320
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2	40	80	<	<	<	40	<	640	320	320
TEST VIRUSES													
A/Netherlands/01624/2019	3C.3a	2019-11-27	hCK1/SIAT1	40	80	<	<	<	40	<	640	320	320
A/Netherlands/10267/2019	3C.3a	2019-12-02	MDCK-MIX2/SIAT1	80	160	40	40	40	40	<	640	320	640
A/Netherlands/01672/2019	3C.3a	2019-12-04	hCK1/SIAT1	<	40	<	<	<	80	<	640	320	320
A/Netherlands/10268/2019	3C.3a	2019-12-10	MDCK-MIX3/SIAT1	80	160	40	80	40	80	<	640	320	640
A/Netherlands/10273/2019	3C.3a	2019-12-16	MDCK-MIX2/SIAT1	80	320	160	160	80	160	40	1280	640	640
A/Netherlands/10279/2019	3C.3a	2019-12-20	MDCK-MIX2/SIAT1	40	160	80	80	<	80	<	640	320	320
A/Lithuania/MB42123/2019	3C.3a	2019-12-20	SIAT1/SIAT1	40	80	<	<	<	40	<	640	320	320

Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) 1 < = <40

Vaccine
NH 2018-19

Vaccine
SH 2020

Vaccine
NH 2020-21

Vaccine
NH 2019-20

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 7-2. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2020-02-21

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/HK	A/Singapore	A/Norway	A/Sth Aus	A/La Rioja	A/HK	A/HK	A/Eng	NYMC X-327	A/Kansas
				5738/14	0019/16	3275/18	34/19	2202/18	2671/19	2669/19	538/18	A/Kansas/14	14/17
				MDCK	Egg 10 ⁻⁴	SIAT	Egg	SIAT	Egg	SIAT	SIAT	Egg	SIAT
Passage history				St Jude's	F13/19 ¹	F03/19 ¹	F45/19 ¹	F26/18 ¹	F44/19 ¹	F04/20 ¹	F31/18 ¹	F16/19 ¹	F17/19 ¹
Ferret number				F60/17 ¹									
Genetic group				3C.2a	3C.2a1	3C.2a1b+T131K	3C.2a1b+T131K	3C.2a1b+T135K-A	3C.2a1b+T135K-B	3C.2a1b+T135K-B	3C.3a	3C.3a	3C.3a
REFERENCE VIRUSES													
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK3/SIAT2	160	320	160	160	160	40	40	160	160	160
A/Singapore/INFIMH-16-0019/2016	3C.2a1	2016-04-14	E5/E2	160	320	40	40	160	80	80	80	40	40
A/South Australia/34/2019	3C.2a1b+T131K	2019-02-06	E6/E1	160	320	1280	1280	80	320	40	80	80	40
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	2019-06-17	E8/E1	40	80	<	80	40	640	160	160	640	80
A/Hong Kong/2669/2019	3C.2a1b+T135K-B	2019-06-18	MDCK1/SIAT5	160	160	160	80	320	160	320	160	80	80
A/England/538/2018	3C.3a	2018-02-26	MDCK1/SIAT4	40	40	<	40	<	40	<	640	160	320
NYMC X-327 (A/Kansas/14/17)	3C.3a	2017-12-14	Ex/E1	<	40	<	<	<	160	<	160	640	160
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2	40	80	<	<	<	40	<	320	160	320
TEST VIRUSES													
A/Catalonia/11962S/2019	No seq	2019-11-27	P0/SIAT1	80	80	<	40	<	80	<	320	160	320
A/Czech Republic/1633/2019	3C.3a	2019-11-19	P3/SIAT1	40	80	40	40	<	40	<	320	160	320
A/Catalonia/11960S/2019	3C.3a	2019-11-27	P0/SIAT1	40	80	<	40	<	80	<	160	160	320
A/Czech Republic/1670/2019	3C.3a	2019-11-28	P3/SIAT1	40	80	40	40	40	320	<	320	160	320
A/Lithuania/MB41285/2019	3C.3a	2019-12-09	SIAT2/SIAT1	40	80	40	40	<	40	<	320	160	320
A/Lithuania/MB40700/2019	3C.3a	2019-12-09	SIAT2/SIAT1	40	80	40	40	<	40	<	320	160	320
A/Lithuania/MB41882/2019	3C.3a	2019-12-18	SIAT1/SIAT1	<	40	<	<	<	40	<	320	160	160
A/Mauritius/I-34/2020	3C.3a	2020-01-05	MDCK1/SIAT1	40	80	40	40	<	40	<	640	160	320
A/Mauritius/I-2000/2019	3C.2a1b+T131K	2019-07-18	MDCK2/SIAT1	160	320	320	320	160	80	80	160	160	80
A/Mauritius/I-3006/2019	3C.2a1b+T131K	2019-11-21	MDCK1/SIAT1	80	160	320	320	160	80	40	160	80	80
A/Friuli Venezia Giulia/233/2019	3C.2a1b+T131K	2019-11-28	SIAT2/SIAT1	160	160	320	160	160	160	80	80	80	80
A/Mauritius/I-19/2020	3C.2a1b+T131K	2020-01-02	MDCK1/SIAT1	160	160	320	320	160	40	80	160	160	80
A/Lithuania/MB41185/2019	3C.2a1b+T135K-A	2019-12-11	SIAT1/SIAT1	40	80	80	<	40	40	160	40	<	40
A/Lithuania/MB41235/2019	3C.2a1b+T135K-A	2019-12-13	SIAT2/SIAT1	40	40	80	<	40	40	160	40	<	<
Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) 1 < = <40				Vaccine NH 2018-19		Vaccine SH 2020		Vaccine NH 2020-21		Vaccine NH 2019-20			

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)

Table 7-3. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2020-02-28

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/HK	A/Singapore	A/Norway	A/Sth Aus	A/La Rioja	A/HK	A/HK	A/Eng	NYMC X-327	A/Kansas
				5738/14	0019/16	3275/18	34/19	2202/18	2671/19	2669/19	538/18	A/Kansas/14	14/17
	Passage history			MDCK	Egg 10 ⁴	SIAT	Egg	SIAT	Egg	SIAT	SIAT	Egg	SIAT
	Ferret number			St Jude's	F13/19 ¹	F03/19 ¹	F45/19 ¹	F26/18 ¹	F44/19 ¹	F04/20 ¹	F31/18 ¹	F16/19 ¹	F17/19 ¹
	Genetic group			3C.2a	3C.2a1	3C.2a1b+T131K	3C.2a1b+T131K	3C.2a1b+T135K-A	3C.2a1b+T135K-B	3C.2a1b+T135K-B	3C.3a	3C.3a	3C.3a
REFERENCE VIRUSES													
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK3/SIAT2	160	320	160	160	160	40	40	320	160	160
A/Singapore/INFIMH-16-0019/2016	3C.2a1	2016-04-14	E8/E2	160	640	80	80	320	80	80	80	80	<
A/South Australia/34/2019	3C.2a1b+T131K	2019-02-06	E6/E1	160	320	640	1280	80	160	<	80	80	40
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	2019-06-17	E8/E1	40	80	<	40	40	640	160	160	320	80
A/Hong Kong/2669/2019	3C.2a1b+T135K-B	2019-06-18	MDCK1/SIAT5	80	160	160	80	160	160	320	160	80	80
A/England/538/2018	3C.3a	2018-02-26	MDCK1/SIAT4	40	40	<	<	<	40	<	640	160	320
NYMC X-327 (A/Kansas/14/17)	3C.3a	2017-12-14	Ex/E1	40	40	<	<	<	160	<	320	1280	320
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2	40	80	<	<	<	40	<	640	160	320
TEST VIRUSES													
A/England/530/2019	No seq	2019-12-02	SIAT1/SIAT1	80	80	40	<	40	40	<	640	320	320
A/England/599/2019	3C.3a	2019-11-13	SIAT1/SIAT1	80	80	<	<	<	40	<	640	160	320
A/England/476/2019	3C.3a	2019-11-20	SIAT1/SIAT1	80	80	<	<	<	40	<	640	320	320
A/England/601/2019	3C.3a	2019-11-24	SIAT2/SIAT1	80	160	80	40	40	40	<	640	320	640
A/England/531/2019	3C.3a	2019-11-25	SIAT1/SIAT1	80	160	40	40	<	40	<	640	320	320
A/England/513/2019	3C.3a	2019-11-26	SIAT1/SIAT1	80	80	<	<	<	40	<	320	160	320
A/England/612/2019	3C.3a	2019-11-29	SIAT1/SIAT1	80	160	40	80	40	40	<	640	320	320
A/England/551/2019	3C.3a	2019-11-29	SIAT1/SIAT1	80	80	<	<	<	40	<	640	160	320
A/England/521/2019	3C.3a	2019-11-29	SIAT1/SIAT1	80	80	40	40	40	40	<	640	160	320
A/England/651/2019	3C.3a	2019-12-05	SIAT1/SIAT1	80	80	<	40	<	40	<	640	320	320
A/England/645/2019	3C.3a	2019-12-05	SIAT1/SIAT1	80	160	40	<	40	<	<	640	320	640
A/England/648/2019	3C.3a	2019-12-07	SIAT1/SIAT1	80	160	80	80	80	80	<	640	320	640
A/England/675/2019	3C.3a	2019-12-08	SIAT1/SIAT1	80	160	40	40	40	40	<	640	320	640
A/England/194840298/2019	3C.2a1b+T131K	2019-11-24	SIAT2/SIAT1	80	160	160	320	160	<	160	160	80	160
A/England/194840192/2019	3C.2a1b+T131K	2019-11-25	SIAT2/SIAT1	160	160	320	320	320	80	160	160	160	160
A/Ghana/3361/2019	3C.2a1b+T135K-A	2019-10-28	SIAT1	80	160	160	80	160	40	320	160	80	40
A/Ghana/3532/2019	3C.2a1b+T135K-A	2019-11-04	SIAT1	80	80	160	40	160	40	160	80	80	40
A/England/194660616/2019	3C.2a1b+T135K-A	2019-11-12	SIAT1/SIAT1	40	40	40	<	40	<	80	<	<	<
A/England/387/2019	3C.2a1b+T135K-A	2019-11-13	MDCK1/SIAT1	40	80	80	<	40	80	40	160	80	40
A/England/194820759/2019	3C.2a1b+T135K-B	2019-11-17	SIAT2/SIAT1	80	160	160	<	80	80	160	40	<	40

Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) 1 < = <40

Vaccine
NH 2018-19

Vaccine
SH 2020

Vaccine
NH 2020-21

Vaccine
NH 2019-20

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 7-4. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2020-03-20

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/HK	A/Singapore	A/Norway	A/Sth Aus	A/La Rioja	A/HK	A/HK	A/Eng	NYMC X-327	A/Kansas
				5738/14	0019/16	3275/18	34/19	2202/18	2671/19	2669/19	538/18	A/Kansas/14	14/17
	Passage history			MDCK	Egg 10 ⁴	SIAT	Egg	SIAT	Egg	SIAT	SIAT	Egg	SIAT
	Ferret number			St Jude's	F13/19 ¹	F03/19 ¹	F45/19 ¹	F26/18 ¹	F44/19 ¹	F04/20 ¹	F31/18 ¹	F16/19 ¹	F17/19 ¹
	Genetic group			3C.2a	3C.2a1	3C.2a1b+T131K	3C.2a1b+T131K	3C.2a1b+T135K-A	3C.2a1b+T135K-B	3C.2a1b+T135K-B	3C.3a	3C.3a	3C.3a
REFERENCE VIRUSES													
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK3/SIAT2	160	160	160	160	80	<	40	160	160	80
A/Singapore/INFIMH-16-0019/2016	3C.2a1	2016-04-14	E5/E2	160	320	40	40	160	80	80	40	40	<
A/South Australia/34/2019	3C.2a1b+T131K	2019-02-06	E6/E1	160	320	640	640	40	160	<	40	40	40
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	2019-06-17	E8/E2	40	160	<	40	40	640	160	160	320	80
A/Hong Kong/2669/2019	3C.2a1b+T135K-B	2019-06-18	MDCK1/SIAT5	160	160	320	160	160	160	320	160	80	160
A/England/538/2018	3C.3a	2018-02-26	MDCK1/SIAT4	40	40	<	<	<	40	<	640	160	320
NYMC X-327 (A/Kansas/14/17)	3C.3a	2017-12-14	Ex/E1	<	<	<	<	<	160	<	320	640	320
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2	40	80	<	<	<	40	<	640	160	320
TEST VIRUSES													
A/Ireland/85561/2019	3C.3a	2019-10-24	SIAT2	40	80	<	<	<	<	<	320	160	320
A/Ireland/88070/2019	3C.3a	2019-11-02	SIAT2	40	80	<	<	<	<	<	320	160	320
A/Ireland/89141/2019	3C.3a	2019-11-06	SIAT2	40	80	40	40	40	40	<	640	320	320
A/Ireland/91781/2019	3C.3a	2019-11-17	SIAT2	40	80	<	<	<	<	<	320	160	320
A/Belgium/G0506/2019	3C.3a	2019-11-28	SIAT2	80	160	40	40	40	40	<	640	320	640
A/Belgium/G0514/2019	3C.3a	2019-12-10	SIAT1	40	80	<	<	<	40	<	640	320	320
A/Belgium/G0528/2019	3C.3a	2019-12-19	SIAT1	80	160	<	40	40	40	<	640	160	320
A/Belgium/G0542/2019	3C.3a	2019-12-23	SIAT1	40	160	<	40	40	40	<	640	320	640
A/Bulgaria/1575/2019	3C.3a	2019-12-23	SIAT1	40	80	<	<	<	40	<	640	160	320
A/South Lebanon_Tyre/NS20/2019	3C.3a	2019-12-23	SIAT1	40	80	<	<	<	<	<	320	160	320
A/Bayern/2/2020	3C.3a	2020-01-13	C1/SIAT1	40	80	<	<	<	40	<	640	160	320
A/Athens.GR/84/2020	3C.3a	2020-01-14	SIAT1	160	320	640	320	40	160	40	640	320	320
A/Sachsen/5/2020	3C.3a	2020-01-21	C1/SIAT1	<	<	<	<	<	<	<	320	160	160
A/Berlin/21/2020	3C.3a	2020-01-28	C1/SIAT1	40	80	<	<	<	<	<	320	160	320
A/Thuringen/11/2020	3C.3a	2020-01-30	C1/SIAT1	40	80	<	<	<	<	<	320	160	160
A/Athens.GR/1848/2019	3C.2a1b+T135K-A	2019-10-02	SIAT1	80	160	320	80	160	80	160	160	80	80
A/Lithuania/MB40701/2019	3C.2a1b+T135K-A	2019-12-07	SIAT1/SIAT2	40	80	80	<	80	<	160	40	<	<
A/Baden-Wuerttemberg/17/2020	3C.2a1b+T135K-A	2020-01-21	C2/SIAT2	80	160	160	40	160	80	160	160	80	160
A/Qatar/10-VI-19-0069058/2019	3C.2a1b+T135K-B	2019-11-06	SIAT1	80	160	80	<	80	80	160	80	<	40
A/Ireland/91453/2019	3C.2a1b+T135K-B	2019-11-17	SIAT1	40	80	80	<	<	80	160	40	<	40
A/Kyrgyzstan/305/2019	3C.2a1b+T135K-B	2019-12-16	SIAT1	80	160	160	80	160	160	320	160	80	80
A/Kyrgyzstan/168/2019	3C.2a1b+T135K-B	2019-12-16	SIAT1	80	160	160	80	160	80	320	80	40	80
A/Kyrgyzstan/166/2019	3C.2a1b+T135K-B	2019-12-16	SIAT1	80	160	160	80	160	80	320	80	80	80
A/Niedersachsen/51/2020	3C.3a	2020-02-11	C1/SIAT1	40	80	<	<	<	<	<	320	160	160
A/Schleswig-Holstein/2/2020	3C.2a1b+T135K-A	2020-02-03	C1/SIAT1	160	320	160	160	160	80	160	160	160	160

Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) 1 <= 40

Vaccine
NH 2018-19

Vaccine
SH 2020

Vaccine
NH 2020-21

Vaccine
NH 2019-20

< 4-fold
4-fold
8-fold
> 8-fold
not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 7-5. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2020-06-03

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				NEW									
				A/HK	A/Singapore	A/Norway	A/Sth Aus	A/La Rioja	A/HK	A/HK	A/Eng	NYMC X-327	A/Kansas
				5738/14	0019/16	3275/18	34/19	2202/18	2671/19	2671/19	2669/19	538/18	14/17
	Passage history			MDCK	Egg 10 ⁻⁴	SIAT	Egg	SIAT	Egg	Cell	SIAT	SIAT	SIAT
	Ferret number			St Jude's F60/17 ¹	F13/19 ¹	F03/19 ¹	F45/19 ¹	F26/18 ¹	F44/19 ¹	St Jude's F21/20 ¹	F04/20 ¹	F31/18 ¹	F16/19 ¹
	Genetic group			3C.2a	3C.2a1	3C.2a1b+T131K	3C.2a1b+T131K	3C.2a1b+T135K-A	3C.2a1b+T135K-B	3C.2a1b+T135K-B	3C.2a1b+T135K-B	3C.3a	3C.3a
REFERENCE VIRUSES													
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK3/SIAT2	320	320	320	160	160	<	160	80	320	160
A/Singapore/INFIMH-16-0019/2016	3C.2a1	2016-04-14	E5/E2	80	320	40	40	160	40	40	40	<	<
A/South Australia/34/2019	3C.2a1b+T131K	2019-02-06	E6/E1	160	320	640	1280	80	160	40	<	80	40
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	2019-06-17	E8/E2	40	160	<	80	40	640	160	160	640	80
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	2019-06-17	MDCK1/SIAT4	160	320	320	320	320	320	640	640	320	160
A/Hong Kong/2669/2019	3C.2a1b+T135K-B	2019-06-18	MDCK1/SIAT5	160	320	320	160	160	160	640	320	160	160
A/England/538/2018	3C.3a	2018-02-26	MDCK1/SIAT4	80	80	<	40	40	80	160	<	640	160
NYMC X-327 (A/Kansas/14/17)	3C.3a	2017-12-14	Ex/E1	<	40	<	<	<	160	80	<	320	640
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2	40	80	<	40	40	40	160	<	640	320
TEST VIRUSES													
A/Bretagne/775/2020	3C.3a	2020-02-03	MDCK2/SIAT1	40	80	<	40	80	<	80	40	640	320
A/England/103/2020	3C.3a	2020-02-05	SIAT1/SIAT1	40	80	<	<	40	40	80	<	640	160
A/Bulgaria/912/2020	3C.3a	2020-02-06	SIAT2/SIAT1	80	80	<	<	80	<	80	<	640	320
A/Bratislava/225/2020	3C.3a	2020-02-06	MDCKx/SIAT1	<	80	<	<	40	40	80	<	640	320
A/Bulgaria/927/2020	3C.3a	2020-02-07	SIAT2/SIAT1	80	160	80	80	80	80	160	<	640	320
A/England/107/2020	3C.3a	2020-02-11	SIAT1/SIAT1	<	40	<	<	<	<	80	<	640	160
A/Sachsen/38/2020	3C.3a	2020-02-14	C1/SIAT1	80	80	<	<	40	<	160	<	640	320
A/Bulgaria/1024/2020	3C.3a	2020-02-16	SIAT2/SIAT1	<	<	40	80	40	40	80	<	640	160
A/Bulgaria/1009/2020	3C.3a	2020-02-17	SIAT2/SIAT1	<	80	<	<	<	<	40	<	640	160
A/Bulgaria/1008/2020	3C.3a	2020-02-17	SIAT2/SIAT1	80	80	<	40	40	40	80	<	640	320
A/Bourgogne/1249/2020	3C.3a	2020-02-17	MDCK1/SIAT1	40	80	<	40	40	<	80	<	640	160
A/L'ubica/314/2020	3C.3a	2020-02-17	MDCK1/SIAT1	<	80	<	<	<	<	80	<	640	320
A/L'ubica/315/2020	3C.3a	2020-02-18	MDCK1/SIAT1	<	80	<	<	<	<	80	<	320	160
A/Bulgaria/1052/2020	3C.3a	2020-02-21	SIAT2/SIAT1	80	160	<	40	40	<	160	<	640	160
A/Bratislava/319/2020	3C.3a	2020-02-21	MDCK1/SIAT1	40	80	<	<	40	40	80	<	640	160
A/Brandenburg/9/2020	3C.3a	2020-02-23	C1/SIAT1	80	80	<	<	40	<	80	<	640	160
A/Rheinland-Pfalz/26/2020	3C.3a	2020-03-02	C1/SIAT1	80	80	<	40	40	<	80	<	640	160
A/Bremen/7/2020	3C.3a	2020-03-02	C1/SIAT1	40	80	<	<	40	<	80	<	640	160
A/Baden-Wuerttemberg/95/2020	3C.2a1b+T135K-A	2020-02-10	C1/SIAT1	40	160	40	<	80	40	160	160	160	<
A/Nord pas de Calais/1108/2020	3C.2a1b+T135K-A	2020-02-10	MDCK2/SIAT1	40	80	80	40	80	80	160	80	320	160
A/Bulgaria/1772/2020	3C.2a1b+T135K-A	2020-03-09	SIAT2/SIAT1	80	320	160	80	160	320	320	160	320	160
A/Bulgaria/906/2020	3C.2a1b+T135K-B	2020-02-04	SIAT2/SIAT1	40	80	40	<	80	40	320	160	40	<
A/Bulgaria/962/2020	3C.2a1b+T135K-B	2020-02-11	SIAT2/SIAT1	80	160	80	<	160	80	320	160	80	<
A/Bulgaria/969/2020	3C.2a1b+T135K-B	2020-02-12	SIAT2/SIAT1	40	40	80	<	40	80	320	160	40	<

Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) 1 < = <40

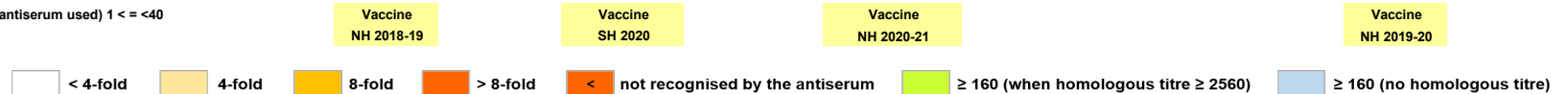


Table 7-6. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2020-06-12

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/HK 5738/14	A/Singapore 0019/16	A/Norway 3275/18	A/Sth Aus 34/19	A/La Rioja 2202/18	A/HK 2671/19	A/HK 2671/19	A/Eng 538/18	NYMC X-327 A/Kansas/14	A/Kansas 14/17
				MDCK	Egg 10 ⁻⁴	SIAT	Egg	SIAT	Cell	Cell	SIAT	Egg	SIAT
	Passage history			St Jude's F60/17 ¹	F13/19 ¹	F03/19 ¹	F45/19 ¹	F26/18 ¹	F44/19 ¹	St Jude's F21/20 ¹	F31/18 ¹	F16/19 ¹	F17/19 ¹
	Ferret number			3C.2a	3C.2a1	3C.2a1b+T131K	3C.2a1b+T131K	3C.2a1b+T135K-A	3C.2a1b+T135K-B	3C.2a1b+T135K-B	3C.3a	3C.3a	3C.3a
	Genetic group												
REFERENCE VIRUSES													
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK3/SIAT2	320	320	320	160	160	<	160	320	160	160
A/Singapore/INFIMH-16-0019/2016	3C.2a1	2016-04-14	E5/E2	80	320	40	40	160	40	40	40	40	<
A/South Australia/34/2019	3C.2a1b+T131K	2019-02-06	E6/E1	160	320	640	1280	80	320	40	80	40	40
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	2019-06-17	E8/E2	40	80	<	80	40	640	160	160	640	80
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	2019-06-17	CK1/SIAT4	160	320	320	160	320	320	640	320	160	160
A/England/538/2018	3C.3a	2018-02-26	MDCK1/SIAT4	40	80	<	40	40	80	160	640	320	640
NYMC X-327 (A/Kansas/14/17)	3C.3a	2017-12-14	Ex/E1	40	40	<	80	<	160	80	320	1280	320
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2	40	80	<	40	40	40	160	640	320	320
TEST VIRUSES													
A/Iceland/23/2020	No seq	2020-03-09	MDCK1/SIAT1	<	80	40	40	40	160	320	80	<	40
A/Belgium/S0751/2020	3C.3a	2020-02-01	SIAT1/SIAT2	<	40	<	40	40	40	40	640	160	320
A/Slovenia/944/2020	3C.3a	2020-02-03	MDCKx/SIAT1	40	80	<	40	<	40	<	640	160	320
A/Belgium/S0571/2020	3C.3a	2020-02-03	SIAT1/SIAT1	40	80	<	40	<	40	40	640	160	320
A/Belgium/G0229/2020	3C.3a	2020-02-03	SIAT1/SIAT1	<	<	<	40	<	<	<	320	80	160
A/Stockholm/16/2020	3C.3a	2020-02-04	SIAT0/SIAT1	40	80	<	80	<	40	80	640	320	640
A/Belgium/G0214/2020	3C.3a	2020-02-05	SIAT1/SIAT2	<	40	<	40	<	40	40	320	160	320
A/Belgium/S1258/2020	3C.3a	2020-02-07	SIAT1/SIAT1	80	80	<	80	<	40	80	640	320	320
A/Belgium/S0710/2020	3C.3a	2020-02-08	SIAT1/SIAT1	40	160	<	80	<	40	80	320	320	320
A/Belgium/H0017/2020	3C.3a	2020-02-09	SIAT1/SIAT1	<	40	<	80	<	40	40	320	160	320
A/Belgium/S0734/2020	3C.3a	2020-02-10	SIAT1/SIAT1	40	80	<	80	<	40	80	640	320	320
A/Belgium/G0275/2020	3C.3a	2020-02-10	SIAT1/SIAT1	40	80	<	40	<	40	40	640	160	320
A/Belgium/G0274/2020	3C.3a	2020-02-10	SIAT1/SIAT1	40	80	<	80	<	40	40	320	160	320
A/Belgium/G0270/2020	3C.3a	2020-02-10	SIAT1/SIAT1	<	40	<	40	<	<	40	320	160	320
A/Belgium/S0900/2020	3C.3a	2020-02-11	SIAT1/SIAT1	<	40	<	40	<	<	40	320	160	160
A/Belgium/S0831/2020	3C.3a	2020-02-11	SIAT1/SIAT1	<	80	<	40	<	40	40	320	160	320
A/Slovenia/1270/2020	3C.3a	2020-02-12	SIATx/SIAT1	40	80	40	80	40	40	80	640	320	320
A/Belgium/S0826/2020	3C.3a	2020-02-12	SIAT1/SIAT1	40	80	<	40	<	40	40	320	160	320
A/Belgium/S0878/2020	3C.3a	2020-02-14	SIAT1/SIAT2	<	40	<	40	<	40	40	320	160	320
A/Stockholm/30/2020	3C.3a	2020-02-17	SIAT0/SIAT1	<	40	<	40	<	40	40	640	160	320
A/Slovenia/1451/2020	3C.3a	2020-02-18	SIATx/SIAT1	40	80	<	40	<	40	80	640	160	320
A/Slovenia/1465/2020	3C.3a	2020-02-19	SIATx/SIAT1	<	40	<	40	<	40	40	640	160	320
A/Belgium/S1003/2020	3C.3a	2020-02-19	SIAT1/SIAT1	40	40	<	40	<	40	40	320	160	320
A/Slovenia/1506/2020	3C.3a	2020-02-21	SIATx/SIAT1	40	80	<	40	<	40	40	640	160	320
A/Slovenia/1650/2020	3C.3a	2020-02-27	SIATx/SIAT1	40	80	80	80	<	40	40	640	160	320
A/Iceland/18/2020	3C.3a	2020-02-29	MDCK1/SIAT1	40	80	<	40	<	40	80	640	320	320
A/Slovenia/1708/2020	3C.3a	2020-03-02	SIATx/SIAT1	40	80	<	80	<	40	80	640	320	320
A/Iceland/25/2020	3C.3a	2020-03-12	MDCK1/SIAT1	40	80	<	40	<	40	80	640	160	320
A/Iceland/30/2020	3C.3a	2020-03-13	MDCK1/SIAT1	<	80	<	40	<	<	40	320	160	320
A/Iceland/26/2020	3C.3a	2020-03-13	MDCK1/SIAT1	80	160	40	80	40	80	80	640	320	640
A/Iceland/29/2020	3C.3a	2020-03-16	MDCK1/SIAT1	<	80	<	40	<	40	40	320	160	320
A/Iceland/37/2020	3C.3a	2020-03-26	MDCK1/SIAT1	40	80	<	80	<	40	80	640	160	320
A/Iceland/05/2020	3C.2a1b+T131K	2020-02-21	MDCK1/SIAT2	80	160	160	160	160	40	80	80	80	80
A/Slovenia/1752/2020	3C.2a1b+T131K	2020-03-05	MDCKx/SIAT1	40	160	160	160	40	<	40	<	40	<
A/Belgium/S1081/2020	3C.2a1b+T135K-A	2020-02-24	SIAT1/SIAT1	40	160	40	40	80	160	160	160	80	40
A/Stavropol/10/2020	3C.2a1b+T135K-A	2020-03-03	MDCK2/SIAT1	<	80	<	40	40	80	80	160	80	80
A/S. Petersburg/RIL-180/2020	3C.2a1b+T135K-B	2020-03-10	MDCK2/SIAT1	<	40	<	40	<	80	160	40	40	40
A/Stockholm/13/2020	3C.2a1b+T135K-B	2020-02-02	SIAT0/SIAT1	40	80	80	40	80	80	320	40	<	40

Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) 1 < = <40

Vaccine
NH 2018-19Vaccine
SH 2020Vaccine
NH 2020-21Vaccine
NH 2019-20

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 7-7. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2020-06-23

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/HK 5738/14	A/Singapore 0019/16	A/Norway 3275/18	A/Sth Aus 34/19	A/La Rioja 2202/18	A/HK 2671/19	A/HK 2671/19	A/Eng 538/18	NYMC X-327 A/Kansas/14	A/Kansas 14/17
				MDCK	Egg 10 ⁻⁴	SIAT	Egg	SIAT	Cell	Cell	SIAT	Egg	SIAT
	Passage history			St Jude's F60/17 ¹	F13/19 ¹	F03/19 ¹	F45/19 ¹	F26/18 ¹	F44/19 ¹	St Jude's F21/20 ¹	F31/18 ¹	F16/19 ¹	F17/19 ¹
	Ferret number			3C.2a	3C.2a1	3C.2a1b+T131K	3C.2a1b+T131K	3C.2a1b+T135K-A	3C.2a1b+T135K-B	3C.2a1b+T135K-B	3C.3a	3C.3a	3C.3a
	Genetic group												
REFERENCE VIRUSES													
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK3/SIAT2	320	320	320	160	160	40	160	320	160	160
A/Singapore/INFIMH-16-0019/2016	3C.2a1	2016-04-14	E5/E2	160	320	40	40	160	80	40	40	40	<
A/South Australia/34/2019	3C.2a1b+T131K	2019-02-06	E6/E1	320	320	1280	1280	80	320	40	80	40	40
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	2019-06-17	E8/E2	40	160	<	80	40	640	160	160	640	80
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	2019-06-17	CK1/SIAT4	160	320	320	160	160	320	640	160	160	160
A/England/538/2018	3C.3a	2018-02-26	MDCK1/SIAT4	40	80	<	40	<	40	80	320	160	320
NYMC X-327 (A/Kansas/14/17)	3C.3a	2017-12-14	Ex/E1	80	80	<	80	40	320	160	320	1280	640
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2	40	80	<	40	40	40	80	320	160	320
TEST VIRUSES													
A/Finland/171/2020	3C.2a1b+T135K-A	2020-01-02	SIAT1/SIAT1	80	80	160	80	160	80	320	80	80	40
A/Finland/180/2020	3C.2a1b+T135K-B	2020-01-14	SIAT1/SIAT1	40	80	80	<	80	80	320	40	<	<
A/Finland/181/2020	3C.3a	2020-02-01	SIAT1/SIAT1	40	80	40	40	<	40	40	320	160	320
A/Denmark/1090/2020	3C.3a	2020-02-13	SIAT2/SIAT1	160	160	80	80	80	80	80	640	320	320
A/Denmark/1138/2020	3C.3a	2020-02-18	SIAT2/SIAT1	80	80	80	80	40	80	80	640	320	320
A/Finland/183/2020	3C.2a1b+T131K	2020-03-02	SIAT1/SIAT1	40	40	40	40	80	<	40	40	<	<
A/Finland/184/2020	3C.2a1b+T131K	2020-03-16	SIAT1/SIAT1	40	40	40	40	80	<	40	40	<	<
A/Denmark/1262/2020	3C.2a1b+T135K-A	2020-02-17	SIAT3/SIAT1	1280	640	640	640	640	320	1280	640	320	320
A/Kyrgyzstan Bishkek/65/2020	3C.2a1b+T135K-B	2020-02-02	MDCKx/SIAT1	160	160	160	80	160	160	640	160	80	80
A/Kyrgyzstan Bishkek/63/2020	3C.2a1b+T135K-B	2020-02-03	MDCKx/SIAT1	160	160	160	80	160	160	640	160	80	80
A/Denmark/1295/2020	3C.2a1b+T135K-B	2020-02-18	SIAT3/SIAT1	80	160	160	40	80	160	640	40	40	40

Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) 1 <= <40

Vaccine
NH 2018-19Vaccine
SH 2020Vaccine
NH 2020-21Vaccine
NH 2019-20

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 7-8. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2020-06-30

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				NEW										
				A/HK 5738/14 MDCK St Jude's F60/17 ¹	A/Singapore 0019/16 Egg 10 ⁴ F13/19 ¹	A/Norway 3275/18 SIAT F03/19 ¹	A/Sth Aus 34/19 Egg F45/19 ¹	A/La Rioja 2202/18 SIAT F26/18 ¹	A/HK 2671/19 Egg F44/19 ¹	A/HK 2671/19 Egg St Jude's F20/20 ¹	A/HK 2671/19 Cell F21/20 ¹	A/Eng 538/18 SIAT F31/18 ¹	NYMC X-327 A/Kansas/14 Egg F16/19 ¹	A/Kansas 14/17 SIAT F17/19 ¹
	Passage history			3C.2a	3C.2a1	3C.2a1b+T131K	3C.2a1b+T131K	3C.2a1b+T135K-A	3C.2a1b+T135K-B	3C.2a1b+T135K-B	3C.2a1b+T135K-B	3C.3a	3C.3a	3C.3a
REFERENCE VIRUSES														
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK3/SIAT2	320	320	320	160	160	40	80	160	320	160	160
A/Singapore/INFIMH-16-0019/2016	3C.2a1	2016-04-14	E5/E2	160	320	80	40	320	80	40	40	40	40	<
A/South Australia/34/2019	3C.2a1b+T131K	2019-02-06	E6/E1	160	320	1280	1280	80	320	40	40	80	40	40
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	2019-06-17	E8/E2	40	160	<	80	40	640	320	160	160	640	80
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	2019-06-17	CK1/SIAT4	160	320	320	160	320	320	160	640	160	160	160
A/England/538/2018	3C.3a	2018-02-26	MDCK1/SIAT4	40	80	<	40	<	40	40	80	320	320	320
NYMC X-327 (A/Kansas/14/17)	3C.3a	2017-12-14	Ex/E1	80	80	<	40	40	320	80	160	320	1280	640
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2	40	40	40	40	40	80	40	80	320	160	320
TEST VIRUSES														
A/Friuli Venezia Giulia/229/2020	3C.3a	2020-02-10	SIAT2/SIAT1	40	80	<	<	40	40	<	40	320	160	160
A/Parma/26/2020	3C.3a	2020-02-12	SIAT2/SIAT1	40	80	<	<	<	40	<	40	320	160	320
A/Switzerland/6529/2020	3C.3a	2020-02-21	SIAT1	40	80	<	<	<	40	<	40	320	160	320
A/Switzerland/4684/2020	3C.3a	2020-02-24	SIAT1	40	80	<	<	<	40	<	40	320	160	320
A/Lorraine/1589/2020	3C.3a	2020-02-24	SIAT1	40	80	<	<	<	40	<	40	320	160	320
A/Centre/2293/2020	3C.3a	2020-03-09	SIAT1	40	80	<	<	<	40	<	40	320	160	160
A/Milano/55/2020	3C.2a1b+T131K	2020-02-10	SIAT2/SIAT1	320	320	320	320	320	40	80	160	160	160	160
A/Bretagne/1323/2020	3C.2a1b+T131K	2020-02-20	SIAT1	40	160	80	160	160	160	<	80	40	<	40
A/Picardie/1840/2020	3C.2a1b+T131K	2020-03-02	SIAT1	40	80	80	160	160	40	<	160	40	40	<
A/Parma/21/2020	3C.2a1b+T135K-A	2020-02-05	SIAT2/SIAT1	80	80	160	80	160	80	80	320	160	80	40
A/Friuli Venezia Giulia/240/2020	3C.2a1b+T135K-A	2020-02-08	SIAT2/SIAT1	40	80	80	<	80	40	40	160	40	<	<
A/Milano/49/2020	3C.2a1b+T135K-A	2020-02-10	SIAT2/SIAT1	40	160	80	<	160	160	80	160	80	80	80
A/Friuli Venezia Giulia/228/2020	3C.2a1b+T135K-A	2020-02-10	SIAT2/SIAT1	40	80	160	40	160	80	80	320	40	<	<
A/Parma/23/2020	3C.2a1b+T135K-A	2020-02-10	SIAT2/SIAT1	80	160	160	80	160	80	80	320	80	40	40
A/Friuli Venezia Giulia/257/2020	3C.2a1b+T135K-A	2020-02-11	SIAT2/SIAT1	40	80	80	<	80	40	40	160	40	<	<
A/Friuli Venezia Giulia/244/2020	3C.2a1b+T135K-A	2020-02-11	SIAT2/SIAT1	40	80	80	<	80	40	40	80	40	<	<
A/Parma/31/2020	3C.2a1b+T135K-A	2020-02-12	SIAT2/SIAT1	40	160	160	80	160	40	80	320	40	<	<
A/Algiers/368/2020	3C.2a1b+T135K-A	2020-02-13	SIAT1	80	160	160	40	160	80	80	320	160	80	<
A/Parma/30/2020	3C.2a1b+T135K-A	2020-02-17	SIAT2/SIAT1	40	160	80	80	160	80	80	320	80	40	40
A/Switzerland/6206/2020	3C.2a1b+T135K-A	2020-03-09	SIAT1	80	80	160	80	160	40	80	160	80	80	40
A/Switzerland/8482/2020	3C.2a1b+T135K-A	2020-03-12	SIAT1	<	80	40	<	40	80	80	40	80	<	<

Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) 1 < = <40

Vaccine
NH 2018-19Vaccine
SH 2020Vaccine
NH 2020-21Vaccine
NH 2020-21Vaccine
NH 2019-20

 < 4-fold
 4-fold
 8-fold
 > 8-fold
 not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 7-9. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2020-07-08

Haemagglutination inhibition titre														
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera										
				NEW										
				A/HK	A/Singapore	A/Norway	A/Sth Aus	A/La Rioja	A/HK	A/HK	A/HK	A/Eng	NYMC X-327	A/Kansas
				5738/14	0019/16	3275/18	34/19	2202/18	2671/19	2671/19	2671/19	538/18	A/Kansas/14	14/17
				MDCK	Eqa 10 ⁻⁴	SIAT	Egg	SIAT	Egg	Cell	Cell	SIAT	Egg	SIAT
	Ferret number													
	Genetic group													
REFERENCE VIRUSES														
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK3/SIAT2	320	320	320	320	160	40	160	80	320	160	320
A/Singapore/INFIMH-16-0019/2016	3C.2a1	2016-04-14	E5/E2	160	320	80	80	160	80	40	<	40	40	<
A/South Australia/34/2019	3C.2a1b+T131K	2019-02-06	E6/E1	320	640	1280	1280	80	320	40	<	80	40	40
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	2019-06-17	E8/E2	40	160	<	80	40	640	160	320	160	640	80
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	2019-06-17	CK1/SIAT4	160	320	320	160	320	320	640	640	160	160	160
A/England/538/2018	3C.3a	2018-02-26	MDCK1/SIAT4	40	80	<	40	<	80	80	<	320	320	320
NYMC X-327 (A/Kansas/14/17)	3C.3a	2017-12-14	Ex/E1	80	80	<	80	40	320	80	<	320	1280	640
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2	40	40	40	80	40	80	80	40	320	160	640
TEST VIRUSES														
A/South Africa/9306/2019	3C.3a	2019-07-29	MDCK1/SIAT1	40	80	40	40	<	40	<	<	320	160	320
A/Lyon/CHU/R19.154.50/2019	3C.3a	2019-12-31	MDCK2/SIAT1	40	80	<	40	<	80	<	<	160	160	320
A/Grenoble/185/2020	3C.3a	2020-01-10	MDCK2/SIAT1	40	80	40	40	40	320	<	<	320	160	320
A/Lyon/162/2020	3C.2a1b+T131K	2020-01-16	MDCK2/SIAT1	40	80	40	40	<	40	<	<	320	160	320
A/South Africa/8914/2019	3C.2a1b+T135K-A	2019-07-18	MDCK1/SIAT1	40	80	40	40	<	40	<	<	320	160	320
A/Lyon/86/2020	3C.2a1b+T135K-A	2020-01-09	MDCK2/SIAT1	<	40	<	<	<	40	<	<	320	160	160
A/Lyon/219/2020	3C.2a1b+T135K-A	2020-01-20	MDCK2/SIAT1	40	80	40	40	<	40	<	<	640	160	320
A/South Africa/0803/2020	3C.2a1b+T135K-A	2020-01-22	MDCK2/SIAT1	160	320	320	320	160	80	80	80	160	160	80
A/South Africa/9493/2019	3C.2a1b+T135K-B	2019-07-30	MDCK2/SIAT1	80	160	320	320	160	80	40	40	160	80	80
A/South Africa/9550/2019	3C.2a1b+T135K-B	2019-08-01	MDCK2/SIAT1	160	160	320	160	160	160	80	80	80	80	80
A/Lyon/CHU/R19.146.85/2019	3C.2a1b+T135K-B	2019-12-19	MDCK4/SIAT1	160	160	320	320	160	160	80	80	160	160	80
A/Beirut/421/2020	No seq	2020-03-05	SIAT1	80	160	160	80	160	80	320	320	160	80	80
A/Cyprus/F835/2020	3C.3a	2020-02-02	SIAT1	40	80	80	<	40	40	160	160	40	<	40
A/Cyprus/F913/2020	3C.3a	2020-02-04	SIAT1	40	40	80	<	40	40	160	160	40	<	<
A/Cyprus/F926/2020	3C.3a	2020-02-05	SIAT1	80	80	<	40	<	80	<	<	320	160	320
A/Cyprus/F960/2020	3C.3a	2020-02-07	SIAT1	80	80	40	80	40	80	80	40	640	320	320
A/Portugal/GG66/2020	3C.3a	2020-02-08	SIAT1	40	80	40	40	40	80	80	40	640	320	640
A/Macedonia/450/2020	3C.3a	2020-02-10	SIAT1	160	160	80	160	80	80	80	40	640	640	320
A/Saida/294/2020	3C.3a	2020-02-13	SIAT1	80	80	40	80	40	40	40	<	640	320	320
A/Cyprus/F1108/2020	3C.3a	2020-02-16	SIAT1	80	160	40	80	40	80	80	40	640	320	640
A/Poland/222/2020	3C.3a	2020-02-17	SIAT1	80	160	80	160	80	80	40	40	160	80	640
A/Poland/220/2020	3C.3a	2020-02-17	SIAT1	80	160	40	80	80	80	40	40	640	320	640
A/Macedonia/509/2020	3C.3a	2020-02-17	SIAT1	160	160	80	160	80	80	80	80	640	320	640
A/Macedonia/505/2020	3C.3a	2020-02-17	SIAT1	80	160	80	80	40	80	80	40	640	320	320
A/Macedonia/501/2020	3C.3a	2020-02-17	SIAT1	80	160	80	80	80	80	80	40	640	320	320
A/Avignon/1286/2020	3C.3a	2020-02-17	MDCK2/SIAT1	80	160	40	80	40	80	80	<	640	320	320
A/Poland/2724/2020	3C.3a	2020-02-18	SIAT1	80	160	40	80	40	80	80	40	640	320	640
A/Bosnia and Herzegovina/318/2020	3C.3a	2020-02-18	SIAT1	160	160	80	80	80	80	80	40	640	320	640
A/Macedonia/593/2020	3C.3a	2020-02-21	SIAT1	80	160	40	80	80	80	80	40	640	320	640
A/Bosnia and Herzegovina/360/2020	3C.3a	2020-02-25	SIAT1	80	80	40	80	40	40	40	<	640	320	320
A/Poland/3278/2020	3C.3a	2020-02-26	SIAT1	80	160	40	80	40	80	80	40	640	320	640
A/Macedonia/648/2020	3C.3a	2020-02-27	SIAT1	80	160	80	80	40	80	80	40	640	320	320
A/Lyon/1631/2020	3C.3a	2020-03-01	MDCK3/SIAT1	80	160	40	80	40	80	80	<	640	320	320
A/Saint-Etienne/2539/2020	3C.2a1b+T131K	2020-02-11	MDCK3/SIAT1	80	160	320	320	80	80	80	<	40	40	40
A/Portugal/MS74/2020	3C.2a1b+T131K	2020-03-21	SIAT1	160	160	160	80	160	80	320	40	320	160	160
A/Macedonia/465/2020	3C.2a1b+T135K-A	2020-02-10	SIAT1	160	320	160	160	160	160	320	320	160	160	160
A/Macedonia/607/2020	3C.2a1b+T135K-A	2020-02-24	SIAT1	160	160	320	160	160	160	160	160	160	160	160
A/Beirut/331/2020	3C.2a1b+T135K-A	2020-02-25	SIAT1	80	80	160	160	160	80	160	160	160	160	80
A/Macedonia/650/2020	3C.2a1b+T135K-A	2020-02-27	SIAT1	160	160	320	160	160	80	160	160	320	160	160

Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) 1 < = <40

Vaccine
NH 2018-19Vaccine
SH 2020Vaccine
NH 2020-21Vaccine
NH 2019-20

< 4-fold

4-fold

8-fold

> 8-fold

< not recognised by the antiserum

≥ 160 (when homologous titre ≥ 2560)

≥ 160 (no homologous titre)

Table 7-10. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2020-07-15

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/HK 5738/14	A/Singapore 0019/16	A/Norway 3275/18	A/Sth Aus 34/19	A/La Rioja 2202/18	A/HK 2671/19	A/HK 2671/19	A/Eng 538/18	NYMC X-327 A/Kansas/14	A/Kansas 14/17
				MDCK	Egg 10 ⁻⁴	SIAT	Egg	SIAT	Egg	Cell	SIAT	Egg	SIAT
	Passage history			St Jude's F60/17 [†]	F13/19 [†]	F03/19 [†]	F45/19 [†]	F26/18 [†]	F44/19 [†]	St Jude's F21/20 [†]	F31/18 [†]	F16/19 [†]	F17/19 [†]
	Ferret number			3C.2a	3C.2a1	3C.2a1b+T131K	3C.2a1b+T131K	3C.2a1b+T135K-A	3C.2a1b+T135K-B	3C.2a1b+T135K-B	3C.3a	3C.3a	3C.3a
	Genetic group												
REFERENCE VIRUSES													
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK3/SIAT2	320	320	320	320	160	40	160	320	160	320
A/Singapore/INFIMH-16-0019/2016	3C.2a1	2016-04-14	E5/E2	160	320	80	80	160	80	40	80	40	<
A/South Australia/34/2019	3C.2a1b+T131K	2019-02-06	E6/E1	320	640	1280	1280	80	160	40	80	40	40
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	2019-06-17	E8/E2	40	160	<	80	40	640	160	160	640	80
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	2019-06-17	CK1/SIAT4	160	320	320	160	320	320	640	160	160	160
A/England/538/2018	3C.3a	2018-02-26	MDCK1/SIAT4	40	80	<	40	<	80	80	320	320	320
NYMC X-327 (A/Kansas/14/17)	3C.3a	2017-12-14	Ex/E1	80	80	<	80	40	320	80	320	1280	640
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2	40	40	40	40	40	80	80	320	160	640
TEST VIRUSES													
A/Estonia/126086/2020	3C.3a	2020-01-30	SIAT1/SIAT1	<	40	<	<	<	<	40	320	160	160
A/Norway/2231/2020	No seq	2020-03-16	SIAT1	80	160	320	160	160	<	80	160	80	40
A/Estonia/126096/2020	3C.3a	2020-02-01	SIAT1/SIAT1	<	80	<	<	<	40	<	320	160	320
A/Norway/749/2020	3C.3a	2020-02-03	SIAT1	40	80	<	<	<	40	40	320	160	320
A/Norway/828/2020	3C.3a	2020-02-06	SIAT1	40	80	<	<	<	40	40	320	160	320
A/Estonia/126500/2020	3C.3a	2020-02-19	SIAT1/SIAT1	<	40	<	<	<	<	<	320	160	160
A/Estonia/126846/2020	3C.3a	2020-03-02	SIAT1/SIAT1	<	80	<	<	<	40	40	320	160	320
A/Estonia/127241/2020	3C.3a	2020-03-05	SIAT1/SIAT2	40	80	<	80	<	80	40	320	320	320
A/Norway/2069/2020	3C.3a	2020-03-12	SIAT1	<	80	<	<	<	<	<	640	160	160
A/Norway/1900/2020	3C.3a	2020-03-12	SIAT1	<	80	<	<	<	40	40	640	160	320
A/Norway/2228/2020	3C.3a	2020-03-15	SIAT1	80	80	40	80	<	80	40	640	160	320
A/Norway/2176/2020	3C.3a	2020-03-18	SIAT2	40	80	<	<	<	40	40	640	160	320
A/Norway/2310/2020	3C.3a	2020-03-28	SIAT1	40	80	<	40	<	40	40	640	160	160
A/Norway/1906/2020	3C.2a1b+T131K	2020-03-12	SIAT1	160	320	320	320	160	40	80	160	80	40
A/Norway/2205/2020	3C.2a1b+T131K	2020-03-15	SIAT1	160	160	320	160	160	40	160	160	80	40
A/Norway/2174/2020	3C.2a1b+T131K	2020-03-17	SIAT1	160	160	320	320	160	80	160	160	80	40
A/Norway/2115/2020	3C.2a1b+T131K	2020-03-18	SIAT1	160	160	640	160	160	40	160	160	80	40
A/Norway/2292/2020	3C.2a1b+T131K	2020-03-20	SIAT1	160	160	320	320	160	40	160	160	80	40
A/Norway/2320/2020	3C.2a1b+T131K	2020-03-24	SIAT1	160	160	320	160	160	<	40	160	160	80
A/Norway/2290/2020	3C.2a1b+T131K	2020-03-24	SIAT1	160	160	320	320	160	80	80	320	80	80
A/Norway/2357/2020	3C.2a1b+T131K	2020-03-28	SIAT1	160	320	320	320	320	40	160	320	160	80
A/Norway/2208/2020	3C.2a1b+T135K-A	2020-03-14	SIAT1	80	160	160	80	160	80	160	160	80	40

Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) 1 < = <40

Vaccine
NH 2018-19Vaccine
SH 2020Vaccine
NH 2020-21Vaccine
NH 2019-20

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 7-11. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2020-07-29

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/HK 5738/14	A/Singapore 0019/16	A/Norway 3275/18	A/Sth Aus 34/19	A/La Rioja 2202/18	A/HK 2671/19	A/HK 2671/19	A/Eng 538/18	NYMC X-327 A/Kansas/14	A/Kansas 14/17
				MDCK	Egg 10 ⁻⁴	SIAT	Egg	SIAT	Egg	Cell	SIAT	Egg	SIAT
	Passage history			St Jude's F60/17 [†]	F13/19 [†]	F03/19 [†]	F45/19 [†]	F26/18 [†]	F44/19 [†]	St Jude's F21/20 [†]	F31/18 [†]	F16/19 [†]	F17/19 [†]
	Ferret number			3C.2a	3C.2a1	3C.2a1b+T131K	3C.2a1b+T131K	3C.2a1b+T135K-A	3C.2a1b+T135K-B	3C.2a1b+T135K-B	3C.3a	3C.3a	3C.3a
	Genetic group												
REFERENCE VIRUSES													
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK3/SIAT2	320	320	320	320	160	40	160	320	160	320
A/Singapore/INFIMH-16-0019/2016	3C.2a1	2016-04-14	E5/E2	160	320	80	80	160	80	40	40	40	<
A/South Australia/34/2019	3C.2a1b+T131K	2019-02-06	E6/E1	320	640	1280	1280	80	320	40	80	40	40
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	2019-06-17	E8/E2	40	160	<	80	80	640	160	160	640	80
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	2019-06-17	CK1/SIAT4	160	320	320	160	320	320	640	160	160	160
A/England/538/2018	3C.3a	2018-02-26	MDCK1/SIAT4	40	80	<	40	<	80	80	320	320	320
NYMC X-327 (A/Kansas/14/17)	3C.3a	2017-12-14	Ex/E1	40	80	<	40	40	320	40	320	1280	640
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2	40	80	40	40	40	80	80	640	160	640
TEST VIRUSES													
A/Cyprus/F411/2020	3C.3a	2020-01-21	SIAT1	160	160	<	40	<	40	80	1280	640	640
A/Cyprus/F968/2020	3C.3a	2020-02-10	SIAT1	40	160	<	40	<	<	40	640	320	640
A/Cyprus/F1123/2020	3C.3a	2020-02-17	SIAT1	40	160	<	<	<	<	40	640	320	640
A/Cyprus/F1190/2020	3C.3a	2020-02-23	SIAT1	160	320	<	80	40	80	80	1280	640	640
A/Cyprus/F1165/2020	3C.3a	2020-02-23	SIAT1	160	320	<	40	40	80	80	1280	320	640
A/Cyprus/F1149/2020	3C.3a	2020-02-23	SIAT1	80	320	80	160	40	80	40	1280	320	640
A/Cyprus/F1218/2020	3C.3a	2020-02-25	SIAT1	160	320	<	40	80	40	80	640	320	640
A/Cyprus/F1267/2020	3C.3a	2020-02-27	SIAT1	80	160	40	40	40	40	80	640	160	640
A/Cyprus/F1305/2020	3C.3a	2020-03-03	SIAT1	80	160	40	40	40	40	80	640	160	640
A/Ireland/23296/2020	3C.2a1b+T135K-A	2020-03-09	SIAT1	160	320	320	320	320	160	320	320	80	160
A/Moscow/119/2020	3C.2a1b+T135K-A	2020-03-25	SIAT1/SIAT1	320	160	320	160	320	160	320	160	40	160

Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) 1 < = <40

Vaccine
NH 2018-19Vaccine
SH 2020Vaccine
NH 2020-21Vaccine
NH 2019-20

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 7-12. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2020-08-14

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre												
				Post-infection ferret antisera												
				NEW			NEW			NEW			NEW			
				A/HK 5738/14 MDCK	A/Singapore 0019/16 Egg 10 ⁻⁴	A/Norway 3275/18 SIAT	A/Sth Aus 34/19 Egg	A/La Rioja 2202/18 SIAT	A/Denmark 3264/19 SIAT	A/Paris 2554/19 Egg	A/HK 2671/19 Egg	A/HK 2671/19 Cell	A/HK 45/19 SIAT	A/Eng 538/18 SIAT	NYMC X-327 A/Kansas/14 Egg	A/Kansas 14/17 SIAT
	Passage history			St Jude's F60/17 ¹	F13/19 ¹	F03/19 ¹	F45/19 ¹	F26/18 ¹	F19/20 ¹	F14/20 ¹	F44/19 ¹	St Jude's F21/20 ¹	F18/20 ¹	F31/18 ¹	F16/19 ¹	F17/19 ¹
	Ferret number															
	Genetic group			3C.2a	3C.2a1	3C.2a1b+T131K	3C.2a1b+T131K	3C.2a1b+T135K-A	3C.2a1b+T135K-A	3C.2a1b+T135K-A	3C.2a1b+T135K-B	3C.2a1b+T135K-B	3C.2a1b+T135K-B	3C.3a	3C.3a	3C.3a
REFERENCE VIRUSES																
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK3/SIAT2	160	160	160	160	80	<	<	<	160	<	160	160	80
A/Singapore/INFIMH-16-0019/2016	3C.2a1	2016-04-14	E5/E2	80	320	40	40	160	80	40	80	<	40	40	40	<
A/South Australia/34/2019	3C.2a1b+T131K	2019-02-06	E6/E1	80	320	640	640	40	<	80	160	<	<	40	40	<
A/Denmark/3264/2019	3C.2a1b+T135K-A	2019-10-25	SIAT3/SIAT2	ND	ND	ND	ND	ND	320	160	ND	ND	ND	ND	ND	ND
A/Paris/2554/2019	3C.2a1b+T135K-A	2019-11-25	E5	40	160	<	40	40	320	320	320	40	40	160	80	80
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	2019-06-17	E8/E2	40	160	<	80	40	160	320	640	160	160	160	320	80
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	2019-06-17	MDCK1/SIAT4	80	160	160	80	160	160	80	160	160	320	80	80	80
A/Hong Kong/45/2019	3C.2a1b+T135K-B	2018-12-24	SIAT3	80	160	160	80	80	160	80	80	160	320	80	40	40
A/England/538/2018	3C.3a	2018-02-26	MDCK1/SIAT4	<	40	<	<	<	40	40	40	<	<	320	160	320
NYMC X-327 (A/Kansas/14/17)	3C.3a	2017-12-14	Ex/E1	<	<	<	<	<	40	80	160	<	40	160	640	320
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2	40	80	<	<	<	40	80	40	<	<	320	160	320
TEST VIRUSES																
A/Hungary/104/2020	3C.3a	2020-01-27	MDCK1/SIAT1	40	80	<	<	<	40	40	<	<	<	320	160	320
A/Hungary/91/2020	No seq	2020-02-18	MDCK1/SIAT1	40	80	<	<	<	40	40	<	<	<	320	160	160
A/Hungary/61/2020	3C.3a	2020-02-05	MDCK1/SIAT1	<	40	<	40	<	40	80	40	<	<	320	160	320
A/Hungary/64/2020	3C.2a1b+T131K	2020-02-08	MDCK1/SIAT1	80	80	160	160	80	80	40	<	80	40	40	40	40

Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) 1 < = <40

ND = Not Done

Vaccine NH 2018-19	Vaccine SH 2020	Vaccine NH 2020-21	Vaccine NH 2019-20
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< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 7-13. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2020-09-03

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/HK 5738/14	A/Singapore 0019/16	A/Norway 3275/18	A/La Rioja 2202/18	A/Denmark 3264/19	A/HK 2671/19	A/HK 2671/19	A/Eng 538/18	NYMC X-327 A/Kansas/14	A/Kansas 14/17
				MDCK	Egg 10 ⁻⁴	SIAT	SIAT	SIAT	Egg	Cell	SIAT	Egg	SIAT
	Passage history			St Jude's F60/17 [†]	F13/19 [†]	F03/19 [†]	F26/18 [†]	F19/20 [†]	F44/19 [†]	St Jude's F21/20 [†]	F31/18 [†]	F16/19 [†]	F17/19 [†]
	Ferret number			3C.2a	3C.2a1	3C.2a1b+T131K	3C.2a1b+T135K-A	3C.2a1b+T135K-A	3C.2a1b+T135K-B	3C.2a1b+T135K-B	3C.3a	3C.3a	3C.3a
	Genetic group												
REFERENCE VIRUSES													
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT2	160	320	160	80	40	<	40	160	160	160
A/Singapore/INFIMH-16-0019/2016	3C.2a1	2016-04-14	E5/E2	160	320	40	160	80	80	<	40	40	<
A/Denmark/3264/2019	3C.2a1b+T135K-A	2019-10-25	SIAT3/SIAT3	160	160	160	160	160	40	80	80	80	40
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	2019-06-17	E8/E2	<	80	<	40	80	640	80	80	320	80
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	2019-06-17	MDCK1/SIAT4	80	320	160	160	320	160	320	160	160	160
A/England/538/2018	3C.3a	2018-02-26	MDCK1/SIAT4	40	40	<	<	40	<	<	320	160	320
NYMC X-327 (A/Kansas/14/17)	3C.3a	2017-12-14	Ex/E1	<	40	<	<	40	160	<	160	1280	320
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2	40	80	<	<	40	40	40	320	160	320
TEST VIRUSES													
A/Bulgaria/1051/2020	3C.3a	2020-02-21	SIAT2/MDCK2	40	80	<	<	40	40	<	320	160	320
A/Melilla/1088/2020	3C.3a	2020-02-24	SIAT1	40	80	<	<	40	40	<	320	160	320
A/Aragon/13141/2020	3C.3a	2020-02-25	SIAT1	40	80	<	<	40	<	<	320	160	320
A/Moldova/416.1/2020	3C.3a	2020-03-03	SIAT1	40	80	<	<	40	40	40	320	320	320
A/Hungary/108/2020	3C.3a	2020-03-06	MDCK1/SIAT1	40	80	<	<	<	<	<	320	160	160
A/Aragon/13140/2020	3C.2a1b+T135K-A	2020-03-05	SIAT1	40	80	<	40	160	80	40	160	80	80

Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) 1 < = <40

Vaccine
NH 2018-19Vaccine
NH 2020-21Vaccine
NH 2019-20

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 7-14. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) - Summary

Haemagglutination inhibition titre													
Viruses	Other information	Post-infection ferret antisera											
		A/HK	A/Singapore	A/Norway	A/Sth Aus	A/La Rioja	A/HK	A/HK	A/HK	A/Eng	NYMC X-327	A/Kansas	
		5738/14	0019/16	3275/18	34/19	2202/18	2671/19	2671/19	2669/19	538/18	A/Kansas/14	14/17	
		Passage history	MDCK	Egg 10 ⁻⁴	SIAT	Egg	SIAT	Egg	Cell	SIAT	SIAT	Egg	SIAT
		Ferret number	St Jude's F60/17 [†]	F13/19 [†]	F03/19 [†]	F45/19 [†]	F26/18 [†]	F44/19 [†]	St Jude's F21/20 [†]	F04/20 [†]	F31/18 [†]	F16/19 [†]	F17/19 [†]
Genetic group	3C.2a	3C.2a1	3C.2a1b+T131K	3C.2a1b+T131K	3C.2a1b+T135K-A	3C.2a1b+T135K-B	3C.2a1b+T135K-B	3C.2a1b+T135K-B	3C.3a	3C.3a	3C.3a		
REFERENCE VIRUSES													
A/Hong Kong/5738/2014	3C.2a	320	320		320	160	160	<	160	80	320	160	160
A/Singapore/INFIMH-16-0019/2016	3C.2a1	80	320		40	40	160	40	40	40	40	<	<
A/South Australia/34/2019	3C.2a1b+T131K	160	320		640	1280	80	160	40	<	80	40	40
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	40	160		<	80	40	640	160	160	160	640	80
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	160	320		320	320	320	320	640	640	320	80	160
A/Hong Kong/2669/2019	3C.2a1b+T135K-B	160	320		320	160	160	160	640	320	160	80	160
A/England/538/2018	3C.3a	80	80		<	40	40	80	160	<	640	160	640
NYMC X-327 (A/Kansas/14/17)	3C.3a	<	40		<	<	<	160	80	<	320	640	320
A/Kansas/14/2017	3C.3a	40	80		<	40	40		160	<	640	160	320
TEST VIRUSES													
Genetic group													
All viruses	No tested	162	162	162*	156	162*	162	160	26	162	162	162	162
	Titre reduced ≤2-fold	28	64	35	1	37	2	22	5	133	9	108	108
	%	17.2	39.5	21.6	0.7	22.8	1.2	13.7	19.2	82.1	5.6	66.7	66.7
	Titre reduced =4-fold	50	79		10		11	23	2	9	53	15	15
	%	30.9	48.8		6.4		6.8	14.4	7.7	5.6	32.7	9.2	9.2
	Titre reduced ≥8-fold	84	19		145		149	115	19	20	100	39	39
	%	51.9	11.7		92.9		92.0	71.9	73.1	12.3	61.7	24.1	24.1
3C.3a	No tested	105	105	105	99	105	105	104	19	105	105	105	105
	Titre reduced ≤2-fold	8	28					1		105	9	102	102
	%	7.7	26.7					1.0		100.0	8.6	97.1	97.1
	Titre reduced =4-fold	35	62					2			48	3	3
	%	33.3	59.0					1.9			45.7	2.9	2.9
	Titre reduced ≥8-fold	62	15		99		105	101	19		48		
	%	59.0	14.3		100.0		100.0	97.1	100.0		45.7		
3C.2a1b+T131K	No tested	18	18	18	18	18	18	18		18	18	18	18
	Titre reduced ≤2-fold	11	14	14		13		1		10		1	1
	%	61.1	77.8	77.8		72.2		5.6		55.6		5.6	5.6
	Titre reduced =4-fold	2	2		8		1	8				2	2
	%	11.1	11.1		44.4		5.6	44.4				11.1	11.1
	Titre reduced ≥8-fold	5	2		10		17	9		8	18	15	15
	%	27.8	11.1		55.6		94.4	50.0		44.4	100.0	83.3	83.3
3C.2a1b+T135K-A	No tested	27	27	27	27	27	27	26	4	27	27	27	27
	Titre reduced ≤2-fold	7	16	16	1	19	2	11	3	12		4	4
	%	26.0	59.3	59.3	3.7	70.4	7.4	42.3		44.4		15.4	15.4
	Titre reduced =4-fold	8	11		2		6	11	1	10	4	8	8
	%	29.6	40.7		7.4		22.2	42.3		37.0	14.8	29.0	29.0
	Titre reduced ≥8-fold	12			24		19	4		5	23	15	15
	%	44.4			88.9		70.4	15.4		18.6	85.2	55.6	55.6
3C.2a1b+T135K-B	No tested	8	8	8	8	8	8	8	3	8	8	8	8
	Titre reduced ≤2-fold	2	4	3		3		7	3	2			
	Titre reduced =4-fold	2	2				3	1				2	2
	Titre reduced ≥8-fold	4	2		8		5			6	8	6	6
No Seq	No tested	4	4	4	4	4	4	4		4	4	4	4
	Titre reduced ≤2-fold		2	2		2		2		3		1	1
	Titre reduced =4-fold	3	2				1	1			1		
	Titre reduced ≥8-fold	1			4		3	2		1	3	3	3
		Vaccine NH 2018-19		Vaccine SH 2020		Vaccine NH 2020-21		NH 2019-20					

* No homologous titre, number with titres of ≥160 indicated

Reference virus results are taken from an individual table as an example. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed

Table 8-1. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2020-08-13

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹ Post-infection ferret antisera												HA1 substitutions for 3C.2a/3a viruses compared to A/Norway/3275/2018: Egg adaptation HA substitutions compared to the corresponding cell culture propagated virus			
				A/Norway 3275/2018 SIAT F03/19		A/SouthAustralia 34/2019 Egg F43/19		A/Denmark 3264/19 SIAT F19/20		A/Paris 2554/19 Egg F14/20		A/Hong Kong 2671/19 SIAT F33/19		A/Hong Kong 2671/19 Egg F44/19			A/Kansas 14/2017 SIAT F17/19		
				3C.2a1b+T131K		3C.2a1b+T131K		3C.2a1b+T135K-A		3C.2a1b+T135K-A		3C.2a1b+T135K-B		3C.2a1b+T135K-B			3C.3a		
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read		2-fold	Read	
REFERENCE VIRUSES																			
A/Norway/3275/2018	3C.2a1b+T131K	2018-10-04	SIAT2 10 ⁻³	640	514	80	75	160	147	<	20	40	55	<	0	<	0	none	
IVR-197 (A/South Australia/34/2019)	3C.2a1b+T131K	2019-02-06	E5/D8 10 ⁻⁸	320	297	1280	1077	<	0	<	0	<	0	40	30	<	0	V106A, T160K(-CHO), G186I, S219F	
A/Denmark/3264/2019	3C.2a1b+T135K-A	2019-10-25	SIAT3/SIAT2 10 ⁻³	80	77	<	10	640	486	40	36	320	257	40	38	<	10	V106A, T128A(-CHO), K131T, T135K(-CHO), A138S, G186D, D190N, F193S, S198P	
A/Paris/2554/2019	3C.2a1b+T135K-A	2019-11-25	E5 10 ⁻⁶	<	0	40	22	320	274	640	722	80	63	160	224	40	29	V106A, T128A(-CHO), K131T, T135K(-CHO), A138S, T160I(-CHO),G186D, D190N, F193S, S198P, D225G	
A/HongKong/2671/2019	3C.2a1b+T135K-B	2019-06-17	MDCK1/SIAT4 10 ⁻³	40	59	40	20	320	350	40	20	320	376	80	61	40	23	V106A, T128A(-CHO), K131T, T135K(-CHO), S137F, A138S, F193S	
A/HongKong/2671/2019	3C.2a1b+T135K-B	2019-06-17	E8/E2 10 ⁻³	<	0	320	450	80	71	320	400	160	137	640	812	40	47	V106A, T128A(-CHO), K131T, T135K(-CHO), S137F, A138S, T160I(-CHO), G186V, F193S, S219X, D225N	
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2 10 ⁻¹	<	0	<	0	<	0	40	49	<	0	<	0	160	146	G62E, S91N, R92K, V106A, K121N, T128A(-CHO), K131T, A138S, S144K, Y159S, T160K(-CHO), K171N, F193S, K326R	
TEST VIRUSES																			
A/Slovenia/1637/2020	3C.2a1b+T131K	2020-02-28	SIATx/SIAT1	640	513	80	105	80	116	<	0	80	80	<	0	<	10	As A/Norway/3275/2018 + K83E, Y94N, V106A	
A/Slovenia/1752 /2020	3C.2a1b+T131K	2020-03-05	MDCKx/SIAT1	160	230	80	93	<	0	<	0	<	0	<	5	<	0	As A/Norway/3275/2018 + K83E, Y94N, V106A, {N158X + T160X(-CHO)}	
A/Thuringen/45/2020	3C.2a1b+T135K-A	2020-02-28	C1/SIAT1	80	85	<	10	320	259	40	40	160	178	40	20	40	20	As A/Denmark/3264/2019	
A/Slovenia/1608 /2020	3C.2a1b+T135K-A	2020-02-26	SIATx/SIAT1	80	74	40	20	320	271	<	0	160	176	40	20	<	5	As A/Denmark/3264/2019 + Q57R	
A/Stavropol/10/2020	3C.2a1b+T135K-A	2020-03-03	MDCK2/SIAT1	<	0	40	20	40	56	160	141	<	0	80	62	40	40	As A/Denmark/3264/2019 + V130I, T160K(-CHO), S219Y	
A/S.Petersburg/RIL-180/2020	3C.2a1b+T135K-B	2020-03-10	MDCK2/SIAT1	<	10	<	0	80	65	40	41	80	78	40	51	<	0	As A/Hong Kong/2671/2019 + D53N, T160K(-CHO)	
A/Slovenia/1708 /2020	3C.3a	2020-03-02	SIATx/SIAT1	<	5	40	20	<	0	40	58	<	0	<	0	160	141	As A/Kansas/14/2017	
A/Bremen/7/2020	3C.3a	2020-03-02	C1/SIAT1	<	0	<	0	<	0	40	20	<	0	<	0	80	106	As A/Kansas/14/2017 + S198P	
A/Iceland/37/2020	3C.3a	2020-03-26	MDCK1/SIAT1	<	0	<	0	<	0	160	196	40	40	40	40	640	592	As A/Kansas/14/2017 + D225G	
						Vaccine SH 2020												Vaccine NH 2020-21	

Vaccine
SH 2020

Vaccine
NH 2020-21

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 8-2. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2020-08-17

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre [†]												HA1 substitutions for 3C.2a/3a viruses compared to A/Norway/3275/2018: Egg adaptation HA substitutions compared to the corresponding cell culture propagated virus		
				Post-infection ferret antisera														
				A/Norway 3275/2018 SIAT F03/19		A/SouthAustralia 34/2019 Egg F43/19		A/Denmark 3264/19 SIAT F19/20		A/Paris 2554/19 Egg F14/20		A/Hong Kong 2671/19 SIAT F33/19		A/Hong Kong 2671/19 Egg F44/19			A/Kansas 14/2017 SIAT F17/19	
				3C.2a1b+T131K		3C.2a1b+T131K		3C.2a1b+T135K-A		3C.2a1b+T135K-A		3C.2a1b+T135K-B		3C.2a1b+T135K-B			3C.3a	
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES																		
A/Norway/3275/2018	3C.2a1b+T131K	2018-10-04	SIAT2 10 ⁻³	640	514	80	75	160	147	<	20	40	55	<	0	<	0	none
IVR-197 (A/South Australia/34/2019)	3C.2a1b+T131K	2019-02-06	E5/D8 10 ⁻⁶	320	297	1280	1077	<	0	<	0	<	0	40	30	<	0	V106A, T160K(-CHO), G186I, S219F
A/Denmark/3264/2019	3C.2a1b+T135K-A	2019-10-25	SIAT3/SIAT2 10 ⁻³	80	77	<	10	640	486	40	36	320	257	40	38	<	10	V106A, T128A(-CHO), K131T, T135K(-CHO), A138S, G186D, D190N, F193S, S198P
A/Paris/2554/2019	3C.2a1b+T135K-A	2019-11-25	E5 10 ⁻⁵	<	0	40	22	320	274	640	722	80	63	160	224	40	29	V106A, T128A(-CHO), K131T, T135K(-CHO), A138S, T160I(-CHO),G186D, D190N, F193S, S198P, D225G
A/HongKong/2671/2019	3C.2a1b+T135K-B	2019-06-17	MDCK1/SIAT4 10 ⁻³	40	59	40	20	320	350	40	20	320	376	80	61	40	23	V106A, T128A(-CHO), K131T, T135K(-CHO), S137F, A138S, F193S
A/HongKong/2671/2019	3C.2a1b+T135K-B	2019-06-17	E8/E2 10 ⁻³	<	0	320	450	80	71	320	400	160	137	640	812	40	47	V106A, T128A(-CHO), K131T, T135K(-CHO), S137F, A138S, T160I(-CHO), G186V, F193S, S219X, D225N
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2 10 ⁻¹	<	0	<	0	<	0	40	49	<	0	<	0	160	146	G62E, S91N, R92K, V106A, K121N, T128A(-CHO), K131T, A138S, S144K, Y159S, T160K(-CHO), K171N, F193S, K326R
TEST VIRUSES																		
A/Finland/183/2020	3C.2a1b+T131K	2020-03-02	SIAT1/SIAT1	80	100	<	0	160	202	40	20	160	135	<	0	<	0	As A/Norway/3275/2018 + K83E, Y94N, V106A, G186S, F193S, Y195F, S198P
A/Finland/184/2020	3C.2a1b+T131K	2020-03-16	SIAT1/SIAT1	640	518	80	114	160	167	40	20	160	124	<	0	<	0	As A/Norway/3275/2018 + I48M, V106A, Q197R, K207R, I214T, S219F
A/Switzerland/8482/2020	3C.2a1b+T135K-A	2020-03-12	SIAT1	40	20	<	10	160	177	320	280	40	20	160	143	40	20	As A/Denmark/3264/2019 + E50K, T160A(-CHO)
A/Iceland/23/2020	3C.2a1b+T135K-B	2020-03-09	MDCK1/SIAT1	40	58	<	0	320	366	80	64	320	304	160	139	<	0	As A/Hong Kong/2671/2019 + E50K, T160X(-CHO)
A/Iceland/18/2020	3C.3a	2020-02-29	MDCK1/SIAT1	<	0	<	0	<	0	40	43	<	0	<	0	80	85	As A/Kansas/14/2017
A/Iceland/25/2020	3C.3a	2020-03-12	MDCK1/SIAT1	40	20	<	0	<	0	40	54	<	0	<	10	160	120	As A/Kansas/14/2017
A/Iceland/30/2020	3C.3a	2020-03-13	MDCK1/SIAT1	40	20	<	0	40	20	80	63	<	0	40	20	160	172	As A/Kansas/14/2017
A/Iceland/29/2020	3C.3a	2020-03-16	MDCK1/SIAT1	<	0	<	0	<	0	40	44	<	0	<	0	80	97	As A/Kansas/14/2017
A/Iceland/26/2020	3C.3a	2020-03-13	MDCK1/SIAT1	40	40	<	0	40	20	80	93	40	20	40	20	160	129	As A/Kansas/14/2017 + R208X, K264E
						Vaccine SH 2020				Vaccine NH 2020-21								

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 8-3. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2020-08-20

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹														HA1 substitutions for 3C.2a/3a viruses compared to A/Norway/3275/2018: Egg adaptation HA substitutions compared to the corresponding cell culture propagated virus	
				Post-infection ferret antisera															
				A/Norway 3275/2018 SIAT F03/19	A/SouthAustralia 34/2019 Egg F43/19	A/Denmark 3264/19 SIAT F19/20	A/Paris 2554/19 Egg F14/20	A/Hong Kong 2671/19 SIAT F33/19	A/Hong Kong 2671/19 Egg F44/19	A/Kansas 14/2017 SIAT F17/19									
				3C.2a1b+T131K 2-fold Read	3C.2a1b+T131K 2-fold Read	3C.2a1b+T135K-A 2-fold Read	3C.2a1b+T135K-A 2-fold Read	3C.2a1b+T135K-B 2-fold Read	3C.2a1b+T135K-B 2-fold Read	3C.3a 2-fold Read									
REFERENCE VIRUSES																			
A/Norway/3275/2018	3C.2a1b+T131K	2018-10-04	SIAT2 10 ⁻³	640	514	80	75	160	147	<	20	40	55	<	0	<	0	none	
IVR-197 (A/South Australia/34/2019)	3C.2a1b+T131K	2019-02-06	E5/D8 10 ⁻⁶	320	297	1280	1077	<	0	<	0	<	0	40	30	<	0	V106A, T160K(-CHO), G186I, S219F	
A/Denmark/3264/2019	3C.2a1b+T135K-A	2019-10-25	SIAT3/SIAT2 10 ⁻³	80	77	<	10	640	486	40	36	320	257	40	38	<	10	V106A, T128A(-CHO), K131T, T135K(-CHO), A138S, G186D, D190N, F193S, S198P	
A/Paris/2554/2019	3C.2a1b+T135K-A	2019-11-25	E5 10 ⁻⁶	<	0	40	22	320	274	640	722	80	63	160	224	40	29	V106A, T128A(-CHO), K131T, T135K(-CHO), A138S, T160I(-CHO), G186D, D190N, F193S, S198P, D225G	
A/HongKong/2671/2019	3C.2a1b+T135K-B	2019-06-17	MDCK1/SIAT4 10 ⁻³	40	59	40	20	320	350	40	20	320	376	80	61	40	23	V106A, T128A(-CHO), K131T, T135K(-CHO), S137F, A138S, F193S	
A/HongKong/2671/2019	3C.2a1b+T135K-B	2019-06-17	E8/E2 10 ⁻³	<	0	320	450	80	71	320	400	160	137	640	812	40	47	V106A, T128A(-CHO), K131T, T135K(-CHO), S137F, A138S, T160I(-CHO), G186V, F193S, S219X, D225N	
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2 10 ⁻¹	<	0	<	0	<	0	40	49	<	0	<	0	160	146	G62E, S91N, R92K, V106A, K121N, T128A(-CHO), K131T, A138S, S144K, Y159S, T160K(-CHO), K171N, F193S, K326R	
TEST VIRUSES																			
A/Switzerland/3278/2020	3C.2a1b+T131K	2020-03-04	SIAT1	1280	1109	320	261	640	817	40	49	160	184	<	0	40	20	As A/Norway/3275/2018 + G62R, V106A, I192L, K207R, S219F	
A/Picardie/1840/2020	3C.2a1b+T131K	2020-03-02	SIAT1	640	640	160	132	160	229	<	0	160	139	<	0	<	0	As A/Norway/3275/2018 + I48M, V106A, Q197R, K207R, I214T, S219F	
A/Portugal/MS74/2020	3C.2a1b+T131K	2020-03-21	SIAT1	320	437	80	119	160	208	40	20	160	145	<	0	<	10	As A/Norway/3275/2018 + I48M, V106A, Q197R, K207R, V213A, I214T, S219F	
A/Zahle/358/2020	3C.2a1b+T135K-A	2020-02-26	SIAT1	80	73	<	0	320	303	<	0	320	240	40	40	<	10	As A/Denmark/3264/2019 + D104G	
A/Switzerland/6206/2020	3C.2a1b+T135K-A	2020-03-09	SIAT1	80	64	<	0	320	452	40	20	640	581	<	0	<	10	As A/Denmark/3264/2019 + S145G, D175E, D186G, N190D	
A/Lyon/1826/2020	3C.2a1b+T135K-A	2020-03-16	MDCK2/SIAT1	40	57	<	0	320	245	40	40	160	176	<	0	<	0	As A/Denmark/3264/2019 + L164X	
A/Beirut/421/2020	3C.2a1b+T135K-B	2020-03-05	SIAT1	40	42	<	0	320	298	<	10	320	421	80	72	40	20	As A/Hong Kong/2671/2019	
A/Saida/361/2020	3C.2a1b+T135K-B	2020-03-02	SIAT1	80	68	<	0	320	240	40	40	160	189	40	20	<	10	As A/Hong Kong/2671/2019 + E50K, S144N(+CHO)	
A/Bosnia and Herzegovina/360/2020	3C.3a	2020-02-25	SIAT1	<	0	<	0	<	0	40	55	<	0	<	0	160	193	As A/Kansas/14/2017	
A/Poland/3278/2020	3C.3a	2020-02-26	SIAT1	<	0	<	0	<	0	40	58	<	0	<	0	160	194	As A/Kansas/14/2017	
A/Macedonia/648/2020	3C.3a	2020-02-27	SIAT1	<	0	<	0	<	0	80	61	<	0	<	0	320	245	As A/Kansas/14/2017 + K264E	
A/Lyon/1631/2020	3C.3a	2020-03-01	MDCK3/SIAT1	<	0	<	0	<	0	80	61	<	0	<	0	160	186	As A/Kansas/14/2017	
A/Estonia/126846/2020	3C.3a	2020-03-02	SIAT1/SIAT1	<	0	<	0	<	0	40	58	<	0	<	0	160	160	As A/Kansas/14/2017 + I58X, Q327X	
A/Centre/2293/2020	3C.3a	2020-03-09	SIAT1	40	20	<	0	<	0	40	50	<	0	<	0	160	158	As A/Kansas/14/2017 + S198P	
A/Norway/2176/2020	3C.3a	2020-03-18	SIAT2	40	20	<	0	<	0	40	50	<	0	<	0	160	193	As A/Kansas/14/2017	
						Vaccine SH 2020										Vaccine NH 2020-21			

Vaccine
SH 2020Vaccine
NH 2020-21¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 8-4. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2020-09-03

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹												HA1 substitutions for 3C.2a/3a viruses compared to A/Norway/3275/2018: Egg adaptation HA substitutions compared to the corresponding cell culture propagated virus			
				Post-infection ferret antisera															
				A/Norway 3275/2018 SIAT F03/19 3C.2a1b+T131K		A/SouthAustralia 34/2019 Egg F43/19 3C.2a1b+T131K		A/Denmark 3264/19 SIAT F19/20 3C.2a1b+T135K-A		A/Paris 2554/19 Egg F14/20 3C.2a1b+T135K-A		A/Hong Kong 2671/19 SIAT F33/19 3C.2a1b+T135K-B		A/Hong Kong 2671/19 Egg F44/19 3C.2a1b+T135K-B			A/Kansas 14/2017 SIAT F17/19 3C.3a		
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read		2-fold	Read	
REFERENCE VIRUSES																			
A/Norway/3275/2018	3C.2a1b+T131K	2018-10-04	SIAT2 10 ⁻³	640	514	80	75	160	147	<	20	40	55	<	0	<	0	none	
IVR-197 (A/South Australia/34/2019)	3C.2a1b+T131K	2019-02-06	E5/D8 10 ⁻⁶	320	297	1280	1077	<	0	<	0	<	0	40	30	<	0	V106A, T160K(-CHO), G186I, S219F	
A/Denmark/3264/2019	3C.2a1b+T135K-A	2019-10-25	SIAT3/SIAT2 10 ⁻³	80	66	<	10	320	376	<	10	160	157	<	10	<	0	V106A, T128A(-CHO), K131T, T135K(-CHO), A138S, G186D, D190N, F193S, S198P	
A/Paris/2554/2019	3C.2a1b+T135K-A	2019-11-25	E5 10 ⁻⁶	<	0	40	22	320	274	640	722	80	63	160	224	40	29	V106A, T128A(-CHO), K131T, T135K(-CHO), A138S, T160K(-CHO), G186D, D190N, F193S, S198P, D225G	
A/HongKong/2671/2019	3C.2a1b+T135K-B	2019-06-17	MDCK1/SIAT4 10 ⁻³	40	49	<	0	320	259	<	<	320	257	40	42	<	0	V106A, T128A(-CHO), K131T, T135K(-CHO), S137F, A138S, F193S	
A/HongKong/2671/2019	3C.2a1b+T135K-B	2019-06-17	E8/E2 10 ⁻³	<	0	320	450	80	71	320	400	160	137	640	812	40	47	V106A, T128A(-CHO), K131T, T135K(-CHO), S137F, A138S, T160K(-CHO), G186V, F193S, S219X, D225N	
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2 10 ⁻¹	<	0	<	0	<	0	40	49	<	0	<	0	160	146	G62E, S91N, R92K, V106A, K121N, T128A(-CHO), K131T, A138S, S144K, Y159S, T160K(-CHO), K171N, F193S, K326R	
TEST VIRUSES																			
A/Norway/2320/2020	3C.2a1b+T131K	2020-03-24	SIAT1	320	361	40	56	160	126	<	0	40	44	<	<	<	0	As A/Norway/3275/2018 + D60E, K83E, Y94N, V106A	
A/Norway/1906/2020	3C.2a1b+T131K	2020-03-12	SIAT1	320	353	80	74	160	179	<	0	80	71	<	<	<	0	As A/Norway/3275/2018 + V106A, Q197R, K207R, S219F	
A/Norway/2205/2020	3C.2a1b+T131K	2020-03-15	SIAT1	320	361	80	70	160	128	<	0	40	54	<	<	<	0	As A/Norway/3275/2018 + I48M, V106A, Q197R, K207R, I214T, S219F	
A/Norway/2292/2020	3C.2a1b+T131K	2020-03-20	SIAT1	320	382	80	76	160	128	<	0	80	70	<	<	<	0	As A/Norway/3275/2018 + I48M, V106A, Q197R, K207R, I214T, S219F	
A/Norway/2357/2020	3C.2a1b+T131K	2020-03-28	SIAT1	320	299	80	67	80	112	<	0	80	62	<	<	<	0	As A/Norway/3275/2018 + S46T, I48M, V106A, Q197R, K207R, I214T, S219F	
A/Parma/21/2020	3C.2a1b+T135K-A	2020-02-05	SIAT2/SIAT1	<	10	<	0	160	124	<	0	80	70	<	<	<	0	As A/Denmark/3264/2019 + I140M, P273S	
A/Friuli Venezia Giulia/240/2020	3C.2a1b+T135K-A	2020-02-08	SIAT2/SIAT1	40	20	<	0	160	121	<	0	80	71	<	<	<	0	As A/Denmark/3264/2019 + E50K, V130I	
A/Friuli Venezia Giulia/228/2020	3C.2a1b+T135K-A	2020-02-10	SIAT2/SIAT1	40	20	<	0	160	128	<	0	80	80	<	<	<	0	As A/Denmark/3264/2019 + D53N	
A/Parma/23/2020	3C.2a1b+T135K-A	2020-02-10	SIAT2/SIAT1	40	20	<	0	160	202	<	10	160	125	<	10	<	0	As A/Denmark/3264/2019 + I140M, P273S	
A/Friuli Venezia Giulia/257/2020	3C.2a1b+T135K-A	2020-02-11	SIAT2/SIAT1	<	10	<	0	160	137	<	0	80	91	<	<	<	0	As A/Denmark/3264/2019 + E50K, V130I	
A/Algiers/368/2020	3C.2a1b+T135K-A	2020-02-13	SIAT1	40	40	<	0	160	154	<	10	80	89	<	<	<	0	As A/Denmark/3264/2019	
A/Norway/2208/2020	3C.2a1b+T135K-A	2020-03-14	SIAT1	40	54	<	0	160	173	<	10	160	144	<	<	<	0	As A/Denmark/3264/2019	
A/Milano/49/2020	3C.2a1b+T135K-A	2020-02-10	SIAT2/SIAT1	<	0	<	0	160	134	160	122	<	0	40	46	<	0	As A/Denmark/3264/2019 + I3T, N158K(-CHO), S199T	
A/Kyrgyzstan Bishkek/65/2020	3C.2a1b+T135K-B	2020-02-02	MDCKx/SIAT1	40	20	<	0	160	182	<	0	320	250	<	<	<	0	As A/Hong Kong/2671/2019 + K2E, A106V, I140K	
A/Kyrgyzstan Bishkek/63/2020	3C.2a1b+T135K-B	2020-02-03	MDCKx/SIAT1	40	20	<	0	160	138	<	0	320	279	<	<	<	0	As A/Hong Kong/2671/2019 + K2E, A106V, I140K, S262N, V309X	
A/Denmark/1295/2020	3C.2a1b+T135K-B	2020-02-18	SIAT3/SIAT1	80	60	<	0	320	336	<	0	320	346	40	49	<	10	As A/Hong Kong/2671/2019 + D53N	
						Vaccine SH 2020								Vaccine NH 2020-21					

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 8-5. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2020-09-07

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹												HA1 substitutions for 3C.2a/3a viruses compared to A/Norway/3275/2018: Egg adaptation HA substitutions compared to the corresponding cell culture propagated virus			
				Post-infection ferret antisera															
				A/Norway 3275/2018 SIAT F03/19	A/SouthAustralia 34/2019 Egg F43/19	A/Denmark 3264/19 SIAT F19/20	A/Paris 2554/19 Egg F14/20	A/Hong Kong 2671/19 SIAT F33/19	A/Hong Kong 2671/19 Egg F44/19	A/Kansas 14/2017 SIAT F17/19									
				3C.2a1b+T131K 2-fold Read	3C.2a1b+T131K 2-fold Read	3C.2a1b+T135K-A 2-fold Read	3C.2a1b+T135K-A 2-fold Read	3C.2a1b+T135K-B 2-fold Read	3C.2a1b+T135K-B 2-fold Read	3C.3a 2-fold Read									
REFERENCE VIRUSES																			
A/Norway/3275/2018	3C.2a1b+T131K	2018-10-04	SIAT2 10 ⁻³	640	514	80	75	160	147	<	20	40	55	<	0	<	0	none	
IVR-197 (A/South Australia/34/2019)	3C.2a1b+T131K	2019-02-06	E5/D8 10 ⁻⁸	320	297	1280	1077	<	0	<	0	<	0	40	30	<	0	V106A, T160K(-CHO), G186I, S219F	
A/Denmark/3264/2019	3C.2a1b+T135K-A	2019-10-25	SIAT3/SIAT2 10 ⁻³	80	66	<	10	320	376	<	10	160	157	<	10	<	0	V106A, T128A(-CHO), K131T, T135K(-CHO), A138S, G186D, D190N, F193S, S198P	
A/Paris/2554/2019	3C.2a1b+T135K-A	2019-11-25	E5 10 ⁻⁵	<	0	40	22	320	274	640	722	80	63	160	224	40	29	V106A, T128A(-CHO), K131T, T135K(-CHO), A138S, T160I(-CHO),G186D, D190N, F193S, S198P, D225G	
A/HongKong/2671/2019	3C.2a1b+T135K-B	2019-06-17	MDCK1/SIAT4 10 ⁻³	40	49	<	0	320	259	<	<	320	257	40	42	<	0	V106A, T128A(-CHO), K131T, T135K(-CHO), S137F, A138S, F193S	
A/HongKong/2671/2019	3C.2a1b+T135K-B	2019-06-17	E8/E2 10 ⁻³	<	0	320	450	80	71	320	400	160	137	640	812	40	47	V106A, T128A(-CHO), K131T, T135K(-CHO), S137F, A138S, T160I(-CHO), G186V, F193S, S219X, D225N	
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2 10 ⁻¹	<	0	<	0	<	0	40	49	<	0	<	0	160	146	G62E, S91N, R92K, V106A, K121N, T128A(-CHO), K131T, A138S, S144K, Y159S, T160K(-CHO), K171N, F193S, K326R	
TEST VIRUSES																			
A/Denmark/1263/2020	3C.2a1b+T131K	2020-02-17	SIAT3/SIAT1	320	413	80	71	160	150	<	0	80	70	<	<	<	0	As A/Norway/3275/2018 + V106A, S219F	
A/Denmark/1136/2020	3C.2a1b+T131K	2020-02-17	SIAT3/SIAT1	640	486	80	83	160	140	<	0	80	77	<	<	<	0	As A/Norway/3275/2018 + V106A, S219F	
A/Hungary/64/2020	3C.2a1b+T131K	2020-02-08	MDCK1/SIAT1	320	252	40	57	160	140	<	0	40	41	<	<	<	0	As A/Norway/3275/2018 + I48M, V106A, Q197R, K207R, I214T, S219F	
A/Milano/55/2020	3C.2a1b+T131K	2020-02-10	SIAT2/SIAT1	320	439	80	82	160	235	<	10	80	99	<	<	<	10	As A/Norway/3275/2018 + V106A, Q197R, K207R, S219F, P239S	
A/Bayern/31/2020	3C.2a1b+T131K	2020-02-13	C1/SIAT1	640	621	80	108	160	156	<	0	80	104	<	<	<	0	As A/Norway/3275/2018 + Q57K, V106A, Q197R, K207R, S219F, R261Q	
A/Halmstad/3/2020	3C.2a1b+T131K	2020-02-14	SIAT0/SIAT1	640	549	80	107	160	160	<	0	80	83	<	<	<	0	As A/Norway/3275/2018 + I25M, V106A, Q197R, K207R, S219F	
A/Halmstad/2/2020	3C.2a1b+T131K	2020-02-16	SIAT0/SIAT1	640	554	80	102	160	231	<	0	160	120	<	<	<	0	As A/Norway/3275/2018 + T37X, Q57K, V106A, Q197R, S198P, K207R, S219F	
A/Saint-Etienne/2539/2020	3C.2a1b+T131K	2020-02-11	MDCK3/SIAT1	80	67	80	63	<	0	<	0	<	0	<	<	<	0	As A/Norway/3275/2018 + V106A, {N158X, T160X(-CHO)}, Q197R, K207R, S219F, K276N	
A/Friuli Venezia Giulia/244/2020	3C.2a1b+T135K-A	2020-02-11	SIAT2/SIAT1	<	0	<	0	80	64	<	0	<	0	<	<	<	0	As A/Denmark/3264/2019 + E50K, V130I, P221L, R261Q	
A/Parma/30/2020	3C.2a1b+T135K-A	2020-02-17	SIAT2/SIAT1	<	0	<	0	80	89	<	10	<	0	<	<	<	0	As A/Denmark/3264/2019 + N158X(-CHO), P221X	
A/Hungary/61/2020	3C.3a	2020-02-05	MDCK1/SIAT1	<	0	<	0	<	0	<	0	<	0	<	<	160	202	As A/Kansas/14/2017 + N96S	
A/Hungary/91/2020	3C.3a	2020-02-18	MDCK1/SIAT1	<	0	<	0	<	0	<	0	<	0	<	<	80	117	As A/Kansas/14/2017 + S270T	
				Vaccine SH 2020								Vaccine NH 2020-21							

Table 8-6. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) - Summary

Viruses		Neutralisation Titre ¹													
		Post-infection ferret antisera													
		A/Norway 3275/2018 SIAT F03/19 3C.2a1b+T131K 2-fold Read		A/SouthAustralia 34/2019 Egg F43/19 3C.2a1b+T131K 2-fold Read		A/Denmark 3264/19 SIAT F19/20 3C.2a1b+T135K-A 2-fold Read		A/Paris 2554/19 Egg F14/20 3C.2a1b+T135K-A 2-fold Read		A/Hong Kong 2671/19 SIAT F33/19 3C.2a1b+T135K-B 2-fold Read		A/Hong Kong 2671/19 Egg F44/19 3C.2a1b+T135K-B 2-fold Read		A/Kansas 14/2017 SIAT F17/19 3C.3a 2-fold Read	
Passage history Ferret number Genetic group															
REFERENCE VIRUSES															
A/Norway/3275/2018	3C.2a1b+T131K	640	514	80	75	160	147	<	20	40	55	<	0	<	0
IVR-197 (A/South Australia/34/2019)	3C.2a1b+T131K	320	297	1280	1077	<	0	<	0	<	0	40	30	<	0
A/Denmark/3264/2019	3C.2a1b+T135K-A	80	77	<	10	640	486	40	36	320	257	40	38	<	10
A/Paris/2554/2019	3C.2a1b+T135K-A	<	0	40	22	320	274	640	722	80	63	160	224	40	29
A/HongKong/2671/2019	3C.2a1b+T135K-B	40	59	40	20	320	350	40	20	320	376	80	61	40	23
A/HongKong/2671/2019	3C.2a1b+T135K-B	<	0	320	450	80	71	320	400	160	137	640	812	40	47
A/Kansas/14/2017	3C.3a	<	0	<	0	<	0	40	49	<	0	<	0	160	146
TEST VIRUSES															
Number of viruses tested		61		61		61		61		61		61		61	
No with titre reduction ≤2-fold		17				31		1		19				17	
%		27.9				50.8		1.6		31.1				27.9	
No with titre reduction =4-fold		1		1		8		3		15		2		5	
%		1.6		1.6		13.1		5.0		24.6		3.3		8.2	
No with titre reduction ≥8-fold		43		60		22		57		27		59		39	
%		70.5		98.4		36.1		93.4		44.3		96.7		63.9	
Subgroup 3C.2a1b viruses + T131K															
Number of viruses tested		20		20		20		20		20		20		20	
No with titre reduction ≤2-fold		17				12				6					
%		85.0	80.0		none	60.0	5.0		none	30.0	none		none	1	none
No with titre reduction =4-fold		1		1		5				9				1	
%		5.0	90.0	5.0	none	25.0	85.0		none	45.0	40.0		none	5.0	none
No with titre reduction ≥8-fold		2		19		3		20		5		20		19	
%		10.0		95.0		15.0		100.0		25.0		100.0		95.0	
Subgroup 3C.2a1b viruses + T135K-A															
Number of viruses tested		17		17		17		17		17		17		17	
No with titre reduction ≤2-fold						13		1		7					
%			none		none	76.5	29.4	5.9	none	41.2	11.8		none		none
No with titre reduction =4-fold						3		2		5		1		3	
%			none		none	17.6	82.4	11.7	none	29.4	47.1	5.9	none	17.6	5.9
No with titre reduction ≥8-fold		17		17		1		14		5		16		14	
%		100.0		100.0		5.9		82.4		29.4		94.1		82.4	
Subgroup 3C.2a1b viruses + T135K-B															
Number of viruses tested		7		7		7		7		7		7		7	
No with titre reduction ≤2-fold						6				6					
%			none		none	85.7	28.6		14.3	85.7	85.7		none		none
No with titre reduction =4-fold										1		1		1	
%			none		none		71.4		14.3	14.3	85.7	14.3	none	14.3	none
No with titre reduction ≥8-fold		7		7		1		7				6		6	
%		100.0		100.0		14.3		100.0				85.7		85.7	
Clade 3C.3a viruses															
Number of viruses tested		17		17		17		17		17		17		17	
No with titre reduction ≤2-fold														17	
%			none		none		none		none		none		none	100.0	100.0
No with titre reduction =4-fold								1							
%			none		none		none	5.9	5.9		none		none		100.0
No with titre reduction ≥8-fold		17		17		17		16		17		17			
%		100.0		100.0		100.0		94.1		100.0		100.0			
				Vaccine SH 2020								Vaccine NH 2020-21			

Figure 6a. HA gene phylogeny of A(H3N2) viruses used in VN assay

Vaccine viruses

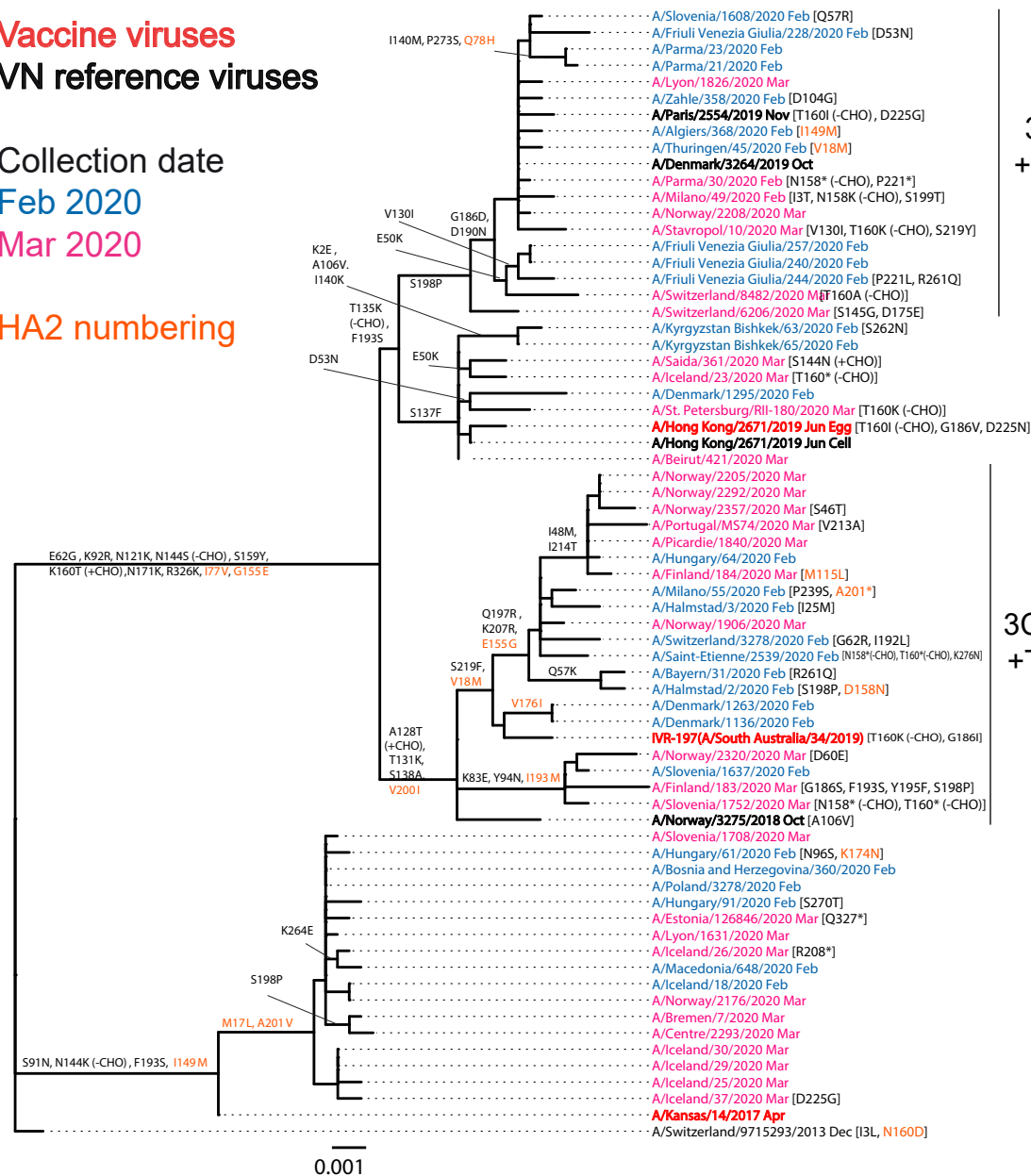
VN reference viruses

Collection date

Feb 2020

Mar 2020

HA2 numbering



3C.2a1b
+T135K-A

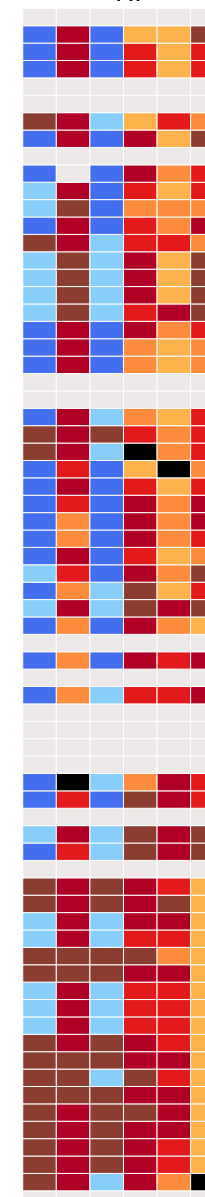
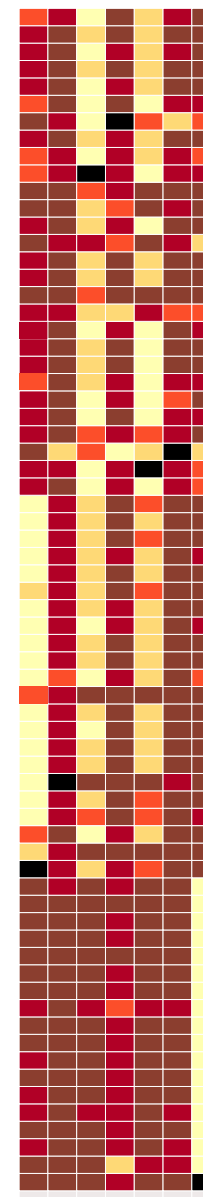
3C.2a1b
+T135K-B

3C.2a1b
+T131K

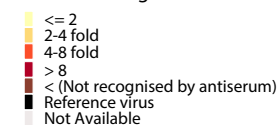
3C.3a

VN

HI



VN fold change



HI fold change

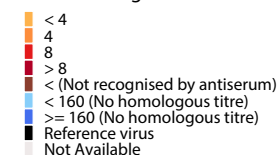


Figure 6b. Phylogenetic comparison of A(H3N2) HA genes

Vaccine viruses

Reference viruses

Collection date

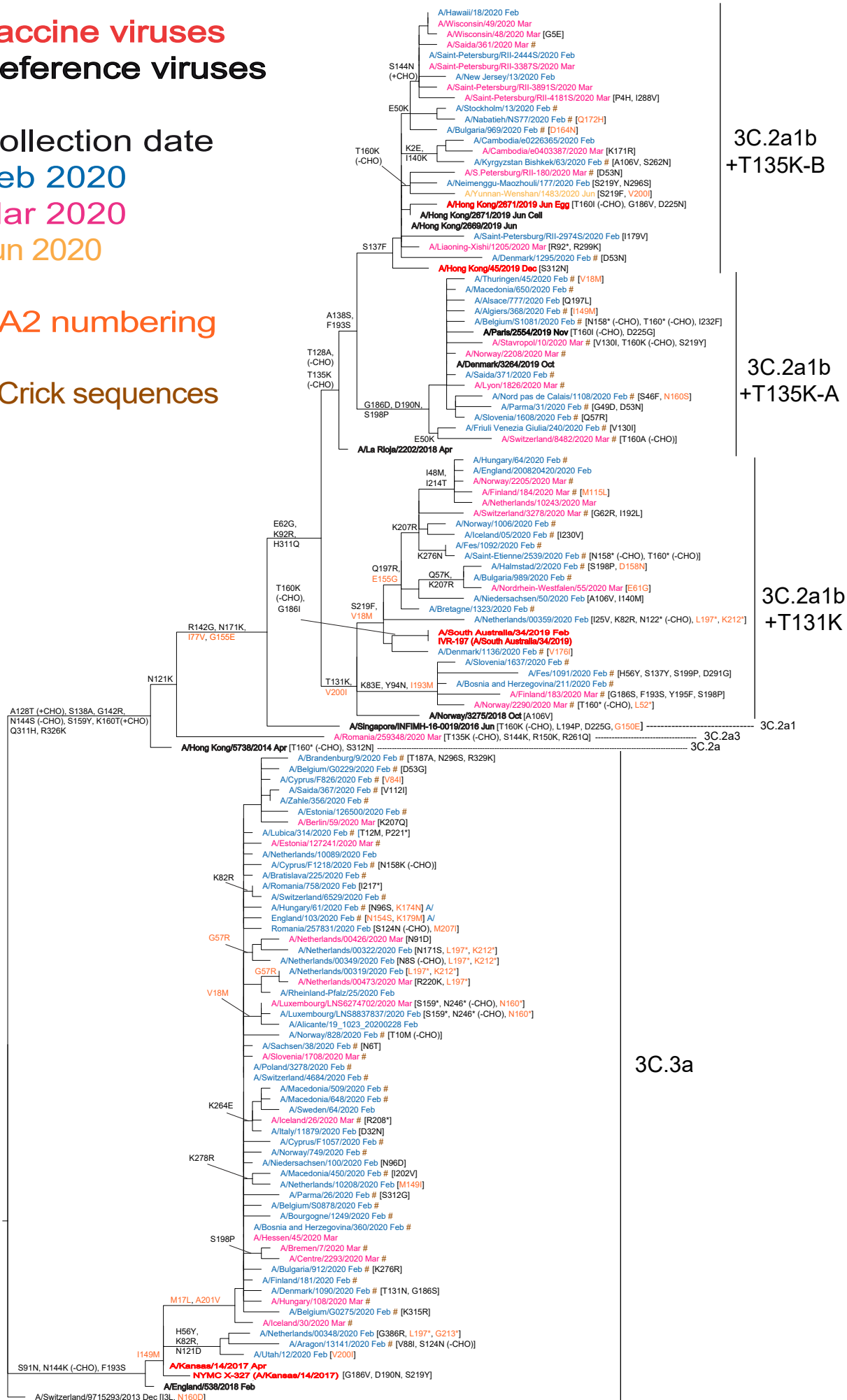
Feb 2020

Mar 2020

Jun 2020

HA2 numbering

Crick sequences



Vaccine virus: **Reference virus:** **Genetic group:** **HA1/HA2 boundary:** **Potential N-linked glycosylation site**



	310	320	330	340	350	360	370	380	390	400	
A/Kansas/14/2017 Apr	TYGACPRYVKQSTLKLATGMNRVPERQTRGIFGAIAGFIENGWEGMVDGWYGFRRHONSEGRGQAADLKSTQAAIDQINGKLNRLIGKTNKPFHQIEKEFS										3C.3a
A/Iceland/37/2020 Mar					L						3C.3a
A/Bremen/7/2020 Mar					L						3C.3a
A/Bulgaria/1009/2020 Feb					L						3C.3a
A/Belgium/H0017/2020 Feb		R.			L						3C.3a
A/Macedonia/505/2020 Feb					L						3C.3a
A/Norway/1900/2020 Mar					L						3C.3a
A/Norway/2069/2020 Mar					L						3C.3a
A/Hong Kong/5738/2014 Apr		HN.	K.								3C.2a
A/Norway/3275/2018 Oct			K.								T131K
A/Finland/183/2020 Mar			K.								T131K
A/Lyon/1826/2020 Mar			K.								T131K
A/South Australia/34/2019 Feb			K.		M.						T131K
A/Bayern/31/2020 Feb			K.		M.						T131K
A/Switzerland/3278/2020 Mar			K.		M.						T131K
A/Fes/1085/2020 Feb			K.		M.						T131K
A/Portugal/MS74/2020 Mar			K.		M.						T131K
A/Norway/2357/2020 Mar			K.		M.						T135KA
A/La Rioja/2202/2018 Apr			K.								T135KA
A/Switzerland/6206/2020 Mar			K.								T135KA
A/Friuli Venezia Giulia/240/20			K.								T135KA
A/Stavropol/10/2020 Mar			K.								T135KA
A/Parma/23/2020 Feb			K.								T135KA
A/Zahle/358/2020 Feb			K.								T135KA
A/Hong Kong/45/2019 Dec		N.	K.								T135KB
A/S.Petersburg/RII-180/2020 Ma			K.								T135KB
A/Kyrgyzstan Bishkek/65/2020 F			K.								T135KB
A/Hong Kong/2671/2019 Jun Egg			K.								T135KB
A/Saida/361/2020 Mar			K.								T135KB
A/Bulgaria/969/2020 Feb			K.								T135KB

	410	420	430	440	450	460	470	480	490	500	
A/Kansas/14/2017 Apr	EVEGRIQDLEKYVEDTKIDLWSYNAELLVALENQHTIDLTDSMNKLFERTKKQLRENAEDMGNGCFKIYHKCDNACMGSI										
A/Iceland/37/2020 Mar											
A/Bremen/7/2020 Mar											
A/Bulgaria/1009/2020 Feb											
A/Belgium/H0017/2020 Feb											
A/Macedonia/505/2020 Feb											
A/Norway/1900/2020 Mar											
A/Norway/2069/2020 Mar											
A/Hong Kong/5738/2014 Apr								I			
A/Norway/3275/2018 Oct		V						I	E		
A/Finland/183/2020 Mar		V						I	E		
A/Lyon/1826/2020 Mar		V						I	E		
A/South Australia/34/2019 Feb		V						I	E		
A/Bayern/31/2020 Feb		V						I	E		
A/Switzerland/3278/2020 Mar		V						I	E		
A/Fes/1085/2020 Feb		V						I	E		
A/Portugal/MS74/2020 Mar		V						I	E		
A/Norway/2357/2020 Mar		V						I	E		
A/La Rioja/2202/2018 Apr		V						I	E		
A/Switzerland/6206/2020 Mar		V						I	E		
A/Friuli Venezia Giulia/240/20		V						I	E		
A/Stavropol/10/2020 Mar		V						I	E		
A/Parma/23/2020 Feb		VH						I	E		
A/Zahle/358/2020 Feb		V						I	E		
A/Hong Kong/45/2019 Dec		V						I	E		
A/S.Petersburg/RII-180/2020 Ma		V						I	E		
A/Kyrgyzstan Bishkek/65/2020 F		V						I	E		
A/Hong Kong/2671/2019 Jun Egg		V						I	E		
A/Saida/361/2020 Mar		V						I	E		
A/Bulgaria/969/2020 Feb		V						I	E	N	

	510	520	530	540	550	
A/Kansas/14/2017 Apr	QIKGVELKSGYKDWILWISFAISCFLLCVALLGFIMWACQKGNIRCNICI					
A/Iceland/37/2020 Mar		V				
A/Bremen/7/2020 Mar		V				
A/Bulgaria/1009/2020 Feb		V				
A/Belgium/H0017/2020 Feb		V				
A/Macedonia/505/2020 Feb		V				
A/Norway/1900/2020 Mar		V				
A/Norway/2069/2020 Mar		V				
A/Hong Kong/5738/2014 Apr						
A/Norway/3275/2018 Oct			I			
A/Finland/183/2020 Mar			M	I		
A/Lyon/1826/2020 Mar						
A/South Australia/34/2019 Feb			I			
A/Bayern/31/2020 Feb			I			
A/Switzerland/3278/2020 Mar			I			
A/Fes/1085/2020 Feb			I			
A/Portugal/MS74/2020 Mar			I			
A/Norway/2357/2020 Mar			I			
A/La Rioja/2202/2018 Apr						
A/Switzerland/6206/2020 Mar						
A/Friuli Venezia Giulia/240/20						
A/Stavropol/10/2020 Mar						
A/Parma/23/2020 Feb						
A/Zahle/358/2020 Feb						
A/Hong Kong/45/2019 Dec						
A/S.Petersburg/RII-180/2020 Ma						
A/Kyrgyzstan Bishkek/65/2020 F						
A/Hong Kong/2671/2019 Jun Egg						
A/Saida/361/2020 Mar						
A/Bulgaria/969/2020 Feb						

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

a) Subclade 3C.2a1b

b) Clade 3C.3a

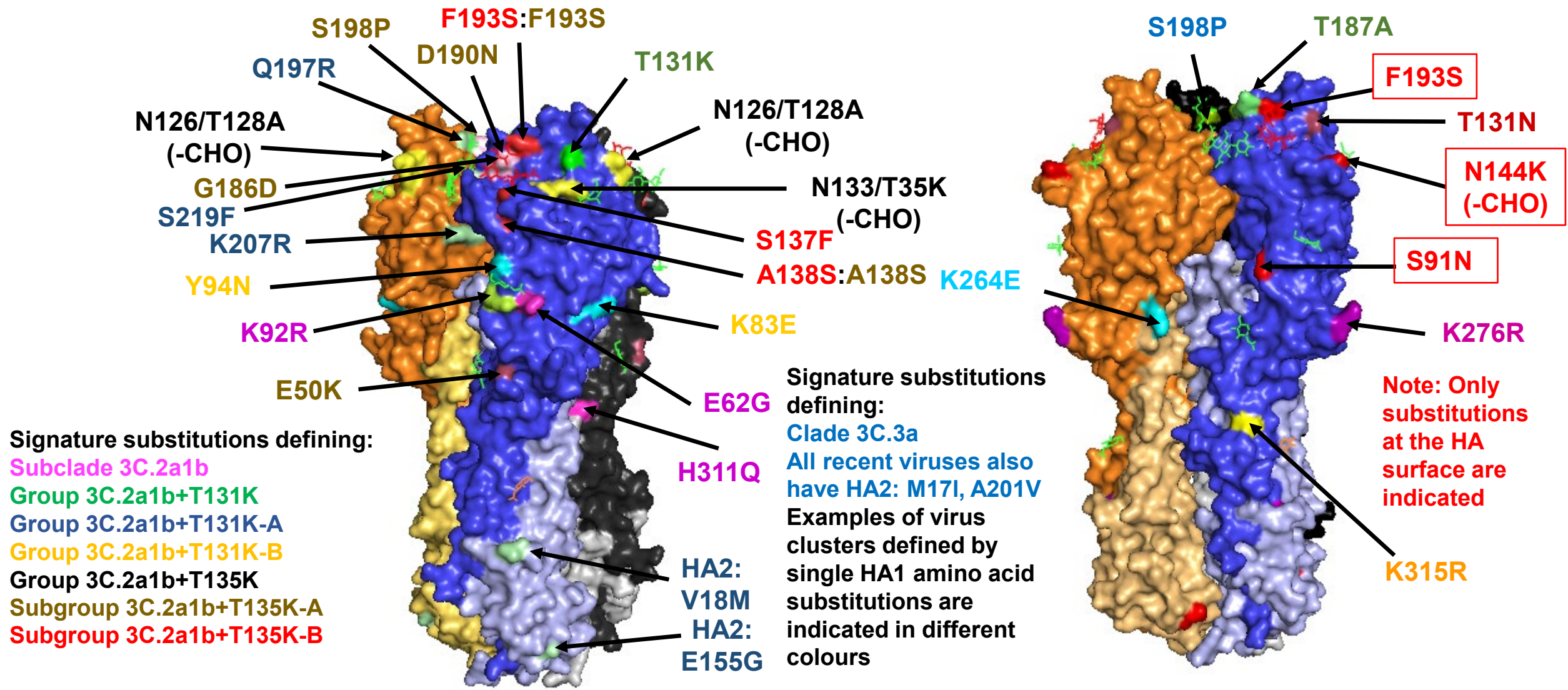


Figure 8. Amino acid substitutions in the HA of (a) recent group 3C.2a1b H3N2 viruses compared with A/Singapore/INFIMH-16-0019/2016 and (b) clade 3C.3a viruses compared with A/Switzerland/9715293/2013

Figure 9. Phylogenetic comparison of A(H3N2) NA genes

Vaccine viruses

Reference viruses

Collection date

Feb 2020

Mar 2020

Jun 2020

Crick sequences

D151* = +CHO if *=N

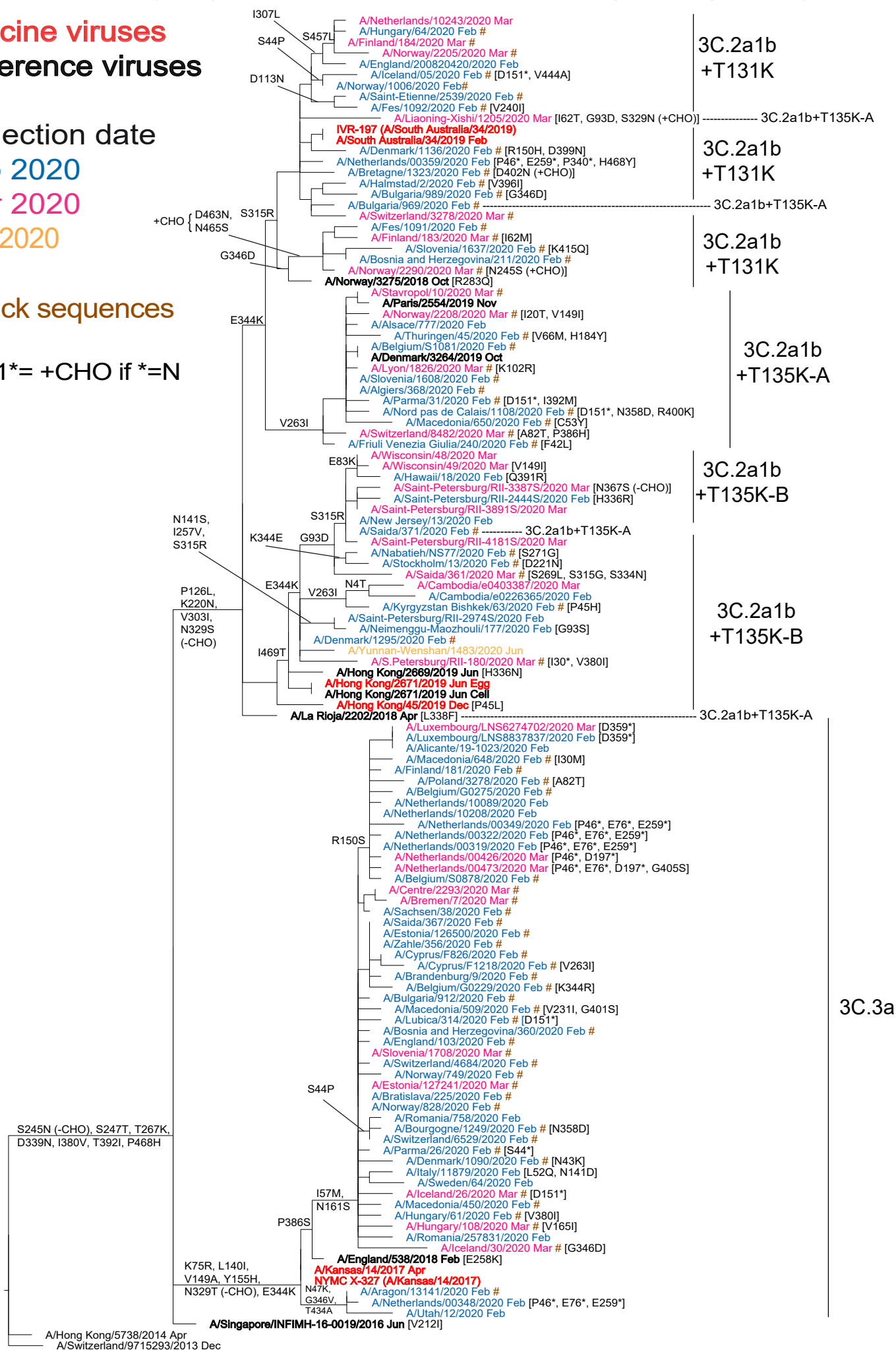
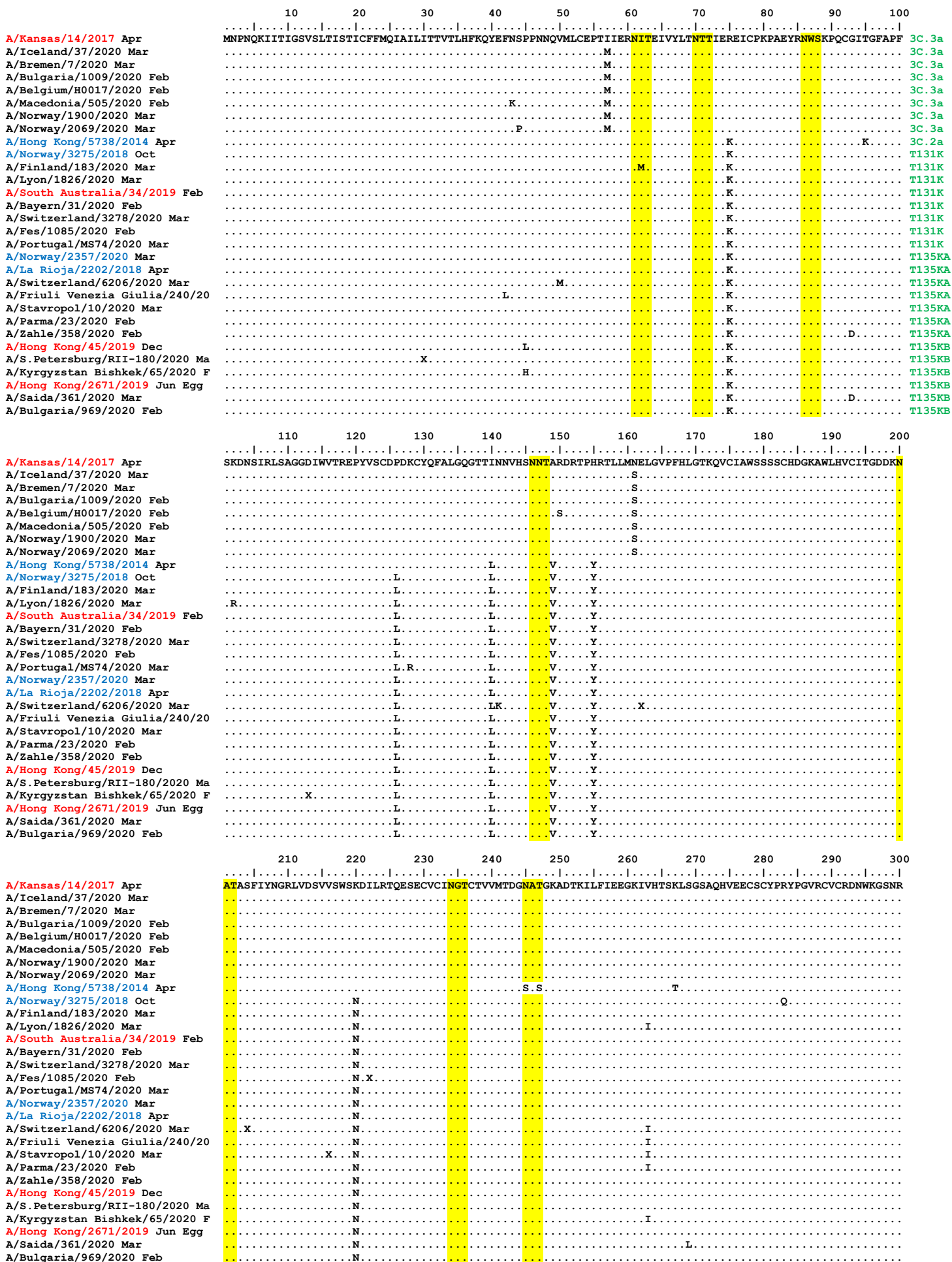


Figure 10. NA protein alignment for A(H3N2) viruses

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site



Influenza B virus analyses.

Influenza B viruses represented 30% of the samples received with collection dates after 2020-01-31 (**Table 1**) and B/Victoria lineage viruses were received from NICs in 29 countries or administrative regions in three WHO Regions: African (3: Algeria, Cote d'Ivoire and South Africa), Eastern Mediterranean (Lebanon), and European (25) (**Table 9**). B/Yamagata lineage viruses were received from just three countries: France, Norway and the Russian Federation (**Table 11**). B/Victoria lineage viruses outnumbered B/Yamagata lineage viruses at a ratio of ~60:1.

B/Victoria lineage.

HI analyses

The collection details and summary results for viruses that were recovered and assayed by HI are shown in **Table 9**.

HI results for the propagated viruses analysed since the February 2020 report are shown in **Tables 10-1 to 10-12**, test viruses in each table are sorted by date of collection and those below the red horizontal lines in **Tables 10-7 to 10-12** are those with collection dates after 2020-01-31. The HA genetic group is indicated for those viruses that have been sequenced. A summary of the recognition of the test viruses collected after 2020-01-31 is shown in **Table 10-13**. Two clearly distinct patterns of antigenicity were detected.

A total of 177 viruses with collection dates after 2020-01-31 were analysed; gene sequencing has been completed for all but seven of the viruses analysed antigenically. The great majority of viruses (168 of 170, 99%) were identified as having HA genes encoding a triple deletion $\Delta 162-164$ in **HA1** and only two viruses had HA genes encoding a double deletion $\Delta 162-163$. The majority of these viruses were recognised well by antisera raised against viruses with the three amino acid deletion in **HA1** ($\Delta 162-164$) (**Table 1**) the cell culture-propagated and egg-propagated cultivars of B/Washington/02/2019, the vaccine virus recommended for use in the Southern Hemisphere 2020 and the Northern Hemisphere 2020-2021 influenza seasons. These antisera recognised 88% and 83%, respectively, of the tested viruses with the ($\Delta 162-164$) deletion at titres within 4-fold of the titres of the antisera for the homologous viruses, and 86% and 66% within 2-fold. These viruses were recognised poorly by antisera raised against both egg- and cell culture-propagated viruses with no deletion in the HA (B/Brisbane/60/2008, B/South Australia/81/2012, B/Ireland/3154/2016, and B/Nordrhein-Westfalen/1/2016). The majority of $\Delta 162-164$ viruses were recognised at titres >4-fold lower than the homologous titres by antisera raised against viruses having HA genes with a two amino acid deletion $\Delta 162-163$ in **HA1**: antisera raised against cell culture-propagated B/Norway/2409/2017 and B/Colorado/06/2017 and the egg-propagated cultivar of B/Colorado/06/2017 recognised 99%, 77% and 91%, respectively of test viruses collected after 2020-01-31 at titres >4-fold lower than the homologous titres.

A proportion of viruses with the triple deletion $\Delta 162-164$ in **HA1** were recognised poorly by one or both of the antisera raised against cell culture- and egg-propagated cultivars of B/Washington/02/2019 with approximately 12% and 17%, respectively, of the test viruses being recognised poorly. Most of these viruses had HA genes that encoded amino acid substitutions in **HA1**: **D129N** most of which also encoded **N233S** resulting in the loss of a potential glycosylation site; **N126K**; **T155A**; **N150K** with **G184E**, **N197D** and **R279K**; or **T155K**; some had additional substitutions.

Two viruses had HA genes with the two amino acid deletion in **HA1** ($\Delta 162-163$). These were recognised best by antisera raised against cell culture-propagated B/Norway/2409/2017 and B/Colorado/06/2017 and the egg-propagated cultivar of B/Colorado/06/2017 which recognised, respectively, both, both and one of the two tested viruses at titres within 4-fold of the titres of the antisera with the homologous viruses. Both these viruses were poorly recognised by antisera raised against viruses with no deletion in the HA or those with the $\Delta 162-164$ deletion in **HA1**.

Genetic analysis

Phylogenetic analysis of the HA genes of representative B/Victoria lineage viruses sequenced at the Crick, and elsewhere, is shown in **Figure 11**. A phylogenetic tree of the HA genes of all viruses in the B/Victoria lineage with collection dates after 2020-01-31 analysed genetically at the Francis Crick Institute is shown in the Annex, **Figure A-5**. An alignment of the HA glycoproteins of a selection of these viruses is shown in **Figure 12**. All recently circulating viruses analysed carried HA genes that fell into genetic clade 1A, the B/Brisbane/60/2008 genetic clade and all fell in a genetic group defined by three **HA1** amino acid substitutions compared with B/Brisbane/60/2008: **I117V**, **N129D** and **V146I**. There are three clusters within clade 1A with deletions in **HA1**. Test viruses with HA genes encoding the two amino acid deletion in **HA1** ($\Delta 162-163$) were in the small minority, and also encoded amino acid substitutions **D129G** and **I180V** in **HA1** and **R151K** in **HA2** (genetic group **1A($\Delta 2$)**), representing only 2% of the samples received since 2020-01-31. Three of the four viruses had HA substitutions **I180V** in **HA1** and **D179G** and **V208I** in **HA2** (**Figure A-5**). The HA genes of viruses collected since 2020-01-31 with a three amino acid deletion in **HA1** ($\Delta 162-164$) fell into a group encoding **K136E** substitution in **HA1** (genetic group **1A($\Delta 3$)B**) and represented the vast majority of viruses (98%) received with collection dates after 2020-01-31. Most viruses in this group also encoded **G133R** substitution in **HA1**. Within this group, substitutions in **HA1** of either **E128K** (~50% of the viruses) or **N233K** (resulting in the loss of a potential glycosylation site; ~30% of the viruses), were both common.

Minor groups within genetic group **1A($\Delta 3$)B** without the **G133R** substitution have been seen. One group had the HA1 substitutions **D129N** and **N233S** (resulting in the loss of a potential glycosylation site), another had the **D129N** substitution with either **R80K**, **P134T** and **E198K**, or **D129N** with **K307R** in **HA1**, a third minor group had **N126K**. A fourth had **HA1** substitutions **N150K**, **G184E**, **N197D** (resulting in the loss of the glycosylation motif at **197-199** in **HA1**) and **R275K**. **Figure 13** shows the locations of amino acid substitutions that define the three groups of deletion variants on a B/Brisbane/60/2008 HA structure.

Figure 14 shows a phylogenetic tree for the NA genes of representative viruses and a phylogenetic tree of the NA genes of all viruses in the B/Victoria lineage with collection dates after 2020-01-31 analysed genetically at the Francis Crick Institute is shown in the Annex, **Figure A-6**. **Figure 15** shows a corresponding amino acid sequence alignment for a selection of NA glycoproteins. The NA gene phylogeny is largely congruent with the HA phylogeny and the vast majority of viruses in the main recently emerged Δ 162-164 genetic group (**genetic group 1A(Δ 3)B**) encode a **A395T** substitution in the NA. Those in the Δ 162-163 group of viruses (**genetic group 1A(Δ 2)**) share an **NA K371Q** amino acid substitution. Minor genetic groups have been seen within the NA and are associated with those observed in the HA.

B/Yamagata lineage.

The collection details and summary results for viruses assayed by HI with collection dates after 2020-01-31 are shown in **Table 11**.

HI analyses

The results of HI analyses for propagated viruses of the B/Yamagata lineage are shown in **Tables 12-1** and **12-2**. Test viruses in each table are sorted by date of collection and the three below the red horizontal line in **Table 12-2** are viruses with collection dates after 2020-01-31. These viruses fell into clade 3, the B/Wisconsin/1/2010 – B/Phuket/3073/2103 clade.

Post-infection ferret antiserum raised against the egg-propagated vaccine virus B/Phuket/3073/2013, recommended for use in quadrivalent vaccines for the Southern Hemisphere 2020 and the Northern Hemisphere 2020-2021 influenza seasons, recognised all three test viruses collected after 2020-01-31 at titres within 2-fold of the titre with the homologous virus. However, the post-infection ferret antiserum raised against the cell culture-propagated cultivar of B/Phuket/3073/2013 recognised the three recently collected test viruses less well with only one of the three being recognised at a titre within 4-fold of the titre with the homologous virus.

Antisera raised against two other viruses from the B/Wisconsin/1/2010-B/Phuket/3073/2013 clade (clade 3), egg-propagated B/Wisconsin/1/2010 and cell culture-propagated B/Mauritius/I-762/2018, recognised the test viruses collected after 2020-01-31 well, at titres equal to or within 2-fold of the homologous titre. However, the antiserum raised against egg-propagated B/Norway/2134/2019, recognised the test viruses poorly.

An antiserum raised against a former vaccine virus, egg-propagated B/Massachusetts/02/2012 (clade 2), recognised B/Norway/993/2020 at a titre equal to the titre of the antiserum with the homologous virus, but the other two test viruses were recognised poorly. However, antisera raised against cell culture-propagated cultivars of two clade 2 viruses, B/Massachusetts/02/2012 and B/Estonia/55669/2011, recognised none of the three recently collected test viruses at titres within 4-fold of their homologous titres.

Genetic analyses

Figure 16 shows a phylogenetic analysis of all HA genes from viruses of the B/Yamagata lineage with collection dates after 2019-12-31, those sequenced at the Crick and those available within the EpiFlu™ database of GISAID, as of 2020-09-06. An amino acid sequence alignment of a selection of the HA glycoproteins is shown in **Figure 17**. All viruses analysed genetically with collection dates after 2020-01-31 had HA gene sequences falling in clade 3, the B/Phuket/3073/2013 clade, and fall into a genetic group defined by two **HA1** amino acid substitutions compared with B/Phuket/3073/2013: **L172Q** and **M251V**.

Figure 18 shows a phylogenetic tree for NA genes of the same viruses represented in the HA phylogeny and **Figure 19** shows a corresponding amino acid sequence alignment for selected NA glycoproteins. As for the HA gene, the vast majority of NA genes of recently collected viruses fell into the B/Phuket/3073/2013 clade (clade 3) falling in a group defined by NA **I49M**, **I171M**, **D342K** and **K373Q** amino acid substitutions compared to the B/Phuket/3073/2013 NA sequence. A number of subgroups have emerged within this group, notably those defined by an **A395S** amino acid substitution or **R65H** and **S402P** substitutions.

The reference virus B/Norway/2134/2019 has seven substitutions in **HA1**, **R69G**, **G141R**, **K149E**, **I150N**, **G183E**, **N196I** (resulting in the loss of the glycosylation motif at residues **196-198** in **HA1**) and **P240Q**, and six in NA, **K69E**, **R188K**, **I248V**, **G346R**, **G378E** and **K435R**.

Antiviral Analysis.

Susceptibility to the neuraminidase inhibitors (NAI) zanamivir and oseltamivir was assessed phenotypically and by NA gene sequencing for 176 B/Victoria-lineage and three B/Yamagata-lineage viruses with collection dates after 2020-01-31 (**Table 13**). All showed normal inhibition (NI: no reduced susceptibility) by both antivirals.

Of the 56 influenza B viruses that yielded full-length PA gene sequences all encoded isoleucine at position 38 (I38) and showed no evidence of other substitutions associated with reduced susceptibility to baloxavir marboxil.

Conclusion.

Viruses of the B/Victoria lineage predominated over those of the B/Yamagata lineage at a ratio of ~60:1. Viruses of the B/Victoria lineage fell into two genetic groups. The predominant group, representing 98% of the viruses sequenced, had HAs with a three amino acid deletion in **HA1** (**Δ162-164**) with the additional **HA1** amino acid substitution **K136E**. Viruses with HA genes encoding a deletion of two amino acids in **HA1** (**Δ162-163**) represented only 2% of those sequenced, but no viruses with collection dates after 2020-01-31 that lacked amino acid deletions were received. Viruses with either deletion are antigenically distinct from the other viruses in clade 1A and the **Δ162-163** and **Δ162-164** variants are antigenically distinct from each other.

Viruses of the B/Yamagata lineage fell within genetic clade 3, the B/Phuket/3073/2013 clade, and the recently collected viruses were recognised well by antisera raised against egg-propagated clade 3 vaccine/reference viruses.

All influenza B viruses showed normal inhibition by the NA inhibitors oseltamivir and zanamivir. Viruses yielding full-length PA gene sequences (n = 56) all encoded proteins likely to be well inhibited by baloxavir marboxil.

Table 9. Summary of influenza B (Victoria-lineage) viruses analysed, with collection dates after 2020-01-31

MONTH	B Victoria lineage														
	Country	Number received ¹	Number propagated ²	Fold-reductions in HI titre compared to homologous titres (post-infection ferret antisera)											
				Clade 1A(Δ2) virus			Clade 1A(Δ2) virus			Clade 1A(Δ3)B virus			Clade 1A(Δ3)B virus		
				≤2 fold ³	4 fold ³	≥8 fold ³	≤2 fold ⁴	4 fold ⁴	≥8 fold ⁴	≤2 fold ⁵	4 fold ⁵	≥8 fold ⁵	≤2 fold ⁶	4 fold ⁶	≥8 fold ⁶
2020															
FEBRUARY															
Algeria	2	2			2			2			2			2	
Belgium	5	5			5			5			5	1	3	1	
Bosnia and Herzegovina	3	3		1	2			3	2	1		2	1		
Bulgaria	3	3			3			3	3			3			
Cyprus	11	5			5			5	5			4		1	
Denmark	3	3			3			3	3			2		1	
Estonia	5	4		4			1	3	4			4			
Finland	2	2			2			2	2			2			
France	9	9		3	6		1	8	9			8		1	
Germany	5	5			5		2	3	5			3	2		
Hungary	2	2	1	1				2	2			1		1	
Iceland	4	4			4			4			4		4		
Ireland	8	6			6			6	6			6			
Italy	6	6		4	2			6	6			5		1	
Lebanon	11	5			5		1	4	4	1		1	2	2	
Macedonia	5	1			1			1	1				1		
Norway	3	0													
Poland	2	0													
Portugal	1	1		1				1	1			1			
Russia	2	2	1	1		1		1	2			2			
Slovakia	2	2			2			2	2			2			
Slovenia	3	3			3			3	3			1		2	
South Africa	4	4			4			4	4					4	
Spain	8	7	1	4	2		2	5	5		2	5		2	
Sweden	5	5			5			5	5			3	1	1	
Switzerland	8	8			8			8	7		1	7		1	
United Kingdom	4	4			4			4	4			4			
MARCH															
Belgium	2	2			2			2			2		1	1	
Bulgaria	6	6		3	3		1	5	6			6			
Cote d'Ivoire	5	0													
Cyprus	7	0													
Estonia	1	1		1				1	1			1			
Finland	2	2	1		1	1		1	1		1	1		1	
France	7	7	1	1	5		1	6	7			7			
Germany	1	1			1			1	1					1	
Iceland	3	3			3		1	2			3	1	1	1	
Ireland	6	4		2	2		1	3	4			4			
Macedonia	2	2			2			2	2			1	1		
Moldova	2	2		1	1			2	2			2			
Norway	14	9	2	3	4	1		8	7	1	1	8		1	
Russia	21	21	1	2	18		1	20	18	1	2	9	9	3	
Slovenia	6	6			6			6	6			5	1		
South Africa	1	1			1			1	1					1	
Spain	1	1			1			1	1					1	
Switzerland	6	6		1	5			6	6			5		1	
United Kingdom	1	1			1			1	1					1	
APRIL															
Iceland	1	1			1			1	1				1		
Norway	1	0													
MAY															
Cote d'Ivoire	1	0													
JUNE															
Cote d'Ivoire	1	0													
JULY															
Cote d'Ivoire	2	0													
AUGUST															
Cote d'Ivoire	2	0													
	228	177	8	33	136	3	12	162	150	4	23	117	28	32	
29 Countries			4.5%	18.6%	76.8%	1.7%	6.8%	91.5%	84.7%	2.3%	13.0%	66.1%	15.8%	18.1%	

1 Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared

2 Propagated to sufficient titre to perform HI assay

3 Compared to homologous titres for ferret antisera raised against tissue culture-propagated B/Colorado/06/2017 - Clade 1A(Δ2)

4 Compared to homologous titres for ferret antisera raised against egg-propagated B/Colorado/06/2017 - Clade 1A(Δ2)

5 Compared to homologous titres for ferret antisera raised against tissue culture-propagated B/Washington/02/2019 - Clade 1A(Δ3)B

6 Compared to homologous titres for ferret antisera raised against egg-propagated B/Washington/02/2019 - Clade 1A(Δ3)B

Table 10-1. Antigenic analyses of influenza B viruses (Victoria lineage) 2020-02-12

Haemagglutination inhibition titre												
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera								
				B/Bris	B/Bris	B/Sth Aus	B/Nord-West	B/Norway	B/Colorado	B/Colorado	B/Wash'ton	B/Wash'ton
				60/08	60/08	81/12	1/16	2409/17	06/17	06/17	02/19	02/19
				Egg	Egg	Egg	MDCK	MDCK	MDCK	Egg	MDCK	Egg
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ⁴	F25/16 ⁴	F38/17 ²	F40/17 ²	F21/18 ⁴	F11/18 ⁴	F37/19 ⁴	F38/19 ⁴
Passage history	Genetic group	1A	1A	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)B	1A(Δ3)B		
REFERENCE VIRUSES												
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	1280	640	640	80	<	<	160	<	160
B/South Australia/81/2012	1A	2012-11-28	E4/E2	2560	1280	1280	160	<	<	160	<	320
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK2	1280	40	<	80	<	<	<	<	<
B/Norway/2409/2017	1A(Δ2)	2017-04-27	MDCK1/MDCK3	80	<	<	<	40	40	80	<	<
B/Colorado/06/2017	1A(Δ2)	2017-02-05	MDCK1/MDCK2	80	<	<	<	40	40	80	<	<
B/Colorado/06/2017	1A(Δ2)	2017-02-05	E5/E2	640	160	<	<	20	40	320	<	160
B/Washington/02/2019	1A(Δ3)B	2019-01-19	C2/MDCK2	160	<	40	<	<	<	80	40	320
B/Washington/02/2019	1A(Δ3)B	2019-01-19	E3/E2	640	80	40	<	<	<	160	40	320
TEST VIRUSES												
B/Lithuania/MB39700/2019	No seq	2019-10-28	MDCK2/MDCK1	160	<	<	<	<	<	40	40	320
B/Beirut/570/2019	No seq	2019-12-02	SIAT3/SIAT1/MDCK1	40	<	<	<	<	<	20	<	80
B/Valladolid/299/2019	1A(Δ3)B	2019-10-23	MDCK1/MDCK1	40	<	<	<	<	<	40	20	320
B/Mount Lebanon-Ain-W-zein/548/2019	1A(Δ3)B	2019-10-28	SIAT3/SIAT1/MDCK1	40	<	<	<	<	<	<	40	160
B/Montenegro/35/2019	1A(Δ3)B	2019-11-19	MDCKx/MDCK1	320	40	40	<	<	<	40	<	80
B/West Kazakhstan/481/2019	1A(Δ3)B	2019-12-03	MDCKx/MDCK1	40	<	<	<	<	<	20	40	320
B/Mount Lebanon-Ain-W-zein/563/2019	1A(Δ3)B	2019-12-04	SIAT3/SIAT1/MDCK1	40	<	<	<	<	<	10	40	320
B/Samara/124326V/2019	1A(Δ3)B	2019-12-10	MDCK1/MDCK1	640	<	<	<	<	<	160	40	160
B/Nur-Sultan/478/2019	1A(Δ3)B	2019-12-10	MDCKx/MDCK1	40	<	<	<	<	<	20	40	320
B/Moscow/24/2019	1A(Δ3)B	2019-12-12	MDCK1/MDCK1	160	<	<	<	<	<	40	40	320
B/Sevastopol/2147V/2019	1A(Δ3)B	2019-12-13	MDCK1/MDCK1	80	<	<	<	<	<	80	20	320
B/Moscow/30/2019	1A(Δ3)B	2019-12-13	MDCK1/MDCK1	160	<	<	<	<	<	10	20	160
B/England/695/2019	1A(Δ3)B	2019-12-13	MDCK1/MDCK1	40	<	<	<	<	<	10	40	160
B/West Kazakhstan/482/2019	1A(Δ3)B	2019-12-13	MDCKx/MDCK1	160	<	<	<	<	<	40	40	320
B/Sevastopol/6V/2019	1A(Δ3)B	2019-12-14	MDCK1/MDCK1	640	<	<	<	<	<	160	40	320
B/Samara/124384V/2019	1A(Δ3)B	2019-12-16	MDCK1/MDCK1	640	<	<	<	<	<	160	20	160
B/West Kazakhstan/483/2019	1A(Δ3)B	2019-12-16	MDCKx/MDCK1	<	<	<	<	<	<	10	40	320
B/Lisboa/37/2019	1A(Δ3)B	2019-12-18	MDCK2/MDCK2	640	80	<	<	<	<	320	<	320
B/Lisboa/32/2019	1A(Δ3)B	2019-12-18	MDCK2/MDCK1	<	<	<	<	<	<	40	40	320
B/Moscow/32/2019	1A(Δ3)B	2019-12-18	MDCK1/MDCK1	320	<	<	<	<	<	40	40	320
B/Lisboa/55/2019	1A(Δ3)B	2019-12-20	MDCK1/MDCK1	80	<	<	<	<	<	80	40	320
B/Parma/8/2019	1A(Δ3)B	2019-12-23	MDCK1/MDCK1	40	<	<	<	<	<	20	40	320
B/Arges/255788/2019	1A(Δ3)B	2019-12-27	MDCK1/MDCK1	<	<	<	<	<	<	<	40	160
B/Lithuania/MB380/2019	1A(Δ3)B	2019-12-31	MDCK1/MDCK1	40	<	<	<	<	<	20	40	320
B/Parma/1/2020	1A(Δ3)B	2020-01-07	MDCK1/MDCK1	640	40	40	<	<	<	160	20	160
B/Valladolid/15/2020	1A(Δ3)B	2020-01-09	MDCK1/MDCK1	40	<	<	<	<	<	20	40	320
B/Valladolid/11/2020	1A(Δ3)B	2020-01-09	MDCK1/MDCK1	80	<	<	<	<	<	20	40	320
B/Montenegro/206/2020	1A(Δ3)B	2020-01-10	MDCKx/MDCK1	<	<	<	<	<	<	10	40	320
B/Arges/256326/2020	1A(Δ3)B	2020-01-13	MDCK1/MDCK1	320	<	<	<	<	<	20	80	320
B/Bucuresti/256302/2020	1A(Δ3)B	2020-01-13	MDCK1/MDCK1	160	<	<	<	<	<	20	<	40
B/Bucuresti/256301/2020	1A(Δ3)B	2020-01-15	MDCK1/MDCK1	80	<	<	<	<	<	<	<	20
B/Palermo/5/2019	1A(Δ2)	2019-12-16	MDCK2/MDCK1	40	<	<	<	40	40	160	<	<
B/Palermo/7/2019	1A(Δ2)	2019-12-23	MDCK2/MDCK1	40	<	<	<	40	40	80	<	<
B/Ashgabat/1/2020	1A	2019-04-15	MDCK1/MDCK1	1280	40	<	160	<	<	<	<	<

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not DoneVaccine
SH 2019
NH 2019-20Vaccine
SH 2020
NH 2020-21

< 4-fold

4-fold

8-fold

> 8-fold

< not recognised by the antiserum

≥ 160 (when homologous titre ≥ 2560)

≥ 160 (no homolog)

Table 10-2. Antigenic analyses of influenza B viruses (Victoria lineage) 2020-02-19

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Sth Aus 81/12 Egg	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Norway 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/Wash'ton 02/19 MDCK	B/Wash'ton 02/19 Egg
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ²	F25/16 ²	F15/16 ²	F38/17 ²	F40/17 ²	F21/18 ²	F11/18 ²	F37/19 ²	F38/19 ²
	Passage history												
	Ferret number												
	Genetic group												
REFERENCE VIRUSES													
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	1280	640	320	80	80	<	10	160	<	160
B/South Australia/81/2012	1A	2012-11-28	E4/E2	2560	640	640	80	80	<	10	160	<	320
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	2560	160	40	160	160	<	<	<	<	<
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK2	1280	80	40	80	160	<	<	<	<	<
B/Norway/2409/2017	1A(Δ2)	2017-04-27	MDCK1/MDCK3	80	<	<	<	<	40	80	160	<	<
B/Colorado/06/2017	1A(Δ2)	2017-02-05	MDCK1/MDCK2	80	<	<	<	<	40	40	160	<	<
B/Colorado/06/2017	1A(Δ2)	2017-02-05	E5/E2	640	320	80	<	<	40	40	640	<	160
B/Washington/02/2019	1A(Δ3)B	2019-01-19	C2/MDCK3	640	40	40	<	<	<	<	160	40	160
B/Washington/02/2019	1A(Δ3)B	2019-01-19	E3/E2	640	160	80	<	<	<	20	320	40	640
TEST VIRUSES													
B/Lisboa/61/2019	No seq	2019-12-06	MDCK2/MDCK1	640	40	10	40	<	<	20	40	40	160
B/Lisboa/49/2019	No seq	2019-12-16	MDCK3/MDCK1	80	20	<	<	<	<	<	40	40	160
B/Lisboa/35/2019	No seq	2019-12-16	MDCK3/MDCK1	160	40	10	<	<	<	<	80	40	160
B/Lisboa/33/2019	No seq	2019-12-16	MDCK2/MDCK1	<	<	<	<	<	<	<	20	40	320
B/La Reunion/2124/2019	1A(Δ3)B	2019-08-06	MDCK2/MDCK1	160	40	10	<	<	<	<	40	20	160
B/La Reunion/2154/2019	1A(Δ3)B	2019-11-06	MDCK2/MDCK1	40	20	10	<	<	<	<	40	40	160
B/Ireland/90048/2019	1A(Δ3)B	2019-11-12	MDCK1	160	40	10	<	<	<	<	40	40	320
B/Ireland/90739/2019	1A(Δ3)B	2019-11-13	MDCK1	80	40	<	<	<	<	<	40	40	320
B/Ireland/92760/2019	1A(Δ3)B	2019-11-21	MDCK1	640	40	10	<	<	<	<	80	80	320
B/Catalonia/3511411NS/2019	1A(Δ3)B	2019-11-22	P0/MDCK1	640	40	10	<	<	<	<	160	20	80
B/Parma/3/2019	1A(Δ3)B	2019-11-25	MDCK1/MDCK1	640	20	20	<	<	<	<	80	<	80
B/Paris/2578/2019	1A(Δ3)B	2019-11-26	MDCK1	320	80	10	<	<	<	<	80	40	160
B/Friuli Venezia Giulia/01/2019	1A(Δ3)B	2019-11-28	MDCK2/MDCK1	80	<	<	<	<	<	<	20	40	160
B/Parma/5/2019	1A(Δ3)B	2019-12-02	MDCK1/MDCK1	320	40	10	<	<	<	<	80	<	40
B/Catalonia/12004S/2020	1A(Δ3)B	2019-12-10	P0/MDCK1	320	80	10	<	<	<	<	80	40	320
B/Lisboa/51/2019	1A(Δ3)B	2019-12-10	MDCK1/MDCK1	320	40	10	<	<	<	<	40	40	160
B/Lisboa/50/2019	1A(Δ3)B	2019-12-10	MDCK1/MDCK1	640	40	10	<	<	<	<	80	40	320
B/Catalonia/12014S/2020	1A(Δ3)B	2019-12-11	P0/MDCK1	160	20	10	<	<	<	<	40	40	160
B/Lisboa/44/2019	1A(Δ3)B	2019-12-12	MDCK1/MDCK1	320	40	<	<	<	<	<	80	40	160
B/Lisboa/40/2019	1A(Δ3)B	2019-12-12	MDCK2/MDCK1	320	<	<	<	<	<	<	10	20	160
B/Lisboa/45/2019	1A(Δ3)B	2019-12-14	MDCK1/MDCK1	80	<	<	<	<	<	<	40	40	160
B/Lisboa/56/2019	1A(Δ3)B	2019-12-16	MDCK2/MDCK1	<	<	<	<	<	<	<	40	20	160
B/Lisboa/52/2019	1A(Δ3)B	2019-12-16	MDCK2/MDCK1	80	<	<	<	<	<	<	20	40	160
B/Lisboa/36/2019	1A(Δ3)B	2019-12-17	MDCK3/MDCK1	640	160	80	<	<	<	<	160	10	160
B/Paris/2862/2019	1A(Δ3)B	2019-12-24	MDCK1	40	<	10	<	<	<	<	40	40	160
B/Paris/004/2020	1A(Δ3)B	2020-01-03	MDCK1	320	80	10	<	<	<	<	40	40	160
B/La Reunion/2152/2019	1A(Δ2)	2019-09-13	MDCK2/MDCK1	40	<	<	<	<	40	80	160	<	<
B/La Reunion/2153/2019	1A(Δ2)	2019-09-20	MDCK2/MDCK1	80	<	10	<	<	40	80	160	<	<
B/Catalonia/2147889NS/2019	1A(Δ2)	2019-10-16	P0/MDCK1	80	<	<	<	<	20	40	80	<	<
B/Paris/2694/2019	1A(Δ2)	2019-12-09	MDCK1	160	<	<	<	<	40	40	160	<	<
B/Catalonia/3512065NS/2019	1A(Δ2)	2019-12-09	P0/MDCK1	40	<	10	<	<	40	80	160	<	<

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not DoneVaccine
SH 2019
NH 2019-20Vaccine
SH 2020
NH 2020-21

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 10-3. Antigenic analyses of influenza B viruses (Victoria lineage) 2020-03-04

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Sth Aus 81/12 Egg	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Norway 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/Wash'ton 02/19 MDCK	B/Wash'ton 02/19 Egg
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ²	F25/16 ²	F15/16 ²	F38/17 ²	F40/17 ²	F21/18 ²	F11/18 ²	F37/19 ²	F38/19 ²
	Passage history												
	Ferret number												
	Genetic group												
REFERENCE VIRUSES													
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	1280	320	640	40	80	<	10	80	<	160
B/South Australia/81/2012	1A	2012-11-28	E4/E2	2560	640	1280	80	160	<	10	160	<	320
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	2560	40	40	80	160	<	<	<	<	<
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK2	1280	40	20	40	160	<	<	<	<	<
B/Norway/2409/2017	1A(Δ2)	2017-04-27	MDCK1/MDCK3	40	<	10	<	<	40	40	80	<	<
B/Colorado/06/2017	1A(Δ2)	2017-02-05	MDCK1/MDCK2	80	<	10	<	<	40	40	80	<	<
B/Colorado/06/2017	1A(Δ2)	2017-02-05	E5/E2	640	160	80	<	<	20	40	320	<	160
B/Washington/02/2019	1A(Δ3)B	2019-01-19	C2/MDCK3	320	80	80	<	<	10	10	160	20	320
B/Washington/02/2019	1A(Δ3)B	2019-01-19	E3/E2	640	160	80	<	<	20	20	320	40	320
TEST VIRUSES													
B/Valladolid/7/2020	No seq	2020-01-07	MDCK1/MDCK1	80	10	20	<	<	<	10	40	20	160
B/Guarda/160/2019	1A(Δ3)B	2019-11-08	MDCK1	80	10	20	<	<	<	<	20	40	160
B/Guarda/161/2019	1A(Δ3)B	2019-11-09	MDCK2	40	10	20	<	<	<	<	40	20	160
B/Lisboa/RL032/2019	1A(Δ3)B	2019-11-09	MDCK1	80	10	20	<	<	<	<	40	20	160
B/Madeira/59/2019	1A(Δ3)B	2019-11-11	MDCK1	80	10	20	<	<	<	<	40	20	160
B/Lisboa/RL045/2019	1A(Δ3)B	2019-11-13	MDCK2	320	80	80	<	<	<	<	40	<	80
B/Lisboa/47/2019	1A(Δ3)B	2019-11-17	MDCK2	40	<	<	<	<	<	<	10	20	160
B/Madeira/64/2019	1A(Δ3)B	2019-11-17	MDCK2	320	10	10	<	<	<	<	20	20	160
B/Coimbra/140/2019	1A(Δ3)B	2019-11-19	MDCK1	320	40	20	<	<	<	<	40	20	160
B/St. Petersburg/2186V/2019	1A(Δ3)B	2019-11-21	MDCK1/MDCK1	40	<	20	<	<	<	<	10	20	160
B/Novosibirsk/4V/2019	1A(Δ3)B	2019-11-26	MDCK1/MDCK1	640	40	80	<	<	<	<	160	10	160
B/Vladivostok/4905V/2019	1A(Δ3)B	2019-11-27	MDCK1/MDCK1	640	40	40	<	<	<	<	160	10	160
B/Ulyanovsk/652V/2019	1A(Δ3)B	2019-11-28	MDCK1/MDCK1	320	20	40	<	<	<	<	40	40	320
B/St. Petersburg/2267V/2019	1A(Δ3)B	2019-12-03	MDCK1/MDCK1	80	<	20	<	<	<	<	40	20	160
B/St. Petersburg/2276V/2019	1A(Δ3)B	2019-12-05	MDCK1/MDCK1	40	10	10	<	<	<	<	40	40	160
B/Valladolid/300/2019	1A(Δ3)B	2019-12-09	MDCK1/MDCK1	320	80	80	<	<	<	<	40	<	80
B/Valladolid/304/2019	1A(Δ3)B	2019-12-26	MDCK1/MDCK1	<	<	10	<	<	<	<	20	20	160
B/Valladolid/308/2019	1A(Δ3)B	2019-12-27	MDCK1/MDCK1	<	<	10	<	<	<	<	10	20	160
B/Valladolid/309/2019	1A(Δ3)B	2019-12-30	MDCK1/MDCK1	40	10	10	<	<	<	<	20	20	160
B/Pays de Loire/006/2020	1A(Δ3)B	2019-12-31	MDCK3	320	10	40	<	<	<	<	40	40	160
B/Valladolid/8/2020	1A(Δ3)B	2020-01-08	MDCK1/MDCK1	320	80	40	<	<	<	<	40	<	80
B/Thuringen/1/2020	1A(Δ3)B	2020-01-14	C1/MDCK1	320	40	80	<	<	<	10	40	40	160
B/Berlin/2/2020	1A(Δ3)B	2020-01-20	C1/MDCK1	<	<	20	<	<	<	<	10	<	10
B/Baden-Wuerttemberg/6/2020	1A(Δ3)B	2020-01-23	C1/MDCK1	<	<	20	<	<	<	<	<	<	<
B/Nordrhein-Westfalen/2/2020	1A(Δ3)B	2020-01-27	C1/MDCK1	160	40	40	<	<	<	10	20	40	160
B/Sachsen/2/2020	1A(Δ3)B	2020-01-29	C1/MDCK1	80	10	20	<	<	<	10	40	20	160
B/Valladolid/303/2019	1A(Δ2)	2019-12-23	MDCK1/MDCK1	40	<	<	<	<	40	40	80	<	<
B/Niedersachsen/1/2020	1A(Δ2)	2020-01-28	C1/MDCK1	80	<	10	<	<	40	40	80	<	<

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not Done

Vaccine SH 2019	Vaccine SH 2020
NH 2019-20	NH 2020-21

< 4-fold	4-fold	8-fold	> 8-fold	< not recognised by the antiserum	≥ 160 (when homologous titre ≥ 2560)	≥ 160 (no homologous titre)
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Table 10-4. Antigenic analyses of influenza B viruses (Victoria lineage) 2020-03-11

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Sth Aus 81/12 Egg	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Norway 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/Wash'ton 02/19 MDCK	B/Wash'ton 02/19 Egg
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ²	F25/16 ²	F15/16 ²	F38/17 ²	F40/17 ²	F21/18 ²	F11/18 ²	F37/19 ²	F38/19 ⁴
	Passage history												
	Ferret number												
	Genetic group			1A	1A	1A	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)B	1A(Δ3)B
REFERENCE VIRUSES													
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	1280	640	640	40	80	<	10	80	<	80
B/South Australia/81/2012	1A	2012-11-28	E4/E2	2560	640	640	80	160	<	10	160	<	160
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	1280	80	40	80	160	<	10	<	<	<
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK2	1280	40	20	40	80	<	<	<	<	<
B/Norway/2409/2017	1A(Δ2)	2017-04-27	MDCK1/MDCK3	40	<	<	<	<	40	40	80	<	<
B/Colorado/06/2017	1A(Δ2)	2017-02-05	MDCK1/MDCK2	40	<	<	<	<	20	20	80	<	<
B/Colorado/06/2017	1A(Δ2)	2017-02-05	E5/E2	640	160	80	<	<	20	40	320	<	80
B/Washington/02/2019	1A(Δ3)B	2019-01-19	C2/MDCK3	640	160	80	<	<	<	10	160	20	160
B/Washington/02/2019	1A(Δ3)B	2019-01-19	E3/E2	640	160	80	<	<	<	10	320	40	320
TEST VIRUSES													
B/Coimbra/138/2019	No seq	2019-11-13	MDCK1	40	20	<	<	<	<	<	20	20	160
B/Athens.GR/72/2020	1A(Δ3)B		MDCK1	160	40	10	<	<	<	<	20	40	160
B/Qatar/47-VI-19-0062540/2019	1A(Δ3)B	2019-10-17	MDCK1	640	320	320	<	<	<	10	320	<	160
B/Qatar/13-VI-19-0063573/2019	1A(Δ3)B	2019-10-20	MDCK1	640	320	320	<	<	<	20	320	10	320
B/Fes/158/2019	1A(Δ3)B	2019-11-25	MDCK1	40	20	<	<	<	<	<	20	<	<
B/Kyrgyzstan/307/2019	1A(Δ3)B	2019-12-14	MDCK1	160	80	20	<	<	<	<	40	20	160
B/Kyrgyzstan/200/2019	1A(Δ3)B	2019-12-16	MDCK1	320	160	80	<	<	<	<	40	<	80
B/Tajikistan/151/2019	1A(Δ3)B	2019-12-19	MDCK1	160	80	20	<	<	<	<	40	20	160
B/South Lebanon-Tyre/NS18/2019	1A(Δ3)B	2019-12-23	MDCK1	320	40	20	<	<	<	<	40	<	40
B/South Lebanon-Tyre/NS14/2019	1A(Δ3)B	2019-12-23	MDCK1	320	40	40	<	<	<	<	40	<	40
B/South Lebanon-Tyre/NS26/2019	1A(Δ3)B	2019-12-24	MDCK1	640	160	10	<	<	<	<	80	<	40
B/Mount Lebanon-Ain-W-zein/3/2019	1A(Δ3)B	2019-12-26	MDCK1	320	80	20	<	<	<	<	40	40	320
B/Mount Lebanon-Ain-W-zein/1/2019	1A(Δ3)B	2019-12-26	MDCK1	160	40	20	<	<	<	<	40	40	320
B/South Lebanon-Saida/57/2020	1A(Δ3)B	2020-01-10	MDCK1	320	40	20	<	<	<	<	10	<	<

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not Done

Vaccine SH 2019 NH 2019-20	Vaccine SH 2020 NH 2020-21
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< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 10-5. Antigenic analyses of influenza B viruses (Victoria lineage) 2020-03-18

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Sth Aus 81/12 Egg	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Norway 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/Wash'ton 02/19 MDCK	B/Wash'ton 02/19 Egg
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ²	F25/16 ²	F15/16 ²	F38/17 ²	F40/17 ²	F21/18 ²	F11/18 ²	F37/19 ²	F38/19 ²
Passage history													
Ferret number													
Genetic group													
REFERENCE VIRUSES													
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	1280	640	640	40	80	<	10	80	<	80
B/South Australia/81/2012	1A	2012-11-28	E4/E2	5120	1280	1280	80	160	<	20	160	<	320
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	2560	80	80	80	160	<	<	<	<	<
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK2	1280	80	40	80	160	<	<	<	<	<
B/Norway/2409/2017	1A(Δ2)	2017-04-27	MDCK1/MDCK3	80	<	10	<	<	40	80	160	<	<
B/Colorado/06/2017	1A(Δ2)	2017-02-05	MDCK1/MDCK2	80	<	<	<	<	40	40	80	<	<
B/Colorado/06/2017	1A(Δ2)	2017-02-05	E5/E2	640	160	40	<	<	20	40	320	<	80
B/Washington/02/2019	1A(Δ3)B	2019-01-19	C2/MDCK3	640	160	80	<	<	<	10	160	20	320
B/Washington/02/2019	1A(Δ3)B	2019-01-19	E3/E2	640	160	80	<	<	<	20	320	40	320
TEST VIRUSES													
B/Kazakhstan/9382/2019	No seq	2019-12-13	MDCK1	80	40	10	<	<	<	10	20	40	320
B/Vladivostok/1/2019	1A(Δ3)B	2019-11-17	MDCK1/MDCK1	640	160	40	<	<	<	10	160	40	320
B/Meknes/127/2019	1A(Δ3)B	2019-11-20	MDCK2	320	80	40	<	<	<	10	80	40	320
B/Vladivostok/3/2019	1A(Δ3)B	2019-11-22	MDCK1/MDCK1	640	160	80	<	<	<	10	160	10	320
B/Vladivostok/5/2019	1A(Δ3)B	2019-11-26	MDCK1/MDCK1	640	160	80	<	<	<	10	80	40	320
B/Vladivostok/9/2019	1A(Δ3)B	2019-11-28	MDCK1/MDCK1	1280	80	40	10	1280	<	10	160	20	320
B/Moscow/18/2019	1A(Δ3)B	2019-12-02	MDCK1/MDCK1	640	80	20	<	<	<	10	40	20	320
B/Moscow/26/2019	1A(Δ3)B	2019-12-09	MDCK1/MDCK1	320	40	20	<	<	<	<	40	20	160
B/Moscow/20/2019	1A(Δ3)B	2019-12-09	MDCK1/MDCK1	320	80	20	<	<	<	10	40	20	320
B/Moscow/22/2019	1A(Δ3)B	2019-12-10	MDCK1/MDCK1	320	80	20	<	<	<	10	40	20	160
B/Arges/255337/2019	1A(Δ3)B	2019-12-10	MDCK1/MDCK1	80	40	10	<	<	<	10	20	40	320
B/Arges/255335/2019	1A(Δ3)B	2019-12-10	SIAT1/MDCK1	80	20	<	<	<	<	<	<	40	160
B/Moscow/28/2019	1A(Δ3)B	2019-12-11	MDCK1/MDCK1	320	80	40	<	<	<	10	80	40	320
B/Cheboksary/35/2019	1A(Δ3)B	2019-12-11	MDCK1/MDCK1	320	80	40	<	<	<	10	80	40	320
B/Tunis/15739/2019	1A(Δ3)B	2019-12-11	MDCK1	160	80	40	<	20	<	10	40	40	320
B/Arges/255625/2019	1A(Δ3)B	2019-12-17	MDCK1/MDCK1	320	80	20	<	10	<	10	20	40	320
B/Kazakhstan/9433/2019	1A(Δ3)B	2019-12-18	MDCK1	160	40	10	<	<	<	10	20	20	320
B/Arges/255693/2019	1A(Δ3)B	2019-12-19	MDCK1/MDCK1	320	80	40	<	<	<	10	40	<	<
B/Arges/255796/2019	1A(Δ3)B	2019-12-22	MDCK1/MDCK1	320	40	20	<	<	<	<	<	80	320
B/Arges/255753/2019	1A(Δ3)B	2019-12-22	MDCK1/MDCK1	320	40	10	<	<	<	<	<	40	320
B/Arges/255749/2019	1A(Δ3)B	2019-12-22	MDCK2/MDCK3	80	40	10	<	<	<	<	10	40	320
B/Kazakhstan/9363/2019	1A(Δ3)B	2019-12-23	MDCK1	80	40	20	<	<	<	<	10	40	320
B/Kazakhstan/9440/2019	1A(Δ3)B	2019-12-29	MDCK1	320	40	20	<	<	<	10	40	20	160
B/Tunis/31/2019	1A(Δ3)B	2019-12-30	MDCK1	160	40	20	<	<	<	10	40	20	320
B/Kazakhstan/9379/2019	1A(Δ3)B	2019-12-10	MDCK1	80	40	10	<	<	<	10	20	40	320
B/Kazakhstan/9374/2019	1A(Δ3)B	2019-12-10	MDCK1	320	40	20	<	<	<	<	20	40	320
B/Kazakhstan/157/2019	1A(Δ3)B	2019-12-10	MDCK1	320	40	20	<	<	<	<	40	10	320
B/Montenegro/139/2019	1A(Δ3)B	2019-12-21	MDCK1	80	40	10	<	<	<	<	10	40	320

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not Done

Vaccine SH 2019	Vaccine SH 2020
NH 2019-20	NH 2020-21

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8-fold
> 8-fold
< not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous ti

Table 10-6. Antigenic analyses of influenza B viruses (Victoria lineage) 2020-03-25

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Sth Aus 81/12 Egg	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Norway 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/Wash'ton 02/19 MDCK	B/Wash'ton 02/19 Egg
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ²	F25/16 ²	F15/16 ²	F38/17 ²	F40/17 ²	F21/18 ²	F11/18 ²	F37/19 ²	F38/19 ²
	Passage history												
	Ferret number												
	Genetic group												
REFERENCE VIRUSES													
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	1280	640	640	40	80	<	10	160	<	160
B/South Australia/81/2012	1A	2012-11-28	E4/E2	2560	1280	1280	160	160	<	20	160	<	640
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	1280	80	40	80	160	<	<	<	<	<
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK2	1280	80	20	80	160	<	<	<	<	<
B/Norway/2409/2017	1A(Δ2)	2017-04-27	MDCK1/MDCK3	80	<	<	<	<	40	80	160	<	<
B/Colorado/06/2017	1A(Δ2)	2017-02-05	MDCK1/MDCK2	80	<	<	<	<	80	40	160	<	<
B/Colorado/06/2017	1A(Δ2)	2017-02-05	E5/E2	640	160	40	<	<	40	40	320	<	160
B/Washington/02/2019	1A(Δ3)B	2019-01-19	C2/MDCK3	640	80	40	<	<	<	10	160	20	320
B/Washington/02/2019	1A(Δ3)B	2019-01-19	E3/E2	640	160	80	<	<	<	20	320	40	640
TEST VIRUSES													
B/Mauritius/I-2823/2019	1A(Δ3)B	2019-10-11	MDCKx/MDCK2	40	<	<	<	<	<	<	<	40	320
B/Mauritius/I-2920/2019	1A(Δ3)B	2019-10-28	MDCK1/MDCK1	1280	160	160	<	<	<	<	320	<	320
B/Mauritius/I-2936/2019	1A(Δ3)B	2019-10-31	MDCK1/MDCK1	40	<	<	<	<	<	<	<	40	320
B/Nur-Sultan/475/2019	1A(Δ3)B	2019-11-01	MDCKx/MDCK1	40	<	<	<	<	<	<	<	40	320
B/Nur-Sultan/476/2019	1A(Δ3)B	2019-11-22	MDCKx/MDCK1	80	40	<	<	<	<	<	<	40	160
B/Karaganda/480/2019	1A(Δ3)B	2019-11-27	MDCKx/MDCK1	80	40	<	<	<	<	<	40	40	320
B/Karaganda/479/2019	1A(Δ3)B	2019-11-28	MDCKx/MDCK1	160	80	<	<	<	<	<	40	40	640
B/Mauritius/I-66/2020	1A(Δ3)B	2020-01-07	MDCK1/MDCK1	40	40	20	<	<	<	<	<	<	<
B/Mauritius/I-85/2020	1A(Δ3)B	2020-01-20	MDCK1/MDCK1	320	80	40	<	<	<	<	160	<	320

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not Done

Vaccine SH 2019 NH 2019-20	Vaccine SH 2020 NH 2020-21
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< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 10-7. Antigenic analyses of influenza B viruses (Victoria lineage) 2020-06-16

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Sth Aus 81/12 Egg	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Norway 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/Wash'ton 02/19 MDCK	B/Wash'ton 02/19 Egg
					Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ²	F25/16 ²	F15/16 ²	F38/17 ²	F40/17 ²	F21/18 ²	F11/18 ²	F37/19 ²	F38/19 ²
		Genetic group			1A	1A	1A	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)B	1A(Δ3)B
REFERENCE VIRUSES														
B/Brisbane/60/2008		1A	2008-08-04	E4/E4	1280	640	640	40	80	<	10	160	<	160
B/South Australia/81/2012		1A	2012-11-28	E4/E2	2560	1280	1280	160	160	<	20	160	<	320
B/Ireland/3154/2016		1A	2016-01-14	MDCK1/MDCK4	1280	80	40	80	160	<	<	<	<	<
B/Nordrhein-Westfalen/1/2016		1A	2016-01-04	C2/MDCK2	1280	80	20	80	160	<	<	<	<	<
B/Norway/2409/2017		1A(Δ2)	2017-04-27	MDCK1/MDCK3	80	<	<	<	<	40	80	160	<	<
B/Colorado/06/2017		1A(Δ2)	2017-02-05	MDCK1/MDCK2	80	<	<	<	<	80	40	160	<	<
B/Colorado/06/2017		1A(Δ2)	2017-02-05	E5/E2	640	160	40	<	<	40	40	320	<	160
B/Washington/02/2019		1A(Δ3)B	2019-01-19	C2/MDCK3	640	160	40	<	<	<	10	160	40	640
B/Washington/02/2019		1A(Δ3)B	2019-01-19	E3/E2	640	160	80	<	<	<	10	160	40	640
TEST VIRUSES														
B/Iceland/32/2020		No seq	2020-03-19	MDCK1/MDCK1	640	80	40	<	<	<	<	80	<	80
B/Slovenia/912/2020		1A(Δ3) B	2020-02-02	SIATx/MDCK1	320	40	<	<	<	<	<	40	80	320
B/Belgium/S1068/2020		1A(Δ3) B	2020-02-08	MDCK1/MDCK1	160	<	<	<	<	<	<	<	<	160
B/Belgium/S1274/2020		1A(Δ3) B	2020-02-10	MDCK1/MDCK1	160	<	<	<	<	<	<	<	<	160
B/Belgium/S1071/2020		1A(Δ3) B	2020-02-12	MDCK1/MDCK1	160	10	<	<	<	<	<	<	<	160
B/Belgium/G0316/2020		1A(Δ3) B	2020-02-13	MDCK1/MDCK1	320	40	<	<	<	<	<	20	<	320
B/Belgium/S1053/2020	T155A	1A(Δ3) B	2020-02-16	MDCK1/MDCK1	160	20	<	<	<	<	<	<	<	<
B/Bulgaria/1004/2020		1A(Δ3) B	2020-02-17	SIAT2/MDCK1	80	20	10	<	<	<	<	20	80	320
B/Iceland/03/2020		1A(Δ3) B	2020-02-19	MDCK1/MDCK1	80	10	<	<	<	<	<	<	<	160
B/Slovenia/1584/2020	N126K	1A(Δ3) B	2020-02-24	SIATx/MDCK1	160	40	<	<	<	<	<	<	40	<
B/Bulgaria/1084/2020		1A(Δ3) B	2020-02-25	SIAT2/MDCK1	80	20	20	<	<	<	20	40	320	320
B/Iceland/17/2020		1A(Δ3) B	2020-02-26	MDCK1/MDCK1	40	<	<	<	<	<	<	<	<	160
B/Iceland/12/2020		1A(Δ3) B	2020-02-26	MDCK1/MDCK1	80	<	<	<	<	<	<	<	<	160
B/Bulgaria/1114/2020		1A(Δ3) B	2020-02-27	SIAT2/MDCK1	80	20	40	<	<	<	20	80	320	320
B/Slovenia/1611/2020	N126K, K345R	1A(Δ3) B	2020-02-27	MDCKx/MDCK1	160	20	<	<	<	<	<	20	<	<
B/Iceland/16/2020		1A(Δ3) B	2020-02-27	MDCK1/MDCK1	40	<	<	<	<	<	<	<	<	160
B/Slovenia/1694/2020		1A(Δ3) B	2020-03-02	MDCKx/MDCK1	160	10	<	<	<	<	<	<	80	320
B/Slovenia/1692/2020		1A(Δ3) B	2020-03-02	SIATx/MDCK1	80	20	<	<	<	<	<	<	40	320
B/S.Petersburg/RIL-170/2020		1A(Δ3) B	2020-03-02	MDCK1/MDCK1	160	20	10	<	<	<	20	40	320	320
B/S.Petersburg/RIL-158/2020		1A(Δ3) B	2020-03-02	MDCK2/MDCK1	160	<	<	<	<	<	<	20	160	160
B/Iceland/21/2020		1A(Δ3) B	2020-03-02	MDCK1/MDCK1	80	20	<	<	<	<	<	<	<	320
B/Bulgaria/1251/2020		1A(Δ3) B	2020-03-04	SIAT2/MDCK1	160	20	10	<	<	<	<	80	320	320
B/Slovenia/1745/2020		1A(Δ3) B	2020-03-04	MDCKx/MDCK1	160	20	<	<	<	<	<	80	320	320
B/Slovenia/1743/2020		1A(Δ3) B	2020-03-04	MDCKx/MDCK1	80	20	<	<	<	<	<	40	160	160
B/S.Petersburg/RIL-156/2020		1A(Δ3) B	2020-03-04	MDCK2/MDCK1	160	20	<	<	<	<	<	40	160	160
B/Belgium/S1228/2020		1A(Δ3) B	2020-03-04	MDCK1/MDCK1	160	20	<	<	<	<	<	<	<	160
B/Belgium/G0472/2020	T155A	1A(Δ3) B	2020-03-04	MDCK1/MDCK1	320	20	<	<	<	<	<	<	<	<
B/Bulgaria/1170/2020		1A(Δ3) B	2020-03-05	SIAT2/MDCK1	160	40	20	<	<	10	40	80	640	640
B/Bulgaria/1308/2020		1A(Δ3) B	2020-03-05	SIAT2/MDCK1	320	80	20	<	<	<	40	80	320	320
B/Bulgaria/1279/2020		1A(Δ3) B	2020-03-05	SIAT2/MDCK1	160	20	20	<	<	<	40	80	320	320
B/S.Petersburg/RIL-174/2020		1A(Δ3) B	2020-03-05	MDCK1/MDCK1	320	10	<	<	<	<	<	20	160	160
B/Bulgaria/1774/2020		1A(Δ3) B	2020-03-09	SIAT2/MDCK1	160	80	20	<	<	10	80	80	640	640
B/Bulgaria/1722/2020		1A(Δ3) B	2020-03-09	SIAT2/MDCK1	160	40	20	<	<	10	40	80	640	640
B/Krasnoyarsk/09/2020		1A(Δ3) B	2020-03-10	MDCK2/MDCK1	640	40	20	<	<	<	<	40	160	160
B/Astrakhan/41/2020		1A(Δ3) B	2020-03-10	MDCK2/MDCK1	320	40	20	<	<	<	<	40	160	160
B/Slovenia/1906/2020		1A(Δ3) B	2020-03-11	MDCKx/MDCK1	160	20	<	<	<	<	40	80	320	320
B/S.Petersburg/RIL-162/2020		1A(Δ3) B	2020-03-11	MDCK1/MDCK1	160	20	<	<	<	<	<	40	320	320
B/S.Petersburg/RIL-160/2020		1A(Δ3) B	2020-03-11	MDCK1/MDCK1	640	80	40	<	<	<	<	40	320	320
B/S.Petersburg/RIL-177/2020		1A(Δ3) B	2020-03-12	MDCK1/MDCK1	80	20	<	<	<	<	<	40	160	160
B/S.Petersburg/RIL-164/2020		1A(Δ3) B	2020-03-13	MDCK1/MDCK1	160	40	<	<	<	<	<	<	40	320
B/Astrakhan/45/2020	V87A, G133R, E136K	1A(Δ3) B	2020-03-13	MDCK2/MDCK1	160	<	<	<	<	<	<	10	80	80
B/Astrakhan/43/2020		1A(Δ3) B	2020-03-13	MDCK2/MDCK1	320	<	10	<	<	<	<	20	160	160
B/S.Petersburg/RIL-166/2020		1A(Δ3) B	2020-03-14	MDCK2/MDCK1	80	20	<	<	<	<	<	40	160	160
B/Slovenia/2232/2020		1A(Δ3) B	2020-03-17	MDCKx/MDCK1	320	40	<	<	<	<	20	80	320	320
B/Iceland/31/2020		1A(Δ3) B	2020-03-17	MDCK1/MDCK1	80	<	<	<	<	<	<	<	<	160
B/Iceland/38/2020		1A(Δ3) B	2020-04-20	MDCK1/MDCK1	80	<	<	<	<	<	<	<	40	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not DoneVaccine
SH 2019
NH 2019-20Vaccine
SH 2020
NH 2020-21

< 4-fold
 4-fold
 8-fold
 > 8-fold
 not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 10-8. Antigenic analyses of influenza B viruses (Victoria lineage) 2020-06-23

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre										
					B/Bris 60/08 Egg Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3} 1A	Post-infection ferret antisera								B/Wash'ton 02/19 Egg MDCK 1A(Δ3)B	B/Wash'ton 02/19 Egg MDCK 1A(Δ3)B
						B/Bris 60/08 Egg F44/17 ² 1A	B/Sth Aus 81/12 Egg F25/16 ² 1A	B/Ireland 3154/16 MDCK F15/16 ² 1A	B/Nord-West 1/16 MDCK F38/17 ² 1A	B/Norway 2409/17 MDCK F40/17 ² 1A(Δ2)	B/Colorado 06/17 MDCK F21/18 ² 1A(Δ2)	B/Colorado 06/17 Egg F11/18 ² 1A(Δ2)			
													B/Wash'ton 02/19 MDCK F37/19 ² 1A(Δ3)B		
REFERENCE VIRUSES															
B/Brisbane/60/2008		1A	2008-08-04	E4/E4	1280	640	640	40	80	<	10	160	<	160	
B/South Australia/81/2012		1A	2012-11-28	E4/E2	2560	1280	1280	80	160	<	20	160	<	320	
B/Ireland/3154/2016		1A	2016-01-14	MDCK1/MDCK4	1280	80	40	80	160	<	<	<	<	<	
B/Nordrhein-Westfalen/1/2016		1A	2016-01-04	C2/MDCK2	1280	80	20	160	160	<	<	<	<	<	
B/Norway/2409/2017		1A(Δ2)	2017-04-27	MDCK1/MDCK3	80	<	<	<	<	80	80	160	<	<	
B/Colorado/06/2017		1A(Δ2)	2017-02-05	MDCK1/MDCK2	80	<	<	<	<	80	80	160	<	<	
B/Colorado/06/2017		1A(Δ2)	2017-02-05	E5/E2	1280	160	40	<	<	40	40	320	<	160	
B/Washington/02/2019		1A(Δ3)B	2019-01-19	C2/MDCK3	640	80	40	<	<	<	10	160	20	320	
B/Washington/02/2019		1A(Δ3)B	2019-01-19	E3/E2	640	80	80	<	<	<	20	160	40	320	
TEST VIRUSES															
B/Denmark/20/2020		1A(Δ3) B	2020-01-22	SIAT2/MDCK1	160	40	10	<	<	<	<	10	<	40	
B/Bayern/6/2020		No seq	2020-02-26	C1/MDCK1	160	20	10	<	<	<	10	80	40	640	
B/Paris/679/2020		1A(Δ3) B	2020-02-03	MDCK1/MDCK1	160	20	<	<	<	<	10	20	80	320	
B/Bratislava/202/2020		1A(Δ3) B	2020-02-03	MDCK1/MDCK1	640	80	<	<	<	10	10	40	80	320	
B/Linköping/1/2020		1A(Δ3) B	2020-02-03	SIAT0/MDCK1	160	40	<	<	<	<	<	40	40	320	
B/Stockholm/7/2020		1A(Δ3) B	2020-02-05	SIAT0/MDCK1	320	40	<	<	<	<	10	40	80	320	
B/Denmark/46/2020	N126K, E198G	1A(Δ3) B	2020-02-05	SIAT2/MDCK1	160	40	<	<	<	10	<	40	10	40	
B/Alsace/923/2020		1A(Δ3) B	2020-02-06	MDCK1/MDCK1	160	20	<	<	<	<	10	20	80	320	
B/England/127/2020		1A(Δ3) B	2020-02-06	MDCK1/MDCK1	640	40	<	<	<	<	10	40	80	320	
B/Eskilstuna/2/2020		1A(Δ3) B	2020-02-07	SIAT0/MDCK1	160	40	<	<	<	<	<	40	40	320	
B/Denmark/65/2020		1A(Δ3) B	2020-02-09	SIAT2/MDCK1	160	20	<	<	<	<	<	20	40	320	
B/Paris/1032/2020		1A(Δ3) B	2020-02-10	MDCK1/MDCK1	320	40	20	<	<	20	20	80	80	320	
B/England/95/2020		1A(Δ3) B	2020-02-10	SIAT1/MDCK1	320	20	<	<	<	10	10	40	40	320	
B/Paris/1065/2020		1A(Δ3) B	2020-02-11	MDCK1/MDCK1	160	20	<	<	<	10	10	20	80	320	
B/Eskilstuna/3/2020	N126K, T182A, R279K, K345R	1A(Δ3) B	2020-02-12	SIAT0/MDCK1	80	20	<	<	<	<	10	20	10	40	
B/Finland/195/2020		1A(Δ3) B	2020-02-12	MDCK1/MDCK1	160	20	<	<	<	<	<	40	40	320	
B/Bremen/1/2020	D129N, G133R, N233S(-CHO), K272R	1A(Δ3) B	2020-02-17	C2/MDCK1	80	40	10	<	<	<	10	40	20	80	
B/Berlin/12/2020		1A(Δ3) B	2020-02-18	C1/MDCK1	320	40	20	<	<	10	10	40	40	320	
B/Saarland/1/2020		1A(Δ3) B	2020-02-18	C1/MDCK1	320	40	20	<	<	10	10	40	80	320	
B/Bretagne/1243/2020		1A(Δ3) B	2020-02-18	MDCK1/MDCK1	160	40	20	<	<	<	10	40	40	320	
B/Centre/1212/2020		1A(Δ3) B	2020-02-18	MDCK1/MDCK1	160	40	20	<	<	<	10	40	40	320	
B/England/130/2020		1A(Δ3) B	2020-02-18	SIAT1/MDCK1	80	10	<	<	<	<	10	20	80	320	
B/Denmark/77/2020		1A(Δ3) B	2020-02-18	SIAT2/MDCK1	320	40	<	<	<	<	<	40	80	320	
B/Finland/197/2020		1A(Δ3) B	2020-02-20	MDCK1/MDCK1	80	20	<	<	<	10	<	40	40	320	
B/England/134/2020		1A(Δ3) B	2020-02-24	SIAT1/MDCK1	320	40	20	<	<	<	10	40	80	320	
B/Nordrhein-Westfalen/6/2020	D129N, G133R, N233S(-CHO), K272R	1A(Δ3) B	2020-02-25	C1/MDCK1	160	20	20	<	<	<	10	80	40	80	
B/Bratislava/439/2020		1A(Δ3) B	2020-02-27	MDCK1/MDCK1	320	40	<	<	<	<	<	20	80	320	
B/Stockholm/10/2020	D129N, G133R, N233S(-CHO), K272R	1A(Δ3) B	2020-02-29	SIAT0/MDCK1	160	40	<	<	<	<	10	40	20	80	
B/Mecklenburg-Vorpommern/1/2020	N126K, E198G	1A(Δ3) B	2020-03-02	C1/MDCK1	160	40	20	<	<	<	<	40	10	40	
B/England/116/2020	R80G, D129N, P134T, E198K, K307R	1A(Δ3) B	2020-03-06	SIAT1/MDCK1	160	20	<	<	<	<	<	20	20	40	
B/Finland/202/2020		1A(Δ3) B	2020-03-06	MDCK1/MDCK1	160	40	<	<	<	10	<	40	40	320	
B/Finland/203/2020		1A(Δ2)	2020-03-12	MDCK1/MDCK1	80	10	<	<	<	40	80	160	<	<	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not Done

Vaccine SH 2019	Vaccine SH 2020
NH 2019-20	NH 2020-21

< 4-fold	4-fold	8-fold	> 8-fold	< not recognised by the antiserum	≥ 160 (when homologous titre ≥ 2560)	≥ 160 (no homologous titre)
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Table 10-9. Antigenic analyses of influenza B viruses (Victoria lineage) 2020-07-10

Haemagglutination inhibition titre														
Viruses	Other information	Passage history	Collection date	Passage history	Post-infection ferret antisera									
					B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Sth Aus 81/12 Egg	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Norway 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/Wash'ton 02/19 MDCK	B/Wash'ton 02/19 Egg
					Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ²	F25/16 ²	F15/16 ²	F38/17 ²	F40/17 ²	F21/18 ²	F11/18 ²	F37/19 ²	F38/19 ²
					Genetic group	1A	1A	1A	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)B
REFERENCE VIRUSES														
B/Brisbane/60/2008		1A	2008-08-04	E4/E4	1280	640	640	40	80	<	10	160	<	160
B/South Australia/81/2012		1A	2012-11-28	E4/E2	2560	1280	1280	80	160	<	20	160	<	320
B/Ireland/3154/2016		1A	2016-01-14	MDCK1/MDCK4	2560	80	40	80	160	<	<	<	<	<
B/Nordrhein-Westfalen/1/2016		1A	2016-01-04	C2/MDCK2	1280	80	40	80	160	<	<	<	<	<
B/Norway/2409/2017		1A(Δ2)	2017-04-27	MDCK1/MDCK3	80	<	<	<	<	80	80	160	<	<
B/Colorado/06/2017		1A(Δ2)	2017-02-05	MDCK1/MDCK2	80	<	<	<	<	80	80	160	<	<
B/Colorado/06/2017		1A(Δ2)	2017-02-05	E5/E2	1280	160	40	<	<	40	40	320	<	160
B/Washington/02/2019		1A(Δ3)B	2019-01-19	C2/MDCK3	1280	80	40	<	<	<	10	160	20	320
B/Washington/02/2019		1A(Δ3)B	2019-01-19	E3/E2	640	160	80	<	<	<	20	160	40	320
TEST VIRUSES														
B/Clermont-Ferrand/2411/2019		1A(Δ3) B	2019-12-28	MDCK2/MDCK1	40	20	10	<	<	<	10	10	40	160
B/Lyon/CHU-R20.2.98/2020		1A(Δ3) B	2020-01-04	MDCK2/MDCK1	160	20	20	<	<	<	10	20	20	20
B/Lyon/156/2020		1A(Δ3) B	2020-01-16	MDCK2/MDCK1	320	40	40	<	<	<	10	20	20	40
B/South Africa/0465/2020		1A(Δ3) B	2020-01-16	MDCK1/MDCK1	160	40	<	<	<	10	10	20	40	320
B/Lyon/225/2020		1A(Δ3) B	2020-01-20	MDCK2/MDCK1	80	20	20	<	<	<	10	20	20	160
B/Estonia/126087/2020		1A(Δ3) B	2020-01-30	SIAT1/MDCK1	80	40	10	<	<	10	10	20	40	320
B/Switzerland/3235/2020		No seq	2020-02-27	MDCK1	160	40	40	<	<	10	10	20	40	320
B/Milano/62/2020	N126K, R279K, K345R	1A(Δ3) B	2020-02-04	MDCK2/MDCK1	<	10	10	<	<	<	10	10	20	10
B/Cyprus/F938/2020	G133R, N233K(-CHO)	1A(Δ3) B	2020-02-06	MDCK1	160	40	40	<	<	<	10	40	40	<
B/Cyprus/F949/2020		1A(Δ3) B	2020-02-07	MDCK1	160	40	20	<	<	<	10	20	40	320
B/Portugal/SU282/2020		1A(Δ3) B	2020-02-08	MDCK1	160	40	20	<	<	10	20	40	40	320
B/Oran/356/2020	R80G, D129N, P134T, E198K	1A(Δ3) B	2020-02-10	MDCK1	<	10	10	<	<	<	<	<	<	10
B/Friuli Venezia Giulia/32/2020		1A(Δ3) B	2020-02-10	MDCK2/MDCK1	160	20	20	<	<	<	10	20	40	160
B/Estonia/126333/2020		1A(Δ3) B	2020-02-11	SIAT1/MDCK1	80	40	20	<	<	10	20	40	40	320
B/Friuli Venezia Giulia/36/2020		1A(Δ3) B	2020-02-12	MDCK2/MDCK1	160	40	20	<	<	10	20	40	40	320
B/Parma/9/2020		1A(Δ3) B	2020-02-12	MDCK2/MDCK1	160	40	20	<	10	<	20	20	40	320
B/South Africa/1632/2020	D129N, G133R, N233S(-CHO), V252L, A276V	1A(Δ3) B	2020-02-12	MDCK1/MDCK1	80	40	<	<	<	10	10	20	20	40
B/South Africa/1739/2020	D129N, G133R, N233S(-CHO), V252L, A276V	1A(Δ3) B	2020-02-13	MDCK1/MDCK1	80	40	<	<	<	10	10	40	20	20
B/Cyprus/F1093/2020		1A(Δ3) B	2020-02-14	MDCK1	640	80	40	<	10	10	10	40	40	320
B/Estonia/126397/2020		1A(Δ3) B	2020-02-14	SIAT1/MDCK1	80	40	10	<	<	10	20	40	40	160
B/Algiers/398/2020	R80G, D129N, P134T, E198K	1A(Δ3) B	2020-02-18	MDCK1	<	10	10	<	<	<	<	<	<	10
B/Cyprus/F1137/2020		1A(Δ3) B	2020-02-18	MDCK1	80	40	20	<	<	<	10	20	20	320
B/Estonia/126489/2020		1A(Δ3) B	2020-02-18	SIAT1/MDCK1	80	40	10	<	<	10	20	40	40	320
B/South Africa/1920/2020	D129N, G133R, N233S(-CHO), V252L, A276V	1A(Δ3) B	2020-02-18	MDCK1/MDCK1	80	40	<	<	<	<	10	20	20	40
B/Switzerland/4531/2020		1A(Δ3) B	2020-02-20	MDCK1	80	20	20	<	<	<	10	20	40	160
B/Saint-Etienne/2541/2020		1A(Δ3) B	2020-02-20	MDCK3/MDCK1	160	20	20	<	<	10	20	20	40	160
B/South Africa/2156/2020	D129N, G133R, N233S(-CHO), V252L, A276V	1A(Δ3) B	2020-02-21	MDCK1/MDCK1	80	40	<	<	<	<	10	40	20	40
B/Cyprus/F1179/2020		1A(Δ3) B	2020-02-23	MDCK1	80	40	20	<	<	<	10	40	40	320
B/Champagne-Ardenne/1586/2020		1A(Δ3) B	2020-02-24	MDCK1	160	40	20	<	<	10	20	40	40	320
B/Switzerland/4546/2020		1A(Δ3) B	2020-02-25	MDCK1	80	20	20	<	<	10	10	20	40	320
B/Switzerland/2663/2020		1A(Δ3) B	2020-02-26	MDCK1	80	40	20	<	<	<	10	40	40	160
B/Switzerland/0521/2020		1A(Δ3) B	2020-02-26	MDCK1	80	40	20	<	<	<	10	40	40	320
B/Switzerland/0325/2020	R80G, D129N, P134T, E198K	1A(Δ3) B	2020-02-26	MDCK1	<	10	10	<	<	<	<	<	<	10
B/Milano/111/2020		1A(Δ3) B	2020-02-26	MDCK3/MDCK1	160	40	20	<	10	10	20	20	40	320
B/Milano/105/2020		1A(Δ3) B	2020-02-26	MDCK3/MDCK1	80	20	20	<	<	10	20	40	40	320
B/Saint-Etienne/2543/2020	N126K	1A(Δ3) B	2020-02-26	MDCK2/MDCK1	80	20	10	<	<	<	10	<	10	10
B/Switzerland/0554/2020		1A(Δ3) B	2020-02-27	MDCK1	80	40	40	<	<	<	10	40	40	160
B/Estonia/126682/2020		1A(Δ3) B	2020-02-27	SIAT1/MDCK1	160	40	10	<	10	10	20	80	40	320
B/Switzerland/3582/2020		1A(Δ3) B	2020-02-29	MDCK1	40	20	10	<	<	<	10	20	20	320
B/Saint-Etienne/2578/2020		1A(Δ3) B	2020-03-01	MDCK2/MDCK1	80	20	10	<	<	<	10	20	40	160
B/Switzerland/6411/2020		1A(Δ3) B	2020-03-02	MDCK1	160	40	20	<	<	<	10	20	40	160
B/Centre/1963/2020		1A(Δ3) B	2020-03-02	MDCK1	160	40	40	<	10	10	20	40	40	320
B/Switzerland/3665/2020		1A(Δ3) B	2020-03-03	MDCK1	160	40	40	<	10	10	10	40	40	320
B/Paris/2113/2020		1A(Δ3) B	2020-03-03	MDCK2	160	40	20	<	<	<	10	40	40	320
B/Saint-Etienne/2550/2020		1A(Δ3) B	2020-03-04	MDCK3/MDCK1	320	40	20	<	<	10	40	40	40	320
B/Switzerland/4074/2020		1A(Δ3) B	2020-03-06	MDCK1	80	40	20	<	<	10	10	20	40	320
B/Estonia/127199/2020		1A(Δ3) B	2020-03-09	SIAT1/MDCK1	160	40	20	<	10	10	20	40	40	160
B/Switzerland/6329/2020		1A(Δ3) B	2020-03-10	MDCK1	160	40	40	<	10	10	10	20	40	320
B/Switzerland/6337/2020		1A(Δ3) B	2020-03-11	MDCK1	80	40	20	<	10	10	20	20	20	320
B/Paris/2560/2020		1A(Δ3) B	2020-03-13	MDCK1	80	40	20	<	<	10	10	40	40	320
B/Switzerland/8873/2020	D129N, G133R, N233S(-CHO), K272R	1A(Δ3) B	2020-03-18	MDCK1	160	40	40	<	<	<	10	40	20	40
B/Paris/3072/2020		1A(Δ3) B	2020-03-18	MDCK2	320	80	40	<	10	10	80	80	40	320
B/Picardie/3414/2020		1A(Δ3) B	2020-03-23	MDCK1	80	20	20	<	<	<	10	20	40	320
B/South Africa/6204/2020	D129N, G133R, N233S(-CHO), V252L, A276V	1A(Δ3) B	2020-03-23	MDCK1/MDCK1	80	40	<	<	<	<	10	20	10	40

Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not DoneVaccine
SH 2019
NH 2019-20Vaccine
SH 2020
NH 2020-21

< 4-fold
 4-fold
 8-fold
 > 8-fold
 not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 10-10. Antigenic analyses of influenza B viruses (Victoria lineage) 2020-07-21

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Sth Aus 81/12 Egg	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Norway 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/Wash'ton 02/19 MDCK	B/Wash'ton 02/19 Egg
					Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ²	F25/16 ²	F15/16 ²	F38/17 ²	F40/17 ²	F21/18 ²	F11/18 ²	F37/19 ²	F38/19 ²
		Genetic group			1A	1A	1A	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)B	1A(Δ3)B
REFERENCE VIRUSES														
B/Brisbane/60/2008		1A	2008-08-04	E4/E4	1280	640	640	40	80	<	10	80	<	160
B/South Australia/81/2012		1A	2012-11-28	E4/E2	2560	1280	1280	80	160	<	20	160	<	320
B/Ireland/3154/2016		1A	2016-01-14	MDCK1/MDCK4	2560	80	40	80	160	<	<	<	<	<
B/Nordrhein-Westfalen/1/2016		1A	2016-01-04	C2/MDCK2	1280	80	40	80	160	<	<	<	<	<
B/Norway/2409/2017		1A(Δ2)	2017-04-27	MDCK1/MDCK3	80	<	<	<	<	80	80	160	<	<
B/Colorado/06/2017		1A(Δ2)	2017-02-05	MDCK1/MDCK2	80	<	<	<	<	80	80	160	<	<
B/Colorado/06/2017		1A(Δ2)	2017-02-05	E5/E2	1280	160	40	<	<	40	80	320	<	160
B/Washington/02/2019		1A(Δ3)B	2019-01-19	C2/MDCK3	1280	80	80	<	<	<	20	320	40	320
B/Washington/02/2019		1A(Δ3)B	2019-01-19	E3/E2	640	160	80	<	<	<	20	160	40	320
TEST VIRUSES														
B/Beirut/NS71/2020		No seq	2020-02-25	MDCK1	160	40	20	<	<	<	10	40	80	320
B/Hermel/NS85/2020		1A(Δ3) B	2020-02-02	MDCK1	80	40	40	<	<	<	10	80	20	80
B/Jbeil/398/2020	D129N, G133K, N233S(-CHO), K272R	1A(Δ3) B	2020-02-07	MDCK1	80	40	20	<	<	<	<	40	10	40
B/Bosnia and Herzegovina/294/2020		1A(Δ3) B	2020-02-14	MDCK1	160	40	20	<	<	<	10	40	10	80
B/Bosnia and Herzegovina/291/2020		1A(Δ3) B	2020-02-14	MDCK1	80	40	10	<	<	<	10	40	40	320
B/Macedonia/576/2020		1A(Δ3) B	2020-02-20	MDCK1	80	40	20	<	<	<	10	20	20	80
B/Beirut/330/2020		1A(Δ3) B	2020-02-20	MDCK1	160	40	20	<	<	<	10	40	40	80
B/Jbeil/406/2020	D129N, G133R, N233S(-CHO), K272R	1A(Δ3) B	2020-02-27	MDCK1	40	20	10	<	<	<	10	20	20	40
B/Macedonia/712/2020		1A(Δ3) B	2020-03-04	MDCK1	80	40	20	<	<	<	10	20	40	320
B/Macedonia/713/2020		1A(Δ3) B	2020-03-06	MDCK1	160	80	40	<	<	<	10	40	20	80

¹ Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not Done

Vaccine SH 2019 NH 2019-20	Vaccine SH 2020 NH 2020-21
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< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 10-11. Antigenic analyses of influenza B viruses (Victoria lineage) 2020-08-12

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Sth Aus 81/12 Egg	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Norway 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/Wash'ton 02/19 MDCK	B/Wash'ton 02/19 Egg
					Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ²	F25/16 ²	F15/16 ²	F38/17 ²	F40/17 ²	F21/18 ²	F11/18 ²	F37/19 ²	F38/19 ²
		Genetic group			1A	1A	1A	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)B	1A(Δ3)B
REFERENCE VIRUSES														
B/Brisbane/60/2008		1A	2008-08-04	E4/E4	1280	640	640	40	80	<	10	160	<	160
B/South Australia/81/2012		1A	2012-11-28	E4/E2	1280	640	320	40	80	<	10	80	<	160
B/Ireland/3154/2016		1A	2016-01-14	MDCK1/MDCK4	1280	80	40	80	160	<	<	<	<	<
B/Nordrhein-Westfalen/1/2016		1A	2016-01-04	C2/MDCK2	1280	80	40	80	160	<	<	<	<	<
B/Norway/2409/2017		1A(Δ2)	2017-04-27	MDCK1/MDCK3	40	10	10	<	<	40	40	160	<	<
B/Colorado/06/2017		1A(Δ2)	2017-02-05	MDCK1/MDCK2	40	10	10	<	<	40	40	80	<	<
B/Colorado/06/2017		1A(Δ2)	2017-02-05	E5/E2	640	160	40	<	<	20	40	320	<	80
B/Washington/02/2019		1A(Δ3)B	2019-01-19	C2/MDCK3	1280	160	160	<	<	<	10	320	40	320
B/Washington/02/2019		1A(Δ3)B	2019-01-19	E3/E2	640	160	80	<	<	<	20	160	40	320
TEST VIRUSES														
B/Orenburg/3/2020		1A(Δ3) B	2020-01-28	MDCK3/MDCK1/MDCK1	40	20	10	<	<	<	10	<	40	160
B/Hungary/67/2020		1A(Δ3) B	2020-02-06	MDCK1/MDCK1	160	40	20	<	<	<	10	40	20	160
B/Hungary/69/2020	V117I, N126K, A127T, A169X	1A(Δ3) B	2020-02-10	MDCK1/MDCK1	80	40	20	<	<	<	20	40	20	40
B/Moscow/29/2020		1A(Δ3) B	2020-02-14	MDCK2/MDCK1	40	20	10	<	<	<	10	10	40	160
B/Moscow/37/2020		1A(Δ3) B	2020-02-28	MDCK2/MDCK1	640	80	40	<	<	<	20	160	40	160
B/Moscow/39/2020		1A(Δ3) B	2020-03-02	MDCK2/MDCK1	80	20	<	<	<	<	10	20	20	160
B/Moscow/42/2020	N150K, V177I, G184E, N197D(-CHO), V220M, P241Q, R279K	1A(Δ3) B	2020-03-03	MDCK2/MDCK1	320	40	40	<	<	<	<	40	<	80
B/Norway/2121/2020		1A(Δ3) B	2020-03-10	MDCK1	640	160	160	<	<	<	<	160	<	160
B/Cheboksary/45/2020		1A(Δ3) B	2020-03-11	MDCK2/MDCK1	80	40	20	<	<	<	20	40	40	320
B/Moscow/44/2020	N150K, V177I, G184E, N197D(-CHO), V220M, P241Q, R279K	1A(Δ3) B	2020-03-11	MDCK2/MDCK1	160	40	40	<	<	<	<	40	<	40
B/Norway/2193/2020		1A(Δ3) B	2020-03-16	MDCK1	160	10	10	<	<	<	<	10	40	160
B/Norway/2192/2020		1A(Δ3) B	2020-03-17	MDCK1	160	40	10	<	<	<	<	20	40	160
B/Norway/2147/2020		1A(Δ3) B	2020-03-17	MDCK1	40	10	10	<	<	<	<	10	40	160
B/NRCM-Orenburg/50/2020		1A(Δ3) B	2020-03-17	MDCK2/MDCK1	80	20	10	<	<	<	10	10	40	160
B/Norway/2116/2020	N126K, E198G	1A(Δ3) B	2020-03-18	MDCK1	160	40	20	<	<	<	10	20	10	40
B/Moscow/48/2020	D129N, N233S(-CHO), K272R	1A(Δ3) B	2020-03-19	MDCK2/MDCK1	160	40	20	<	<	<	<	40	20	40
B/Norway/2324/2020		1A(Δ3) B	2020-03-23	MDCK1	160	20	20	<	<	<	10	20	40	160
B/Norway/2319/2020		1A(Δ3) B	2020-03-23	MDCK1	160	40	20	<	<	<	10	20	40	160
B/Norway/2284/2020		1A(Δ3) B	2020-03-23	MDCK1	160	40	20	<	<	<	20	40	40	320
B/Norway/2340/2020		1A(Δ3) B	2020-03-26	MDCK2	160	20	20	10	<	<	20	20	40	160
B/Moscow/52/2020		1A(Δ3) B	2020-03-25	MDCK2/MDCK1	640	40	40	<	<	<	<	40	20	160
B/Moscow/49/2020		1A(Δ3) B	2020-03-26	MDCK1/MDCK1	640	40	40	<	<	<	<	80	20	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not Done

Vaccine SH 2019	Vaccine SH 2020
NH 2019-20	NH 2020-21

< 4-fold	4-fold	8-fold	> 8-fold	< not recognised by the antiserum	≥ 160 (when homologous titre ≥ 2560)	≥ 160 (no homologous titre)
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Table 10-12. Antigenic analyses of influenza B viruses (Victoria lineage) 2020-08-25

Haemagglutination inhibition titre															
Viruses	Other information	Passage history	Collection date	Passage history	Post-infection ferret antisera										
					NEW										
					B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Sth Aus 81/12 Egg	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Norway 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/Wash'ton 02/19 MDCK	B/Wash'ton 02/19 Egg	B/Wash'ton 02/19 Egg
					Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ²	F25/16 ²	F15/16 ²	F38/17 ²	F40/17 ²	F21/18 ²	F11/18 ²	F37/19 ²	F38/19 ⁴	F20/20 ²
					Genetic group	1A	1A	1A	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)B	1A(Δ3)B
REFERENCE VIRUSES															
B/Brisbane/60/2008		1A	2008-08-04	E4/E4	2560	640	640	40	160	<	10	160	<	320	160
B/South Australia/81/2012		1A	2012-11-28	E4/E2	2560	640	640	40	80	<	10	160	<	160	80
B/Ireland/3154/2016		1A	2016-01-14	MDCK1/MDCK4	2560	160	40	80	160	<	<	<	<	<	<
B/Nordrhein-Westfalen/1/2016		1A	2016-01-04	C2/MDCK3	2560	160	40	80	160	<	<	<	<	<	<
B/Norway/2409/2017		1A(Δ2)	2017-04-27	MDCK1/MDCK3	80	<	10	<	<	80	80	160	<	<	<
B/Colorado/06/2017		1A(Δ2)	2017-02-05	MDCK1/MDCK2	80	<	10	<	<	40	40	160	<	<	<
B/Colorado/06/2017		1A(Δ2)	2017-02-05	E5/E2	640	80	40	<	<	20	40	320	<	160	80
B/Washington/02/2019		1A(Δ3)B	2019-01-19	C2/MDCK3	640	80	80	<	<	<	10	160	20	320	160
B/Washington/02/2019		1A(Δ3)B	2019-01-19	E3/E2	640	80	80	<	<	<	20	160	20	320	160
TEST VIRUSES															
B/Castilla La Mancha/1181/2020		No seq	2020-02-18	MDCK1	160	10	20	<	<	<	10	40	20	160	80
B/Ireland/19600/2020		No seq	2020-02-28	MDCK1	160	<	<	<	<	<	<	10	40	160	80
B/Moldova/1750.13/2020		No seq	2020-03-23	SIAT1	160	40	20	<	<	<	10	40	40	320	ND
B/Ireland/10670/2020		1A(Δ3) B	2020-02-03	MDCK2	80	<	<	<	<	<	<	20	20	160	80
B/Ireland/14667/2020		1A(Δ3) B	2020-02-11	MDCK1	80	<	<	<	<	<	<	10	40	160	80
B/Ireland/14994/2020		1A(Δ3) B	2020-02-12	MDCK1	160	<	<	<	<	<	<	20	40	160	80
B/Ireland/14343/2020		1A(Δ3) B	2020-02-13	MDCK2	80	<	<	<	<	<	<	20	40	160	80
B/Galicia/1171/2020		1A(Δ3) B	2020-02-16	MDCK1	320	40	44	<	<	<	10	80	40	320	ND
B/Pais Vasco/1082/2020		1A(Δ3) B	2020-02-18	MDCK1	160	<	10	<	<	<	10	40	20	160	80
B/Pais Vasco/1078/2020		1A(Δ3) B	2020-02-18	MDCK1	160	<	20	<	<	<	10	40	20	160	80
B/Navarra/1052/2020	N150K, G184E, N197D(-CHO), V220M, P241Q, R279K	1A(Δ3) B	2020-02-19	MDCK1	320	40	40	<	<	<	<	40	<	<	40
B/Castilla La Mancha/2093/2020		1A(Δ3) B	2020-02-21	MDCK1	80	<	10	<	<	<	<	40	10	160	80
B/Bosnia and Herzegovina/358/2020		1A(Δ3) B	2020-02-24	MDCK1	320	40	20	<	<	<	10	40	40	160	80
B/Ireland/19535/2020		1A(Δ3) B	2020-02-28	MDCK1	160	<	<	<	<	<	<	40	40	320	160
B/Castilla La Mancha/1477/2020	N126K	1A(Δ3) B	2020-03-02	MDCK1	80	<	<	<	<	<	<	40	10	<	20
B/Moldova/415.1/2020		1A(Δ3) B	2020-03-03	MDCK1	80	<	<	<	<	<	<	40	20	160	80
B/Ireland/23990/2020		1A(Δ3) B	2020-03-06	MDCK3	160	<	10	<	<	<	<	20	20	160	80
B/Ireland/25800/2020		1A(Δ3) B	2020-03-12	MDCK2	80	<	<	<	<	<	<	10	40	160	80
B/Ireland/27281/2020		1A(Δ3) B	2020-03-13	MDCK2	320	40	10	<	<	<	10	40	40	160	80
B/Ireland/40549/2020		1A(Δ3) B	2020-03-23	MDCK1	320	40	20	<	<	<	10	80	40	160	80
B/Castilla La Mancha/1090/2020		1A(Δ2)	2020-02-22	MDCK1	80	<	<	<	<	40	40	80	<	<	<
^ Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):															
^1 < = <40; ^2 < = <10; ^3 hyperimmune sheep serum; ^4 < = <20; ND = Not Done															
												Vaccine SH 2019 NH 2019-20		Vaccine SH 2020 NH 2020-21	
		< 4-fold	4-fold	8-fold	> 8-fold	<	not recognised by the antiserum				≥ 160 (when homologous titre ≥ 2560)		≥ 160 (no homologous titre)		

¹Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not Done

Vaccine
SH 2019
NH 2019-20

Vaccine
SH 2020
NH 2020-21

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)

Table 10-13. Antigenic analyses of influenza B viruses (Victoria lineage) - Summary

Viruses	Other information	Post-infection ferret antisera								
		B/Bris 60/08	B/Sth Aus 81/12	B/Ireland 3154/16	B/Nord-West 1/16	B/Norway 2409/17	B/Colorado 06/17	B/Colorado 06/17	B/Wash'ton 02/19	B/Wash'ton 02/19
		Egg	Egg	MDCK	MDCK	MDCK	MDCK	Egg	MDCK	Egg
		F44/17 ^{*2}	F25/16 ^{*2}	F15/16 ^{*2}	F38/17 ^{*2}	F40/17 ^{*2}	F21/18 ^{*2}	F11/18 ^{*2}	F37/19 ^{*2}	F38/19 ^{*4}
	Passage history									
	Ferret number									
	Genetic group	1A	1A	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)B	1A(Δ3)B
REFERENCE VIRUSES										
B/Brisbane/60/2008	1A	640	640	40	160	<	10	160	<	320
B/South Australia/81/2012	1A	640	640	40	80	<	10	160	<	160
B/Ireland/3154/2016	1A	160	40	80	160	<	<	<	<	<
B/Nordrhein-Westfalen/1/2016	1A	160	40	80	160	<	<	<	<	<
B/Norway/2409/2017	1A(Δ2)	<	10	<	<	80	80	160	<	<
B/Colorado/06/2017	1A(Δ2)	<	10	<	<	40	40	160	<	<
B/Colorado/06/2017	1A(Δ2)	80	40	<	<	20	40	320	<	160
B/Washington/02/2019	1A(Δ3)B	80	80	<	<	<	10	160	20	320
B/Washington/02/2019	1A(Δ3)B	80	80	<	<	<	20	160	20	320
TEST VIRUSES										
Genetic group										
All viruses	No tested	177	177	177	177	177	177	177	177	177
	Titre reduced ≤2-fold					2	8	3	150	117
	%					1.1	4.6	1.7	84.7	66.1
	Titre reduced =4-fold	1	1			1	33	12	4	28
	%	0.6	0.6			0.6	18.6	6.8	2.3	15.8
	Titre reduced ≥8-fold	176	176	177	177	174	136	162	23	32
	%	99.4	99.4	100.0	100.0	98.3	76.8	91.5	13.0	18.1
1A(Δ3) B	No tested	168	168	168	168	168	168	168	168	168
	Titre reduced ≤2-fold						6	2	144	111
	%						3.5	1.1	85.7	66.1
	Titre reduced =4-fold	1	1			1	31	9	4	28
	%	0.6	0.6			0.6	18.5	5.4	2.4	16.6
	Titre reduced ≥8-fold	167	167	168	168	167	131	157	20	29
	%	99.4	99.4	100.0	100.0	99.4	78.0	93.5	11.9	17.3
1A(Δ2)	No tested	2	2	2	2	2	2	2	2	2
	Titre reduced ≤2-fold					2	2	1		
	Titre reduced =4-fold							1		
	Titre reduced ≥8-fold	2	2	2	2				2	2
No Seq	No tested	7	7	7	7	7	7	7	7	7
	Titre reduced ≤2-fold								6	6
	Titre reduced =4-fold						2	2		
	Titre reduced ≥8-fold	7	7	7	7	7	5	5	1	1
								Vaccine SH 2019 NH 2019-20	Vaccine SH 2020 NH 2020-21	

Reference virus results are taken from an individual table as an example. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed

Figure 11. Phylogenetic comparison of influenza B (Victoria-lineage) HA genes

Vaccine viruses

Reference viruses

Collection date

Feb 2020

Mar 2020

Apr 2020

May 2020

HA2 numbering

Crick sequences

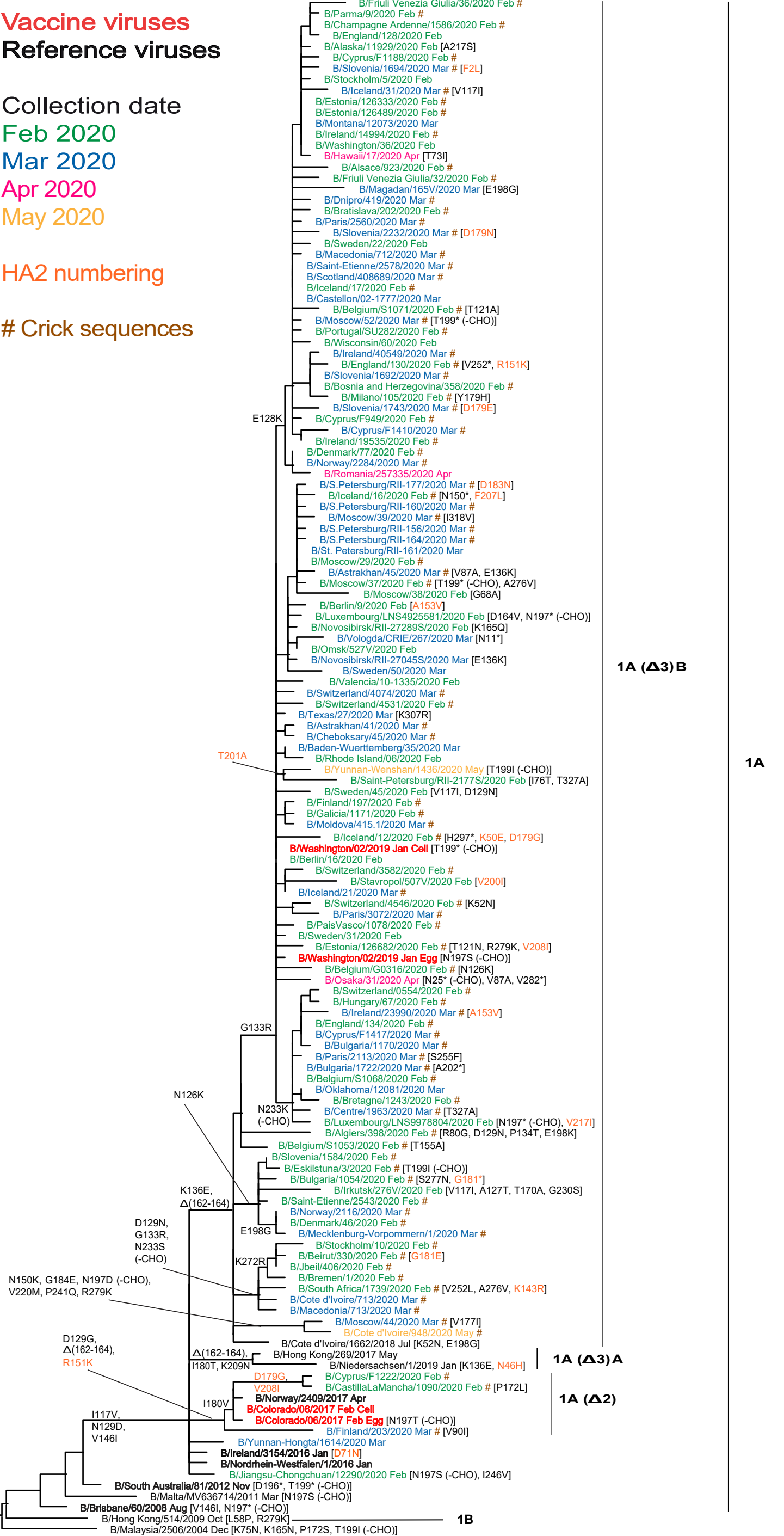
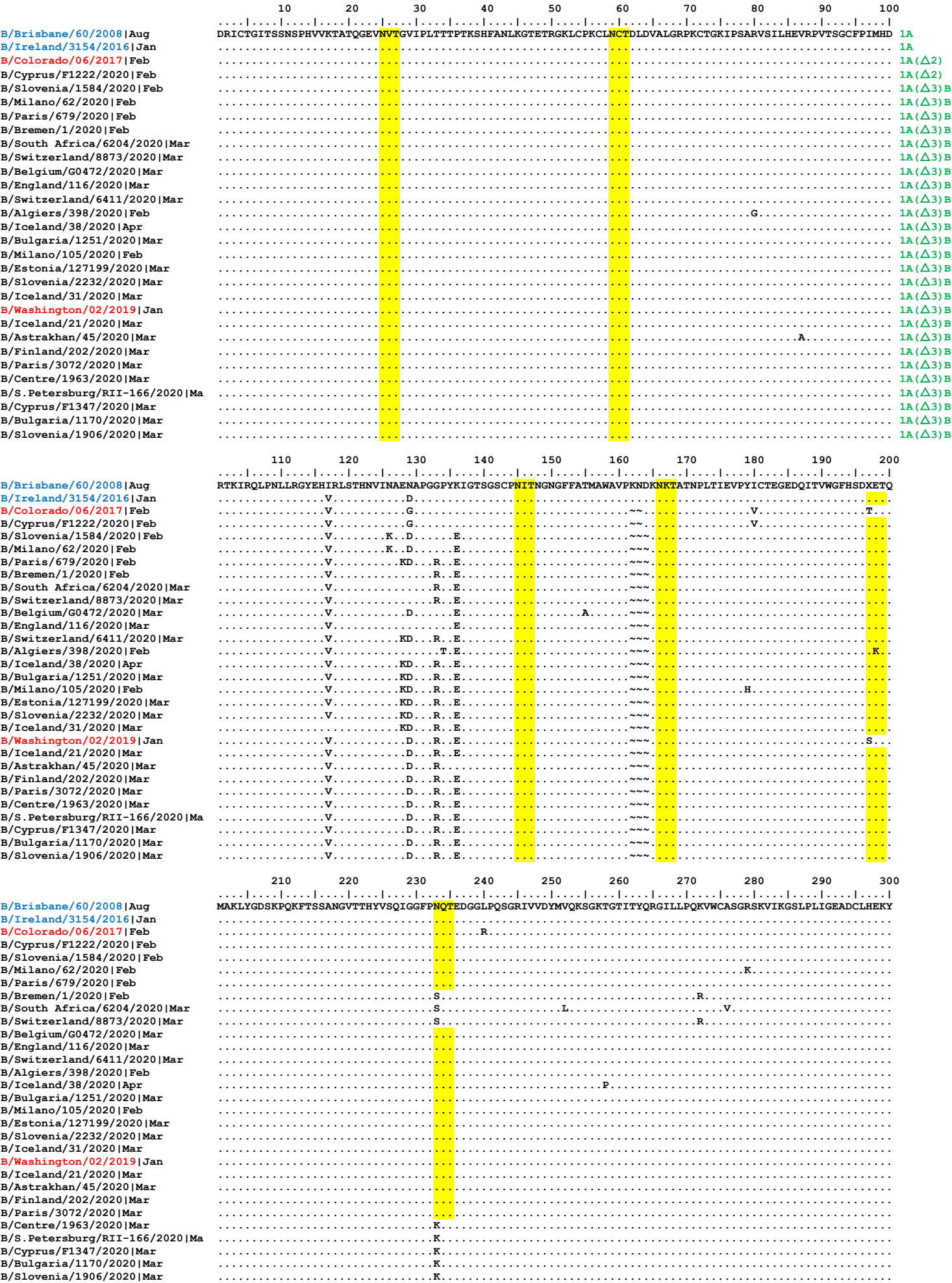


Figure 12. HA protein alignment for B (Victoria-lineage) viruses

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site



	310	320	330	340	350	360	370	380	390	400												
B/Brisbane/60/2008 Aug	GGLN	KSKPY	TGEHAKA	IGNCPI	VWVTK	PLKLAN	GT	KYRPP	AKLLKE	RGFFG	AIAGF	LEGGW	EAGMI	AGWHG	YTS	HGAHG	VAVAAD	LKSTQ	EAINK	ITKNLN	1A	
B/Ireland/3154/2016 Jan																					1A	
B/Colorado/06/2017 Feb																					1A(Δ2)	
B/Cyprus/F1222/2020 Feb																					1A(Δ2)	
B/Slovenia/1584/2020 Feb																					1A(Δ3)B	
B/Milano/62/2020 Feb																					1A(Δ3)B	
B/Paris/679/2020 Feb																					1A(Δ3)B	
B/Bremen/1/2020 Feb																					1A(Δ3)B	
B/South Africa/6204/2020 Mar																					1A(Δ3)B	
B/Switzerland/8873/2020 Mar																					1A(Δ3)B	
B/Belgium/G0472/2020 Mar																					1A(Δ3)B	
B/England/116/2020 Mar																					1A(Δ3)B	
B/Switzerland/6411/2020 Mar																					1A(Δ3)B	
B/Algiers/398/2020 Feb																					1A(Δ3)B	
B/Iceland/38/2020 Apr																					1A(Δ3)B	
B/Bulgaria/1251/2020 Mar																					1A(Δ3)B	
B/Milano/105/2020 Feb																					1A(Δ3)B	
B/Estonia/127199/2020 Mar																					1A(Δ3)B	
B/Slovenia/2232/2020 Mar																					1A(Δ3)B	
B/Iceland/31/2020 Mar																					1A(Δ3)B	
B/Washington/02/2019 Jan																					1A(Δ3)B	
B/Iceland/21/2020 Mar																					1A(Δ3)B	
B/Astrakhan/45/2020 Mar																					1A(Δ3)B	
B/Finland/202/2020 Mar																					1A(Δ3)B	
B/Paris/3072/2020 Mar																					1A(Δ3)B	
B/Centre/1963/2020 Mar																					1A(Δ3)B	
B/S. Petersburg/RII-166/2020 Ma																					1A(Δ3)B	
B/Cyprus/F1347/2020 Mar																					1A(Δ3)B	
B/Bulgaria/1170/2020 Mar																					1A(Δ3)B	
B/Slovenia/1906/2020 Mar																					1A(Δ3)B	
	410	420	430	440	450	460	470	480	490	500												
B/Brisbane/60/2008 Aug	SLSE	LEVKN	LQRLS	GAMDEL	HNEILE	LDERK	VDDL	LRAD	TISSQ	IELAV	LLSNE	GIIN	SEDEH	LLAL	ERLKK	MLGPS	AVEIG	NGCF	FETK	HKCN	QTCLD	RIA
B/Ireland/3154/2016 Jan																						
B/Colorado/06/2017 Feb																						
B/Cyprus/F1222/2020 Feb																						
B/Slovenia/1584/2020 Feb																						
B/Milano/62/2020 Feb																						
B/Paris/679/2020 Feb																						
B/Bremen/1/2020 Feb																						
B/South Africa/6204/2020 Mar																						
B/Switzerland/8873/2020 Mar																						
B/Belgium/G0472/2020 Mar																						
B/England/116/2020 Mar																						
B/Switzerland/6411/2020 Mar																						
B/Algiers/398/2020 Feb																						
B/Iceland/38/2020 Apr																						
B/Bulgaria/1251/2020 Mar																						
B/Milano/105/2020 Feb																						
B/Estonia/127199/2020 Mar																						
B/Slovenia/2232/2020 Mar																						
B/Iceland/31/2020 Mar																						
B/Washington/02/2019 Jan																						
B/Iceland/21/2020 Mar																						
B/Astrakhan/45/2020 Mar																						
B/Finland/202/2020 Mar																						
B/Paris/3072/2020 Mar																						
B/Centre/1963/2020 Mar																						
B/S. Petersburg/RII-166/2020 Ma																						
B/Cyprus/F1347/2020 Mar																						
B/Bulgaria/1170/2020 Mar																						
B/Slovenia/1906/2020 Mar																						
	510	520	530	540	550	560	570															
B/Brisbane/60/2008 Aug	AGT	FDAGE	FSLP	TFDS	LN	ITAA	SLNDD	GLDN	HT	ILLY	STA	ASS	SLAV	TL	MTA	IFV	VY	MV	SR	DN	VSC	SICL
B/Ireland/3154/2016 Jan																						
B/Colorado/06/2017 Feb																						
B/Cyprus/F1222/2020 Feb																						
B/Slovenia/1584/2020 Feb																						
B/Milano/62/2020 Feb																						
B/Paris/679/2020 Feb																						
B/Bremen/1/2020 Feb																						
B/South Africa/6204/2020 Mar																						
B/Switzerland/8873/2020 Mar																						
B/Belgium/G0472/2020 Mar																						
B/England/116/2020 Mar																						
B/Switzerland/6411/2020 Mar																						
B/Algiers/398/2020 Feb																						
B/Iceland/38/2020 Apr																						
B/Bulgaria/1251/2020 Mar																						
B/Milano/105/2020 Feb																						
B/Estonia/127199/2020 Mar																						
B/Slovenia/2232/2020 Mar																						
B/Iceland/31/2020 Mar																						
B/Washington/02/2019 Jan																						
B/Iceland/21/2020 Mar																						
B/Astrakhan/45/2020 Mar																						
B/Finland/202/2020 Mar																						
B/Paris/3072/2020 Mar																						
B/Centre/1963/2020 Mar																						
B/S. Petersburg/RII-166/2020 Ma																						
B/Cyprus/F1347/2020 Mar																						
B/Bulgaria/1170/2020 Mar																						
B/Slovenia/1906/2020 Mar																						

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Substitutions defining
subclade 1A(Δ 3)B viruses
shown in Black

Viruses displaying low
reactivity with antisera raised
against B/Washington/02/2019:

Virus group with 4 substitutions

Individual viruses with
N126K or **D129R** substitution,
sometimes with additional
substitutions

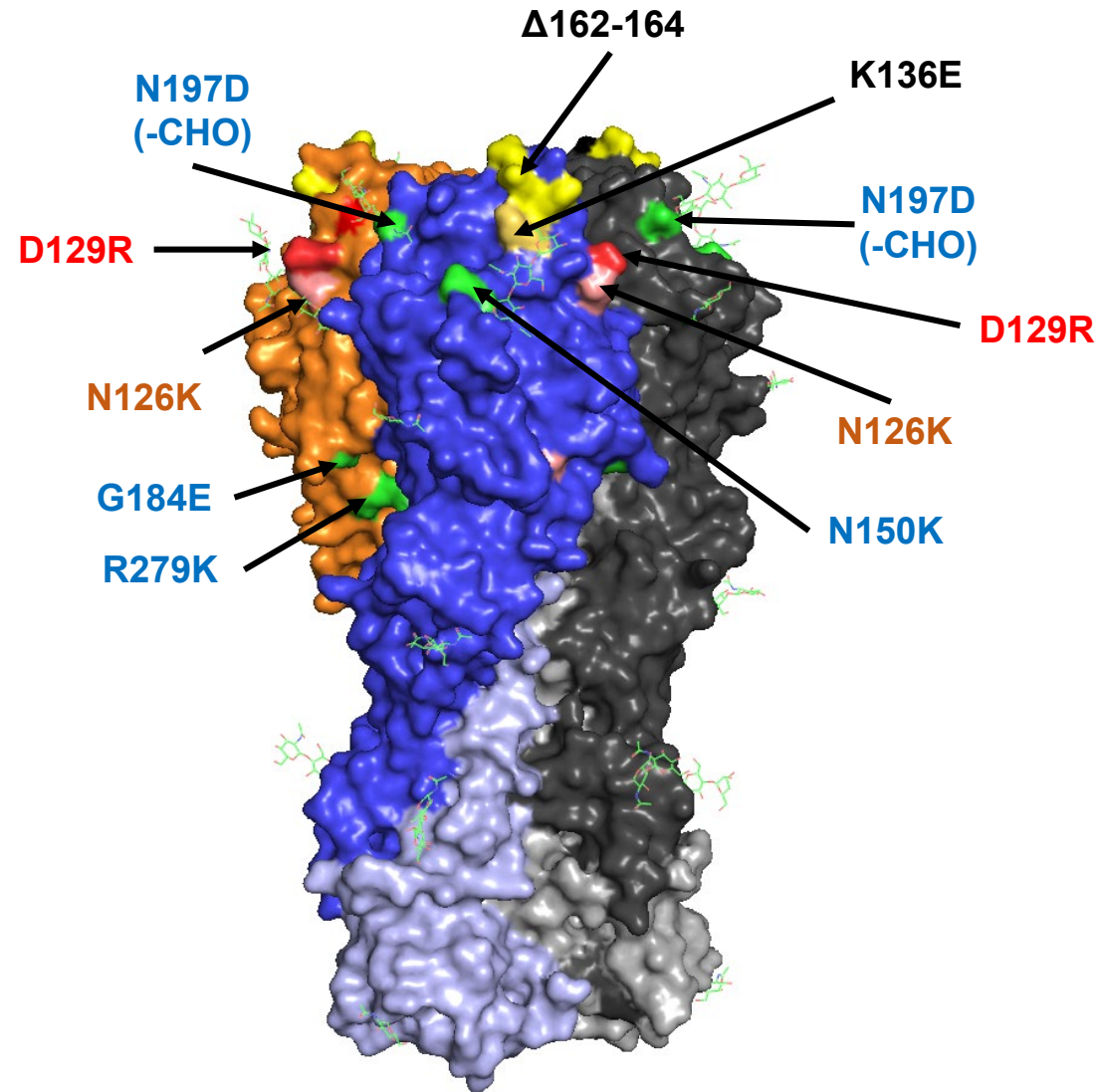


Figure 13. Amino acid substitutions and deletions in the HA of recently circulating B/Victoria lineage viruses mapped onto B/Brisbane/60/2008

Figure 14. Phylogenetic comparison of influenza B (Victoria-lineage) NA genes

Vaccine viruses

Reference viruses

Collection date

Feb 2020

Mar 2020

Apr 2020

May 2020

Crick sequences

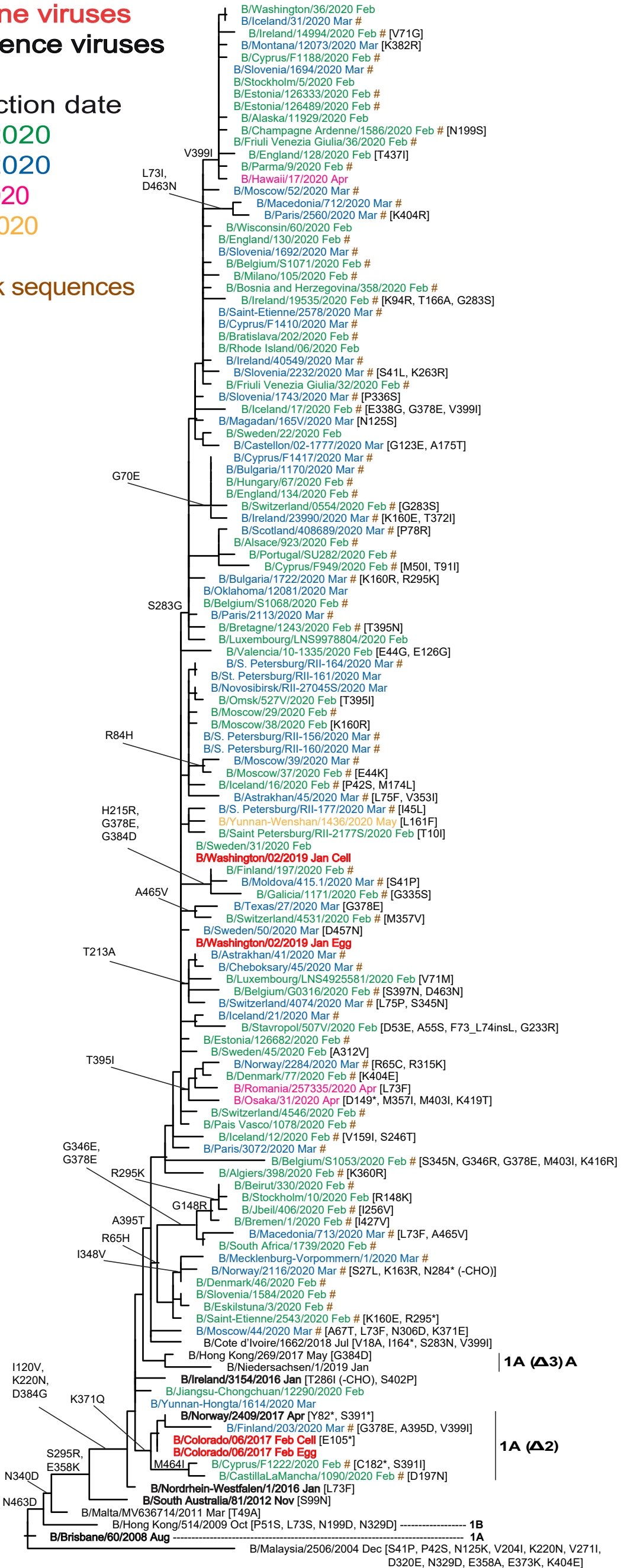
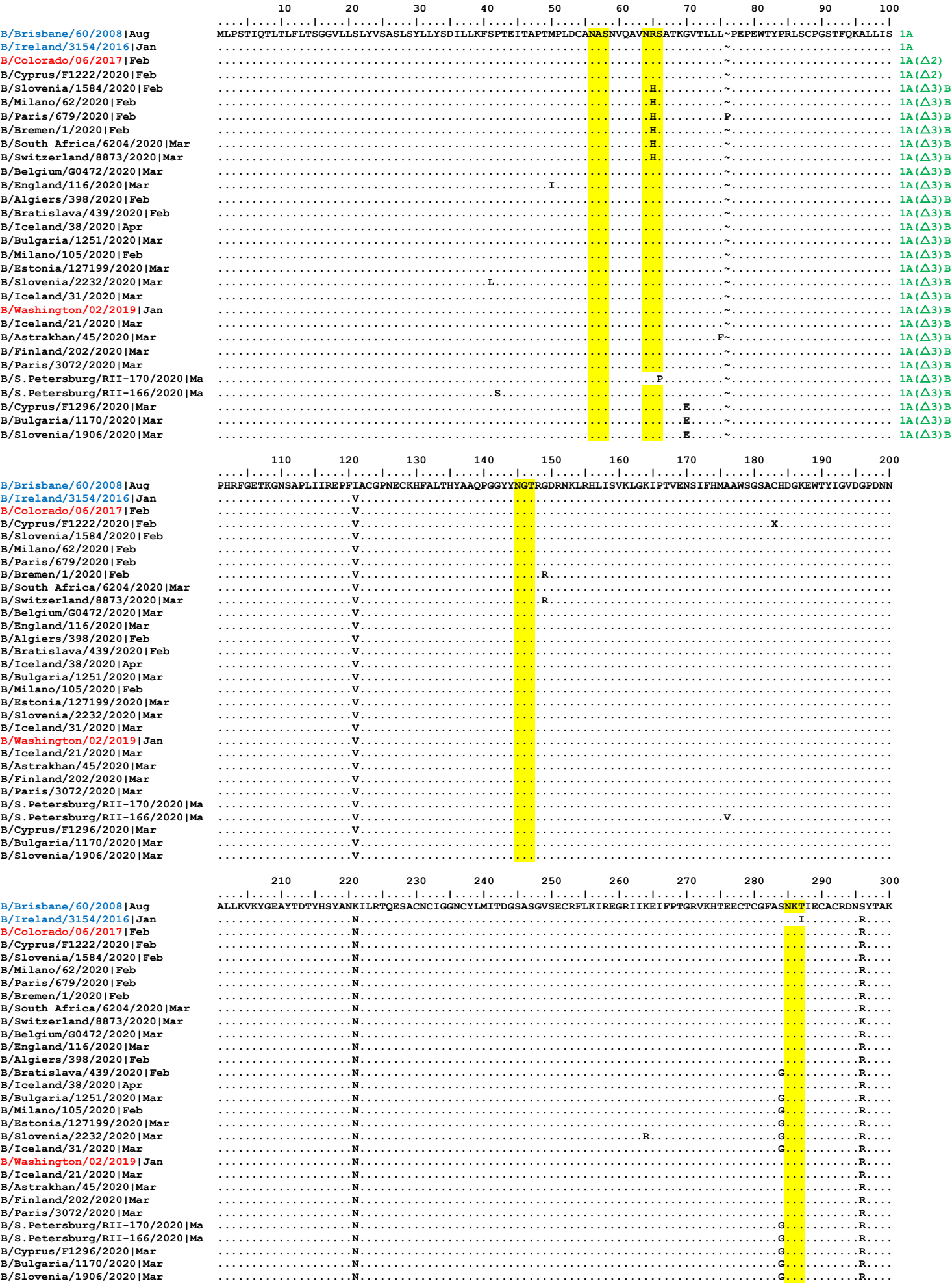


Figure 15. NA protein alignment for B (Victoria-lineage) viruses

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site



[illegible]

Table 11. Summary of influenza B (Yamagata-lineage) viruses analysed, with collection dates after 2020-01-31

MONTH	B Yamagata lineage								
Country	Number received	Number propagated ¹	Fold-reductions in HI titre compared to homologous titres (post-infection ferret antisera)						
			≤2 fold ²	4 fold ²	≥8 fold ²		≤2 fold ³	4 fold ³	≥8 fold ³
2020 FEBRUARY									
	France	1	1			1		1	
	Norway	1	1		1			1	
MARCH									
	Russia	1	1			1		1	
3 Countries	3	3	0	1	2		3	0	0
			0.0%	33.3%	66.7%		100.0%	0.0%	0.0%

1. Propagated to sufficient titre to perform HI assay

2. Fold reduction in HI titre with ferret antiserum raised against B/Phuket/3073/2013 (tissue culture-propagated vaccine virus)

3. Fold reduction in HI titre with ferret antiserum raised against B/Phuket/3073/2013 (egg-propagated vaccine virus)

Table 12-1. Antigenic analyses of influenza B viruses (Yamagata lineage) 2020-03-18

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				B/Phuket 3073/13	B/Estonia 55669/11	B/Mass 02/12	B/Mass 02/12	B/Wis 1/10	B/Stock 12/11	B/Phuket 3073/13	B/Phuket 3073/13	B/Maur 1791/17	B/Maur I-762/18	B/Nor 2134/19	
				Egg	MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	MDCK	MDCK	Egg	
				SH614 ^{*1,3}	F39/17 ^{*2}	F10/16 ^{*2}	F06/17 ^{*2}	F36/15 ^{*2}	F05/17 ^{*2}	F27/15 ^{*2}	F25/17 ^{*2}	F04/18 ^{*2}	F05/19 ^{*2}	F48/19 ^{*2}	
	Passage history			3	2	2	2	3	3	3	3	3	3	3	
	Ferret number														
	Genetic Group														
REFERENCE VIRUSES															
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	320	80	40	80	40	20	20	20	20	40	<	
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK4	1280	320	320	640	320	160	160	320	80	320	10	
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	320	20	20	160	40	40	10	40	<	10	20	
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	1280	<	20	320	80	80	40	160	20	80	80	
B/Stockholm/12/2011	3	2011-03-28	E4/E2	640	<	20	160	80	80	20	80	10	40	20	
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK3	2560	160	320	320	320	160	160	160	160	640	20	
B/Phuket/3073/2013	3	2013-11-21	E4/E3	640	<	<	160	80	80	20	80	20	40	20	
B/Mauritius/1791/2017	3	2017-09-20	MDCK1/MDCK4	1280	10	20	80	80	20	40	80	80	80	<	
B/Mauritius/I-762/2018	3	2018-09-02	MDCK1/MDCK3	1280	20	20	80	80	20	40	80	80	160	<	
B/Norway/2134/2019	3	2019-06-06	E2/E2	160	<	<	<	<	10	<	<	<	<	640	
TEST VIRUSES															
B/Tunis/15474/2019	No seq	2019-11-28	MDCK1	1280	20	40	80	80	20	40	80	80	80	<	
B/Tunis/15812/2019	3	2019-12-08	MDCK1	1280	10	40	160	80	20	80	80	80	80	<	
B/Tunis/16026/2019	3	2019-12-12	MDCK1	640	<	20	80	40	20	40	40	40	80	<	
B/Tunis/16025/2019	3	2019-12-12	MDCK1	1280	20	40	80	40	20	40	80	40	80	<	
B/Tunis/285/2020	3	2020-01-08	MDCK1	640	10	40	160	80	20	40	80	80	160	<	
B/Berlin/5/2020	3	2020-01-24	C1/MDCK1	640	10	40	80	40	20	40	80	40	80	<	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

Vaccine[#]

1 <= <40; 2 <= <10; 3 hyperimmune sheep serum

[#] B/Yamagata-lineage virus recommended for use in quadravalent vaccines SH 2019, NH 2019-20, SH 2020 and NH 2020-21

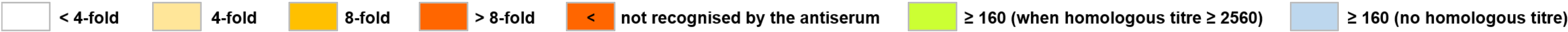


Table 12-2. Antigenic analyses of influenza B viruses (Yamagata lineage) 2020-07-21 + 2020-08-26

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antisera								
				B/Phuket 3073/13	B/Estonia 55669/11	B/Mass 02/12	B/Mass 02/12	B/Wis 1/10	B/Phuket 3073/13	B/Phuket 3073/13	B/Maur I-762/18	B/Nor 2134/19
				Egg SH614 ^{*1,3}	MDCK F39/17 ^{*2}	MDCK F10/16 ^{*2}	Egg F06/17 ^{*4}	Egg F36/15 ^{*2}	MDCK F27/15 ^{*2}	Egg F25/17 ^{*2}	MDCK F05/19 ^{*2}	Egg F48/19 ^{*2}
				Genetic Group	3	2	2	2	3	3	3	3
REFERENCE VIRUSES												
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	320	40	40	80	40	20	20	<	<
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK4	1280	160	320	320	320	80	160	80	<
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	320	10	<	160	40	<	40	<	<
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	1280	<	<	160	80	10	160	40	<
B/Stockholm/12/2011	3	2011-03-28	E4/E2	640	<	<	80	80	10	80	<	<
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK3	2560	160	160	160	320	160	160	320	<
B/Phuket/3073/2013	3	2013-11-21	E4/E3	1280	<	<	80	80	20	80	<	<
B/Mauritius/I-762/2018	3	2018-09-02	MDCK1/MDCK3	1280	<	20	80	80	20	40	80	<
B/Norway/2134/2019	3	2019-06-06	E2/E2	160	<	<	<	<	<	<	<	320
TEST VIRUSES												
B/Norway/993/2020	3	2020-02-11	MDCK1	1280	<	40	160	80	40	80	80	<
B/Pays de Loire/1150/2020	3	2020-02-12	MDCK1/MDCK1	1280	<	10	<	40	20	40	40	<
B/S. Petersburg/RII-181/2020	3	2020-03-17	MDCK1/MDCK1	1280	<	10	<	40	20	40	40	<

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 < = <40; 2 < = <10; 3 hyperimmune sheep serum

B/Yamagata-lineage virus recommended for use in quadravalent vaccines SH 2019, NH 2019-20, SH 2020 and NH 2020-21

Sequences in phylogenetic trees

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)

Vaccine[#]

Figure 16. Phylogenetic comparison of influenza B (Yamagata-lineage) HA genes

Vaccine virus
Reference viruses

Collection date
Jan 2020
Feb 2020
Mar 2020

HA2 numbering

Crick sequences

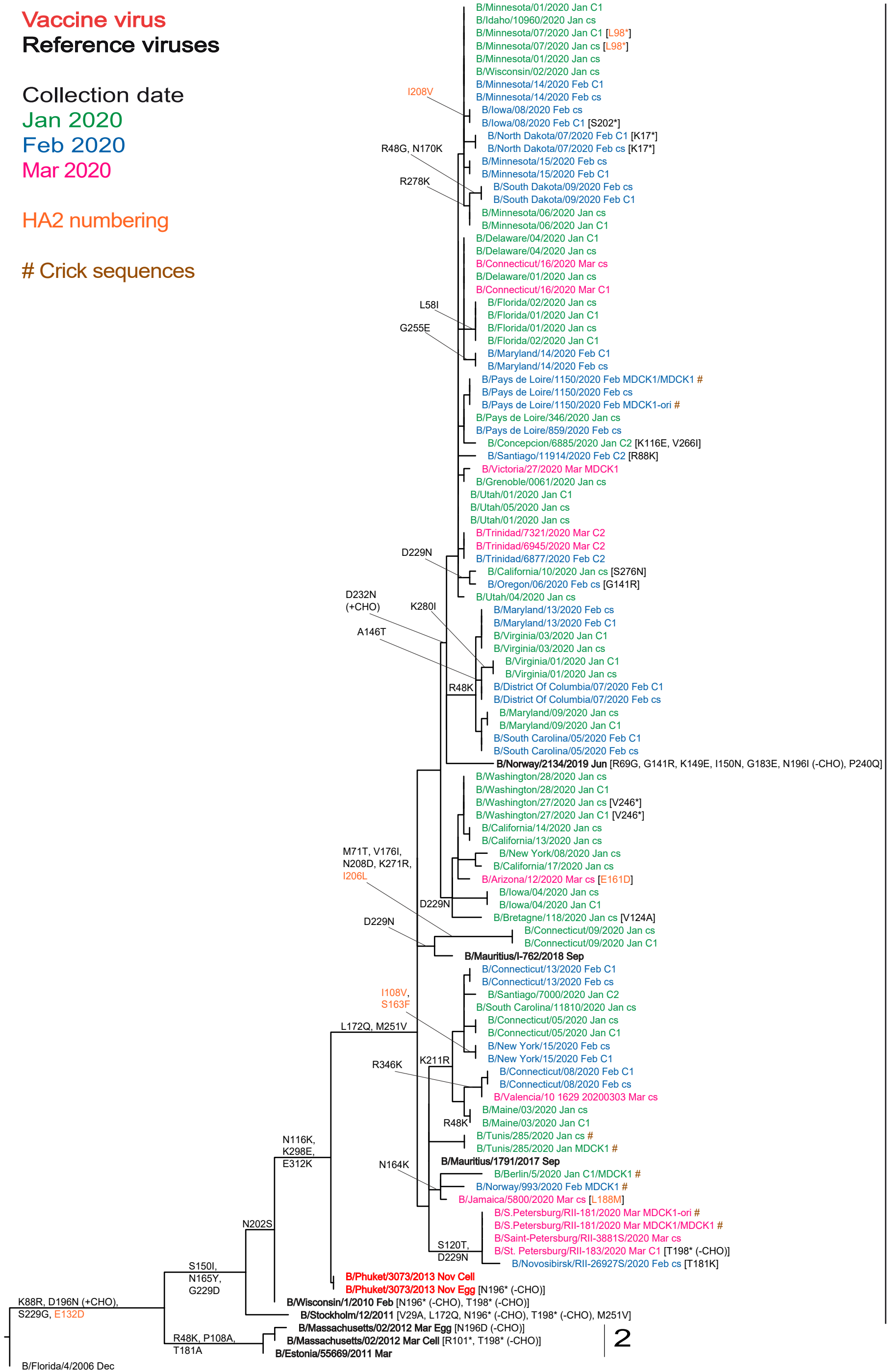


Figure 17. HA protein alignment for B (Yamagata-lineage) viruses

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

	10	20	30	40	50	60	70	80	90	100
B/Phuket/3073/2013 Nov	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/Novosibirsk/RII-26927S/2020 F	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/St. Petersburg/RII-183/2020 M	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/Jamaica/5800/2020 Mar	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/Berlin/5/2020 Jan	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/Mauritius/1791/2017 Sep	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/Tunis/285/2020 Jan	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/Connecticut/08/2020 Feb	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/Valencia/10-1629/2020 Mar	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/Connecticut/13/2020 Feb	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/Santiago/7000/2020 Jan	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/Mauritius/I-762/2018 Sep	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/Connecticut/09/2020 Jan	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/Arizona/12/2020 Mar	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/California/14/2020 Jan	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/South Carolina/05/2020 Feb	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/Maryland/13/2020 Feb	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/Virginia/01/2020 Jan	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/Norway/2134/2019 Jun	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/Oregon/06/2020 Feb	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/Florida/02/2020 Jan	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/Connecticut/16/2020 Mar	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/Grenoble/0061/2020 Jan	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/Victoria/27/2020 Mar	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/Santiago/11914/2020 Feb	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/Trinidad/6945/2020 Mar	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/Pays de Loire/1150/2020 Feb	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/Iowa/08/2020 Feb	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/South Dakota/09/2020 Feb	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/Minnesota/14/2020 Feb	DR	IC	TG	IT	SS	NS	PH	VK	TAT	Q
B/Phuket/3073/2013 Nov	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/Novosibirsk/RII-26927S/2020 F	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/St. Petersburg/RII-183/2020 M	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/Jamaica/5800/2020 Mar	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/Berlin/5/2020 Jan	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/Mauritius/1791/2017 Sep	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/Tunis/285/2020 Jan	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/Connecticut/08/2020 Feb	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/Valencia/10-1629/2020 Mar	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/Connecticut/13/2020 Feb	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/Santiago/7000/2020 Jan	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/Mauritius/I-762/2018 Sep	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/Connecticut/09/2020 Jan	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/Arizona/12/2020 Mar	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/California/14/2020 Jan	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/South Carolina/05/2020 Feb	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/Maryland/13/2020 Feb	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/Virginia/01/2020 Jan	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/Norway/2134/2019 Jun	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/Oregon/06/2020 Feb	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/Florida/02/2020 Jan	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/Connecticut/16/2020 Mar	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/Grenoble/0061/2020 Jan	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/Victoria/27/2020 Mar	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/Santiago/11914/2020 Feb	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/Trinidad/6945/2020 Mar	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/Pays de Loire/1150/2020 Feb	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/Iowa/08/2020 Feb	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/South Dakota/09/2020 Feb	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/Minnesota/14/2020 Feb	RT	KI	RQ	LP	NL	LR	GY	EK	IR	LS
B/Phuket/3073/2013 Nov	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/Novosibirsk/RII-26927S/2020 F	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/St. Petersburg/RII-183/2020 M	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/Jamaica/5800/2020 Mar	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/Berlin/5/2020 Jan	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/Mauritius/1791/2017 Sep	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/Tunis/285/2020 Jan	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/Connecticut/08/2020 Feb	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/Valencia/10-1629/2020 Mar	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/Connecticut/13/2020 Feb	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/Santiago/7000/2020 Jan	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/Mauritius/I-762/2018 Sep	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/Connecticut/09/2020 Jan	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/Arizona/12/2020 Mar	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/California/14/2020 Jan	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/South Carolina/05/2020 Feb	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/Maryland/13/2020 Feb	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/Virginia/01/2020 Jan	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/Norway/2134/2019 Jun	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/Oregon/06/2020 Feb	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/Florida/02/2020 Jan	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/Connecticut/16/2020 Mar	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/Grenoble/0061/2020 Jan	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/Victoria/27/2020 Mar	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/Santiago/11914/2020 Feb	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/Trinidad/6945/2020 Mar	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/Pays de Loire/1150/2020 Feb	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/Iowa/08/2020 Feb	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/South Dakota/09/2020 Feb	KS	LY	GS	NP	QK	FT	SS	ANG	V	T
B/Minnesota/14/2020 Feb	KS	LY	GS	NP	QK	FT	SS	ANG	V	T

	310	320	330	340	350	360	370	380	390	400
B/Phuket/3073/2013 Nov	G	L	N	K	S	K	P	Y	T	G
B/Novosibirsk/RII-26927S/2020 F	K	I	N	K	S	K	P	Y	T	G
B/St. Petersburg/RII-183/2020 M	K	I	N	K	S	K	P	Y	T	G
B/Jamaica/5800/2020 Mar	K	I	N	K	S	K	P	Y	T	G
B/Berlin/5/2020 Jan	K	I	N	K	S	K	P	Y	T	G
B/Mauritius/1791/2017 Sep	K	I	N	K	S	K	P	Y	T	G
B/Tunis/285/2020 Jan	K	I	N	K	S	K	P	Y	T	G
B/Connecticut/08/2020 Feb	K	I	N	K	S	K	P	Y	T	G
B/Valencia/10-1629/2020 Mar	K	I	N	K	S	K	P	Y	T	G
B/Connecticut/13/2020 Feb	K	I	N	K	S	K	P	Y	T	G
B/Santiago/7000/2020 Jan	K	I	N	K	S	K	P	Y	T	G
B/Mauritius/I-762/2018 Sep	K	I	N	K	S	K	P	Y	T	G
B/Connecticut/09/2020 Jan	K	I	N	K	S	K	P	Y	T	G
B/Arizona/12/2020 Mar	K	I	N	K	S	K	P	Y	T	G
B/California/14/2020 Jan	K	I	N	K	S	K	P	Y	T	G
B/South Carolina/05/2020 Feb	K	I	N	K	S	K	P	Y	T	G
B/Maryland/13/2020 Feb	K	I	N	K	S	K	P	Y	T	G
B/Virginia/01/2020 Jan	K	I	N	K	S	K	P	Y	T	G
B/Norway/2134/2019 Jun	K	I	N	K	S	K	P	Y	T	G
B/Oregon/06/2020 Feb	K	I	N	K	S	K	P	Y	T	G
B/Florida/02/2020 Jan	K	I	N	K	S	K	P	Y	T	G
B/Connecticut/16/2020 Mar	K	I	N	K	S	K	P	Y	T	G
B/Grenoble/0061/2020 Jan	K	I	N	K	S	K	P	Y	T	G
B/Victoria/27/2020 Mar	K	I	N	K	S	K	P	Y	T	G
B/Santiago/11914/2020 Feb	K	I	N	K	S	K	P	Y	T	G
B/Trinidad/6945/2020 Mar	K	I	N	K	S	K	P	Y	T	G
B/Pays de Loire/1150/2020 Feb	K	I	N	K	S	K	P	Y	T	G
B/Iowa/08/2020 Feb	K	I	N	K	S	K	P	Y	T	G
B/South Dakota/09/2020 Feb	K	I	N	K	S	K	P	Y	T	G
B/Minnesota/14/2020 Feb	K	I	N	K	S	K	P	Y	T	G

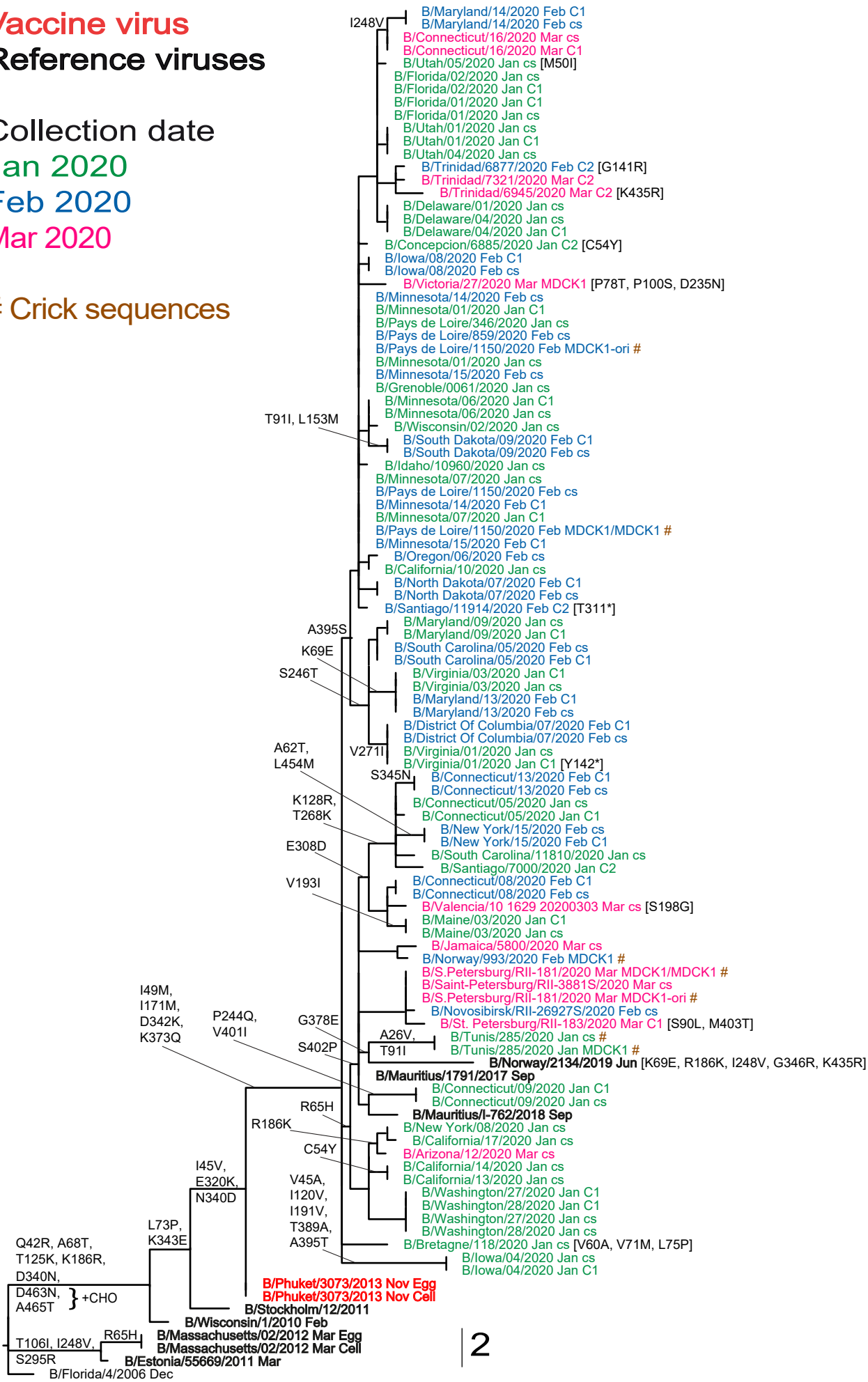
	410	420	430	440	450	460	470	480	490	500
B/Phuket/3073/2013 Nov	L	S	E	L	V	K	N	L	Q	R
B/Novosibirsk/RII-26927S/2020 F	L	S	E	L	V	K	N	L	Q	R
B/St. Petersburg/RII-183/2020 M	L	S	E	L	V	K	N	L	Q	R
B/Jamaica/5800/2020 Mar	L	S	E	L	V	K	N	L	Q	R
B/Berlin/5/2020 Jan	L	S	E	L	V	K	N	L	Q	R
B/Mauritius/1791/2017 Sep	L	S	E	L	V	K	N	L	Q	R
B/Tunis/285/2020 Jan	L	S	E	L	V	K	N	L	Q	R
B/Connecticut/08/2020 Feb	L	S	E	L	V	K	N	L	Q	R
B/Valencia/10-1629/2020 Mar	L	S	E	L	V	K	N	L	Q	R
B/Connecticut/13/2020 Feb	L	S	E	L	V	K	N	L	Q	R
B/Santiago/7000/2020 Jan	L	S	E	L	V	K	N	L	Q	R
B/Mauritius/I-762/2018 Sep	L	S	E	L	V	K	N	L	Q	R
B/Connecticut/09/2020 Jan	L	S	E	L	V	K	N	L	Q	R
B/Arizona/12/2020 Mar	L	S	E	L	V	K	N	L	Q	R
B/California/14/2020 Jan	L	S	E	L	V	K	N	L	Q	R
B/South Carolina/05/2020 Feb	L	S	E	L	V	K	N	L	Q	R
B/Maryland/13/2020 Feb	L	S	E	L	V	K	N	L	Q	R
B/Virginia/01/2020 Jan	L	S	E	L	V	K	N	L	Q	R
B/Norway/2134/2019 Jun	L	S	E	L	V	K	N	L	Q	R
B/Oregon/06/2020 Feb	L	S	E	L	V	K	N	L	Q	R
B/Florida/02/2020 Jan	L	S	E	L	V	K	N	L	Q	R
B/Connecticut/16/2020 Mar	L	S	E	L	V	K	N	L	Q	R
B/Grenoble/0061/2020 Jan	L	S	E	L	V	K	N	L	Q	R
B/Victoria/27/2020 Mar	L	S	E	L	V	K	N	L	Q	R
B/Santiago/11914/2020 Feb	L	S	E	L	V	K	N	L	Q	R
B/Trinidad/6945/2020 Mar	L	S	E	L	V	K	N	L	Q	R
B/Pays de Loire/1150/2020 Feb	L	S	E	L	V	K	N	L	Q	R
B/Iowa/08/2020 Feb	L	S	E	L	V	K	N	L	Q	R
B/South Dakota/09/2020 Feb	L	S	E	L	V	K	N	L	Q	R
B/Minnesota/14/2020 Feb	L	S	E	L	V	K	N	L	Q	R

	510	520	530	540	550	560
B/Phuket/3073/2013 Nov	G	T	F	N	A	G
B/Novosibirsk/RII-26927S/2020 F	G	T	F	N	A	G
B/St. Petersburg/RII-183/2020 M	G	T	F	N	A	G
B/Jamaica/5800/2020 Mar	G	T	F	N	A	G
B/Berlin/5/2020 Jan	G	T	F	N	A	G
B/Mauritius/1791/2017 Sep	G	T	F	N	A	G
B/Tunis/285/2020 Jan	G	T	F	N	A	G
B/Connecticut/08/2020 Feb	G	T	F	N	A	G
B/Valencia/10-1629/2020 Mar	G	T	F	N	A	G
B/Connecticut/13/2020 Feb	G	T	F	N	A	G
B/Santiago/7000/2020 Jan	G	T	F	N	A	G
B/Mauritius/I-762/2018 Sep	G	T	F	N	A	G
B/Connecticut/09/2020 Jan	G	T	F	N	A	G
B/Arizona/12/2020 Mar	G	T	F	N	A	G
B/California/14/2020 Jan	G	T	F	N	A	G
B/South Carolina/05/2020 Feb	G	T	F	N	A	G
B/Maryland/13/2020 Feb	G	T	F	N	A	G
B/Virginia/01/2020 Jan	G	T	F	N	A	G
B/Norway/2134/2019 Jun	G	T	F	N	A	G
B/Oregon/06/2020 Feb	G	T	F	N	A	G
B/Florida/02/2020 Jan	G	T	F	N	A	G
B/Connecticut/16/2020 Mar	G	T	F	N	A	G
B/Grenoble/0061/2020 Jan	G	T	F	N	A	G
B/Victoria/27/2020 Mar	G	T	F	N	A	G
B/Santiago/11914/2020 Feb	G	T	F	N	A	G
B/Trinidad/6945/2020 Mar	G	T	F	N	A	G
B/Pays de Loire/1150/2020 Feb	G	T	F	N	A	G
B/Iowa/08/2020 Feb	G	T	F	N	A	G
B/South Dakota/09/2020 Feb	G	T	F	N	A	G
B/Minnesota/14/2020 Feb	G	T	F	N	A	G

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 18. Phylogenetic comparison of influenza B (Yamagata-lineage) NA genes

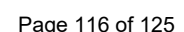
Vaccine virus
Reference viruses
Collection date
Jan 2020
Feb 2020
Mar 2020
Crick sequences



3

2

Vaccine virus: **Reference virus:** **Genetic group:** **Potential N-linked glycosylation site**



	310	320	330	340	350	360	370	380	390	400
B/Phuket/3073/2013 Nov	PFVKLN	VETDTAE	IRLMCTK	TYLDT	PRPN	DGSGIT	GPCESD	GGDEG	SGGKGG	FVHQ
B/Novosibirsk/RII-26927S/2020 F				K.			Q.			3
B/St. Petersburg/RII-183/2020 M							Q.			3
B/Jamaica/5800/2020 Mar				K.			Q.			3
B/Mauritius/1791/2017 Sep				K.			Q.			3
B/Tunis/285/2020 Jan				K.			Q.	E.		3
B/Connecticut/08/2020 F		D.		K.			Q.			3
B/Valencia/10-1629/2020 Mar		D.		K.			Q.			3
B/Connecticut/13/2020 Feb		D.		K.	N.		Q.			3
B/Santiago/7000/2020 Jan		D.		K.			Q.			3
B/Mauritius/I-762/2018 Sep				K.			Q.			3
B/Connecticut/09/2020 Jan				K.			Q.			3
B/Arizona/12/2020 Mar				K.			Q.			3
B/California/14/2020 Jan				K.			Q.			3
B/South Carolina/05/2020 Feb				K.			Q.		S.	3
B/Maryland/13/2020 Feb				K.			Q.		S.	3
B/Virginia/01/2020 Jan				K.			Q.		S.	3
B/Norway/2134/2019 Jun				K.	R.		Q.	E.		3
B/Oregon/06/2020 Feb				K.			Q.		S.	3
B/Florida/02/2020				K.			Q.		S.	3
B/Connecticut/16/2020 Mar				K.			Q.		S.	3
B/Grenoble/0061/2020 Jan				K.			Q.		S.	3
B/Victoria/27/2020 Mar				K.			Q.		S.	3
B/Santiago/11914/2020 Feb		X.		K.			Q.		S.	3
B/Trinidad/6945/2020 Mar				K.			Q.		S.	3
B/Pays de Loire/1150/2020 Feb				K.			Q.		S.	3
B/Iowa/08/2020 Feb				K.			Q.		S.	3
B/South Dakota/09/2020 Feb				K.			Q.		S.	3
B/Minnesota/14/2020 Feb				K.			Q.		S.	3

	410	420	430	440	450	460
B/Phuket/3073/2013 Nov	VSME	EPGW	YSFG	FEIK	DKKCD	VPCIG
B/Novosibirsk/RII-26927S/2020 F	P.					
B/St. Petersburg/RII-183/2020 M	PT.					
B/Jamaica/5800/2020 Mar	P.					
B/Mauritius/1791/2017 Sep	P.					
B/Tunis/285/2020 Jan	P.					
B/Connecticut/08/2020 F	P.					
B/Valencia/10-1629/2020 Mar	P.					
B/Connecticut/13/2020 Feb	P.					
B/Santiago/7000/2020 Jan	P.					
B/Mauritius/I-762/2018 Sep	P.					
B/Connecticut/09/2020 Jan	IP.					
B/Arizona/12/2020 Mar						
B/California/14/2020 Jan						
B/South Carolina/05/2020 Feb						
B/Maryland/13/2020 Feb						
B/Virginia/01/2020 Jan						
B/Norway/2134/2019 Jun	P.		R.			
B/Oregon/06/2020 Feb						
B/Florida/02/2020		X.				
B/Connecticut/16/2020 Mar						
B/Grenoble/0061/2020 Jan						
B/Victoria/27/2020 Mar						
B/Santiago/11914/2020 Feb						
B/Trinidad/6945/2020 Mar			R.			
B/Pays de Loire/1150/2020 Feb						
B/Iowa/08/2020 Feb						
B/South Dakota/09/2020 Feb						
B/Minnesota/14/2020 Feb						

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Table 13. Antiviral (NAI) testing of isolates with collection dates after 2020-01-31*

Month of Collection	Number of viruses tested for phenotype ¹								Totals
	A(H1N1)pdm09		A(H3N2)		Influenza B-Victoria		Influenza B-Yamagata		
	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	
2020									
February	206	1	130		100		2		439
March	69		49		75		1		194
April	1		0		1		0		2
May	0		0		0		0		0
June	0		0		0		0		0
July	0		0		0		0		0
Totals	276	1	179	0	176	0	3	0	635

¹ Phenotype: NI (normal inhibition), RI (reduced inhibition), HRI (highly reduced inhibition) as defined in WHO Wkly. Epidemiol. Rec. (2012) 87, 369-374

* As of 2020-09-09

Virus name	Subtype/ Lineage	Collection date	Fold difference		NA substitution
			Oseltamivir	Zanamivir	
A/Lund/4/2020 [§]	A(H1N1)pdm09	2020-02-17	418 (HRI)	0.64 (NI)	H275Y
Outliers	none				

[§] Clinical sample not available but NIC Sweden reported H275X

WORLDWIDE INFLUENZA CENTRE
WHO CC for Reference & Research on Influenza
THE FRANCIS CRICK INSTITUTE
LONDON NW1 1AT
United Kingdom

**Report prepared for the WHO annual consultation on the
composition of influenza vaccines for the Southern
Hemisphere 2021**

16th – 24th September 2020

Annex

**Phylogenetic analyses of HA and NA genes of all
types A and B/Victoria-lineage seasonal influenza
viruses, with collection dates after 2020-01-31,
characterised at the Worldwide influenza Centre as
of 2020-09-11.**

Figure A-1. Influenza A(H1N1)pdm09 HA genes

Vaccine viruses

Reference viruses

Collection date

Feb 2020

Mar 2020

Apr 2020

May 2020



6B.1A5A

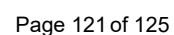
6B.1A

6B.1

6B.1A5B

6B.1A7

Jun 2020



Reference viruses

Mar 2020



Figure A-4 . Influenza A(H3N2) NA genes

Vaccine viruses

Reference viruses

Collection date

Feb 2020

Mar 2020

D151*= +CHO if *=N

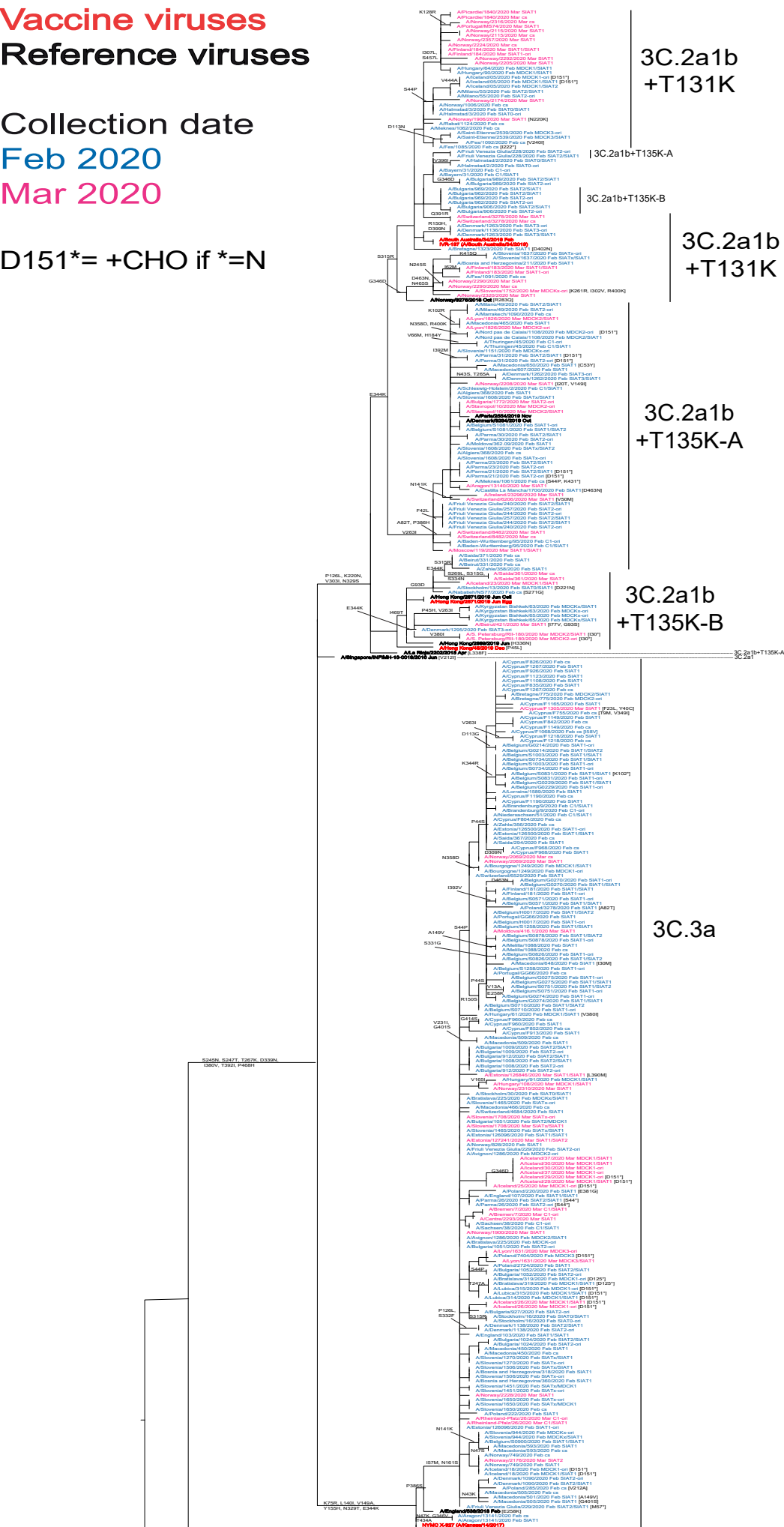


Figure A-5. Influenza B (Victoria-lineage) HA genes

Vaccine viruses

Reference viruses

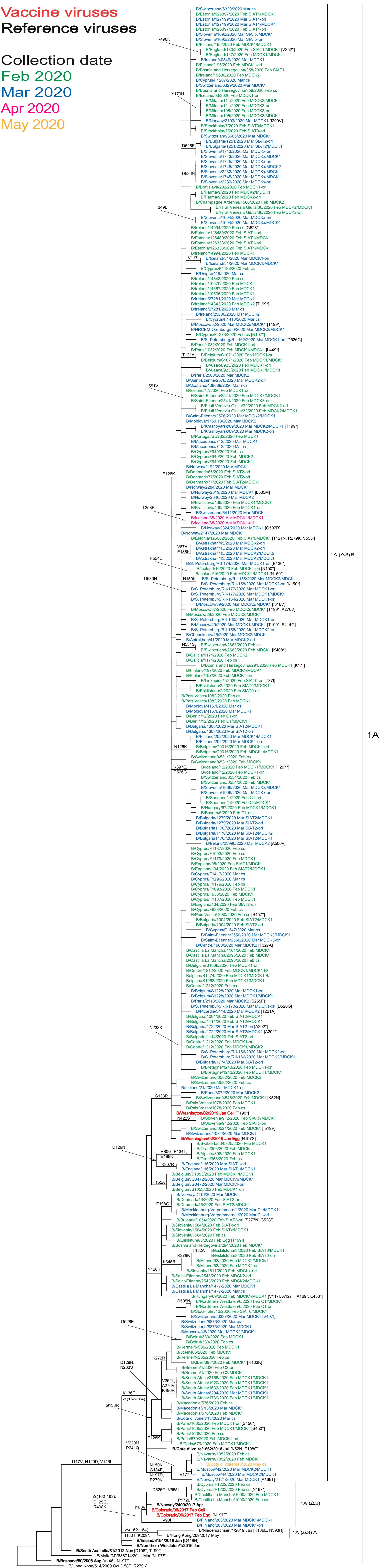
Collection date

Feb 2020

Mar 2020

Apr 2020

May 2020



Reference viruses

Apr 2020

